Mexicali Urban Liveability Index

Technical manual: a cookbook of indicators, domains and index

Versión en español

PDF: `mexicali_uli_manual_en.pdf`

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About this manual

Mexicali Urban Liveability Index — Technical manual: a cookbook of indicators, domains and index

Project	Mexicali Urban Liveability Index (*Índice de entornos vivibles para Mexicali*), Mexicali and the Condesa district, Baja California, Mexico
Project team	Carl Higgs (RMIT University), Eugen Resendiz, Juan Ernesto Díaz Noguez and the team from Tecnológico de Monterrey
Funding	Tecnológico de Monterrey, 2026
Software	Global Healthy and Sustainable City Indicators (GHSCI) 4.15.2 (environment 4.15.0), `mexicali` branch (`f039096`)
Analysis run	2026-09-26; composite goalposts resolved 2026-09-26 from 154,239 sample points
Manual built	2026-11-11 from the ULI repository at `c453d78`
Status	Draft for review by the project team. Spanish text drafted for review by Eugen Resendiz and Juan Ernesto Díaz Noguez; where the indicator register gives a rationale only in English, the Spanish edition marks its translation as preliminary.
Licence	This manual (text, maps and tables): CC BY 4.0. The code that builds it, and the rest of the repository: MIT. Input datasets keep their own licences (Appendix D), and maps and values derived from them remain subject to those terms where they impose conditions (for example, UTCI was modelled with FABDEM terrain, CC BY-NC-SA 4.0).

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What this manual is for

The Mexicali Urban Liveability Index (ULI) summarises how well the everyday environment of each location in Mexicali and the Condesa district supports a healthy, liveable life. It combines 28 indicators across six domains of the Adapted Urban Liveability Framework into one composite measure, for sample points along the pedestrian network, and summarises them for the 100 m grid, census areas (AGEB), blocks (manzanas), and Condesa's lots and subdivisions.

This manual is the **cookbook** for that work. It is written for a team re-implementing the methods, in particular in Reimagina Urbano, where indicators would be recalculated on the fly as environmental conditions change. It also serves as the project's technical record. For each indicator, sub-indicator, domain score and the index itself it gives:

- **identity:** names, variables at each scale, domains and weights;
- **rationale and interpretation:** why it matters, according to the project register, and how to read it;
- **data sources:** the configured records of every dataset it reads, with cross-references to upstream entries;
- **method:** the theory (formulae) and the practice (configuration, GHSCI functions, recalculation);
- **results:** a concise summary of the distribution for the study region (Mexicali, Condesa included) and for Condesa;
- **map:** manzanas across the study region, with Condesa's lots, a legend, scale bar and north arrow.

How to read an entry

Recipes (R1–R14, section 3) describe each shared method once; entries cite them. Code references link to the exact line on GitHub at the commit given above. Configuration references point to the copy of the region configuration published with this manual (`data/MX_Mexicali_2025_ULI.yml`), whose comments record why each choice was made.

Scores are points above or below the study region average (0): each indicator is re-scaled so that the study region's mean, over all sample points, scores 0, and 30 points span its observed range on either side (R11). Domain and index scores combine those scores, less a penalty for imbalance (R12).

Nothing in this manual is estimated by hand. Labels, descriptions and readings come from the region configuration. Goalposts and weights come from the parameters the analysis wrote. Classes and units come from the dashboard export, values from the full-precision results GeoPackage, and rationale from the project's indicator register. Where none of these records something, the manual says *Pending*.

1. Study design

Study region

The study region is the union of the Mexicali urban area (the locality `NOMGEO = 'Mexicali'` of INEGI's 2025 geostatistical framework) and the Condesa development, dissolved to a single feature. Only 82% of Condesa falls within the administrative urban boundary, which alone would have clipped 2,419 of its 14,989 lots (`MX_Mexicali_2025_ULI.yml:43`). The study region covers 211.2 km², of which Condesa is 3.12 km².

Areas are summarised only where they lie within the study region (a point on the area's surface inside it). The network is extracted for the study region itself rather than its 1,600 m buffer, so that streets and destinations across the border with the United States do not enter accessibility calculations (`MX_Mexicali_2025_ULI.yml:55`; GHSCI issue #610).

Coordinate reference system

EPSG:6366, Mexico ITRF2008 / UTM zone 11N, in metres (`MX_Mexicali_2025_ULI.yml:42`).

Scales

Scale	Units	Source	How values reach it
Sample points	154,239	Every 30 m along the pedestrian network, plus one per Condesa lot (R1)	Calculated here
100 m grid	21,076 cells	GHS-POP 2025, reprojected to EPSG:6366 by sum resampling	Mean of the cell's sample points
AGEB	436	INEGI 2020 census areas	From the grid, weighted by grid population
Manzanas (blocks)	14,252	INEGI 2020 census blocks with geometries updated by CFC	Mean of sample points within 30 m
Condesa lots	14,988	CFC lots of 400 m² or less	Mean of sample points within 15 m
Condesa subdivisions	40	CFC fraccionamientos	From the lots, 3.2 residents assumed per lot
Region: study region	1	Study region boundary	From the grid, weighted by population
Region: Condesa	1	Condesa boundary	From the lots

Manzanas are summarised from sample points rather than from the grid because a manzana (median 0.71 ha) is smaller than a 100 m cell; aggregating from the grid would just copy the containing cell (`MX_Mexicali_2025_ULI.yml:121`). Lots are summarised within 15 m so that a lot set deep inside a gated subdivision is not credited with access through a gate it cannot use (`MX_Mexicali_2025_ULI.yml:81`).

Population

- **100 m grid:** GHS-POP R2023A for 2025 (Schiavina et al. 2023), Mollweide, reprojected with sum resampling to conserve counts (`MX_Mexicali_2025_ULI.yml:61`). The grid is deliberately not redirected to census counts, so that the 100 m grid exists as an output.
- **Manzanas and AGEBs:** INEGI Censo 2020 (`POBTOT`). 3,497 manzanas have no recorded population and are excluded from population-weighted summaries.
- **Condesa:** the development is too new for the census, so each lot is assumed to house 3.2 residents (Mexicali's 2020 average household size, INEGI). Sample points are placed in unpopulated areas too (`sampling.sample_unpopulated_areas`), so that Condesa is measured at all.

Network

The routable pedestrian network is retrieved with OSMnx from OpenStreetMap, at the replication timestamp of the Geofabrik Mexico extract of 15 August 2026, which holds the Condesa streets contributed that day (`MX_Mexicali_2025_ULI.yml:396`). The definition retains private and gated access, which the GHSCI default excludes, because Condesa's internal streets and those of Mexicali's gated communities are precisely the areas of interest (`MX_Mexicali_2025_ULI.yml:414`). Intersections are OSMnx consolidated intersections with a 12 m tolerance.

Distances and bands

Walking access is judged at 300 m, the five-minute walk Mexicali's collector-road walkability policy targets, rather than the 500 m used by GHSCI's globally comparable indicators, which are also produced unchanged. Distances are measured in bands of 300, 500, 1,000 and 1,500 m and censored at 1,500 m: a sample point with nothing within 1,500 m has no distance, and is counted as having no access (`MX_Mexicali_2025_ULI.yml:461`).

Two regions, one analysis

Mexicali and Condesa are analysed in one run, sharing the network, population grid and destinations, so that differences between them are real rather than artefacts of separate analyses. The study region's reference, the mean over all sample points against which every score is measured, includes Condesa.

2. Reproducing the analysis

This section is the runbook used for Mexicali and Condesa, from prepared inputs to the published dashboard and this manual. Commands run either in the GHSCI Earth Engine container (`ghsci-ee`, image `globalhealthyliveablecities/global-indicators:v4.15.0.ee`, working directory `/home/ghsci/process`) or on the host, as marked.

Overview

```
inputs (INEGI, DENUE, IMIP, OSM, GHS-POP, CFC, work packages)
  │  preparation scripts (CFC, reference areas)
  ▼
region configuration  MX_Mexicali_2025_ULI.yml
  │  analysis (_00 … _12)  ──────────────►  PostGIS database mx_mexicali_2025_uli
  │     _07 sample points, _11 walking access, euclidean catchments,
  │     cycling LTS, Earth Engine (GUHVI, LPUGS), _12 aggregation →
  │     linked indicators → walkability variants → composite indices
  ▼
generate  ──────►  GeoPackage, CSVs, data dictionary, metadata, reports
export_dashboard ─►  manifest / indicators / stats / distributions + layers
  │  build_tiles.sh (tippecanoe) → deploy.sh
  ▼
dashboard (static site)    build_results_geopackage.py → results GeoPackage
technical_report.py → PDFs   build/manual/manual.py → this manual
```

1. Prepare inputs (host or container)

Step	Script	Produces
Study region and region layers	`process/data/MX/CFC/build_uli_boundaries.py`	the dissolved study region (`INEGI-02l-NOMGEO-Mexicali-Condesa-Fraccionamientos.geojson`) and `mexicali_uli_boundaries.gpkg` (`region_mexicali`, `region_condesa`)
Manzanas	`process/data/MX/CFC/build_manzanas.py`	`mexicali_manzanas.gpkg`: INEGI 2020 blocks with CFC geometries, synthetic `S` keys for 16 unkeyed Condesa blocks
Condesa lot sample points	`process/data/MX/CFC/build_condesa_sample_points.py`	`condesa_lot_sample_points.gpkg`: one point per lot, for snapping to the network
Reference areas	`Mexicali Liveability/build/build_reference_areas.py`	`geography/mexicali_reference_areas.gpkg`: the six reference layers the work packages and linkage use

DENUE (May 2026), the IMIP facilities layer, the GHS-POP 2025 tile, the Geofabrik extract, the UTCI GeoPackage (work package 2) and `final_indicators.gpkg` (work packages) are used as delivered; their paths are in the configuration (Appendix D).

2. Run the analysis (container)

The region configuration lives beside its data, at `process/data/MX/MX_Mexicali_2025_ULI.yml`. In the container, pass its path rather than the codename, since a loose configuration file in `data/MX` is not discovered by codename:

```
python analysis.py data/MX/MX_Mexicali_2025_ULI.yml
```

or, in Python:

```
import ghsci
r = ghsci.Region('data/MX/MX_Mexicali_2025_ULI.yml')
r.analysis()
```

The pipeline runs `_00` to `_12` in sequence (`process/analysis.py:84` `analysis()`). For this region, it adds:

- **Sample points** in unpopulated areas and one per Condesa lot (R1).
- **Walking access** to every configured destination at 300, 500, 1,000 and 1,500 m, the diversity sets and activity centres (R2–R4).
- **Straight-line catchments** to police and fire stations (R5).
- **Cycling level of traffic stress** (R6).
- **Earth Engine:** GUHVI and large public green space (R8), which need the `.ee` container and an authenticated Earth Engine account.
- **In `_12`, after aggregation:**
- linked indicators: UTCI and the work-package indicators (R7);
- walkability variants (R9);
- the composite indices `uli` and `uli_plain` (R10–R13).

A full run takes several hours. Progress is logged to `process/data/_study_region_outputs/MX_Mexicali_2025_ULI/__Mexicali__MX_Mexicali_2025_ULI_processing_log.txt`.

The composite index can be re-scored without re-running the pipeline, for example against frozen goalposts (R14):

```
r.composite_index(goalposts='…/MX_Mexicali_2025_ULI_composite_uli_parameters.yml')
```

3. Generate outputs (container)

```
r.generate()
```

writes the GeoPackage, per-scale CSVs, data dictionary, ISO 19115 metadata, maps and the standard GHSCI reports to the region's output folder.

4. Export the dashboard (container, then host)

```
# container: layers, manifest, stats, vocabulary (combined and general dashboards)
/env/bin/python subprocesses/_export_dashboard.py "data/MX/MX_Mexicali_2025_ULI.yml"
/env/bin/python subprocesses/_export_dashboard.py "data/MX/MX_Mexicali_2025_ULI.yml" --type=general
# host, in Mexicali Liveability/dashboard: PMTiles archives and deployment
docker build -t tippecanoe:local build/
bash build/build_tiles.sh mexicali mexicali_general
bash build/deploy.sh mexicali mexicali_general
```

Large intermediates stay outside OneDrive (`/tmp/ghsci-dashboard`). Label, theme, class or text changes need only `--no-layers`. All on-screen wording is editable in `dashboard/data/mexicali/text.json`.

5. Results GeoPackage and reports

```
# container: full-precision values per reference scale, with a bilingual dictionary
python "data/MX/Mexicali Liveability/build/build_results_geopackage.py" /export/mexicali
# host: the dashboard's technical report (headless Edge)
python dashboard/build/technical_report.py
# host: this manual
python build/manual/manual.py
```

6. Back up and restore the database

```
r.dump(path='E:/backups')               # custom-format dump with a .dump.yml sidecar
ghsci.restore('…/mx_mexicali_2025_uli_YYYYMMDD.dump')
```

The 26 September 2026 analysis database is backed up as `mx_mexicali_2025_uli_20260926.dump` (231 MB; restore steps in the `.dump.txt` beside it).

Pitfalls met in this project

- **Network date.** Until 26 September 2026 the Overpass query was pinned to the start of the extract's publication date, which missed Condesa streets mapped that morning (26 streets; about 1,200 lots without estimates). `_03` now uses the extract's replication timestamp.
- **Region path.** In the container, `ghsci.Region('MX_Mexicali_2025_ULI')` does not resolve the loose file in `data/MX`; pass its path.
- **Parameters file.** If the output folder is not writable by the container user, the composite parameters are still recorded as the comment of the `sample_points_composite` table.
- **OneDrive.** Keep tile scratch and database dumps outside the synced folder.
- **Identifier length.** PostgreSQL truncates identifiers beyond 63 characters, so derived names (for example of diversity sets) are kept short: `community_culture_rec`.

3. Method recipes

Each recipe states a shared method once, in theory (what is computed) and in practice (where GHSCI computes it). It ends with what a re-implementation needs to hold to recalculate it. Indicator entries cite these recipes by number.

R1. Sample points and the pedestrian network

Theory. Liveability is measured at locations people live and walk from: points every 30 m along the pedestrian network. Each point lies on an edge between two nodes `n1` and `n2`, at along-edge distances `d1` and `d2` from them. A network distance from a point is the lesser of the two routes through its end nodes.

Practice.

- The network is built with OSMnx from the configured pedestrian query, which retains private and gated access in Mexicali, with `retain_all` so that disconnected gated communities are kept. Dangles shorter than half the sampling interval are pruned (`process/subprocesses/_03_create_network_resources.py:776` `prune_short_dangles()`).
- Points are generated every `point_sampling_interval` = 30 m along each edge, from its start, excluding trunk roads and ways with `foot = no` (`process/subprocesses/_07_locate_origins_destinations.py:16` `nearest_node_locations()`).
- Duplicate points at junctions are dropped, points inside open space removed, and the set restricted to the urban study region.
- Because `sampling.sample_unpopulated_areas` is set, points are kept where no population is recorded (Condesa). One further point per Condesa lot is snapped to its nearest edge within 100 m (`MX_Mexicali_2025_ULI.yml:108`). Lots further than 100 m from a mapped street lie in subdivisions whose streets are not in OpenStreetMap, and are left without a value rather than credited with access they do not have.

The analysis has 154,239 sample points.

Recalculation. Hold the points with their `n1`, `n2`, `d1`, `d2` and the network graph (edges with lengths). A network edit changes the graph, and the points on new edges.

R2. Network distance to the nearest destination, censored

Theory. For a destination set `D`, attached to the network at the end nodes of its nearest edge with offsets, the node distance is `D(n) = min over destinations (network distance(n, node) + offset)`. The point distance is

```
d(p) = min( D(n1) + d1 ,  D(n2) + d2 )
```

Distances are searched in bands (300, 500, 1,000, 1,500 m) and censored at the largest: a node with no destination within 1,500 m has no distance, so neither does the point. A point's value can exceed 1,500 m by up to its offset along the edge.

Practice.

- Routing is undirected on edge length: pgRouting `pgr_drivingDistance` or, in memory, one Dijkstra from a super-source per destination set, limited to the largest band (`process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`, `process/subprocesses/_accessibility_spec.py:1116` `_nearest_distances_inmemory()`). Only origins not yet within reach of every destination are re-routed at each larger band.
- Point distances are taken from the node distances (`process/subprocesses/setup_sp.py:888` `create_full_nodes()`, `process/subprocesses/setup_sp.py:951` `process_distant_nodes()`).
- Each destination is a configured specification (`MX_Mexicali_2025_ULI.yml:464`). Its distance is `sp_walk_nearest_node_<name>` at points and `avg_walk_dist_<name>` in areas.

Recalculation. Adding or removing a destination changes `d` only where it becomes, or stops being, the nearest. Run a Dijkstra from the changed destinations, limited to 1,500 m, and take the minimum with the stored distance; a removal needs the next-nearest, so store the second-nearest or re-route locally.

R3. Banded access, and distance from a disamenity

Theory. Binary access within distance `D`: `a = 1` if `d ≤ D`, else 0, where a missing distance (nothing within 1,500 m) is 0. For a disamenity (`direction: avoid`), the indicator is distance *from* it: `b = 1 − a`, so nothing within reach scores 1. For an area, the percentage of its sample points with access, `pct_access_walk_<name>_<D>m` (or `pct_beyond_…`).

Practice. `process/subprocesses/_accessibility_spec.py:1272` `sample_point_access()`, `process/subprocesses/setup_sp.py:1057` `binary_access_score()`. A specification may set its own distances, as the petrol station does (250 m).

R4. Diversity of what is reachable

Theory. For a set of `k` configured sub-types, `c_g` is the number of distinct destinations of sub-type `g` reachable within distance `D`. Then

```
p_g = c_g / Σ c          H = − Σ_{p_g > 0} p_g · ln p_g  /  ln k          (0 … 1)
R   = #{g : c_g > 0} / k                                                  (richness)
```

`k` is the number of configured sub-types, not the number present, so the score is comparable between places. Reaching nothing, or one sub-type only, scores 0. Counts at a point are interpolated from its end nodes, `c(p) = Σ_i (1 − d_i/(d1 + d2)) · c(n_i)`, so they may be fractional.

Practice. `process/subprocesses/_accessibility_spec.py:299` `diversity_sets()`, `process/subprocesses/_accessibility_spec.py:929` `_banded_counts()`, `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`, `process/subprocesses/_accessibility_spec.py:493` `richness()`. Sets are configured under `MX_Mexicali_2025_ULI.yml:595`. The ULI uses `sp_walk_diversity_<set>_500m`.

Recalculation. A destination added within `D` of a point adds 1 to its sub-type's count; recompute `H` at the points affected.

R5. Straight-line catchments

Theory. `d(p)` is the straight-line distance to the nearest station, searched to `max_distance`. Beyond it there is no value.

Practice.

- Each specification reads its own file and filter, clipped to the study region buffered by `max_distance` and outside the exclusion region (`process/subprocesses/_euclidean_accessibility.py:66` `resolve_spec()`, `process/subprocesses/_euclidean_accessibility.py:177` `import_destinations()`).
- The nearest station is found by a KNN search (`process/subprocesses/_euclidean_accessibility.py:247` `nearest_distances()`).
- Outputs are `sp_euclid_dist_<name>` and `avg_euclid_dist_<name>` (`MX_Mexicali_2025_ULI.yml:679`).

R6. Level of traffic stress and the low-stress cycling network

Theory. Each link gets a level of traffic stress (LTS) from 1 (suitable for all) to 4, from its cycling facility, speed and assumed traffic. The low-stress network is the links with LTS ≤ 2 where cycling (or walking the bike) is permitted. Distance to the nearest destination along it follows R2, with bands of 500, 1,000, 2,000 and 5,000 m. The rules are in the entry Cycling network by level of traffic stress.

Practice. `process/subprocesses/_cycling_lts_network.py:546` `assign_lts()`, `process/subprocesses/_cycling_lts_network.py:607` `add_impedance()`, `process/subprocesses/_cycling_accessibility.py:469` `cycling_poi_distance()`. Output: `sp_cycle_safe_nearest_node_<dest>`.

R7. Linking externally prepared indicators

Theory. A value prepared outside GHSCI for a set of areas is attached to the matching reference areas and then replicated to the sample points the areas contain, so that it can be scored like any other indicator.

Practice.

- **Match by identifier** (`match: id`, UTCI) or **by geometry** (`match: geometry`, the work-package file, which has no identifiers) (`process/subprocesses/_linkage_indicators.py:403` `match_by_geometry()`). By geometry, an exact normalised match is tried first; failing that, the reference area holding the largest share, if at least half.
- Parts matched to one area are combined as an area-weighted mean (`process/subprocesses/_linkage_indicators.py:479` `combine_matched()`).
- `null_as` fills missing values, then `mask` blanks unreliable ones (`process/subprocesses/_linkage_indicators.py:512` `apply_rules()`).
- **To sample points:** from the first configured layer holding a value, by containment or within a distance (`process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`), giving `sp_<prefix><column>`. These are replicated values: every point in an area has that area's value.

R8. Earth Engine: heat vulnerability and large green space

Theory.

- **GUHVI** (Turner et al. 2025) combines exposure, sensitivity and adaptive capability, each a mean of inputs min–max normalised to 0–100 within the region. Details are in the entry GUHVI and its inputs.
- **Large public urban green space** is public open space of at least 1 ha with annual mean NDVI ≥ 0.2.

Practice. `process/subprocesses/_earth_engine_indicators.py:846` `guhvi_analysis()`, `process/subprocesses/_earth_engine_indicators.py:1105` `normalise_band()`. Sample points take the value of the GUHVI cell containing them. Access to large green space follows R2 and R3 at 500 m. These need the Earth Engine container and `gee: true`.

R9. Walkability, and its attenuation by thermal comfort

Theory.

```
DL   = a_fresh_food_market(300) + a_convenience(300) + a_pt_any(300)          (0 … 3)
W    = z(DL) + z(population density) + z(intersection density)
       z(x) = (x − mean(x)) / sd(x), over all urban sample points (sample SD)
h′   = clip( (UTCI − 30) / (50 − 30), 0, 1 )
W_tm = percentile_rank(W) · (1 − λ · h′),   λ = 0.5
```

The densities are neighbourhood averages within 1,000 m along the network. The percentile rank averages ties and lies in (0, 1]. Attenuation by multiplication means shade cannot make up for having nothing to walk to: a place with nothing within reach scores low however cool it is.

Practice. `process/subprocesses/_11_neighbourhood_analysis.py:115` `compute_nodes_pop_intersect_density()`, `process/subprocesses/_walkability_variants.py:275` `compute_variants()`, `process/subprocesses/_walkability_variants.py:270` `percentile_rank()`, `process/subprocesses/_walkability_variants.py:261` `rescale()` (`MX_Mexicali_2025_ULI.yml:909`). The bounds each heat measure was re-scaled between are recorded in the comment of the `sample_points_walkability` table. The observed 5th and 95th percentiles of daytime UTCI were 43.8 and 46.0 °C.

R10. Transforms: soft thresholds and stepped ladders

Theory. A distance enters the index through a transform, so that being nearer counts, but less so far past a threshold:

```
soft threshold:  s(d) = 1 / (1 + exp( k · (d − t) / t )),   k = 5
                 s(0) ≈ 0.993,  s(t) = 0.5,  s(2t) ≈ 0.007;  missing d → 0
stepped ladder:  s(d) = score of the first step with d ≤ distance;  beyond or missing → beyond score
```

For example, with a 300 m threshold, 150 m scores 0.92, 300 m scores 0.50 and 450 m scores 0.08. GHSCI warns if the censoring distance is below `2t`. That is why distances are searched to 1,500 m (five times 300 m) and police stations to 45 km.

Practice. `process/subprocesses/_composite_index.py:1457` `soft_threshold()`, `process/subprocesses/_composite_index.py:423` `stepped_score()`, `process/subprocesses/_composite_index.py:1504` `prepare()`. Each indicator's `transform` is in `MX_Mexicali_2025_ULI.yml:1019`.

R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI)

Theory. Each prepared indicator `x` (transformed where it has a transform) is re-scaled against goalposts:

```
r = 100 ± 60 · (x − Ref) / (Max − Min)        + where higher is better, − where lower is better
Ref = mean of x over all sample points of the study region
Min, Max = Ref ∓ (sup − inf) / 2              inf, sup = observed minimum and maximum
```

So `Max − Min` is the observed range, and `r` spans 70–130 about the reference at 100. An indicator with configured goalposts (the GUHVI sub-indices: 0 and 100) uses those. An indicator that does not vary is dropped with a warning. Polarity is configured, or inferred from the variable's name.

Practice. `process/subprocesses/_composite_index.py:1602` `resolve_parameters()`, `process/subprocesses/_composite_index.py:1819` `normalise()`, `process/subprocesses/_composite_index.py:234` `infer_polarity()`. The goalposts used are written to `MX_Mexicali_2025_ULI_composite_<index>_parameters.yml`, published with this manual (`data/`) and tabulated in Appendix C. Each indicator's skewness, kurtosis and tied share are recorded there too (`distribution`).

R12. Aggregation: domains, shared indicators and the index

Theory. A domain score is the weighted Mazziotta–Pareto composite of its indicators' scores `r_i`. An indicator listed in `k` domains counts `1/k` of its weight in each (`share`), so that in all it weighs no more than an indicator in one domain:

```
w_i = weight_i · share_{i,d} / Σ_j weight_j · share_{j,d}       (over indicators with a value)
M   = Σ w_i r_i        S = √( Σ w_i (r_i − M)² )        cv = S / M
domain = M − S · cv                        (penalty S·cv: imbalance lowers the score)
index  = the same over the six scored domains, each weight 1
reported score = score − 100               (centre: 0; the penalty is reported as is)
```

The penalty divides by the mean, which is why scores are calculated at 100 and reported less 100. A point is scored only where all of a domain's indicators and all domains have a value (`min_valid`: all). An indicator's **effective weight** is its share of the index's mean level:

```
e_i = Σ_d (W_d / Σ W) · (weight_i · share_{i,d} / Σ_j weight_j · share_{j,d})
```

For Mexicali these range from 1.7% to 6.2%.

Practice. `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`, `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:783` `share_of()`, `process/subprocesses/_composite_index.py:1057` `effective_weights()`.

Variants. `uli_plain` replaces attenuated walkability with walkability alone and activates thermal comfort as an ambient environment indicator, against its own goalposts (`process/subprocesses/_composite_index.py:857` `expand_variants()`).

R13. From sample points to areas

Theory. Scores are computed at sample points and then averaged. An area's index score is the mean of its points' index scores, not a composite of the area's mean domain scores.

Area	Value
100 m grid	unweighted mean of the sample points in the cell
AGEB, region (study region)	mean of grid cells weighted by `pop_est × area share`
Manzanas	unweighted mean of the sample points within 30 m
Condesa lots	unweighted mean of the sample points within 15 m
Subdivisions, region (Condesa)	mean of lots (each 3.2 residents) by area share

Access columns (`pct_*`) are multiplied by 100 on aggregation.

Practice. `process/subprocesses/_12_aggregation.py:1036` `_propagate_sample_point_columns()`, `process/subprocesses/_12_aggregation.py:626` `custom_aggregation()`, `process/subprocesses/_12_aggregation.py:940` `_custom_area_query()`, `process/subprocesses/_composite_index.py:2062` `propagate()`. The aggregation plans are recorded in the database's `aggregation_plans` table.

R14. Frozen goalposts and recalculating a scenario

The goalposts make scores comparable: a scenario scored against the baseline's goalposts is on the same scale, and its scores may move beyond 70–130 if it improves on the best observed. To recalculate a sample point after a change, for example a new park, a shaded street or a new school:

```
params = yaml.safe_load(open('MX_Mexicali_2025_ULI_composite_uli_parameters.yml'))

def indicator_score(i, raw):                    # raw: the point's measured value
    spec = params['indicators'][i]
    x = transform(spec['transform'], raw)        # R10: soft threshold, steps, or as is
    sign = 1 if spec['polarity'] == 'positive' else -1
    return 100 + sign * 60 * (x - spec['reference']) / (spec['max'] - spec['min'])

def mpi(scores, weights):                        # R12
    w = {i: weights[i] / sum(weights[j] for j in scores) for i in scores}
    M = sum(w[i] * scores[i] for i in scores)
    S = (sum(w[i] * (scores[i] - M) ** 2 for i in scores)) ** 0.5
    return M - S * S / M

r = {i: indicator_score(i, raw[i]) for i in params['indicators']}
domains = {d: mpi({i: r[i] for i in r if d in params['indicators'][i]['domains']},
                  {i: params['indicators'][i]['weight'] * params['indicators'][i]['domains'][d]
                   for i in r if d in params['indicators'][i]['domains']})
           for d in params['domains']}
uli = mpi(domains, params['domains']) - 100     # report as points from the average
```

Then average to areas as in R13. A useful check of a re-implementation: re-scoring the baseline's measured values against these goalposts should reproduce the published `sp_index_uli` (this manual has not run that check, which needs the sample point tables in the database). GHSCI does this with `r.composite_index(goalposts=…)` (`process/subprocesses/_composite_index.py:1785` `load_goalposts()`).

What changes when:

- **A destination** (shop, school, stop): recompute its distances (R2), diversity counts (R4) and walkability's daily living score (R9) near it.
- **Shade, vegetation or materials:** the linked UTCI and NDVI of the areas affected (R7), then walkability's attenuation (R9).
- **Streets or crossings:** the network, hence every distance near the change (R2), intersection density (R9) and, for cycling, LTS (R6).
- **New residents:** population density (R9) and population weights in areas (R13).

Mexicali Urban Liveability Index

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> Average of the Mexicali Urban Liveability Index's domain scores, each summarising its normalised sub-indicators, less a penalty for variation across domains that rewards balanced performance. Expressed as points above or below the study region average (0).

The Urban Liveability Index draws on the Adjusted Mazziotta-Pareto Index (AMPI): each sub-indicator is re-scaled so that 0 is the study region average (its mean over all sample points), and scores are points above or below it. Each domain is the weighted mean of its sub-indicators less a penalty for imbalance between them, and the index is likewise the mean of the domains less a penalty for imbalance between domains. A sub-indicator bearing on several domains counts an equal share of its weight in each (a third in each of three), so that in all it weighs no more than one in a single domain. Scored at sample points along the pedestrian network, and averaged to grid cells and, weighted by population, to larger areas.

The published index: walkability enters attenuated by daytime thermal comfort (`sp_walk_idx_300_tm`), and thermal comfort is not scored separately. See also the variant Mexicali Urban Liveability Index: Walkability (300 m), without thermal comfort attenuation.

Domains

Domains	n	Region score: Study region (Mexicali incl. Condesa)	Condesa
Ambient environment	8	−2.6	+2.3
Built environment	8	+2.0	−17.8
Economic development	6	+1.9	−14.8
Housing	–	–	–
Mobility and transport	10	+2.4	−15.4
Safety and protection	5	+2.2	−20.7
Social infrastructure	13	+2.4	−12.1
Sociodemographic characteristics	–	–	–

Method: theory

```
r_i      = 100 ± 60 · (x_i − Ref_i) / (Max_i − Min_i)                (R11)
domain_d = M_d − S_d · (S_d / M_d),  w_i ∝ weight_i · share_{i,d}      (R12)
index    = M − S · (S / M)  over the six scored domains, equal weights
reported = index − 100   (0 = study region average)
```

Recipes: R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Method: practice

Configuration: `MX_Mexicali_2025_ULI.yml:1019`; `MX_Mexicali_2025_ULI_composite_uli_parameters.yml`

Code (GHSCI): `process/subprocesses/_composite_index.py:1602` `resolve_parameters()`, `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`, `process/subprocesses/_composite_index.py:2062` `propagate()`

Recalculation (Reimagina Urbano)

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

R14. Frozen goalposts and recalculating a scenario

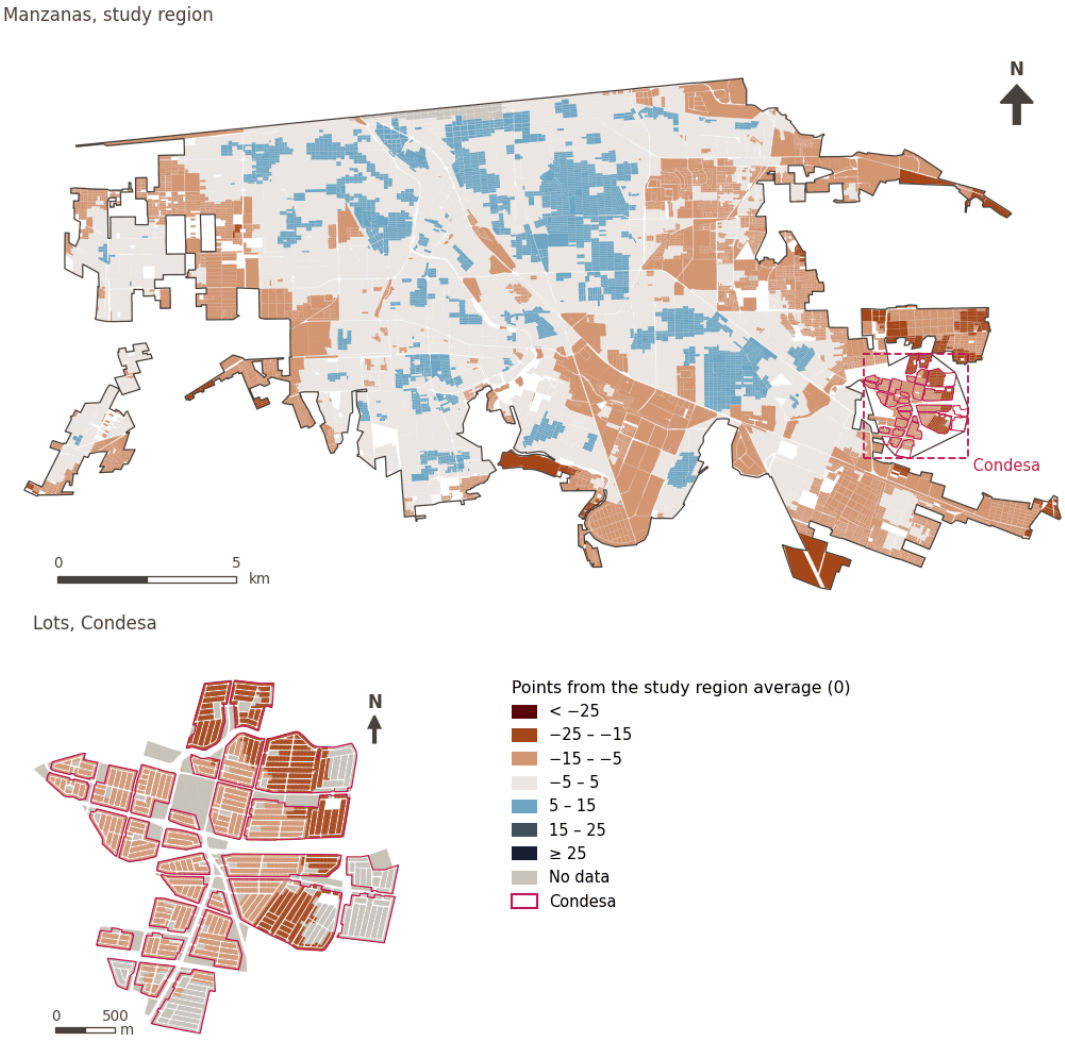
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	+1.0	−13.8
Region score (`uli_mean`)	+1.4	−12.9
Region score (`uli_penalty`)	0.39	0.96
Areas with a value	14,145 of 14,252 manzanas	12,186 of 14,988 lots
Median (P25–P75), weighted	+0.9 (−3.1 – +4.1)	−12.6 (−17.8 – −11.1)
Range	−20.3 – +14.9	−20.7 – −3.7

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region (Mexicali with Condesa) scores +1.0, and Condesa −13.8: 13.8 points below the study region average. The penalty measures imbalance between domains (`_penalty`, not centred).

Map



Map. Mexicali Urban Liveability Index: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Sources declared in the configuration

- Mazziotta, M. & Pareto, A. (2018) Measuring well-being over time: the Adjusted Mazziotta-Pareto Index versus other non-compensatory indices. Social Indicators Research 136, 967-976.
- Mazziotta, M. & Pareto, A. (2022) Weighted composite indices: the case of the Mazziotta-Pareto Index. Rivista Italiana di Economia Demografia e Statistica 76(4), 17-26.
- Higgs, C. et al. (2019) The Urban Liveability Index: developing a policy-relevant urban liveability composite measure and evaluating associations with transport mode choice. International Journal of Health Geographics 18, 14.
- OECD & JRC (2008) Handbook on Constructing Composite Indicators: Methodology and User Guide. OECD Publishing, Paris.
- Population: GHS-POP R2023A (JRC) and INEGI Censo 2020; Condesa lots at an assumed 3.2 residents each.

Caveats and open questions

- The index is provisional: the indicators in each domain and how it is calculated are still under review.

Mexicali Urban Liveability Index: Walkability (300 m), without thermal comfort attenuation

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> Average of the Mexicali Urban Liveability Index's domain scores, each summarising its normalised sub-indicators, less a penalty for variation across domains that rewards balanced performance. Expressed as points above or below the study region average (0).

The Urban Liveability Index draws on the Adjusted Mazziotta-Pareto Index (AMPI): each sub-indicator is re-scaled so that 0 is the study region average (its mean over all sample points), and scores are points above or below it. Each domain is the weighted mean of its sub-indicators less a penalty for imbalance between them, and the index is likewise the mean of the domains less a penalty for imbalance between domains. A sub-indicator bearing on several domains counts an equal share of its weight in each (a third in each of three), so that in all it weighs no more than one in a single domain. Scored at sample points along the pedestrian network, and averaged to grid cells and, weighted by population, to larger areas.

A variant of the index in which walkability is not attenuated by thermal comfort (`sp_walk_idx_300`) and daytime thermal comfort (`sp_utci_day_mean`) is scored as an ambient environment indicator instead, so that thermal comfort counts exactly once. It is scored against its own goalposts.

Domains

Domains	n	Region score: Study region (Mexicali incl. Condesa)	Condesa
Ambient environment	9	−1.8	+1.0
Built environment	8	+1.5	−16.5
Economic development	6	+1.9	−14.8
Housing	–	–	–
Mobility and transport	10	+2.0	−14.4
Safety and protection	5	+2.2	−20.7
Social infrastructure	13	+2.3	−11.6
Sociodemographic characteristics	–	–	–

Method: theory

```
r_i      = 100 ± 60 · (x_i − Ref_i) / (Max_i − Min_i)                (R11)
domain_d = M_d − S_d · (S_d / M_d),  w_i ∝ weight_i · share_{i,d}       (R12)
index    = M − S · (S / M)  over the six scored domains, equal weights
reported = index − 100   (0 = study region average)
```

Recipes: R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Method: practice

Configuration: `MX_Mexicali_2025_ULI.yml:1019`; `MX_Mexicali_2025_ULI_composite_uli_plain_parameters.yml`

Code (GHSCI): `process/subprocesses/_composite_index.py:1602` `resolve_parameters()`, `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`, `process/subprocesses/_composite_index.py:2062` `propagate()`

Recalculation (Reimagina Urbano)

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

R14. Frozen goalposts and recalculating a scenario

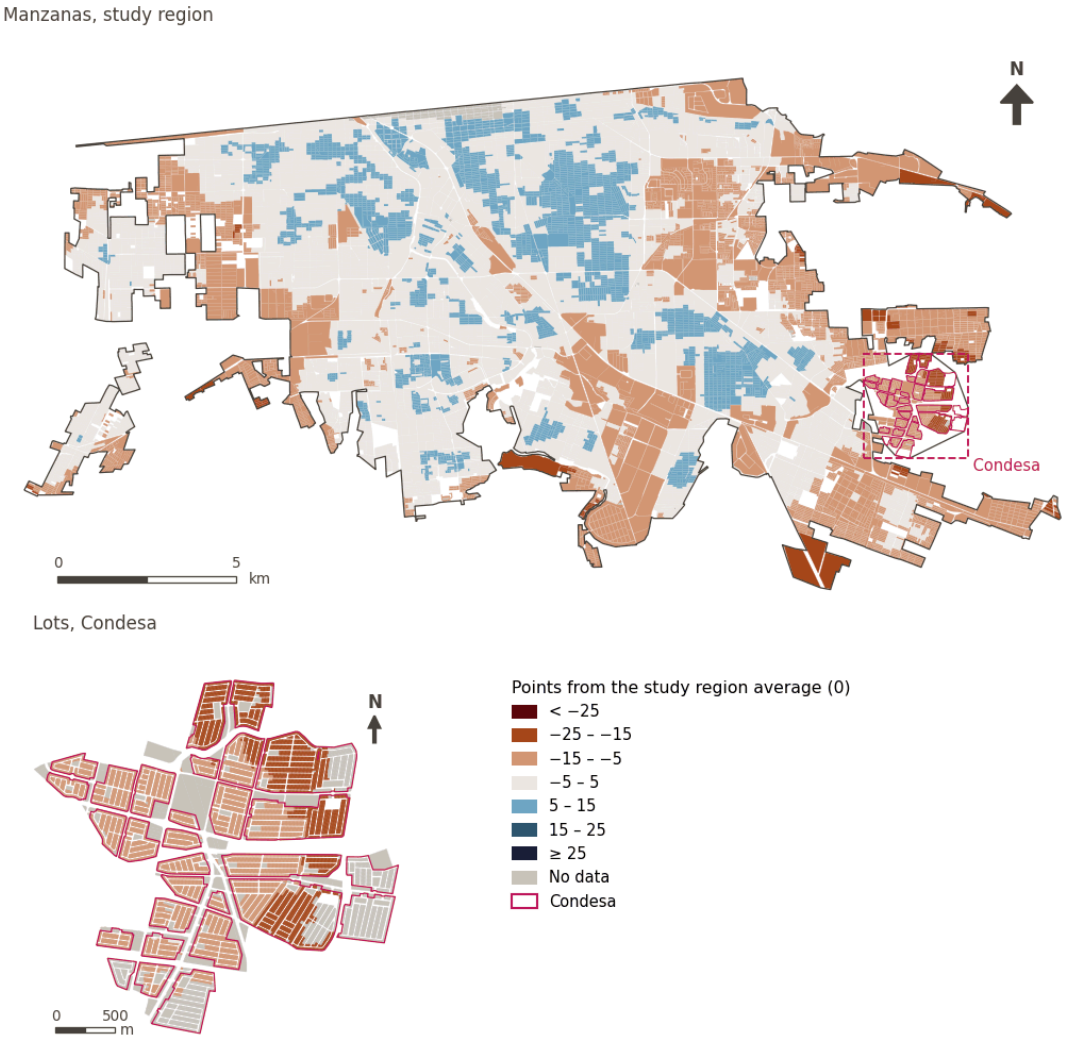
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli_plain`)	+1.0	−13.5
Region score (`uli_plain_mean`)	+1.3	−12.6
Region score (`uli_plain_penalty`)	0.36	0.86
Areas with a value	14,145 of 14,252 manzanas	12,186 of 14,988 lots
Median (P25–P75), weighted	+0.8 (−3.1 – +4.1)	−12.3 (−17.5 – −10.7)
Range	−20.1 – +14.3	−20.5 – −3.3

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region (Mexicali with Condesa) scores +1.0, and Condesa −13.5: 13.5 points below the study region average. The penalty measures imbalance between domains (`_penalty`, not centred).

Map



Map. Mexicali Urban Liveability Index: Walkability (300 m), without thermal comfort attenuation: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Caveats and open questions

- The index is provisional: the indicators in each domain and how it is calculated are still under review.

Ambient environment

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> The quality of environmental conditions to which residents are exposed every day. Examples include air, water, noise and temperature.

Subdomains

- **Climate risks.** The effects of climate change at neighbourhood scale and the capacity to adapt to and mitigate them. Examples include heat islands, flood risk and extreme weather events.
- **Ambient.** Biophysical and environmental conditions of the immediate surroundings, encompassing factors such as noise, water, soil, vegetation and thermal comfort, which directly affect residents' health, comfort and physical well-being.
- **Natural resources.** Availability, access, preservation and sustainable management of natural resources and essential basic services, such as water, land and solid-waste management, which are fundamental to ecological sustainability and the community's continued functioning.

Indicators of the domain

Indicator	Subdomain	Share	Effective weight	Better when
Natural blue space nearby	Ambient	100%	2.8%	higher
Heat adaptive capability (GUHVI)	Climate risks	100%	2.8%	higher
Heat exposure (GUHVI)	Climate risks	100%	2.8%	lower
Heat sensitivity (GUHVI)	Climate risks	100%	2.8%	lower
Daytime thermal comfort *(inactive in `uli`)*	Ambient	100%	–	lower
Parks within 300 m	Ambient	50%	4.0%	higher
Large public green space within 500 m	Ambient	50%	4.0%	higher
Vegetation cover (NDVI)	Ambient	50%	4.0%	higher
Proximity to major roads	Ambient	50%	5.9%	lower

Method: theory

The domain score is the weighted Mazziotta–Pareto composite of its indicators' normalised scores (R12): each indicator weighs `weight × share`, normalised over the indicators with a value; `M` is their weighted mean, `S` their weighted deviation, and the score is `M − S·(S/M)`, less 100. In the `uli_plain` variant walkability is not attenuated and, in the ambient environment, thermal comfort is activated.

Recipes: R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Code (GHSCI): `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

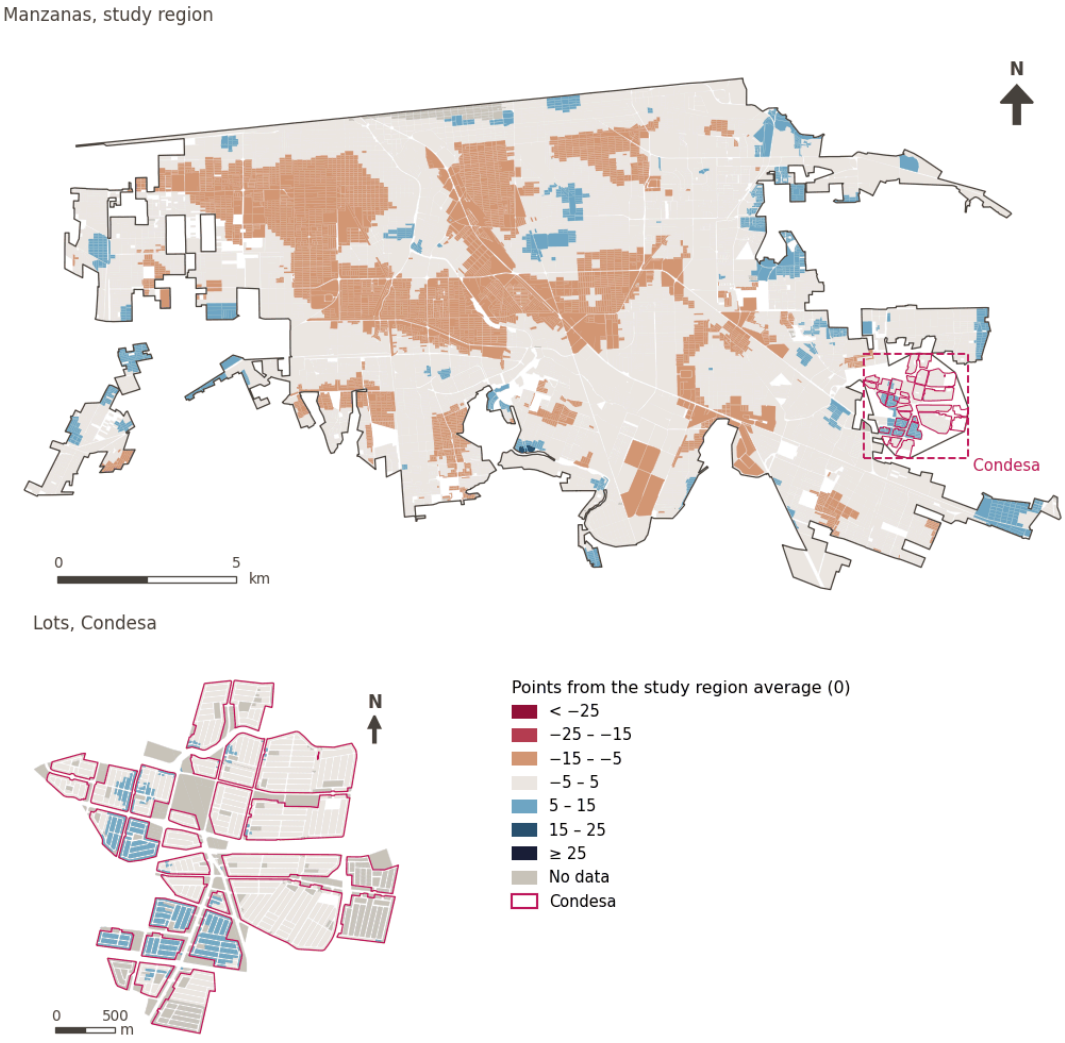
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	−2.6	+2.3
Region score (`uli_plain`)	−1.8	+1.0
Areas with a value	14,148 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	−2.0 (−5.7 – +1.1)	+3.0 (+1.0 – +4.2)
Range	−12.3 – +16.3	−4.5 – +10.2

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region scores −2.6; Condesa lies 2.3 points above the study region average.

Map



Map. Ambient environment: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Built environment

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> The physical environment created by people, whose density, mix of land uses and connectivity shape everyday life. Examples include buildings, streets and public spaces.

Subdomains

- **Urban morphology.** The physical form, structure and design of the urban fabric, which determine how space is organised and how this influences the experience of living in the city.
- **Urban design.** Configuration and quality of the spaces, streets and elements that make up the city, including pedestrian, cycling and road infrastructure, as well as the urban landscape, which influence mobility, accessibility, safety, connectivity and people's everyday experience.
- **Green and blue infrastructure.** Network of green spaces and water features, such as parks, trees, rivers, canals, wetlands and stormwater-management systems, that provide ecosystem services, contribute to climate regulation, support physical activity and recreation, and promote mental health.

Indicators of the domain

Indicator	Subdomain	Share	Effective weight	Better when
Parks within 300 m	Green and blue infrastructure	50%	4.0%	higher
Large public green space within 500 m	Green and blue infrastructure	50%	4.0%	higher
Vegetation cover (NDVI)	Green and blue infrastructure	50%	4.0%	higher
Food within 1.5 km by low-stress cycling	Urban design	33%	6.2%	higher
Block size (perimeter)	Urban morphology	33%	6.2%	lower
Fire station within 3 km		33%	6.2%	higher
Police station within 2 km		33%	6.2%	higher
Walkability (300 m), attenuated by thermal comfort	Urban design	33%	3.7%	higher

Method: theory

The domain score is the weighted Mazziotta–Pareto composite of its indicators' normalised scores (R12): each indicator weighs `weight × share`, normalised over the indicators with a value; `M` is their weighted mean, `S` their weighted deviation, and the score is `M − S·(S/M)`, less 100. In the `uli_plain` variant walkability is not attenuated and, in the ambient environment, thermal comfort is activated.

Recipes: R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Code (GHSCI): `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

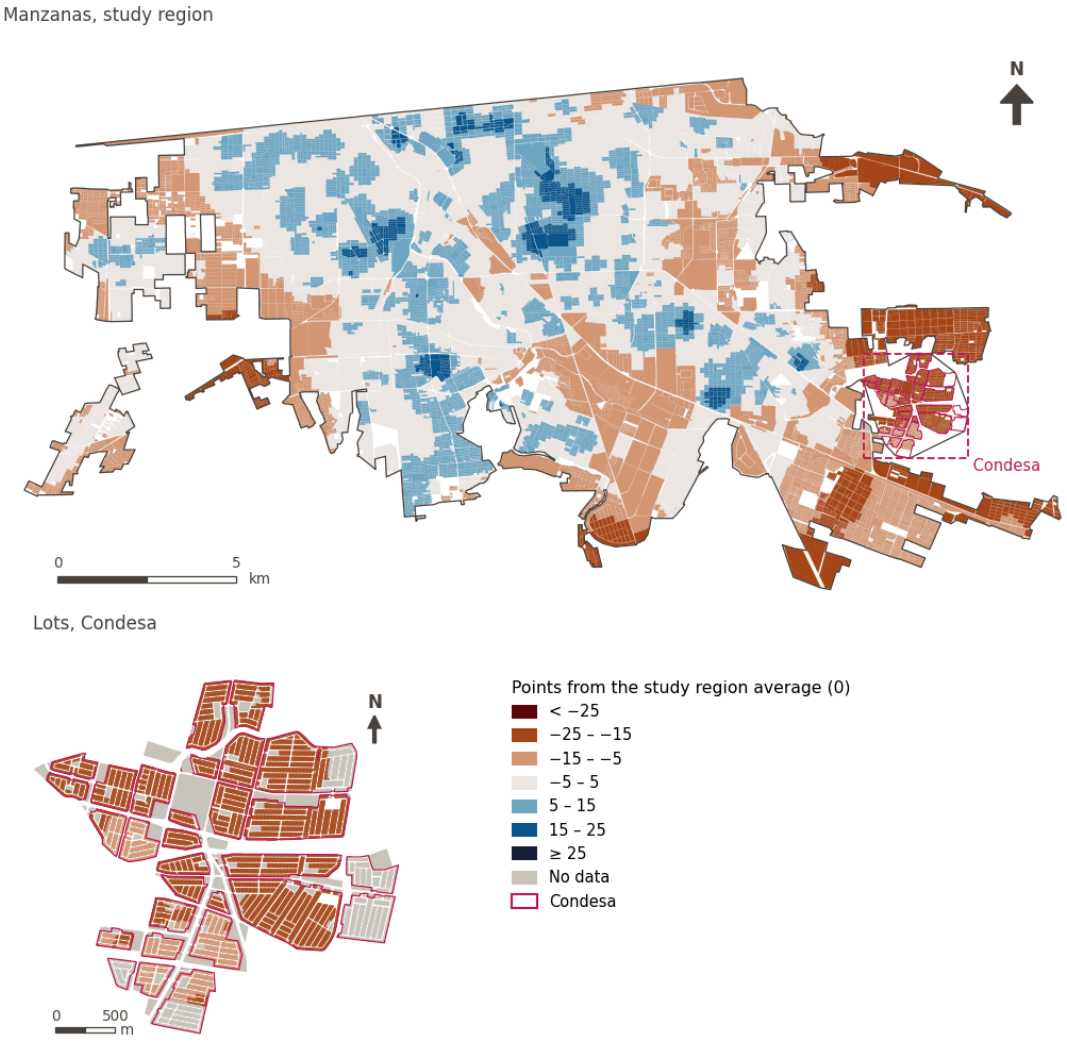
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	+2.0	−17.8
Region score (`uli_plain`)	+1.5	−16.5
Areas with a value	14,216 of 14,252 manzanas	12,862 of 14,988 lots
Median (P25–P75), weighted	+1.8 (−2.9 – +6.3)	−18.9 (−19.4 – −17.7)
Range	−21.2 – +23.5	−20.1 – −7.3

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region scores +2.0; Condesa lies 17.8 points below the study region average.

Map



Map. Built environment: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Economic development

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> Local economic conditions that offer residents opportunities for productive employment, fair income and economic participation.

Subdomains

- **Food environment.** Presence, proximity and diversity of food retail establishments, such as supermarkets and convenience destinations, that determine the ease and equity with which residents access food supplies in their immediate surroundings.
- **Resources.** Availability, access, coverage and continuous supply of essential energy services and inputs, such as electricity and fuels, needed for productive, commercial and household activities.
- **Economy.** Dynamism and structure of local productive activity, including the presence and distribution of retail, industry and employment sources, which generate job opportunities, income and economic support for the urban population.

Indicators of the domain

Indicator	Subdomain	Share	Effective weight	Better when
250 m or more from petrol stations	Resources	33%	3.6%	higher
Convenience stores within 300 m	Food environment	33%	3.6%	higher
Fresh food diversity within 500 m	Food environment	33%	3.6%	higher
Fresh food within 300 m	Food environment	33%	3.6%	higher
Job diversity	Economy	100%	5.0%	higher
Manufacturing within 1 km	Economy	100%	5.0%	higher

Method: theory

The domain score is the weighted Mazziotta–Pareto composite of its indicators' normalised scores (R12): each indicator weighs `weight × share`, normalised over the indicators with a value; `M` is their weighted mean, `S` their weighted deviation, and the score is `M − S·(S/M)`, less 100. In the `uli_plain` variant walkability is not attenuated and, in the ambient environment, thermal comfort is activated.

Recipes: R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Code (GHSCI): `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

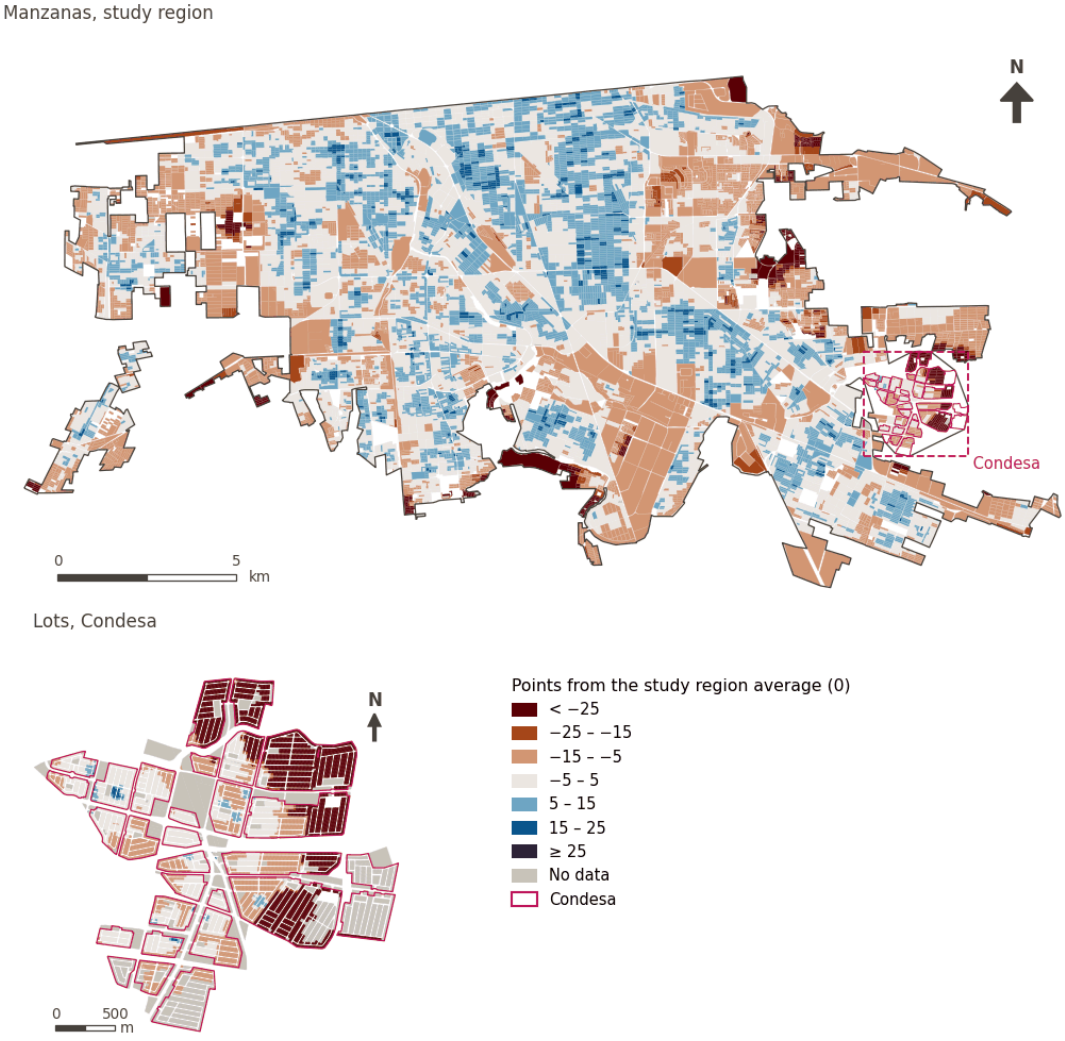
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	+1.9	−14.8
Region score (`uli_plain`)	+1.9	−14.8
Areas with a value	14,213 of 14,252 manzanas	12,589 of 14,988 lots
Median (P25–P75), weighted	+1.7 (−3.3 – +6.8)	−7.6 (−31.3 – −3.4)
Range	−36.9 – +21.0	−35.3 – +18.1

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region scores +1.9; Condesa lies 14.8 points below the study region average.

Map



Map. Economic development: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Housing

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> The availability, affordability and suitability of housing options that meet the diverse needs of neighbourhood residents.

This domain has no indicator yet: it is described but not scored, so the index is the composite of six domains (Appendix G).

Appendix G. Excluded and pending indicators

Mobility and transport

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> Networks and systems that enable efficient travel between home, work and services.

Subdomains

- **Public transport.** Public transport systems that connect residents with work, education and services.
- **Active mobility.** Urban environment conditions that facilitate non-motorised travel, on foot and by bicycle, such as connected, permeable and well-designed street networks.

Indicators of the domain

Indicator	Subdomain	Share	Effective weight	Better when
Food within 1.5 km by low-stress cycling	Active mobility	33%	6.2%	higher
Block size (perimeter)	Active mobility	33%	6.2%	lower
Fire station within 3 km		33%	6.2%	higher
Police station within 2 km		33%	6.2%	higher
Walkability (300 m), attenuated by thermal comfort	Active mobility	33%	3.7%	higher
250 m or more from petrol stations		33%	3.6%	higher
Convenience stores within 300 m	Active mobility	33%	3.6%	higher
Fresh food diversity within 500 m	Active mobility	33%	3.6%	higher
Fresh food within 300 m	Active mobility	33%	3.6%	higher
Public transport within 300 m	Public transport	100%	4.2%	higher

Method: theory

The domain score is the weighted Mazziotta–Pareto composite of its indicators' normalised scores (R12): each indicator weighs `weight × share`, normalised over the indicators with a value; `M` is their weighted mean, `S` their weighted deviation, and the score is `M − S·(S/M)`, less 100. In the `uli_plain` variant walkability is not attenuated and, in the ambient environment, thermal comfort is activated.

Recipes: R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Code (GHSCI): `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

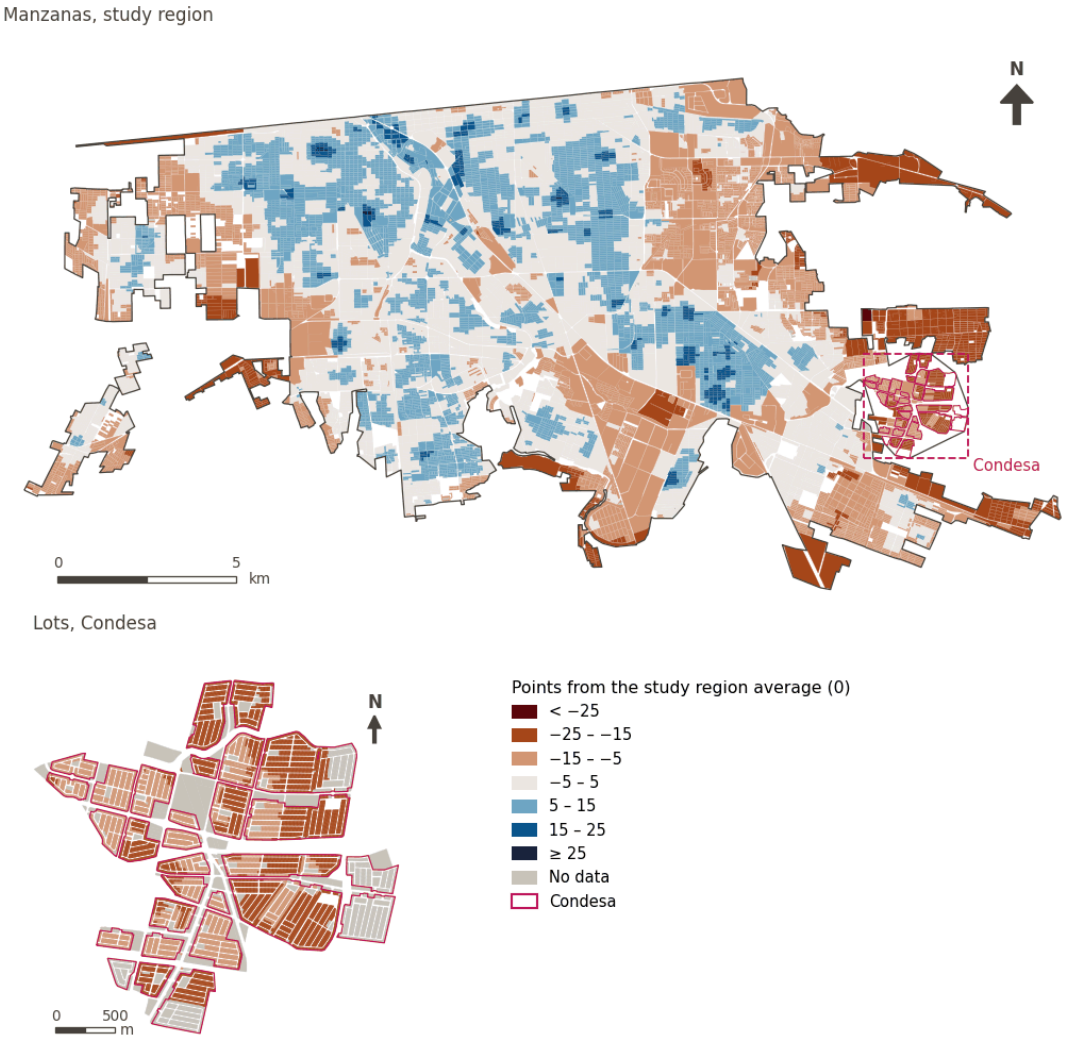
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	+2.4	−15.4
Region score (`uli_plain`)	+2.0	−14.4
Areas with a value	14,216 of 14,252 manzanas	12,862 of 14,988 lots
Median (P25–P75), weighted	+2.4 (−3.1 – +6.4)	−15.5 (−17.1 – −12.9)
Range	−26.4 – +26.7	−21.3 – −6.6

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region scores +2.4; Condesa lies 15.4 points below the study region average.

Map



Map. Mobility and transport: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Safety and protection

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> Conditions in the urban environment that allow residents to move freely and participate in public life without fear of crime, violence or traffic accidents.

Subdomains

- **Personal safety.** Conditions in the urban environment related to crime incidence and the presence of public and surveillance services that determine the degree of people's physical protection, reduce exposure to crime and promote peace of mind in public space.
- **Road safety.** Characteristics and conditions of road and pedestrian infrastructure designed to prevent traffic crashes, protecting the physical integrity of pedestrians, cyclists and drivers travelling through the city.
- **Risks.** Population exposure and vulnerability to natural or human-caused hazards, such as earthquakes, flooding, extreme heat, fire and adverse climate events, as well as the urban environment's capacity to mitigate their impacts and safeguard the community.

Indicators of the domain

Indicator	Subdomain	Share	Effective weight	Better when
Proximity to major roads	Road safety	50%	5.9%	lower
Food within 1.5 km by low-stress cycling	Road safety	33%	6.2%	higher
Block size (perimeter)	Road safety	33%	6.2%	lower
Fire station within 3 km	Risks	33%	6.2%	higher
Police station within 2 km	Personal safety	33%	6.2%	higher

Method: theory

The domain score is the weighted Mazziotta–Pareto composite of its indicators' normalised scores (R12): each indicator weighs `weight × share`, normalised over the indicators with a value; `M` is their weighted mean, `S` their weighted deviation, and the score is `M − S·(S/M)`, less 100. In the `uli_plain` variant walkability is not attenuated and, in the ambient environment, thermal comfort is activated.

Recipes: R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Code (GHSCI): `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

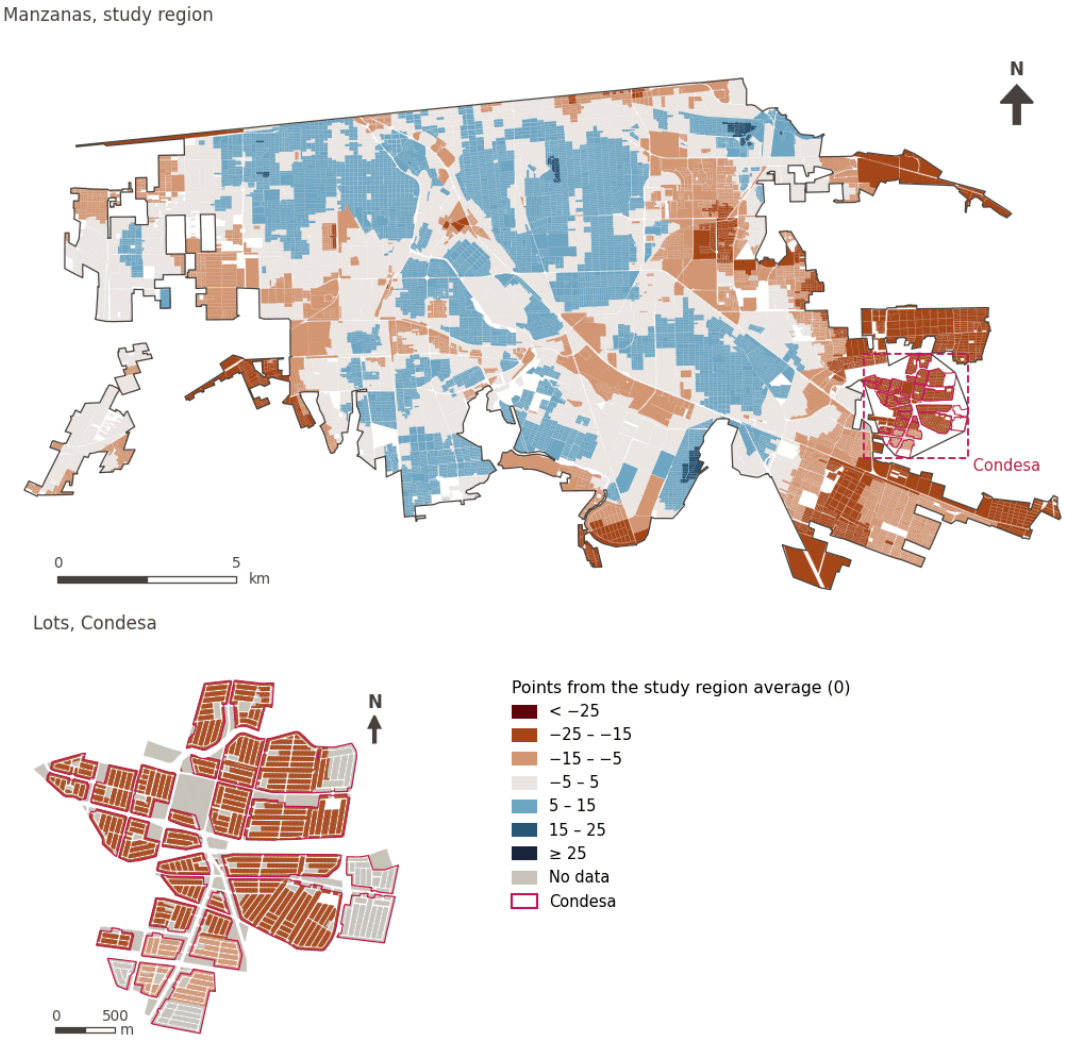
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	+2.2	−20.7
Region score (`uli_plain`)	+2.2	−20.7
Areas with a value	14,216 of 14,252 manzanas	12,862 of 14,988 lots
Median (P25–P75), weighted	+3.3 (−4.1 – +8.2)	−21.4 (−21.8 – −21.2)
Range	−24.1 – +16.1	−23.7 – −10.7

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region scores +2.2; Condesa lies 20.7 points below the study region average.

Map



Map. Safety and protection: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Social infrastructure

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> The network of public facilities that supports social interaction, cohesion and collective well-being.

Subdomains

- **Public services.** Set of essential government or community facilities, equipment, networks and services oriented toward well-being, urban management and support for daily life; their availability, coverage and proper operation ensure dignified, safe and equitable conditions for the population in the urban environment.
- **Health and healthcare.** Facilities and health services that contribute to residents' physical and mental well-being.
- **Community, culture and recreation.** Cultural spaces, heritage sites and community events that foster identity and a sense of belonging.
- **Education.** Educational institutions, from early childhood through higher education, that support human development and social mobility.

Indicators of the domain

Indicator	Subdomain	Share	Effective weight	Better when
Walkability (300 m), attenuated by thermal comfort	Public services	33%	3.7%	higher
250 m or more from petrol stations	Public services	33%	3.6%	higher
Convenience stores within 300 m	Public services	33%	3.6%	higher
Fresh food diversity within 500 m	Public services	33%	3.6%	higher
Fresh food within 300 m	Public services	33%	3.6%	higher
Community, culture and recreation diversity within 500 m	Community, culture and recreation	100%	1.7%	higher
Community, culture and recreation within 300 m	Community, culture and recreation	100%	1.7%	higher
Care service diversity within 500 m	Public services	100%	1.7%	higher
Care services within 300 m	Public services	100%	1.7%	higher
Education diversity within 500 m	Education	100%	1.7%	higher
Health service diversity within 500 m	Health and healthcare	100%	1.7%	higher
Primary healthcare within 300 m	Health and healthcare	100%	1.7%	higher
Primary schools within 300 m	Education	100%	1.7%	higher

Method: theory

The domain score is the weighted Mazziotta–Pareto composite of its indicators' normalised scores (R12): each indicator weighs `weight × share`, normalised over the indicators with a value; `M` is their weighted mean, `S` their weighted deviation, and the score is `M − S·(S/M)`, less 100. In the `uli_plain` variant walkability is not attenuated and, in the ambient environment, thermal comfort is activated.

Recipes: R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Code (GHSCI): `process/subprocesses/_composite_index.py:1874` `score()`, `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

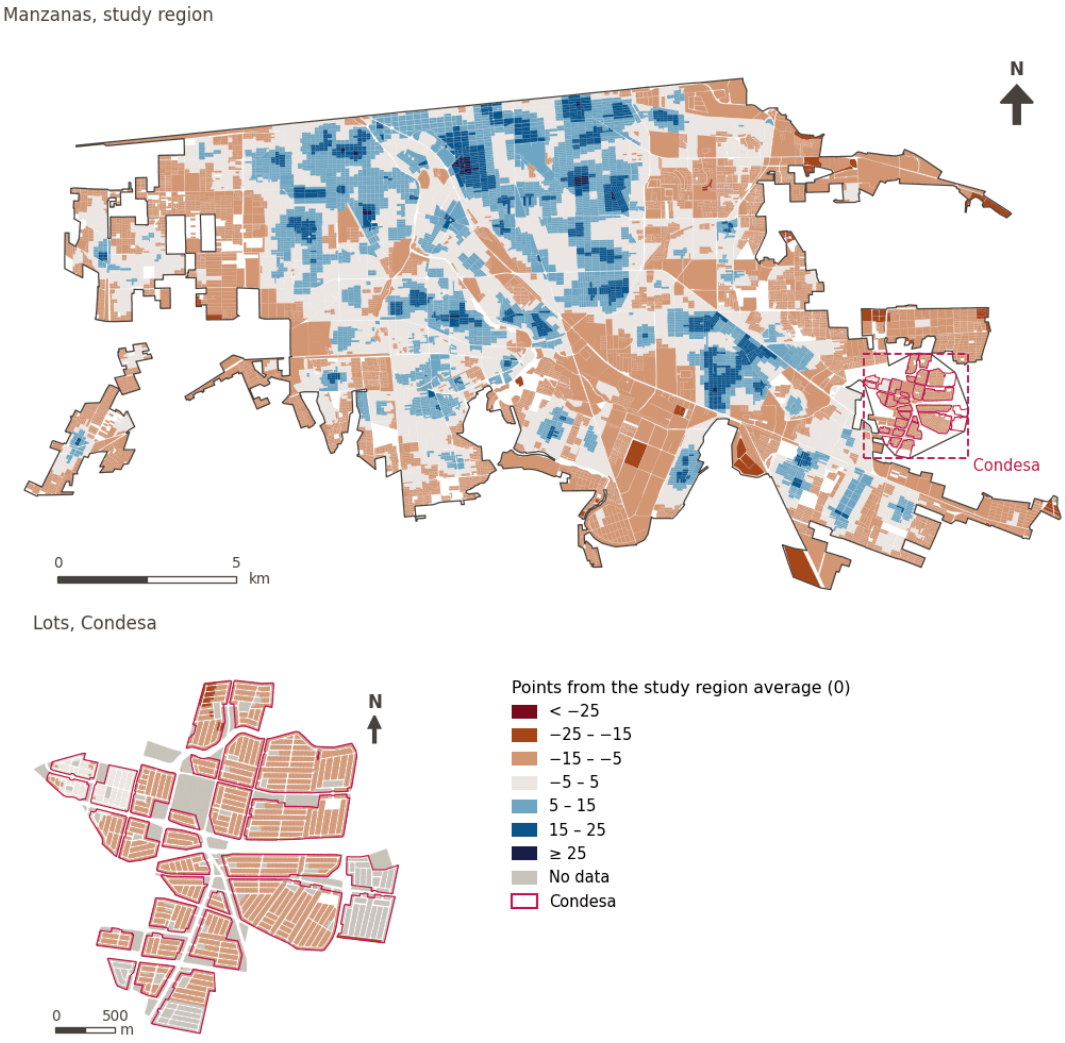
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region score (`uli`)	+2.4	−12.1
Region score (`uli_plain`)	+2.3	−11.6
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	+0.4 (−6.3 – +7.0)	−12.6 (−14.4 – −11.6)
Range	−17.4 – +33.7	−15.2 – +8.5

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- The study region scores +2.4; Condesa lies 12.1 points below the study region average.

Map



Map. Social infrastructure: score by manzana across the study region and by lot in Condesa (inset), in points from the study region average (0); blue is better.

Sociodemographic characteristics

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> Characteristics of the population, such as age, income, educational attainment and household composition, that determine how quality of life is perceived across different groups.

Subdomains

- **Population.** Characteristics of people living in an area, such as age, gender, income, educational attainment, occupation and population distribution, which make it possible to identify differences in quality of life between groups.
- **Household.** Characteristics of households, such as their composition, size, income, housing conditions and members' educational attainment, which influence the perception of quality of life.
- **City.** Sociodemographic characteristics of the population and households living in a city, as well as their spatial distribution and socioeconomic conditions, which influence the perception of quality of life across its different areas.

This domain has no indicator yet: it is described but not scored, so the index is the composite of six domains (Appendix G).

Appendix G. Excluded and pending indicators

Natural blue space nearby

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> Walking distance to the nearest water body, scored by the project's scale for arid cities (100 within 200 m, 90 within 500 m, 80 within 1 km, and 60 beyond).

Identity

Identifier	`blue_space`
Core measure (variable sheet)	Available natural blue spaces
Sheet / register ID	10 · #10
Sample point variable	`sp_walk_nearest_node_blue_space`
Area variable (measured value)	`avg_walk_dist_blue_space` (m)
Normalised score	`index_uli__blue_space`
Domains (share of weight)	Ambient environment
Subdomain	Ambient
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 2.8% · `uli_plain` 2.4%
Status	Scored in the published index

Rationale and interpretation

> Proximity to available water resources is a critical liveability indicator in arid urban environments, as access to water is fundamental to human health, comfort, and survival.
>
> *— Indicator register #10: Access to blue space (Ambient Environment / Resources)*

- Workbook citation: No external citations provided for this indicator.
- Article list: #28: Chen (2023), 'Evaluation of the livability of arid urban environments under global warming: A multi-parameter approach'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 500 m of a water body; on average the nearest is … away (lower is better).

Available natural blue spaces (Available natural blue spaces; ID 10) - *How to calculate (variable sheet):* Based on Landsat OLI_TIRS multispectral data, thenormalized differential water index (MNDWI) was calculated, where: MNDWI = G - MIR/G + MIR where MIR is the reflectance value of the infrared band in Landsat and Gis the reflectance value of the infrared band. Single-factor scores were evaluated in combination with the effects of water bodies on the livability of urban environments within a certain radiation distance. When the radiation range of water bodies was < 200 m, the areas surrounding water bodies (e.g., wetlands and parks) were the most livable,reaching scores of 100. Water bodies could influence the livability of the urban environment within 200–500 m, where AWR was evaluated as having a score of 90. Areas 500–1000 m away from water bodies might be influenced by other buildings, where AWR had a score of 80. Areas 1000–1500 m away from water bodies require travel by foot, bike orother means, and their AWR score was 60. Since water resources are extremely important to arid cities, the lowest limit was determined as 60. That means the lowest level of the livability in this area is the Generally livable when only water resources are considered

Data sources

Sources declared in the configuration:

- OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026).

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_blue_space`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf

Inputs: Natural blue space

Method: theory

1. Distance along the pedestrian network to the nearest destination, `d` (R2), censored at 1,500 m.
2. Stepped ladder (R10): 100 within 200 m, 90 within 500 m, 80 within 1,000 m, 60 within 1,500 m; no value (nothing within 1,500 m) → 60.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1285`):

```
- variable: sp_walk_nearest_node_blue_space
  domains: [ambient_environment]
  transform: {steps: [[200, 100], [500, 90], [1000, 80], [1500, 60]], beyond: 60}
  group: {es: 'Espacios azules naturales disponibles', en: 'Available natural blue spaces'}
  label: {es: 'Espacios azules naturales cercanos', en: 'Natural blue space nearby'}
  reading: {es: '{access} de la población vive a menos de 500 m de un cuerpo de agua; en promedio, el más cercano está a {value}', en: '{access} of the
population live within 500 m of a water body; on average the nearest is {value} away'}
  reading_columns: {access: pct_access_walk_blue_space_500m}
  subdomain: {es: 'Medioambiente', en: 'Ambient'}
  about:
    es: Distancia a pie al cuerpo de agua más cercano, puntuada con la escala del proyecto para ciudades áridas (100 a 200 m, 90 a 500 m, 80 a 1 km y 60
más allá).
    en: Walking distance to the nearest water body, scored by the project's scale for arid cities (100 within 200 m, 90 within 500 m, 80 within 1 km, and
60 beyond).
  sources:
    - 'OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026).'
```

`MX_Mexicali_2025_ULI.yml:489`:

```
- {name: blue_space, category: blue_space, variant: any, layer: blue_space_nodes_30m_line}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:423` `stepped_score()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_blue_space	stepped ladder	56.9 · 76.9 · 96.9	60.0 – 100.0	−0.01 / −1.37 / 36.7%	none
`uli_plain`	sp_walk_nearest_node_blue_space	stepped ladder	56.9 · 76.9 · 96.9	60.0 – 100.0	−0.01 / −1.37 / 36.7%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	797 m	590 m
Region score (`uli`)	−3.9	+7.8
Areas with a value	12,412 of 14,252 manzanas	12,866 of 14,988 lots
Median (P25–P75), weighted	771 m (454 m – 1,111 m)	472 m (338 m – 665 m)
Range	0 m – 1,937 m	8 m – 1,848 m
Nothing within the distance searched	13.5%	3.0%
Score: median (P25–P75) (`uli`)	+3.5 (−25.4 – +12.5)	+19.6 (+4.6 – +19.6)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 23.7% of the population live within 500 m of a water body; on average the nearest is 797 m away.
- Condesa: 53.0% of the population live within 500 m of a water body; on average the nearest is 590 m away.
- For the study region as a whole the value is 797 m; in Condesa, 590 m. Condesa's score lies 7.8 points above the study region average (−3.9 for the study region).

Map

Map. Natural blue space nearby: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The sheet's source method derives water from Landsat (MNDWI) and scores by radiation distance; here the ladder is applied to walking distance to water features mapped in OpenStreetMap, polygons and lines, so that Mexicali's canals and drains count (configuration, accessibility).

Heat adaptive capability (GUHVI)

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> The GUHVI's adaptive capability sub-index (0-100), from the child dependency ratio, the subnational human development index and infant mortality; higher is better.

Identity

Identifier	`adaptive_capability`
Core measure (variable sheet)	Global Urban Heat Vulnerability Index (GUHVI)
Sheet / register ID	–
Sample point variable	`sp_urban_heat_adaptive_capability_index`
Area variable (measured value)	`urban_heat_adaptive_capability_index` (0–100)
Normalised score	`index_uli__adaptive_capability`
Domains (share of weight)	Ambient environment
Subdomain	Climate risks
Lens	Quality
Better when	higher
Effective weight in the index	`uli` 2.8% · `uli_plain` 2.4%
Status	Scored in the published index

Rationale and interpretation

Pending: Not recorded in the project register.

How to read it. Adaptive capability index of … (0-100; higher is better).

Adaptive capability (Global Urban Heat Vulnerability Index (GUHVI); ID –) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Child dependency ratio, Subnational Human Development Index, Infant Mortality Rates)

Data sources

Sources declared in the configuration:

- Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.

Inputs: Global Urban Heat Vulnerability Index (GUHVI) and its inputs

Method: theory

1. A sub-index of the Global Urban Heat Vulnerability Index (GUHVI; Turner et al. 2025), computed in Google Earth Engine on an analysis grid and min–max normalised to 0–100 within the region (R8).
2. Each sample point takes the value of the cell containing it.
3. AMPI normalisation (R11) with fixed goalposts of 0 and 100 (the sub-index's own scale), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R8. Earth Engine: heat vulnerability and large green space, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1247`):

```
- variable: sp_urban_heat_adaptive_capability_index
  name: adaptive_capability
  domains: [ambient_environment]
  polarity: positive
  goalposts: {min: 0, max: 100}
  lens: quality
  group: {es: 'Índice Global de Vulnerabilidad Urbana al Calor (GUHVI)', en: 'Global Urban Heat Vulnerability Index (GUHVI)'}
  label: {es: 'Capacidad de adaptación al calor (GUHVI)', en: 'Heat adaptive capability (GUHVI)'}
  reading: {es: 'Índice de capacidad de adaptación de {value} (0-100; mayor es mejor)', en: 'Adaptive capability index of {value} (0-100; higher is
better)'}
  subdomain: {es: 'Riesgos climáticos', en: 'Climate risks'}
  about:
    es: Subíndice de capacidad de adaptación del GUHVI (0-100), a partir de la razón de dependencia infantil, el índice de desarrollo humano subnacional y
la mortalidad infantil; mayor es mejor.
    en: The GUHVI's adaptive capability sub-index (0-100), from the child dependency ratio, the subnational human development index and infant mortality;
higher is better.
  sources:
    - 'Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.'
```

Code (GHSCI):

- `process/subprocesses/_earth_engine_indicators.py:846` `guhvi_analysis()`

- `process/subprocesses/_earth_engine_indicators.py:1105` `normalise_band()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_urban_heat_adaptive_capability_index	as measured	0 · 22 · 100	0 – 30	−2.81 / 7.51 / 2.6%	skewed
`uli_plain`	sp_urban_heat_adaptive_capability_index	as measured	0 · 22 · 100	0 – 30	−2.81 / 7.51 / 2.6%	skewed

Recalculation (Reimagina Urbano)

GUHVI sub-indices are normalised within the region by their own minima and maxima; a scenario must use the baseline's minima and maxima so as not to re-scale the whole city.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

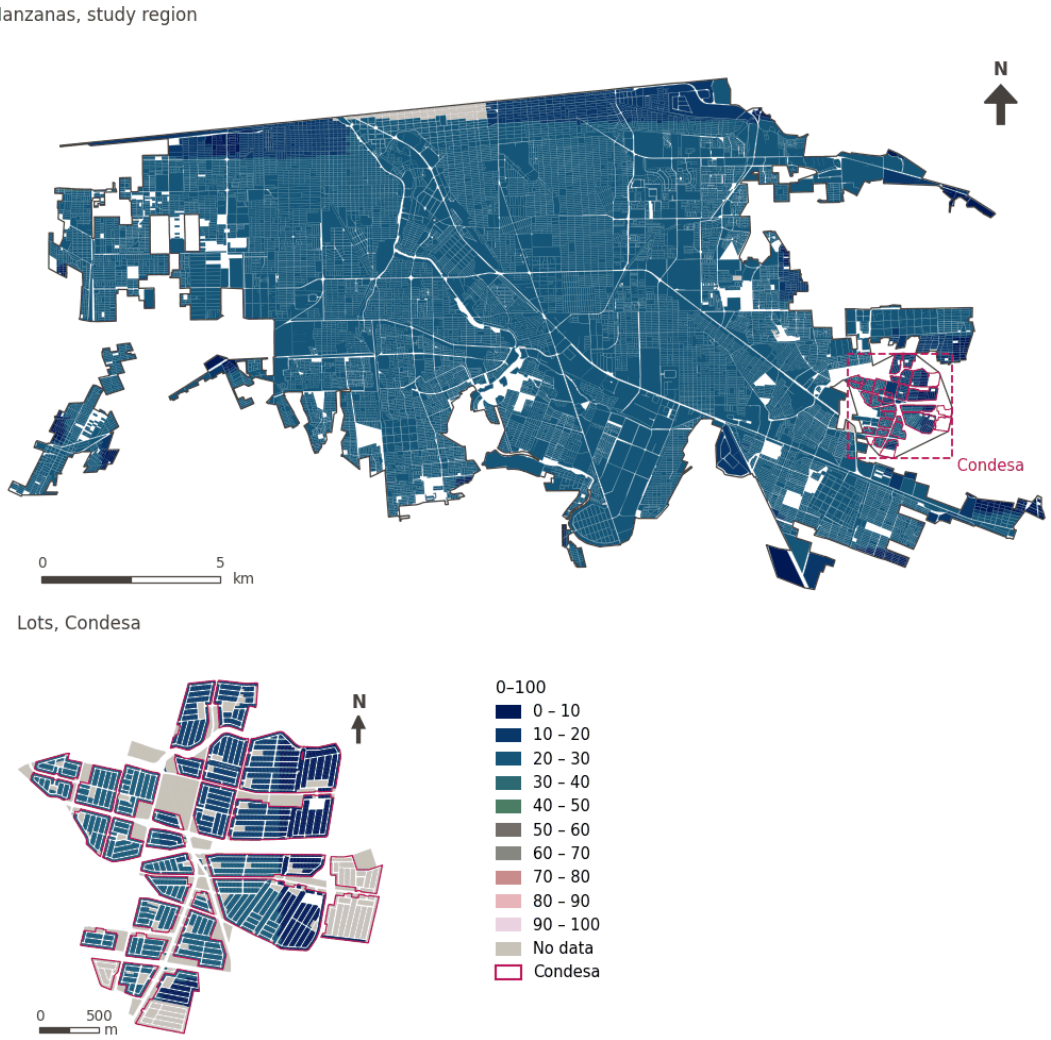
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	22.2	15.0
Region score (`uli`)	+0.6	−4.0
Areas with a value	14,148 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	23.1 (23.1 – 23.2)	13.0 (11.3 – 23.1)
Range	0.0 – 30.3	0.0 – 23.9
Score: median (P25–P75) (`uli`)	+0.9 (+0.9 – +0.9)	−5.2 (−6.2 – +0.9)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Adaptive capability index of 22.2 (0-100; higher is better).
- Condesa: Adaptive capability index of 15.0 (0-100; higher is better).
- For the study region as a whole the value is 22.2; in Condesa, 15.0. Condesa's score lies 4.0 points below the study region average (+0.6 for the study region).

Map



Map. Heat adaptive capability (GUHVI): measured value (0–100) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- Several of its inputs (the subnational human development index, the infant mortality rate) are constant across a single city, so its variation within Mexicali comes mainly from the child dependency ratio (configuration, dashboard.hide).

Heat exposure (GUHVI)

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> The GUHVI's heat exposure sub-index (0-100), from land surface temperature; lower is better.

Identity

Identifier	`heat_exposure`
Core measure (variable sheet)	Global Urban Heat Vulnerability Index (GUHVI)
Sheet / register ID	382 · #382
Sample point variable	`sp_urban_heat_exposure_index`
Area variable (measured value)	`urban_heat_exposure_index` (0–100)
Normalised score	`index_uli__heat_exposure`
Domains (share of weight)	Ambient environment
Subdomain	Climate risks
Lens	Quality
Better when	lower
Effective weight in the index	`uli` 2.8% · `uli_plain` 2.4%
Status	Scored in the published index

Rationale and interpretation

> Mitigating the urban heat island effect by reducing localized temperature increases promotes thermal comfort and supports the physical and mental well-being of city dwellers.
>
> *— Indicator register #382: Urban heat island (Ambient Environment / Risks)*

- Workbook citation: Leya et al., 2022, 'Spatial Variations of Urban Heat Island Development in Khulna City, Bangladesh: Implications for Urban Planning and Development'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Heat exposure index of … (0-100; lower is better).

Heat exposure (Global Urban Heat Vulnerability Index (GUHVI); ID 382) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Land Surface Temperature)

Data sources

Sources declared in the configuration:

- Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.

Inputs: Global Urban Heat Vulnerability Index (GUHVI) and its inputs

Method: theory

1. A sub-index of the Global Urban Heat Vulnerability Index (GUHVI; Turner et al. 2025), computed in Google Earth Engine on an analysis grid and min–max normalised to 0–100 within the region (R8).
2. Each sample point takes the value of the cell containing it.
3. AMPI normalisation (R11) with fixed goalposts of 0 and 100 (the sub-index's own scale), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R8. Earth Engine: heat vulnerability and large green space, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1217`):

```
- variable: sp_urban_heat_exposure_index
  name: heat_exposure
  domains: [ambient_environment]
  polarity: negative
  goalposts: {min: 0, max: 100}
  lens: quality
  group: {es: 'Índice Global de Vulnerabilidad Urbana al Calor (GUHVI)', en: 'Global Urban Heat Vulnerability Index (GUHVI)'}
  label: {es: 'Exposición al calor (GUHVI)', en: 'Heat exposure (GUHVI)'}
  reading: {es: 'Índice de exposición al calor de {value} (0-100; menor es mejor)', en: 'Heat exposure index of {value} (0-100; lower is better)'}
  subdomain: {es: 'Riesgos climáticos', en: 'Climate risks'}
  about:
    es: Subíndice de exposición al calor del GUHVI (0-100), a partir de la temperatura de la superficie terrestre; menor es mejor.
    en: The GUHVI's heat exposure sub-index (0-100), from land surface temperature; lower is better.
  sources:
    - 'Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.'
```

Code (GHSCI):

- `process/subprocesses/_earth_engine_indicators.py:846` `guhvi_analysis()`
- `process/subprocesses/_earth_engine_indicators.py:1105` `normalise_band()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_urban_heat_exposure_index	as measured	0 · 47 · 100	0 – 100	0.58 / 1.97 / 2.6%	none
`uli_plain`	sp_urban_heat_exposure_index	as measured	0 · 47 · 100	0 – 100	0.58 / 1.97 / 2.6%	none

Recalculation (Reimagina Urbano)

GUHVI sub-indices are normalised within the region by their own minima and maxima; a scenario must use the baseline's minima and maxima so as not to re-scale the whole city.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

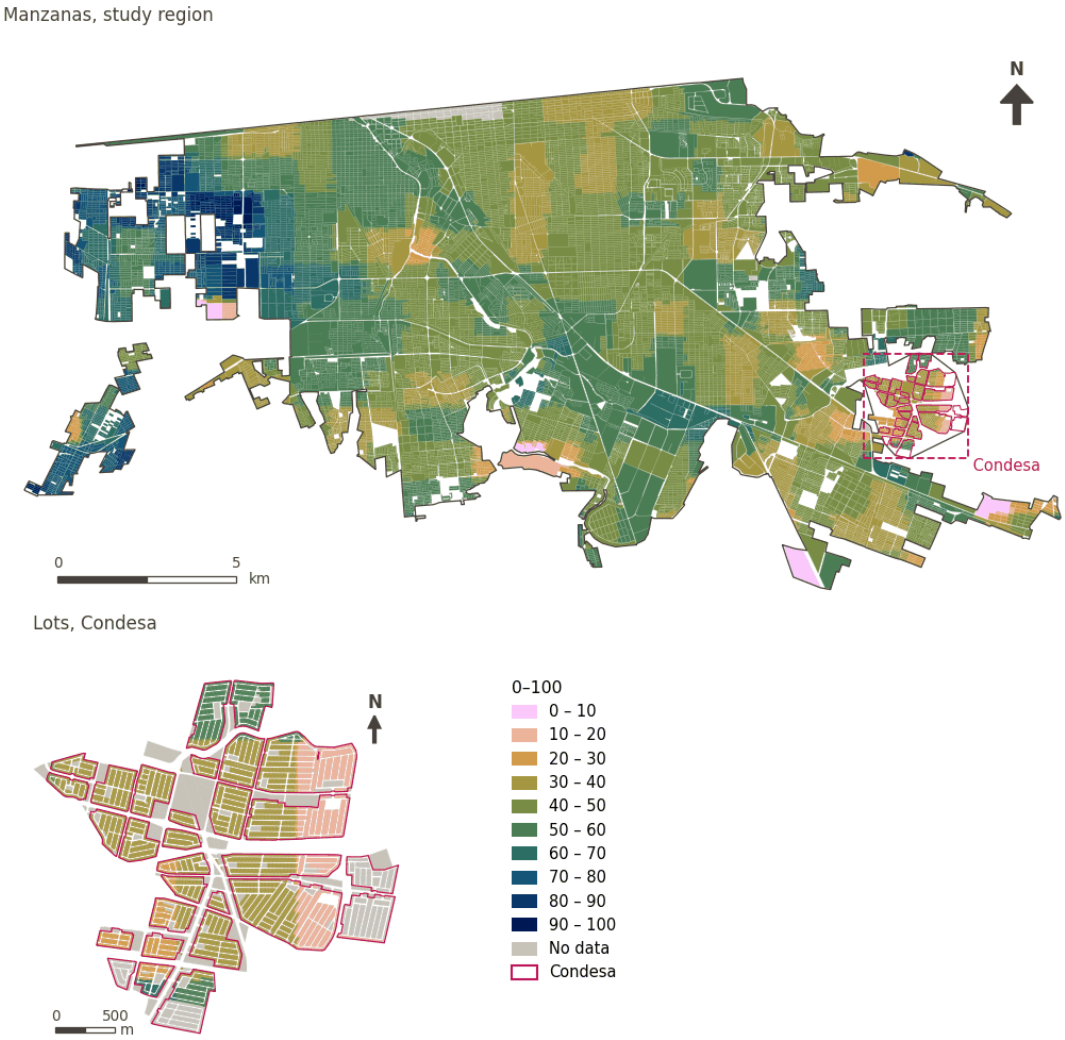
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	49.0	33.3
Region score (`uli`)	+0.2	+8.5
Areas with a value	14,148 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	45.6 (40.2 – 51.4)	33.1 (23.1 – 38.5)
Range	0.0 – 100.0	17.0 – 68.8
Score: median (P25–P75) (`uli`)	+1.1 (−2.4 – +4.3)	+8.6 (+5.4 – +14.6)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Heat exposure index of 49.0 (0-100; lower is better).
- Condesa: Heat exposure index of 33.3 (0-100; lower is better).
- For the study region as a whole the value is 49.0; in Condesa, 33.3. Condesa's score lies 8.5 points above the study region average (+0.2 for the study region).

Map



Map. Heat exposure (GUHVI): measured value (0–100) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The GUHVI sub-indices are min–max normalised within the region, so 0 and 100 are the least and most exposed places in Mexicali, not absolute thresholds (configuration, composite_indices).
- Heat exposure is derived from land surface temperature alone; the sheet's separate heat exposure core measure (land surface temperature, extreme heat frequency, night-time heat retention) awaits delivery (configuration).

Heat sensitivity (GUHVI)

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> The GUHVI's heat sensitivity sub-index (0-100), from albedo, vegetation, local climate zones, population density and the vulnerable population; lower is better.

Identity

Identifier	`heat_sensitivity`
Core measure (variable sheet)	Global Urban Heat Vulnerability Index (GUHVI)
Sheet / register ID	–
Sample point variable	`sp_urban_heat_sensitivity_index`
Area variable (measured value)	`urban_heat_sensitivity_index` (0–100)
Normalised score	`index_uli__heat_sensitivity`
Domains (share of weight)	Ambient environment
Subdomain	Climate risks
Lens	Quality
Better when	lower
Effective weight in the index	`uli` 2.8% · `uli_plain` 2.4%
Status	Scored in the published index

Rationale and interpretation

Pending: Not recorded in the project register.

How to read it. Heat sensitivity index of … (0-100; lower is better).

Heat sensitivity (Global Urban Heat Vulnerability Index (GUHVI); ID –) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Land Surface Albedo, Normalised Difference Vegetation Index, Local Climate Zones, Population Density, Vulnerable Population Percentage)

Data sources

Sources declared in the configuration:

- Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.

Inputs: Global Urban Heat Vulnerability Index (GUHVI) and its inputs

Method: theory

1. A sub-index of the Global Urban Heat Vulnerability Index (GUHVI; Turner et al. 2025), computed in Google Earth Engine on an analysis grid and min–max normalised to 0–100 within the region (R8).
2. Each sample point takes the value of the cell containing it.
3. AMPI normalisation (R11) with fixed goalposts of 0 and 100 (the sub-index's own scale), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R8. Earth Engine: heat vulnerability and large green space, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1232`):

```
- variable: sp_urban_heat_sensitivity_index
  name: heat_sensitivity
  domains: [ambient_environment]
  polarity: negative
  goalposts: {min: 0, max: 100}
  lens: quality
  group: {es: 'Índice Global de Vulnerabilidad Urbana al Calor (GUHVI)', en: 'Global Urban Heat Vulnerability Index (GUHVI)'}
  label: {es: 'Sensibilidad al calor (GUHVI)', en: 'Heat sensitivity (GUHVI)'}
  reading: {es: 'Índice de sensibilidad al calor de {value} (0-100; menor es mejor)', en: 'Heat sensitivity index of {value} (0-100; lower is better)'}
  subdomain: {es: 'Riesgos climáticos', en: 'Climate risks'}
  about:
    es: Subíndice de sensibilidad al calor del GUHVI (0-100), a partir del albedo, la vegetación, las zonas climáticas locales, la densidad de población y
la población vulnerable; menor es mejor.
    en: The GUHVI's heat sensitivity sub-index (0-100), from albedo, vegetation, local climate zones, population density and the vulnerable population;
lower is better.
  sources:
    - 'Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.'
```

Code (GHSCI):

- `process/subprocesses/_earth_engine_indicators.py:846` `guhvi_analysis()`
- `process/subprocesses/_earth_engine_indicators.py:1105` `normalise_band()`

- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_urban_heat_sensitivity_index	as measured	0 · 58 · 100	0 – 81	−0.79 / −0.15 / 2.6%	none
`uli_plain`	sp_urban_heat_sensitivity_index	as measured	0 · 58 · 100	0 – 81	−0.79 / −0.15 / 2.6%	none

Recalculation (Reimagina Urbano)

GUHVI sub-indices are normalised within the region by their own minima and maxima; a scenario must use the baseline's minima and maxima so as not to re-scale the whole city.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

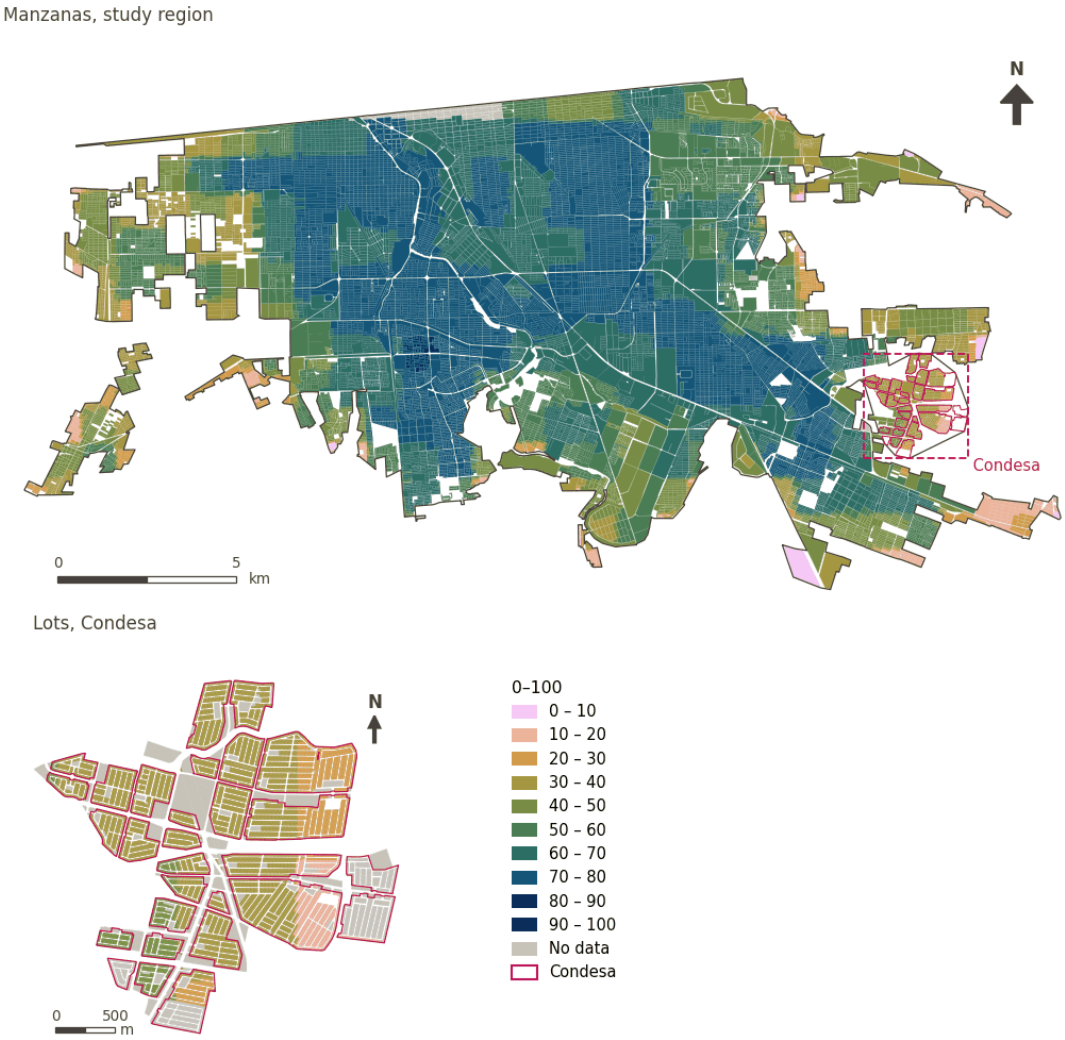
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	59.7	33.8
Region score (`uli`)	−4.7	+14.7
Areas with a value	14,148 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	67.3 (57.3 – 74.2)	35.1 (30.3 – 39.1)
Range	4.2 – 81.2	10.5 – 48.5
Score: median (P25–P75) (`uli`)	−5.4 (−9.5 – +0.6)	+13.9 (+11.6 – +16.8)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Heat sensitivity index of 59.7 (0-100; lower is better).
- Condesa: Heat sensitivity index of 33.8 (0-100; lower is better).
- For the study region as a whole the value is 59.7; in Condesa, 33.8. Condesa's score lies 14.7 points above the study region average (−4.7 for the study region).

Map



Map. Heat sensitivity (GUHVI): measured value (0–100) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Min–max normalised within the region (see heat exposure).

Daytime thermal comfort

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> Mean Universal Thermal Climate Index (UTCI) over sunlit hours on pedestrian surfaces, modelled for the hottest day of 2023; lower is better.

Identity

Identifier	`thermal_comfort`
Core measure (variable sheet)	Thermal comfort
Sheet / register ID	371, 337 · #371, #337, #372
Sample point variable	`sp_utci_day_mean`
Area variable (measured value)	`utci_day_mean` (°C)
Score (unattenuated variant)	`index_uli_plain__thermal_comfort`
Domains (share of weight)	Ambient environment
Subdomain	Ambient
Lens	Quality
Better when	lower
Effective weight in the index	`uli_plain` 2.4%
Status	Listed but not scored in the published index: Included through walkability's thermal comfort attenuation. Choose walkability without attenuation to score it here.

Rationale and interpretation

> Maintaining urban temperatures within a comfortable range is vital for physical health and preventing heat-related illnesses and thermal discomfort.
> *— Indicator register #371: Temperature (Ambient Environment / Ambient)*

- Workbook citation: Kashi et al., 2024, 'Effects of extreme weather events and climate change on cities' livability'; Alijani et al., 2020, 'A new approach of urban livability in Tehran: Thermal comfort as a primitive indicator'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'

> Appropriate humidity levels are necessary for human thermal comfort and can affect respiratory health and the perceived quality of the urban environment.
> *— Indicator register #337: Humidity (Ambient Environment / Ambient)*

- Workbook citation: Kashi et al., 2024, 'Effects of extreme weather events and climate change on cities' livability'; Liang and Li, 2020, 'Resilience and sustainable development goals based social-ecological indicators and assessment of coastal urban areas'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'

> The difference from optimal temperature in July indicates thermal comfort; excessive summer heat is linked to physiological stress and reduced outdoor activity.
> *— Indicator register #372: The difference between the average temperature in July and the optimal temperature for human body (Ambient Environment / Ambient)*

- Workbook citation: Zhan et al., 2018, 'Assessment and determinants of satisfaction with urban livability in China'; Cao et al., 2019, 'The framework of relationship between built environment and residents healthy based on activity perspective'
- Article list: #17: Wang (2023), 'The impact of economic development on urban livability: Evidence from 40 large and medium-sized cities of China'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Mean daytime UTCI of … on pedestrian surfaces on the hottest day of 2023 (lower is better).

Dry bulb air temperature (Thermal Comfort; ID 371) - *Doubts (sheet):* Rossano, please review according to what is needed by PyThermal comfort (https://pythermalcomfort.readthedocs.io/en/latest/documentation/models.html#universal-thermal-climate-index-utci)

Mean radiant temperature (Thermal Comfort; ID 371)

Wind speed 10m above ground level (Thermal Comfort; ID –)

Relative humidity (Thermal Comfort; ID 337) - *Notes (sheet):* Could we also calculate it as "The difference between the relative humidity in the hottest month of the year and the optimum humidity for human body"?

Data sources

Sources declared in the configuration:

- Rossano Schifanella (WP02): SOLWEIG, forced by Mexicali airport observations for 21 July 2023; UTCI operational polynomial (Bröde et al. 2012).

Source recorded by GHSCI (data dictionary) (`utci_day_mean`):

- Outdoor thermal comfort (UTCI): Rossano Schifanella (WP02); SOLWEIG and the UTCI operational polynomial (Bröde et al. 2012), Terrain from FABDEM v1 (CC BY-NC-SA 4.0, non-commercial); other inputs CC BY 4.0 or public domain

Inputs: Outdoor thermal comfort (UTCI)

Method: theory

1. Daytime mean UTCI (Universal Thermal Climate Index), modelled with SOLWEIG for 21 July 2023 over pedestrian surfaces and summarised by area (work package 2); units with fewer than 100 pedestrian pixels are masked (R7).
2. Replicated to each sample point from its 100 m cell.
3. In the published index it enters only through walkability's attenuation (R9). In `uli_plain` it is normalised (R11, lower is better) and aggregated (R12, R13).

This indicator is not scored in the published index; the unattenuated variant (uli_plain) scores it.

Method: practice

Recipes: R1. Sample points and the pedestrian network, R7. Linking externally prepared indicators, R9. Walkability, and its attenuation by thermal comfort, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1264`):

```
- variable: sp_utci_day_mean
  name: thermal_comfort
  domains: [ambient_environment]
  polarity: negative
  lens: quality
  active: false
  inactive_reason:
    es: Incluido mediante la atenuación de la caminabilidad por confort térmico. Elija la caminabilidad sin atenuación para puntuarlo aquí.
    en: Included through walkability's thermal comfort attenuation. Choose walkability without attenuation to score it here.
  group: {es: 'Confort térmico', en: 'Thermal comfort'}
  label: {es: 'Confort térmico diurno', en: 'Daytime thermal comfort'}
  reading: {es: 'UTCI diurno medio de {value} en superficies peatonales en el día más caluroso de 2023', en: 'Mean daytime UTCI of {value} on pedestrian
surfaces on the hottest day of 2023'}
  subdomain: {es: 'Medioambiente', en: 'Ambient'}
  about:
    es: Índice Universal de Clima Térmico (UTCI) medio durante las horas de sol en superficies peatonales, modelado para el día más caluroso de 2023; menor
es mejor.
    en: Mean Universal Thermal Climate Index (UTCI) over sunlit hours on pedestrian surfaces, modelled for the hottest day of 2023; lower is better.
  sources:
    - 'Rossano Schifanella (WP02): SOLWEIG, forced by Mexicali airport observations for 21 July 2023; UTCI operational polynomial (Bröde et al. 2012).'
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:512` `apply_rules()`
- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli_plain`	sp_utci_day_mean	as measured	41.938 · 44.949 · 47.960	40.831 – 46.853	−0.16 / −0.39 / 0.2%	none

Recalculation (Reimagina Urbano)

The value comes from an external work package: a scenario (for example, more vegetation) replaces the area's value, which is replicated to its sample points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

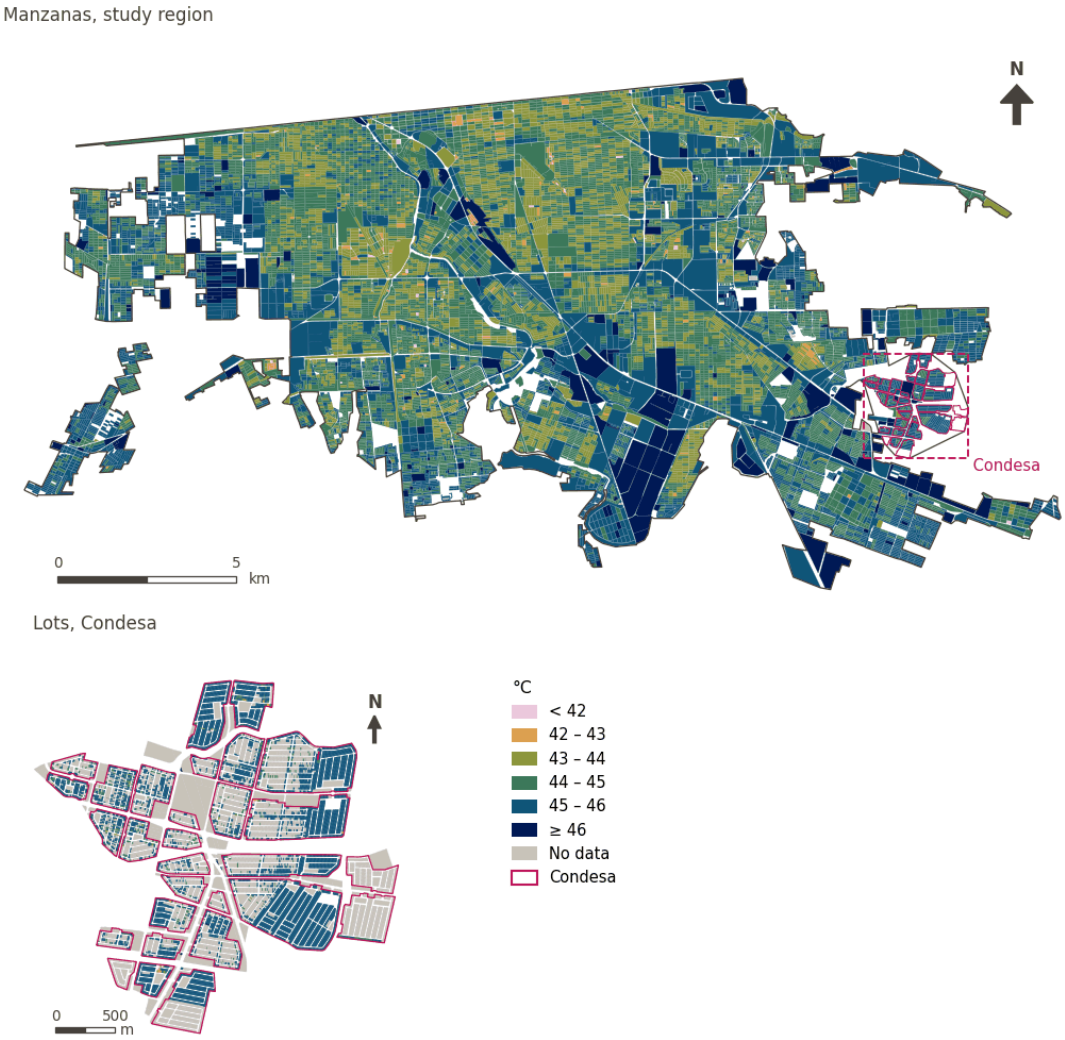
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	45.1 °C	45.5 °C
Region score (`uli_plain`)	+3.8	−5.6
Areas with a value	14,200 of 14,252 manzanas	7,766 of 14,988 lots
Median (P25–P75), weighted	44.3 °C (43.9 °C – 44.8 °C)	45.9 °C (45.6 °C – 46.0 °C)
Range	40.4 °C – 46.7 °C	40.4 °C – 46.4 °C
Score: median (P25–P75) (`uli_plain`)	+3.3 (−0.4 – +6.2)	−5.9 (−8.3 – −3.4)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean daytime UTCI of 45.1 °C on pedestrian surfaces on the hottest day of 2023.
- Condesa: Mean daytime UTCI of 45.5 °C on pedestrian surfaces on the hottest day of 2023.
- For the study region as a whole the value is 45.1 °C; in Condesa, 45.5 °C. Condesa's score lies 5.6 points below the study region average (+3.8 for the study region).

Map



Map. Daytime thermal comfort: measured value (°C) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Air temperature, humidity and wind are spatially uniform in the model, so only radiant temperature (sun and shade) varies across the city; most values exceed the UTCI polynomial's fitted range; comparisons within a neighbourhood are sound where rankings of neighbourhoods kilometres apart are not (configuration, linkage_indicators.utci; README_mexicali_utci.md).
- Only daytime thermal comfort is used: night-time UTCI varies by fractions of a degree across the city (30.3–30.6 °C) (configuration).

Parks within 300 m

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> Walking distance to the nearest public open space, scored with a 300 m soft threshold.

Identity

Identifier	`parks`
Core measure (variable sheet)	Access to parks
Sheet / register ID	162 · #162, #187
Sample point variable	`sp_walk_nearest_node_public_open_space_any`
Area variable (measured value)	`avg_walk_dist_public_open_space_any` (m)
Normalised score	`index_uli__parks`
Domains (share of weight)	Ambient environment (50%), Built environment (50%)
Subdomain	Ambient environment: Ambient; Built environment: Green and blue infrastructure
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 4.0% · `uli_plain` 3.8%
Status	Scored in the published index

Rationale and interpretation

> Source article not available for review.
> *— Indicator register #162: Access to parks (local, neighbourhood, district, regional) (Built Environment / Green & Blue Infrastructure)*

- Workbook citation: Not reported.
- Article list: #19: Wu (2022), 'Urban greenness, mixed land-use, and life satisfaction: Evidence from residential locations and workplace settings in Beijing'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Availability of open spaces provides urban residents with areas for comfort, enjoyment, and relaxation, directly contributing to their overall quality of life.
> *— Indicator register #187: Access to public open space (Built Environment / Green & Blue Infrastructure)*

- Workbook citation: Zhan et al., 2018, 'Assessment and Determinants of Satisfaction with Urban Livability in China'; Paul & Sen, 2018, 'Livability Assessment within a Metropolis Based on the Impact of Integrated Urban Geographic Factors (IUGFs) on Clustering Urban Centers of Kolkata'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 300 m of a park or public open space; on average the nearest is … away (lower is better).

Access to parks (local, neighbourhood, district, regional) (Access to parks; ID 162) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE

Population with access to parks (Access to parks; ID 162) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE

Data sources

Sources declared in the configuration:

- OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); public open space identified by GHSCI.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_public_open_space_any`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf

Inputs: Public open space and large public green space

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1315`):

```
- variable: sp_walk_nearest_node_public_open_space_any
  name: parks
  domains: [ambient_environment, built_environment]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a parques', en: 'Access to parks'}
  label: {es: 'Parques a menos de 300 m', en: 'Parks within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de un parque o espacio público abierto; en promedio, el más cercano está a {value}', en:
'{access} of the population live within 300 m of a park or public open space; on average the nearest is {value} away'}
  subdomain:
    ambient_environment: {es: 'Medioambiente', en: 'Ambient'}
    built_environment: {es: 'Infraestructura verde y azul', en: 'Green and blue infrastructure'}
  about:
    es: Distancia a pie al espacio público abierto más cercano, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest public open space, scored with a 300 m soft threshold.
  sources:
    - 'OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); public open space identified by GHSCI.'
```

`MX_Mexicali_2025_ULI.yml:475`:

```
- {name: public_open_space_any,   category: pos,  variant: lenient, layer: aos_public_any_nodes_30m_line}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_public_open_space_any	soft access, 300 m threshold, k = 5	−0.197 · 0.299 · 0.796	0.000 – 0.993	0.80 / −1.10 / 12.9%	none
`uli_plain`	sp_walk_nearest_node_public_open_space_any	soft access, 300 m threshold, k = 5	−0.197 · 0.299 · 0.796	0.000 – 0.993	0.80 / −1.10 / 12.9%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

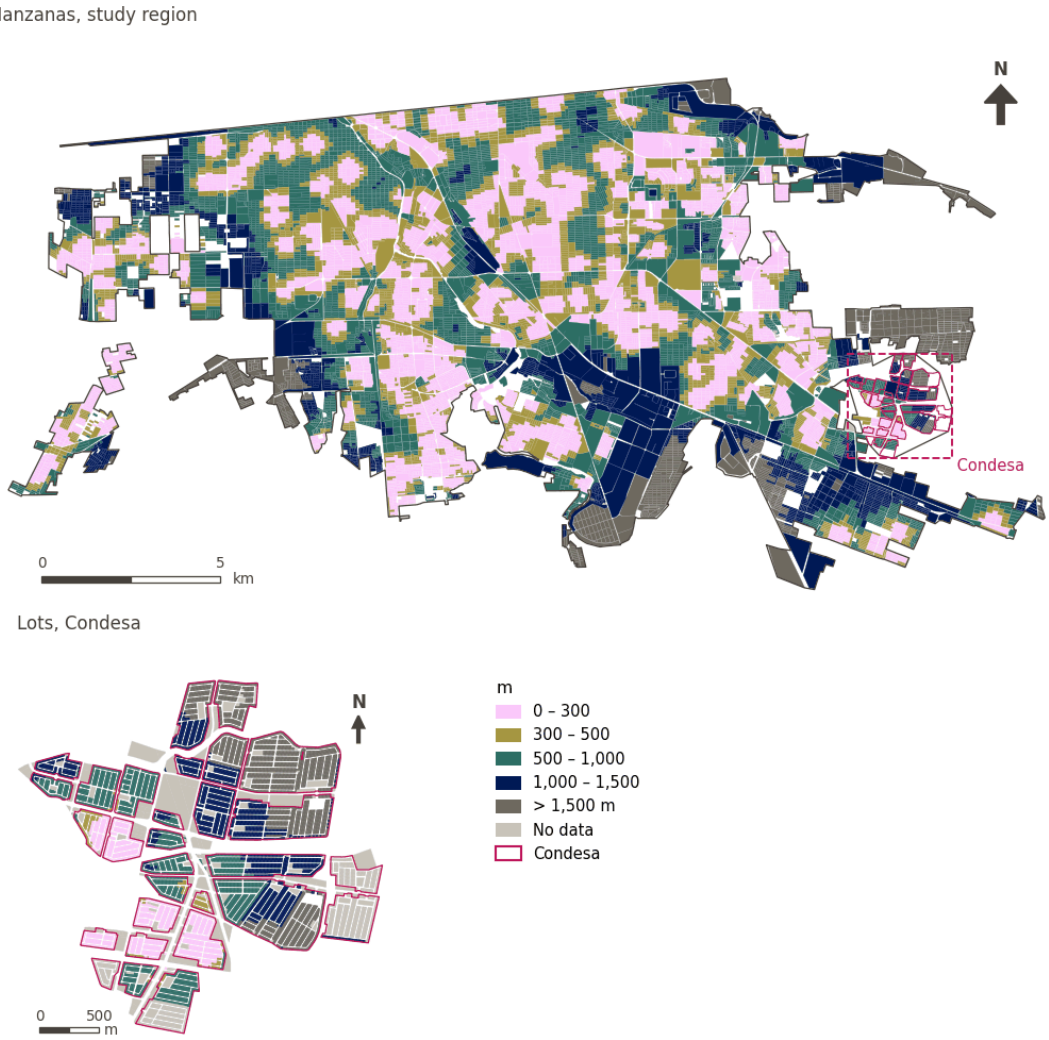
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	469 m	849 m
Region score (`uli`)	+3.7	−9.9
Areas with a value	13,268 of 14,252 manzanas	9,054 of 14,988 lots
Median (P25–P75), weighted	357 m (205 m – 637 m)	811 m (469 m – 1,309 m)
Range	5 m – 2,271 m	11 m – 2,609 m
Nothing within the distance searched	5.0%	31.7%
Score: median (P25–P75) (`uli`)	−1.0 (−17.9 – +27.4)	−18.1 (−18.1 – −18.0)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 36.9% of the population live within 300 m of a park or public open space; on average the nearest is 469 m away.
- Condesa: 14.8% of the population live within 300 m of a park or public open space; on average the nearest is 849 m away.
- For the study region as a whole the value is 469 m; in Condesa, 849 m. Condesa's score lies 9.9 points below the study region average (+3.7 for the study region).

Map



Map. Parks within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Where the sheet pairs access with the percentage of the population with access, one indicator serves both: at a sample point they are the same measure (configuration).

Large public green space within 500 m

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> Walking access within 500 m to a large public green space.

Identity

Identifier	`lpugs`
Core measure (variable sheet)	Large public open space accessibility
Sheet / register ID	91, 187 · #91, #187
Sample point variable	`sp_access_large_public_green_space_score`
Area variable (measured value)	`pct_access_500m_large_public_green_space_score` (%)
Normalised score	`index_uli__lpugs`
Domains (share of weight)	Ambient environment (50%), Built environment (50%)
Subdomain	Ambient environment: Ambient; Built environment: Green and blue infrastructure
Lens	Accessibility
Better when	higher
Effective weight in the index	`uli` 4.0% · `uli_plain` 3.8%
Status	Scored in the published index

Rationale and interpretation

> Source article not available for review.
> *— Indicator register #91: Access to public green space (Built Environment / Green & Blue Infrastructure)*

- Workbook citation: Not reported.

> Availability of open spaces provides urban residents with areas for comfort, enjoyment, and relaxation, directly contributing to their overall quality of life.
> *— Indicator register #187: Access to public open space (Built Environment / Green & Blue Infrastructure)*

- Workbook citation: Zhan et al., 2018, 'Assessment and Determinants of Satisfaction with Urban Livability in China'; Paul & Sen, 2018, 'Livability Assessment within a Metropolis Based on the Impact of Integrated Urban Geographic Factors (IUGFs) on Clustering Urban Centers of Kolkata'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 500 m of a large public green space (higher is better).

% of total land area within 500 m pedestrian network distance of LPUGS (LPUGS Accesibility) (Large Public Open Spaces Accesibility; ID 91, 187) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE - *Notes (sheet):* Exclude inaccessible/private land. - *Doubts (sheet):* Do we want to include plazas here?

% of population with access to LPUGS within 500 m pedestrian network distance (LPUGS Accesibility) (Large Public Open Spaces Accesibility; ID 91, 187) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE - *Notes (sheet):* Exclude inaccessible/private land. - *Doubts (sheet):* Do we want to include plazas here?

Data sources

Sources declared in the configuration:

- GHSCI large public urban green space indicator (Google Earth Engine), with OpenStreetMap public open space.

Source recorded by GHSCI (data dictionary) (`pct_access_500m_large_public_green_space_score`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf

Inputs: Public open space and large public green space

Method: theory

1. Large public green spaces: OpenStreetMap public open spaces of at least 1 ha with annual mean NDVI ≥ 0.2 (Sentinel-2), identified in Google Earth Engine (R8).
2. Distance along the pedestrian network to the nearest entry point (R2), and binary access within 500 m (R3).
3. For areas, the percentage of points with access. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R8. Earth Engine: heat vulnerability and large green space, R2. Network distance to the nearest destination, censored, R3. Banded access, and distance from a disamenity, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1330`):

```
- variable: sp_access_large_public_green_space_score
  name: lpugs
  domains: [ambient_environment, built_environment]
  lens: accessibility
  group: {es: 'Accesibilidad a grandes espacios públicos abiertos', en: 'Large public open space accessibility'}
  label: {es: 'Gran espacio verde público a menos de 500 m', en: 'Large public green space within 500 m'}
  reading: {es: '{value} de la población vive a menos de 500 m de un gran espacio verde público', en: '{value} of the population live within 500 m of a
large public green space'}
  subdomain:
    ambient_environment: {es: 'Medioambiente', en: 'Ambient'}
    built_environment: {es: 'Infraestructura verde y azul', en: 'Green and blue infrastructure'}
  about:
    es: Acceso a pie, a 500 m, a un espacio verde público grande.
    en: Walking access within 500 m to a large public green space.
  sources:
    - 'GHSCI large public urban green space indicator (Google Earth Engine), with OpenStreetMap public open space.'
```

Code (GHSCI):

- `process/subprocesses/_earth_engine_indicators.py:282` `lpugs_analysis()`
- `process/subprocesses/_earth_engine_indicators.py:664` `lpugs_indicators()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_access_large_public_green_space_score	as measured	−0.419 · 0.081 · 0.581	0.000 – 1.000	3.08 / 7.51 / 92.0%	skewed, tied
`uli_plain`	sp_access_large_public_green_space_score	as measured	−0.419 · 0.081 · 0.581	0.000 – 1.000	3.08 / 7.51 / 92.0%	skewed, tied

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

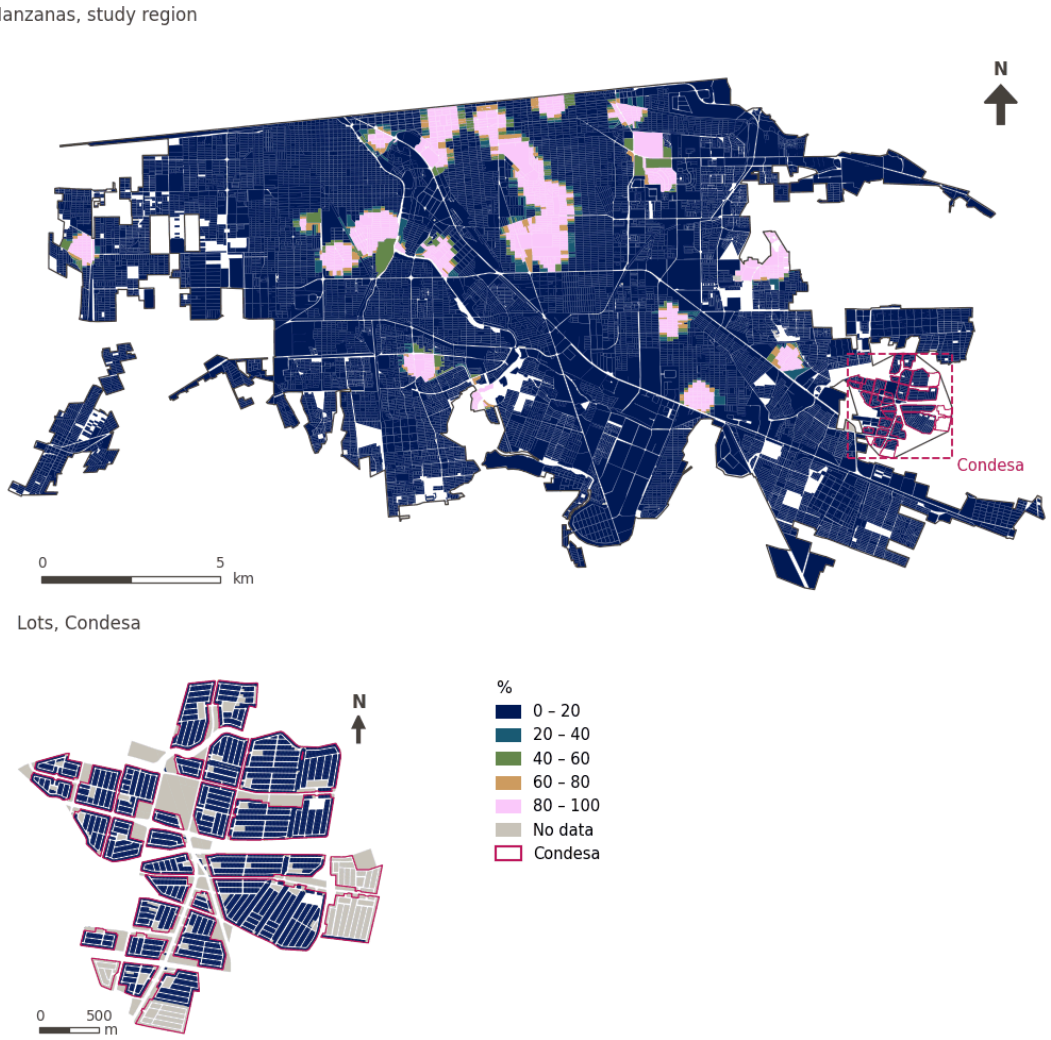
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	8.2%	0.0%
Region score (`uli`)	+0.1	−4.8
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.0% (0.0% – 0.0%)	0.0% (0.0% – 0.0%)
Range	0.0% – 100.0%	0.0% – 0.0%
Score: median (P25–P75) (`uli`)	−4.8 (−4.8 – −4.8)	−4.8 (−4.8 – −4.8)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: 8.2% of the population live within 500 m of a large public green space.
- Condesa: 0.0% of the population live within 500 m of a large public green space.
- For the study region as a whole the value is 8.2%; in Condesa, 0.0%. Condesa's score lies 4.8 points below the study region average (+0.1 for the study region).

Map



Map. Large public green space within 500 m: measured value (%) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- This is the accessibility sub-variable, calculated by the core GHSCI workflow. The sheet's separate availability sub-variable (the share of an area's land that is large public green space) describes an area rather than a sample point's surroundings, so it is not scored in the index as it stands (configuration). The sheet asks whether plazas should be included.

Vegetation cover (NDVI)

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> Mean normalised difference vegetation index (NDVI); higher is better.

Identity

Identifier	`vegetation`
Core measure (variable sheet)	Vegetation and shade
Sheet / register ID	135, 378, 389 · #135, #389, #385
Sample point variable	`sp_ext_ndvi`
Area variable (measured value)	`ext_ndvi` (NDVI)
Normalised score	`index_uli__vegetation`
Domains (share of weight)	Ambient environment (50%), Built environment (50%)
Subdomain	Ambient environment: Ambient; Built environment: Green and blue infrastructure
Lens	Quality
Better when	higher
Effective weight in the index	`uli` 4.0% · `uli_plain` 3.8%
Status	Scored in the published index

Rationale and interpretation

> The Normalised Difference Vegetation Index (NDVI) is used to track city-wide greenness, which supports mental health and environmental sustainability.
> — *Indicator register #135: NDVI (Ambient Environment / Ambient)*

- Workbook citation: Sachs et al., 2019, 'Six transformations to achieve the sustainable development goals'; Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Mean vegetation coverage across the city helps mitigate the urban heat island effect and provides psychological health benefits through exposure to nature.
> — *Indicator register #389: Vegetation percent (Ambient Environment / Ambient)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'; Sachs et al., 2019, 'Six transformations to achieve the sustainable development goals'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Vegetation as a land use type is the optimal land category for human settlement, directly supporting liveability and wellbeing through ecosystem services and green space provision.
> — *Indicator register #385: Vegetation (Ambient Environment / Resources)*

- Workbook citation: No external citations provided for this indicator.
- Article list: #28: Chen (2023), 'Evaluation of the livability of arid urban environments under global warming: A multi-parameter approach'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Mean vegetation index (NDVI) of ....

Vegetation coverage percent (Vegetation and shade; ID 135, 378; 389) - *How to calculate (variable sheet):* Mean NDVI, fractional vegetation, or percent vegetated land. Vegetation Percent (mean); Vegetation Percent (standard deviation) - *Notes (sheet):* Do not equate with accessible cooling.

Data sources

Sources declared in the configuration:

- Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).

Source recorded by GHSCI (data dictionary) (`ext_ndvi`):

- Work package indicators: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)

Inputs: Work package indicators (linked)

Method: theory

1. A value prepared outside GHSCI by a work package and linked to the reference areas (R7: by geometry against the reference areas).

2. Replicated to each sample point from the area containing it (`grid`, `condesa_lotes (15 m)`, `manzanas (30 m)`).

3. AMPI normalisation (R11) of the value as measured, aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R7. Linking externally prepared indicators, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1299`):

```
- variable: sp_ext_ndvi
  name: vegetation
  domains: [ambient_environment, built_environment]
  polarity: positive
  lens: quality
  group: {es: 'Vegetación y sombra', en: 'Vegetation and shade'}
  label: {es: 'Cobertura vegetal (NDVI)', en: 'Vegetation cover (NDVI)'}
  reading: {es: 'Índice de vegetación (NDVI) medio de {value}', en: 'Mean vegetation index (NDVI) of {value}'}
  subdomain:
    ambient_environment: {es: 'Medioambiente', en: 'Ambient'}
    built_environment: {es: 'Infraestructura verde y azul', en: 'Green and blue infrastructure'}
  about:
    es: Índice de vegetación de diferencia normalizada (NDVI) medio; mayor es mejor.
    en: Mean normalised difference vegetation index (NDVI); higher is better.
  sources:
    - 'Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).'
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:403` `match_by_geometry()`
- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_ext_ndvi	as measured	−0.251 · 0.082 · 0.415	−0.031 – 0.636	2.37 / 12.56 / 0.1%	skewed
`uli_plain`	sp_ext_ndvi	as measured	−0.251 · 0.082 · 0.415	−0.031 – 0.636	2.37 / 12.56 / 0.1%	skewed

Recalculation (Reimagina Urbano)

The value comes from an external work package: a scenario (for example, more vegetation) replaces the area's value, which is replicated to its sample points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

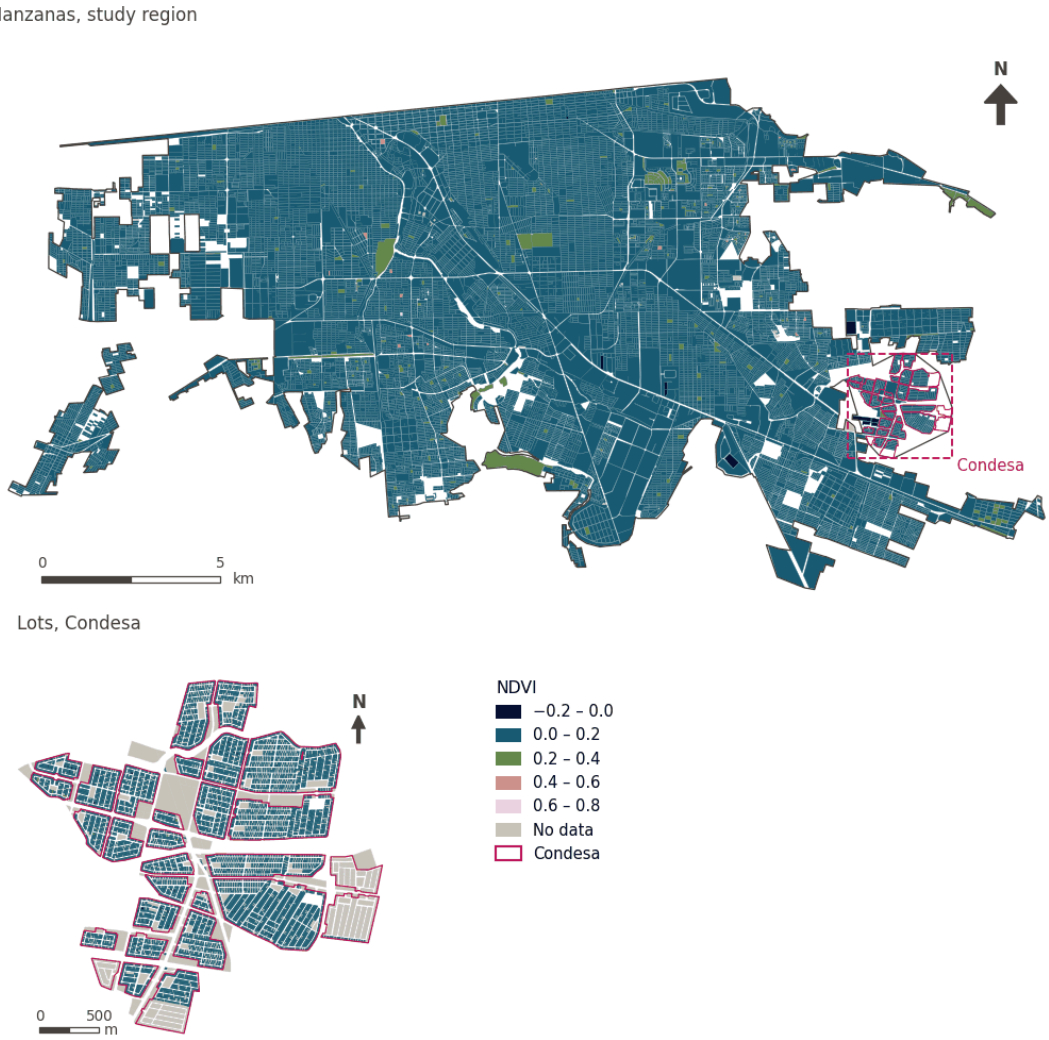
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.08	0.05
Region score (`uli`)	−0.2	−1.8
Areas with a value	14,200 of 14,252 manzanas	11,279 of 14,988 lots
Median (P25–P75), weighted	0.07 (0.06 – 0.08)	0.04 (0.03 – 0.06)
Range	−0.03 – 0.61	−0.08 – 0.34
Score: median (P25–P75) (`uli`)	−0.4 (−1.4 – +0.8)	−2.3 (−3.3 – −0.8)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean vegetation index (NDVI) of 0.08.
- Condesa: Mean vegetation index (NDVI) of 0.05.
- For the study region as a whole the value is 0.08; in Condesa, 0.05. Condesa's score lies 1.8 points below the study region average (−0.2 for the study region).

Map



Map. Vegetation cover (NDVI): measured value (NDVI) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- Not to be equated with accessible cooling (sheet note). Tree canopy, a better proxy for shade, covers only 210 manzanas and is not used (configuration).

Proximity to major roads

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> Length of major roads within 300 m, a proxy for exposure to traffic noise, pollution and danger; lower is better.

Identity

Identifier	`major_roads`
Core measure (variable sheet)	Proximity to high levels of traffic
Sheet / register ID	125 · #125
Sample point variable	`sp_ext_major_road_length_300m`
Area variable (measured value)	`ext_major_road_length_300m` (m)
Normalised score	`index_uli__major_roads`
Domains (share of weight)	Ambient environment (50%), Safety and protection (50%)
Subdomain	Ambient environment: Ambient; Safety and protection: Road safety
Lens	Proximity
Better when	lower
Effective weight in the index	`uli` 5.9% · `uli_plain` 5.7%
Status	Scored in the published index

Rationale and interpretation

> Major roads are a source of air and noise pollution associated with negative health impacts
> *— Indicator register #125: Access to major road (Built Environment / Urban Morphology)*

- Workbook citation: Zhan et al., 2018, 'Assessment and Determinants of Satisfaction with Urban Livability in China'; Tao et al., 2015, 'Housing Choices of Migrant Workers in China: Beyond the Hukou Perspective'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. On average, … of major roads within 300 m (lower is better).

Proximity to major roads (Proximity to high levels of traffic; ID 125) - *How to calculate (variable sheet):* Length of major roads within a 500m radius of each block/ageb

Data sources

Sources declared in the configuration:

- Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).

Source recorded by GHSCI (data dictionary) (`ext_major_road_length_300m`):

- Work package indicators: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)

Inputs: Work package indicators (linked)

Method: theory

1. A value prepared outside GHSCI by a work package and linked to the reference areas (R7: by geometry against the reference areas).
2. Replicated to each sample point from the area containing it (`grid`, `condesa_lotes (15 m)`, `manzanas (30 m)`).
3. AMPI normalisation (R11) of the value as measured, aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R7. Linking externally prepared indicators, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1346`):

```yaml
- variable: sp_ext_major_road_length_300m
  name: major_roads
  domains: [ambient_environment, safety]
  polarity: negative
  lens: proximity
  group: {es: 'Proximidad a altos niveles de tráfico', en: 'Proximity to high levels of traffic'}
  label: {es: 'Proximidad a vialidades principales', en: 'Proximity to major roads'}
  reading: {es: 'En promedio, {value} de vialidades principales a 300 m', en: 'On average, {value} of major roads within 300 m'}
  subdomain:
    ambient_environment: {es: 'Medioambiente', en: 'Ambient'}
    safety: {es: 'Seguridad vial', en: 'Road safety'}
  about:
    es: Longitud de vialidades principales a 300 m, una aproximación a la exposición al ruido, la contaminación y el riesgo del tráfico; menor es mejor.
    en: Length of major roads within 300 m, a proxy for exposure to traffic noise, pollution and danger; lower is better.
  sources:
    - 'Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).'
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:403` `match_by_geometry()`
- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_ext_major_road_length_300m	as measured	−2,429 · 1,756 · 5,940	0 – 8,368	0.62 / 0.16 / 20.7%	none
`uli_plain`	sp_ext_major_road_length_300m	as measured	−2,429 · 1,756 · 5,940	0 – 8,368	0.62 / 0.16 / 20.7%	none

Recalculation (Reimagina Urbano)

The value comes from an external work package: a scenario (for example, more vegetation) replaces the area's value, which is replicated to its sample points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

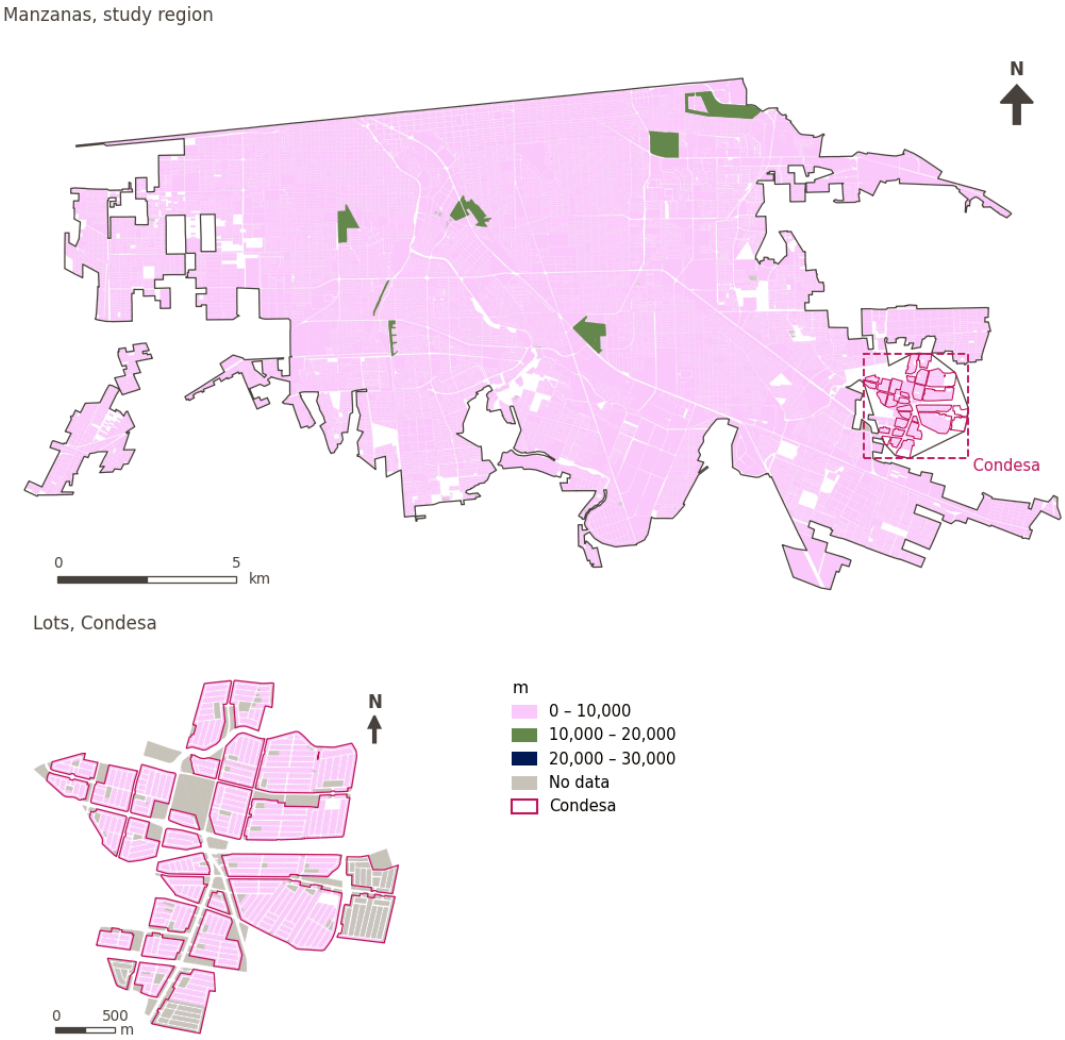
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	1,996 m	414 m
Region score (`uli`)	−1.7	+8.5
Areas with a value	14,200 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	1,848 m (1,075 m – 2,890 m)	0 m (0 m – 949 m)
Range	0 m – 16,859 m	0 m – 2,115 m
Score: median (P25–P75) (`uli`)	+0.4 (−6.5 – +5.5)	+12.6 (+3.0 – +12.6)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: On average, 1,996 m of major roads within 300 m.
- Condesa: On average, 414 m of major roads within 300 m.
- For the study region as a whole the value is 1,996 m; in Condesa, 414 m. Condesa's score lies 8.5 points above the study region average (−1.7 for the study region).

Map



Map. Proximity to major roads: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The sheet describes major road length within 500 m; the delivered variable is within 300 m (final_indicators.gpkg).

Food within 1.5 km by low-stress cycling

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> Distance to the nearest fresh food or convenience store using only low-stress cycling streets (LTS 1-2), scored with a 1.5 km soft threshold.

Identity

Identifier	`cycle_low_stress_food`
Core measure (variable sheet)	Level of traffic stress for cycling
Sheet / register ID	#62
Sample point variable	`sp_cycle_safe_nearest_node_fresh_food_pooled`
Area variable (measured value)	`avg_cycle_dist_safe_fresh_food_pooled` (m)
Normalised score	`index_uli__cycle_low_stress_food`
Domains (share of weight)	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)
Subdomain	Safety and protection: Road safety; Mobility and transport: Active mobility; Built environment: Urban design
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 6.2% · `uli_plain` 6.2%
Status	Scored in the published index

Rationale and interpretation

> Designing side alleys to promote walking and cycling integrates residents into the urban fabric and supports a healthy, human-oriented environment.
> *— Indicator register #62: Level of cycling traffic stress (Mobility & Transport / Active Travel)*

- Workbook citation: Skalicky & Čerpes, 2019, 'Comprehensive assessment methodology for liveable residential environment'
- Article list: #23: Sarkar (2020), 'Socio-physical liveability through socio-spatiality in low-income resettlement archetypes - A case of slum rehabilitation housing in Mumbai, India.'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. On average the nearest fresh food or convenience store is … away by low-stress streets (lower is better).

Average distance to nearest (Level of Traffic Stress for Cycling; ID –) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE - *Doubts (sheet):* This is a suite of variables that we need to disaggregate.

Data sources

Sources declared in the configuration:

- OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); level of traffic stress classified by GHSCI.
- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_cycle_dist_safe_fresh_food_pooled`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf

Inputs: Cycling network by level of traffic stress (LTS)

Method: theory

1. Cycling network classified by level of traffic stress (LTS 1–4) and restricted to low-stress links (LTS ≤ 2) where cycling or walking the bike is permitted (R6).
2. Distance along that network to the nearest destination, `d` (R2), searched to 5,000 m.
3. Soft access (R10): `t = 1,500 m`, `k = 5`; no value → 0. Then R11–R13.

Method: practice

Recipes: R1. Sample points and the pedestrian network, R6. Level of traffic stress and the low-stress cycling network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1440`):

```
- variable: sp_cycle_safe_nearest_node_fresh_food_pooled
  name: cycle_low_stress_food
  domains: [safety, mobility_transport, built_environment]
  transform: {soft_threshold: 1500}
  group: {es: 'Nivel de estrés del tráfico para ciclismo', en: 'Level of traffic stress for cycling'}
  label: {es: 'Alimentos a menos de 1.5 km en bicicleta de bajo estrés', en: 'Food within 1.5 km by low-stress cycling'}
  reading: {es: 'En promedio, la tienda de alimentos más cercana está a {value} por calles de bajo estrés', en: 'On average the nearest fresh food or
convenience store is {value} away by low-stress streets'}
  subdomain:
    safety: {es: 'Seguridad vial', en: 'Road safety'}
    mobility_transport: {es: 'Movilidad activa', en: 'Active mobility'}
    built_environment: {es: 'Diseño urbano', en: 'Urban design'}
  about:
    es: Distancia a la tienda de alimentos frescos o de conveniencia más cercana usando solo calles de bajo estrés para bicicleta (LTS 1-2), puntuada con
un umbral suave de 1.5 km.
    en: Distance to the nearest fresh food or convenience store using only low-stress cycling streets (LTS 1-2), scored with a 1.5 km soft threshold.
  sources:
    - 'OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); level of traffic stress classified by GHSCI.'
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

Code (GHSCI):

- `process/subprocesses/_cycling_lts_network.py:546` `assign_lts()`
- `process/subprocesses/_cycling_accessibility.py:469` `cycling_poi_distance()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_cycle_safe_nearest_node_fresh_food_pooled	soft access, 1,500 m threshold, k = 5	0.059 · 0.555 · 1.052	0.000 – 0.993	−0.34 / −1.76 / 28.6%	none
`uli_plain`	sp_cycle_safe_nearest_node_fresh_food_pooled	soft access, 1,500 m threshold, k = 5	0.059 · 0.555 · 1.052	0.000 – 0.993	−0.34 / −1.76 / 28.6%	none

Recalculation (Reimagina Urbano)

Reclassifying a link's LTS (for example, a new protected cycle lane) changes the low-stress subnetwork; distances must be recomputed for points connected through that link.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

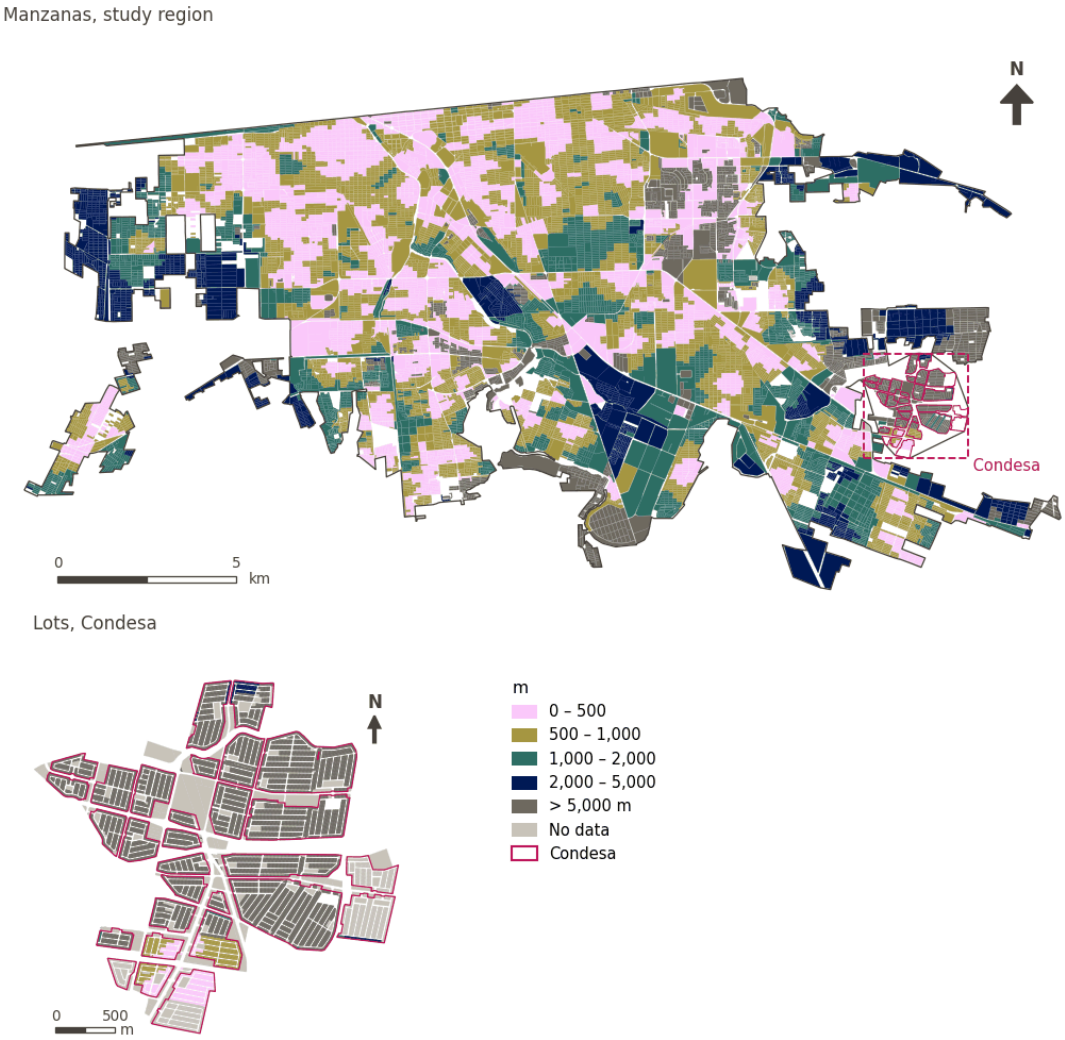
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	826 m	1,025 m
Region score (`uli`)	+10.4	−27.9
Areas with a value	12,510 of 14,252 manzanas	1,530 of 14,988 lots
Median (P25–P75), weighted	677 m (418 m – 1,062 m)	550 m (402 m – 743 m)
Range	10 m – 5,552 m	57 m – 5,142 m
Nothing within the distance searched	9.3%	88.5%
Score: median (P25–P75) (`uli`)	+20.5 (−2.4 – +24.7)	−33.5 (−33.5 – −33.5)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: On average the nearest fresh food or convenience store is 826 m away by low-stress streets.
- Condesa: On average the nearest fresh food or convenience store is 1,025 m away by low-stress streets.
- For the study region as a whole the value is 826 m; in Condesa, 1,025 m. Condesa's score lies 27.9 points below the study region average (+10.4 for the study region).

Map



Map. Food within 1.5 km by low-stress cycling: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The sheet describes level of traffic stress as a suite of variables still to be disaggregated; the destination (fresh food or convenience, from OpenStreetMap) and the 1.5 km threshold are a provisional single choice from that suite.

Block size (perimeter)

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> Mean block perimeter (m); smaller blocks are more walkable; lower is better.

Identity

Identifier	`block_size`
Core measure (variable sheet)	Pedestrian infrastructure presence
Sheet / register ID	–
Sample point variable	`sp_ext_block_perimeter_m`
Area variable (measured value)	`ext_block_perimeter_m` (m)
Normalised score	`index_uli__block_size`
Domains (share of weight)	Built environment (33%), Mobility and transport (33%), Safety and protection (33%)
Subdomain	Built environment: Urban morphology; Mobility and transport: Active mobility; Safety and protection: Road safety
Lens	Presence
Better when	lower
Effective weight in the index	`uli` 6.2% · `uli_plain` 6.2%
Status	Scored in the published index

Rationale and interpretation

Pending: Not recorded in the project register.

How to read it. Blocks average … in perimeter (lower is better).

Block size (Pedestrian infrastructure presence; ID –) - *How to calculate (variable sheet):* Average block length/perimeter - *Notes (sheet):* Smaller blocks are generally more walkable. - *Doubts (sheet):* This one is very important if we consider that Condesa is comprised of multiple gated communities

Data sources

Sources declared in the configuration:

- Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).

Source recorded by GHSCI (data dictionary) (`ext_block_perimeter_m`):

- Work package indicators: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)

Inputs: Work package indicators (linked)

Method: theory

1. A value prepared outside GHSCI by a work package and linked to the reference areas (R7: by geometry against the reference areas).
2. Replicated to each sample point from the area containing it (`grid`, `condesa_lotes (15 m)`, `manzanas (30 m)`).
3. AMPI normalisation (R11) of the value as measured, aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R7. Linking externally prepared indicators, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1390`):

```
- variable: sp_ext_block_perimeter_m
  name: block_size
  domains: [built_environment, mobility_transport, safety]
  polarity: negative
  lens: presence
  group: {es: 'Presencia de infraestructura peatonal', en: 'Pedestrian infrastructure presence'}
  label: {es: 'Tamaño de manzana (perímetro)', en: 'Block size (perimeter)'}
  reading: {es: 'Las manzanas miden en promedio {value} de perímetro', en: 'Blocks average {value} in perimeter'}
  subdomain:
    built_environment: {es: 'Morfología urbana', en: 'Urban morphology'}
    mobility_transport: {es: 'Movilidad activa', en: 'Active mobility'}
    safety: {es: 'Seguridad vial', en: 'Road safety'}
  about:
    es: Perímetro medio de manzana (m); las manzanas más pequeñas son más caminables; menor es mejor.
    en: Mean block perimeter (m); smaller blocks are more walkable; lower is better.
  sources:
    - 'Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).'
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:403` `match_by_geometry()`

- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_ext_block_perimeter_m	as measured	−3,860 · 856 · 5,572	101 – 9,533	3.67 / 20.36 / 0.2%	skewed
`uli_plain`	sp_ext_block_perimeter_m	as measured	−3,860 · 856 · 5,572	101 – 9,533	3.67 / 20.36 / 0.2%	skewed

Recalculation (Reimagina Urbano)

The value comes from an external work package: a scenario (for example, more vegetation) replaces the area's value, which is replicated to its sample points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

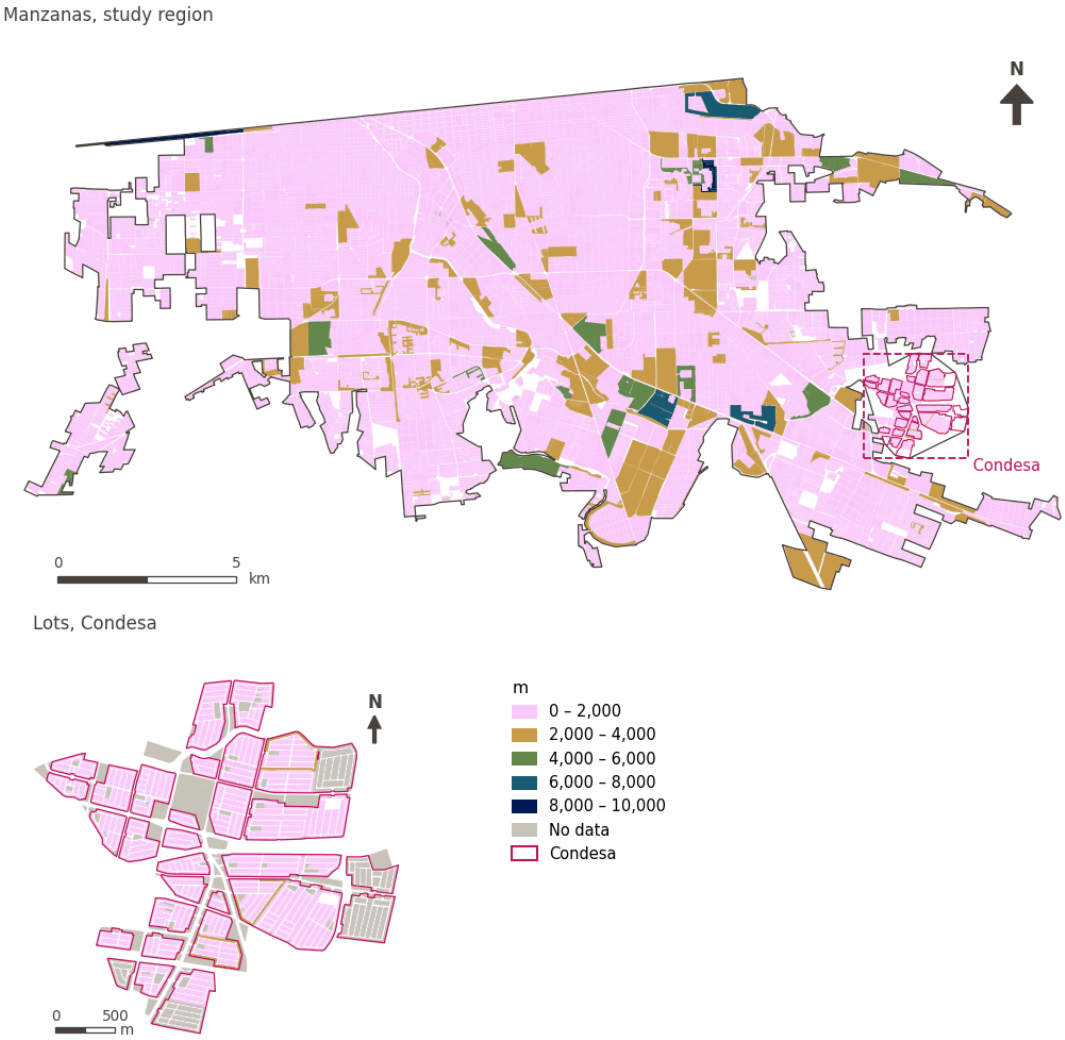
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	677 m	712 m
Region score (`uli`)	+1.1	+1.1
Areas with a value	14,200 of 14,252 manzanas	12,690 of 14,988 lots
Median (P25–P75), weighted	457 m (358 m – 571 m)	509 m (399 m – 837 m)
Range	53 m – 9,533 m	101 m – 2,485 m
Score: median (P25–P75) (`uli`)	+2.2 (+1.1 – +2.7)	+1.3 (+0.2 – +2.2)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Blocks average 677 m in perimeter.
- Condesa: Blocks average 712 m in perimeter.
- For the study region as a whole the value is 677 m; in Condesa, 712 m. Condesa's score lies 1.1 points above the study region average (+1.1 for the study region).

Map



Map. Block size (perimeter): measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The sheet notes this is very important given that Condesa comprises multiple gated communities.

Fire station within 3 km

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> Straight-line distance to the nearest municipal fire station, scored with a 3 km soft threshold (NFPA 1710, about four minutes at 45 km/h).

Identity

Identifier	`fire_3km`
Core measure (variable sheet)	Personal and road safety
Sheet / register ID	69, 174 · #69
Sample point variable	`sp_euclid_dist_imip_fire_station`
Area variable (measured value)	`avg_euclid_dist_imip_fire_station` (m)
Normalised score	`index_uli__fire_3km`
Domains (share of weight)	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)
Subdomain	Safety and protection: Risks
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 6.2% · `uli_plain` 6.2%
Status	Scored in the published index

Rationale and interpretation

> Close proximity to fire stations ensures rapid emergency response times, which is essential for protecting residents from physical harm and property damage.
> *— Indicator register #69: Access to fire station (Safety / Personal Safety)*

- Workbook citation: Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. On average the nearest fire station is … away in a straight line (lower is better).

% of the population with coverage of of emergency services (Personal and road safety; ID 69, 174) - *How to calculate (variable sheet):* % of the population that is within a network buffer of 70 km around a fire station - *Notes (sheet):* Threshold change from 70km to 3km (NFPA 1710 standard, ~4 min at 45 km/h; 70 km covered the entire muncipality with no variation). ~65% of grid cells within 3 km. Condesa: 40/40.

Data sources

Sources declared in the configuration:

- IMIP equipamiento layer via CFC (10 municipal fire stations); threshold revised by the work package (Alejandro).

Source recorded by GHSCI (data dictionary) (`avg_euclid_dist_imip_fire_station`):

- Municipal fire stations (IMIP): IMIP / CFC, Provided by IMIP via CFC

Inputs: Police and fire stations (straight line)

Method: theory

1. Straight-line distance from each sample point to the nearest station, `d` (R5), searched to 75 km; beyond it, no value.
2. Soft access (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, `t = 3,000 m`, `k = 5`; no value → 0.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R5. Straight-line catchments, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1423`):

```
- variable: sp_euclid_dist_imip_fire_station
  name: fire_3km
  domains: [safety, mobility_transport, built_environment]
  transform: {soft_threshold: 3000}
  group: {es: 'Seguridad personal y vial', en: 'Personal and road safety'}
  label: {es: 'Estación de bomberos a menos de 3 km', en: 'Fire station within 3 km'}
  reading: {es: 'En promedio, la estación de bomberos más cercana está a {value} en línea recta', en: 'On average the nearest fire station is {value} away
in a straight line'}
  subdomain:
    safety: {es: 'Riesgos', en: 'Risks'}
  about:
    es: Distancia en línea recta a la estación de bomberos municipal más cercana, puntuada con un umbral suave de 3 km (NFPA 1710, unos cuatro minutos a 45
km/h).
    en: Straight-line distance to the nearest municipal fire station, scored with a 3 km soft threshold (NFPA 1710, about four minutes at 45 km/h).
  sources:
    - 'IMIP equipamiento layer via CFC (10 municipal fire stations); threshold revised by the work package (Alejandro).'
```

Code (GHSCI):

- `process/subprocesses/_euclidean_accessibility.py:247` `nearest_distances()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_euclid_dist_imip_fire_station	soft access, 3,000 m threshold, k = 5	0.096 · 0.592 · 1.089	0.000 – 0.993	−0.53 / −1.28 / 0.1%	none
`uli_plain`	sp_euclid_dist_imip_fire_station	soft access, 3,000 m threshold, k = 5	0.096 · 0.592 · 1.089	0.000 – 0.993	−0.53 / −1.28 / 0.1%	none

Recalculation (Reimagina Urbano)

A new station changes `d` at points nearer it than any other: `d = min(d_previous, straight-line distance to the new station)`.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

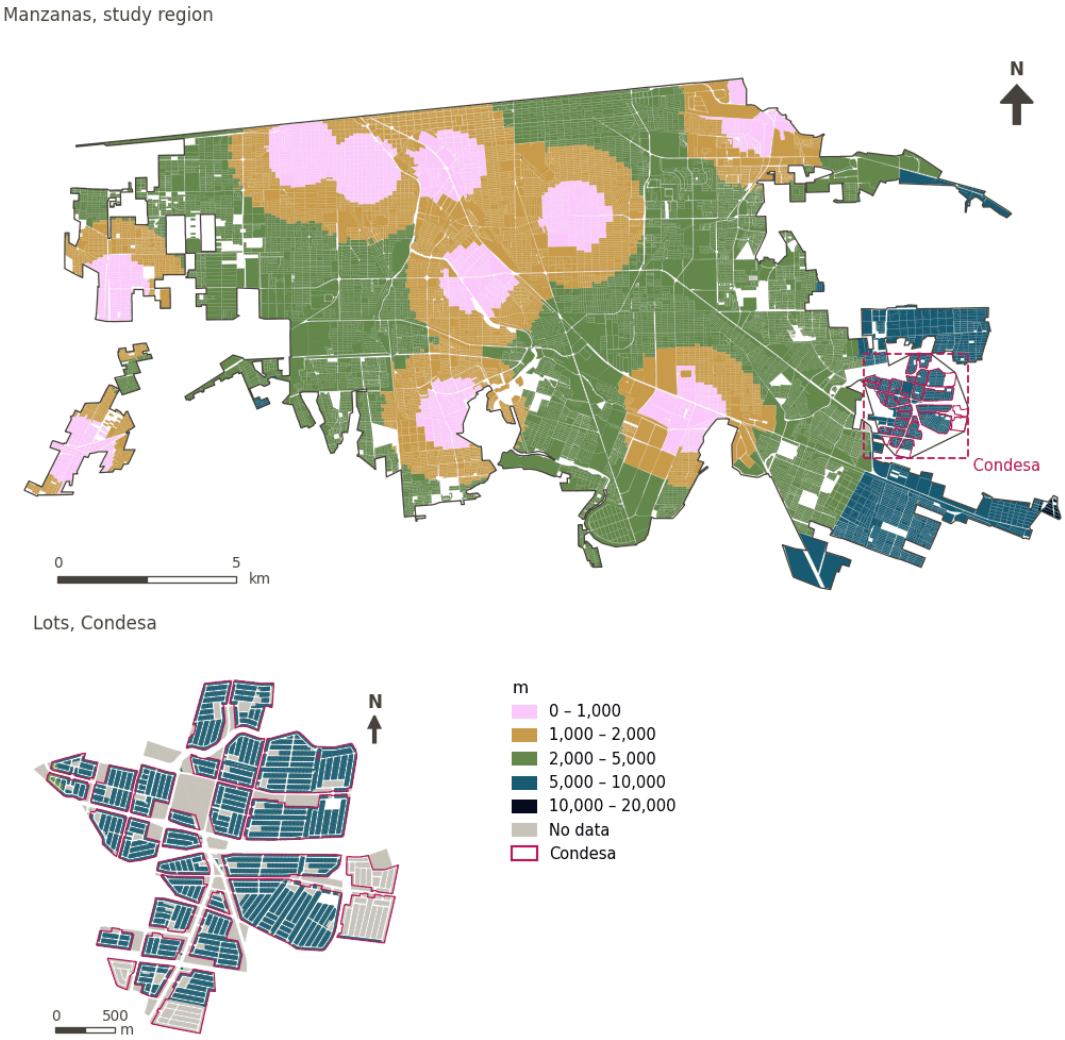
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	2,297 m	6,371 m
Region score (`uli`)	+5.8	−35.4
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	2,293 m (1,501 m – 3,251 m)	6,440 m (5,918 m – 6,830 m)
Range	71 m – 10,582 m	4,920 m – 7,586 m
Score: median (P25–P75) (`uli`)	+10.4 (−11.8 – +20.0)	−35.6 (−35.7 – −35.3)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: On average the nearest fire station is 2,297 m away in a straight line.
- Condesa: On average the nearest fire station is 6,371 m away in a straight line.
- For the study region as a whole the value is 2,297 m; in Condesa, 6,371 m. Condesa's score lies 35.4 points below the study region average (+5.8 for the study region).

Map



Map. Fire station within 3 km: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Threshold revised from 70 km to 3 km (NFPA 1710, about four minutes at 45 km/h); 70 km covered the entire municipality (sheet note). The straight-line search for fire stations has no cap below its 75 km catchment.

Police station within 2 km

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> Straight-line distance to the nearest police facility, scored with a 2 km soft threshold.

Identity

Identifier	`police_2km`
Core measure (variable sheet)	Personal and road safety
Sheet / register ID	69, 174 · #174
Sample point variable	`sp_euclid_dist_imip_police_station`
Area variable (measured value)	`avg_euclid_dist_imip_police_station` (m)
Normalised score	`index_uli__police_2km`
Domains (share of weight)	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)
Subdomain	Safety and protection: Personal safety
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 6.2% · `uli_plain` 6.2%
Status	Scored in the published index

Rationale and interpretation

> Neighborhoods located near police stations experience enhanced security and stability, allowing residents to live without feeling threatened or harmed.
> *— Indicator register #174: Access to police station (Safety / Personal Safety)*

- Workbook citation: Fu et al., 2019, 'The Livable Urban Landscape: GIS and Remote Sensing Extracted Land Use Assessment for Urban Livability in Changchun Proper, China'; Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'; Paul & Sen, 2018, 'Livability Assessment within a Metropolis Based on the Impact of Integrated Urban Geographic Factors (IUGFs) on Clustering Urban Centers of Kolkata'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'; #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'; #8: Onnom (2018), 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. On average the nearest police station is … away in a straight line (lower is better).

% of the population with coverage of police stations (Personal and road safety; ID 69, 174) - *How to calculate (variable sheet):* % of the population that is within a network buffer of 15 km around a police station - *Notes (sheet):* Threshold change from 15km to 2km (road network unavailable; 15km covered 100% of municipality with no variation). Euclidean distance. ~64% of grid cells within 2 km. Condesa: 40/40. - *Doubts (sheet):* For this one we need to define a redius that covers multiple neighborhoods (review SEDESOL)

Data sources

Sources declared in the configuration:

- IMIP equipamiento layer via CFC (27 police facilities); threshold revised by the work package (Alejandro).

Source recorded by GHSCI (data dictionary) (`avg_euclid_dist_imip_police_station`):

- Police stations (IMIP): IMIP / CFC, Provided by IMIP via CFC

Inputs: Police and fire stations (straight line)

Method: theory

1. Straight-line distance from each sample point to the nearest station, `d` (R5), searched to 45 km; beyond it, no value.
2. Soft access (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, `t = 2,000 m`, `k = 5`; no value → 0.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R5. Straight-line catchments, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1409`):

```
- variable: sp_euclid_dist_imip_police_station
  name: police_2km
  domains: [safety, mobility_transport, built_environment]
  transform: {soft_threshold: 2000}
  group: {es: 'Seguridad personal y vial', en: 'Personal and road safety'}
  label: {es: 'Estación de policía a menos de 2 km', en: 'Police station within 2 km'}
  reading: {es: 'En promedio, la estación de policía más cercana está a {value} en línea recta', en: 'On average the nearest police station is {value} away
in a straight line'}
  subdomain:
    safety: {es: 'Seguridad personal', en: 'Personal safety'}
  about:
    es: Distancia en línea recta a la instalación policial más cercana, puntuada con un umbral suave de 2 km.
    en: Straight-line distance to the nearest police facility, scored with a 2 km soft threshold.
  sources:
    - 'IMIP equipamiento layer via CFC (27 police facilities); threshold revised by the work package (Alejandro).'
```

Code (GHSCI):

- `process/subprocesses/_euclidean_accessibility.py:247` `nearest_distances()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_euclid_dist_imip_police_station	soft access, 2,000 m threshold, k = 5	0.115 · 0.611 · 1.108	0.000 – 0.993	−0.63 / −1.26 / 0.1%	none
`uli_plain`	sp_euclid_dist_imip_police_station	soft access, 2,000 m threshold, k = 5	0.115 · 0.611 · 1.108	0.000 – 0.993	−0.63 / −1.26 / 0.1%	none

Recalculation (Reimagina Urbano)

A new station changes `d` at points nearer it than any other: `d = min(d_previous, straight-line distance to the new station)`.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

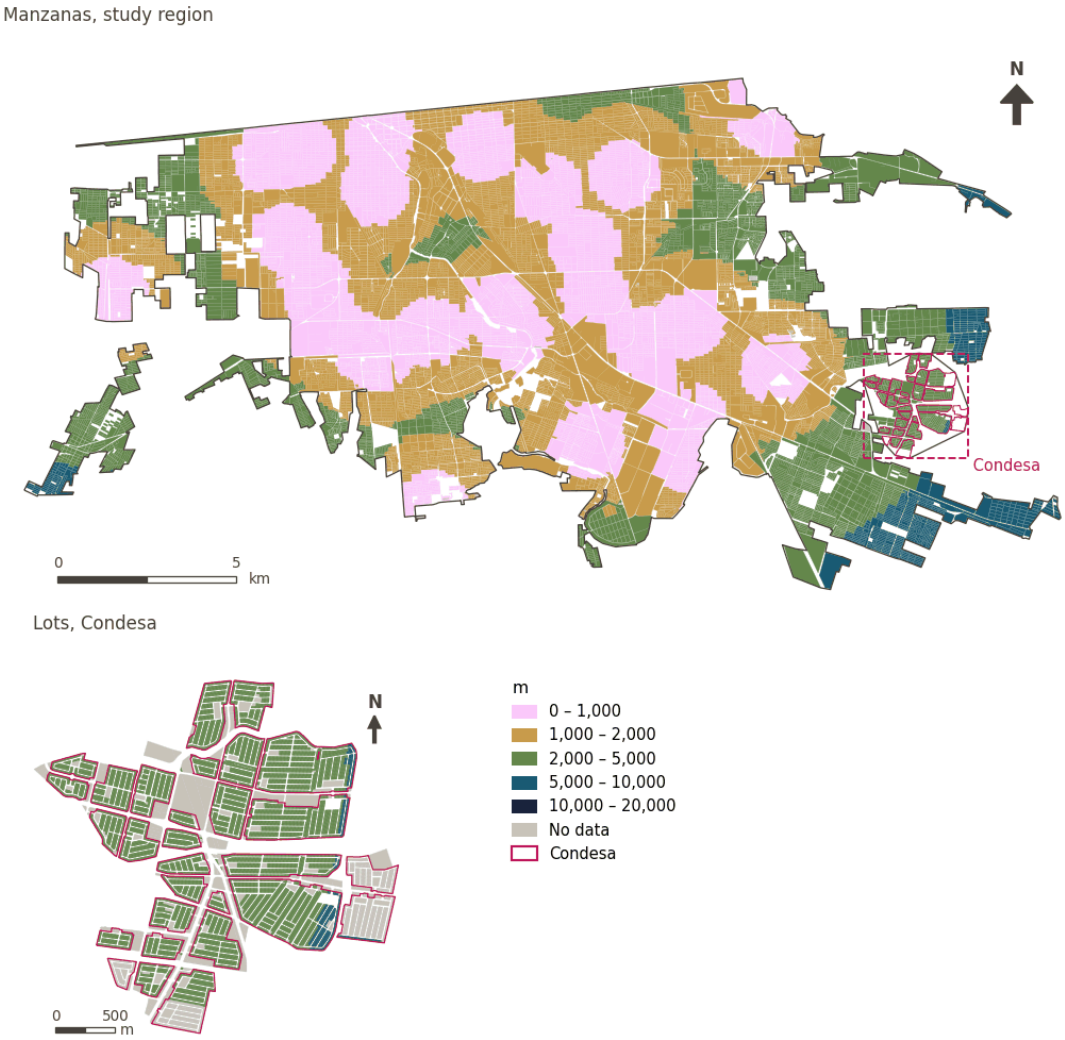
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	1,406 m	4,156 m
Region score (`uli`)	+7.9	−36.1
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	1,284 m (763 m – 2,014 m)	4,220 m (3,813 m – 4,593 m)
Range	53 m – 8,906 m	2,537 m – 5,548 m
Score: median (P25–P75) (`uli`)	+14.8 (−7.3 – +20.8)	−36.7 (−36.8 – −36.3)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: On average the nearest police station is 1,406 m away in a straight line.
- Condesa: On average the nearest police station is 4,156 m away in a straight line.
- For the study region as a whole the value is 1,406 m; in Condesa, 4,156 m. Condesa's score lies 36.1 points below the study region average (+7.9 for the study region).

Map



Map. Police station within 2 km: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The threshold was revised from 15 km to 2 km by the work package: 15 km covered the whole municipality with no variation. Straight-line distance, as the road network was unavailable to the work package (sheet note).

Walkability (300 m), attenuated by thermal comfort

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> Sum of z-scores of access to a fresh food market, convenience store and public transport within 300 m, population density and intersection density. By default its percentile rank is attenuated by daytime thermal comfort, by up to half at the hottest locations.

Identity

Identifier	`walkability`
Core measure (variable sheet)	Walkability index
Sheet / register ID	362, 74, 230, 120, 116 · #74, #362
Sample point variable	`sp_walk_idx_300_tm`
Area variable (measured value)	`walk_idx_300_tm` (0–1)
Normalised score	`index_uli__walkability`
Score (unattenuated variant)	`index_uli_plain__walkability`
Domains (share of weight)	Built environment (33%), Mobility and transport (33%), Social infrastructure (33%)
Subdomain	Built environment: Urban design; Mobility and transport: Active mobility; Social infrastructure: Public services
Lens	Accessibility
Better when	higher
Effective weight in the index	`uli` 3.7% · `uli_plain` 3.7%
Status	Scored in the published index

Rationale and interpretation

> Walking-friendly urban environments support safety and promote healthy lifestyles through human-oriented design.
> *— Indicator register #74: Walkability Index (Safety / Road Safety)*

- Workbook citation: Skalicky & Čerpes, 2019, 'Comprehensive assessment methodology for liveable residential environment'
- Article list: #23: Sarkar (2020), 'Socio-physical liveability through socio-spatiality in low-income resettlement archetypes - A case of slum rehabilitation housing in Mumbai, India.'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> A well-connected pedestrian network promotes a safe environment and directly supports the physical health of residents by encouraging walking.
> *— Indicator register #362: Street connectivity (Built Environment / Urban Morphology)*

- Workbook citation: Ahmed, 2012, 'Urban social sustainability: A study of the Emirati local communities in Al Ain'
- Article list: #23: Sarkar (2020), 'Socio-physical liveability through socio-spatiality in low-income resettlement archetypes - A case of slum rehabilitation housing in Mumbai, India.'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Average walkability score of … (higher is more walkable).

Street connectivity (Walkability Index; ID 362,74) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Average walkable neighbourhood population density (population per km¬≤) of sample points in this area) or Intersection density defined as the number of 3- and 4-way intersections in a buffer area/total buffer area in squared kilometers - *Notes (sheet):* Use pedestrian-relevant network.

Net residential density (Walkability Index; ID 230, 120, 74) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE

Commercial land use proportion (Walkability Index; ID –) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE or Squared kilometers designated to commercial land use in buffer area/total squared kilometers of buffer area.

Land use mix (Walkability Index; ID 116, 74) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE or −[Σi (pi)(ln pi)]/(ln k) where p = proportion of total land uses, i = land use category, ln = natural logarithm, k = number of land uses. Range is 0 to 1. Categoriesb are defined as low (reference)

Data sources

Sources declared in the configuration:

- Frank, L. D. et al. (2010) The development of a walkability index. British Journal of Sports Medicine 44, 924-933.
- OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); INEGI DENUE (May 2026); IMIP / CFC public transport stops; GHS-POP R2023A.
- Thermal comfort: Rossano Schifanella (WP02), SOLWEIG modelled UTCI for 21 July 2023 (Bröde et al. 2012).

Source recorded by GHSCI (data dictionary) (`walk_idx_300_tm`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; Global Human Settlements population data: 2025, Mollweide (EU JRC, 2023): CC BY 4.0, https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/GHSL/GHS_POP_GLOBE_R2023A/GHS_POP_E2025_GLOBE_R2023A_54009_100/V1-0/tiles/GHS_POP_E2025_GLOBE_R2023A_54009_100_V1_0_R6_C8.zip, Schiavina, M., Freire, S., Carioli, A., MacManus, K. (2023). GHS-POP R2023A - GHS population grid multitemporal (1975-2030). European Commission, Joint Research Centre (JRC). https://doi.org/10.2905/2FF68A52-5B5B-4A22-8F40-C41DA8332CFE; Outdoor thermal comfort (UTCI): Rossano Schifanella (WP02); SOLWEIG and the UTCI operational polynomial (Bröde et al. 2012), Terrain from

FABDEM v1 (CC BY-NC-SA 4.0, non-commercial); other inputs CC BY 4.0 or public domain; IMIP / CFC: 2026-05-18, Provided by IMIP via CFC, As per advice from Juan Ernesto Díaz Noguez

Destination filter:

- `pt_any` (pt_any): `MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb` — Destination filter: `"equipamiento_category='transp' AND TIPO != 'TERMINAL INTERIOR DE CARGA (CENTRAL DE SERVICIOS DE CARGA)'"`

Inputs: Walkability within 300 m and its components, Outdoor thermal comfort (UTCI)

Method: theory

1. Daily living within 300 m: `DL = a_market + a_convenience + a_transport` (binary access within 300 m, R3; OpenStreetMap destinations and IMIP/OSM stops).
2. Walkability: `W = z(population density) + z(intersection density) + z(DL)`, z-scores over all urban sample points (R9).
3. Thermal comfort attenuation (R9): `W_tm = percentile_rank(W) · (1 − λ·h')`, with `λ = 0.5` and `h' = clip((UTCI − 30)/(50 − 30), 0, 1)` from the daytime mean UTCI (°C).
4. AMPI normalisation (R11) of `W_tm`, aggregation (R12) and areas (R13). In the unattenuated variant (`uli_plain`) `W` is used and thermal comfort is scored separately.

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R3. Banded access, and distance from a disamenity, R9. Walkability, and its attenuation by thermal comfort, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1368`):

```
- variable: sp_walk_idx_300_tm
  name: walkability
  domains: [built_environment, mobility_transport, social_infrastructure]
  polarity: positive
  lens: accessibility
  group: {es: 'Índice de caminabilidad', en: 'Walkability index'}
  label: {es: 'Caminabilidad (300 m), atenuada por confort térmico', en: 'Walkability (300 m), attenuated by thermal comfort'}
  reading: {es: 'Puntuación media de caminabilidad de {value} (mayor es más caminable)', en: 'Average walkability score of {value} (higher is more
walkable)'}
  subdomain:
    built_environment: {es: 'Diseño urbano', en: 'Urban design'}
    mobility_transport: {es: 'Movilidad activa', en: 'Active mobility'}
    social_infrastructure: {es: 'Servicios públicos', en: 'Public services'}
  about:
    es: Suma de puntuaciones z del acceso a mercado de alimentos frescos, tienda de conveniencia y transporte público a 300 m, la densidad de población y
la densidad de intersecciones. Por defecto, su rango percentil se atenúa según el confort térmico diurno, hasta la mitad en los lugares más calurosos.
    en: Sum of z-scores of access to a fresh food market, convenience store and public transport within 300 m, population density and intersection density.
By default its percentile rank is attenuated by daytime thermal comfort, by up to half at the hottest locations.
  sources:
    - 'Frank, L. D. et al. (2010) The development of a walkability index. British Journal of Sports Medicine 44, 924-933.'
    - 'OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); INEGI DENUE (May 2026); IMIP / CFC public transport stops; GHS-POP R2023A.'
    - 'Thermal comfort: Rossano Schifanella (WP02), SOLWEIG modelled UTCI for 21 July 2023 (Bröde et al. 2012).'
```

Code (GHSCI):

- `process/subprocesses/_walkability_variants.py:275` `compute_variants()`
- `process/subprocesses/_walkability_variants.py:270` `percentile_rank()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_idx_300_tm	as measured	−0.022 · 0.315 · 0.651	0.000 – 0.673	0.00 / −1.21 / 0.1%	none
`uli_plain`	sp_walk_idx_300	as measured	−10.0 · 0.0 · 10.0	−8.9 – 11.1	0.23 / 0.11 / 0.2%	none

Recalculation (Reimagina Urbano)

Walkability uses z-scores and percentile ranks over all points; for a scenario, freeze each component's baseline mean and standard deviation and the baseline distribution of `W` (from the `sample_points_walkability` table and its inputs in the database), so that only the affected points change. These values are not published in the results files: they must be extracted from the database.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

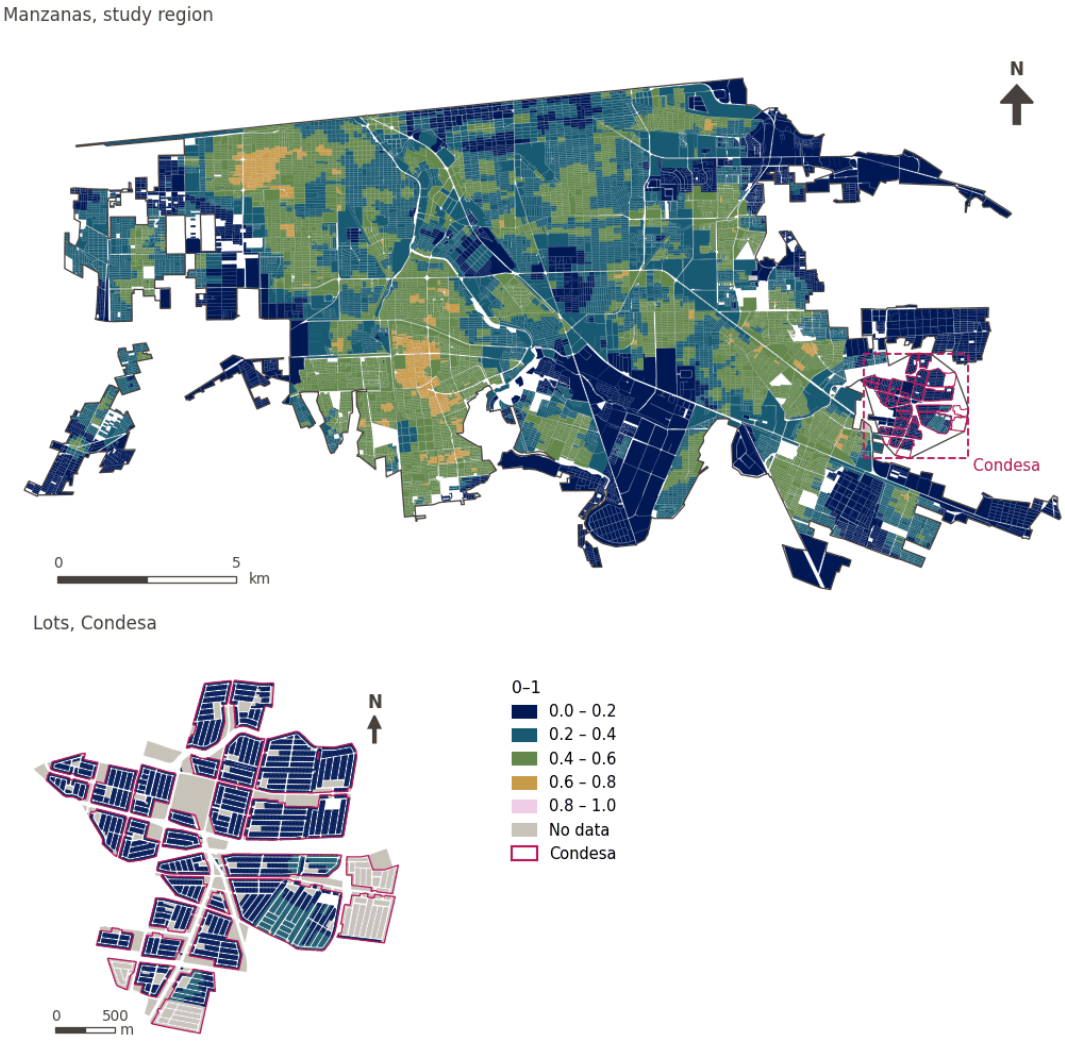
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.41	0.12
Region score (`uli`)	+8.9	−17.5
Region score (`uli_plain`)	+3.1	−5.8

Measure	Study region (Mexicali incl. Condesa)	Condesa
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.42 (0.30 – 0.52)	0.11 (0.08 – 0.15)
Range	0.00 – 0.65	0.01 – 0.26
Score: median (P25–P75) (`uli`)	+9.1 (−0.9 – +18.3)	−18.2 (−20.9 – −14.8)
Score: median (P25–P75) (`uli_plain`)	+2.5 (−0.4 – +5.8)	−6.1 (−7.1 – −4.7)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Average walkability score of 0.41 (higher is more walkable).
- Condesa: Average walkability score of 0.12 (higher is more walkable).
- For the study region as a whole the value is 0.41; in Condesa, 0.12. Condesa's score lies 17.5 points below the study region average (+8.9 for the study region).

Map



Map. Walkability (300 m), attenuated by thermal comfort: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- Net residential density, commercial land use and land use mix (final_indicators.gpkg) are the components of the walkability core measure, which walkability itself represents: scoring them as well would count density twice (configuration).
- The fixed 30–50 °C range judges heat absolutely: most of the city lies between 43.8 and 46.0 °C on the modelled day, all of it very hot, and re-scaling between those percentiles would have treated the least hot as comfortable (configuration, decided 25 September 2026).

250 m or more from petrol stations

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> Whether the nearest petrol station is 250 m or more away on foot (a disamenity when closer).

Identity

Identifier	`petrol_250m`
Core measure (variable sheet)	Daily essential-services access
Sheet / register ID	169 · #169
Sample point variable	`sp_walk_beyond_denue_petrol_station_250m`
Area variable (measured value)	`pct_beyond_walk_denue_petrol_station_250m` (%)
Normalised score	`index_uli__petrol_250m`
Domains (share of weight)	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)
Subdomain	Social infrastructure: Public services; Economic development: Resources
Lens	Proximity
Better when	higher
Effective weight in the index	`uli` 3.6% · `uli_plain` 3.6%
Status	Scored in the published index

Rationale and interpretation

> Living at a sufficient distance from gas stations improves residents' safety by reducing the risks of fire incidents and accidents associated with hazardous fuel locations.
>
> *— Indicator register #169: Access to petrol station (>= 250m away) (Economic Development / Resources)*

- Workbook citation: Basu et al., 2021, 'Urban Livability Index Assessment Based on Land-Use Changes in an Indian Medium-Sized City (Raiganj)'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live 250 m or more from a petrol station (higher is better).

Access to petrol station (>= 250m away) (Daily essential-services access; ID 169) - *How to calculate (variable sheet):* Average distance access to petrol stations - *Notes (sheet):* CH: the evidence for this one might be complex -- I can imagine there might be a sweet spot --- yes, you don't want to be too close (how close is that, maybe that is our indicator --- not within XXX distance m, if we can link with citable evidence.) But the reality is people do drive, and need petrol or perhaps electrical charging; living too far from one could impact some aspects of life. Lets be guided by evidence here. Here's a good basis for an indicator: Petrol station within 250m is associated with increased relative risk of childhood leukemia. Another study: https://link.springer.com/article/10.1007/s00420-021-01767-y - *Doubts (sheet):* I am not sure these are relevant, I added them considering that most people use cars in Mexicali for their daily commute

Percentage of population with access to petrol stations (Daily essential-services access; ID 169) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Average distance (m) along the pedestrian network to the nearest 'denue petrol station' (custom destination) (population weighted)) - *Doubts (sheet):* I am not sure these are relevant, I added them considering that most people use cars in Mexicali for their daily commute

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`pct_beyond_walk_denue_petrol_station_250m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_petrol_station` (Petrol stations (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('468411','468413')"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network to the nearest petrol station, `d` (R2).
2. Binary distance-from indicator (R3, `direction: avoid`): `b = 1` if `d > 250 m` or none was found within the distance searched; `b = 0` if one lies within 250 m.
3. For areas, the percentage of points with `b = 1`. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R3. Banded access, and distance from a disamenity, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1523`):

```
- variable: sp_walk_beyond_denue_petrol_station_250m
  name: petrol_250m
  domains: [social_infrastructure, mobility_transport, economic_development]
  lens: proximity
  group: {es: 'Acceso a servicios esenciales cotidianos', en: 'Daily essential-services access'}
  label: {es: 'A 250 m o más de gasolineras', en: '250 m or more from petrol stations'}
  reading: {es: '{value} de la población vive a 250 m o más de una gasolinera', en: '{value} of the population live 250 m or more from a petrol station'}
  subdomain:
    social_infrastructure: {es: 'Servicios públicos', en: 'Public services'}
    economic_development: {es: 'Recursos', en: 'Resources'}
  about:
    es: Si la gasolinera más cercana está a 250 m o más a pie (una desventaja cuando está más cerca).
    en: Whether the nearest petrol station is 250 m or more away on foot (a disamenity when closer).
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:533`:

```
- {name: denue_petrol_station, category: petrol_station, variant: any, layer: destinations, where: "dest_name = 'denue_petrol_station'", distances: [250], direction: avoid}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_accessibility_spec.py:1272` `sample_point_access()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_beyond_denue_petrol_station_250m	as measured	0.421 · 0.921 · 1.421	0.000 – 1.000	−3.13 / 7.77 / 92.1%	skewed, tied
`uli_plain`	sp_walk_beyond_denue_petrol_station_250m	as measured	0.421 · 0.921 · 1.421	0.000 – 1.000	−3.13 / 7.77 / 92.1%	skewed, tied

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

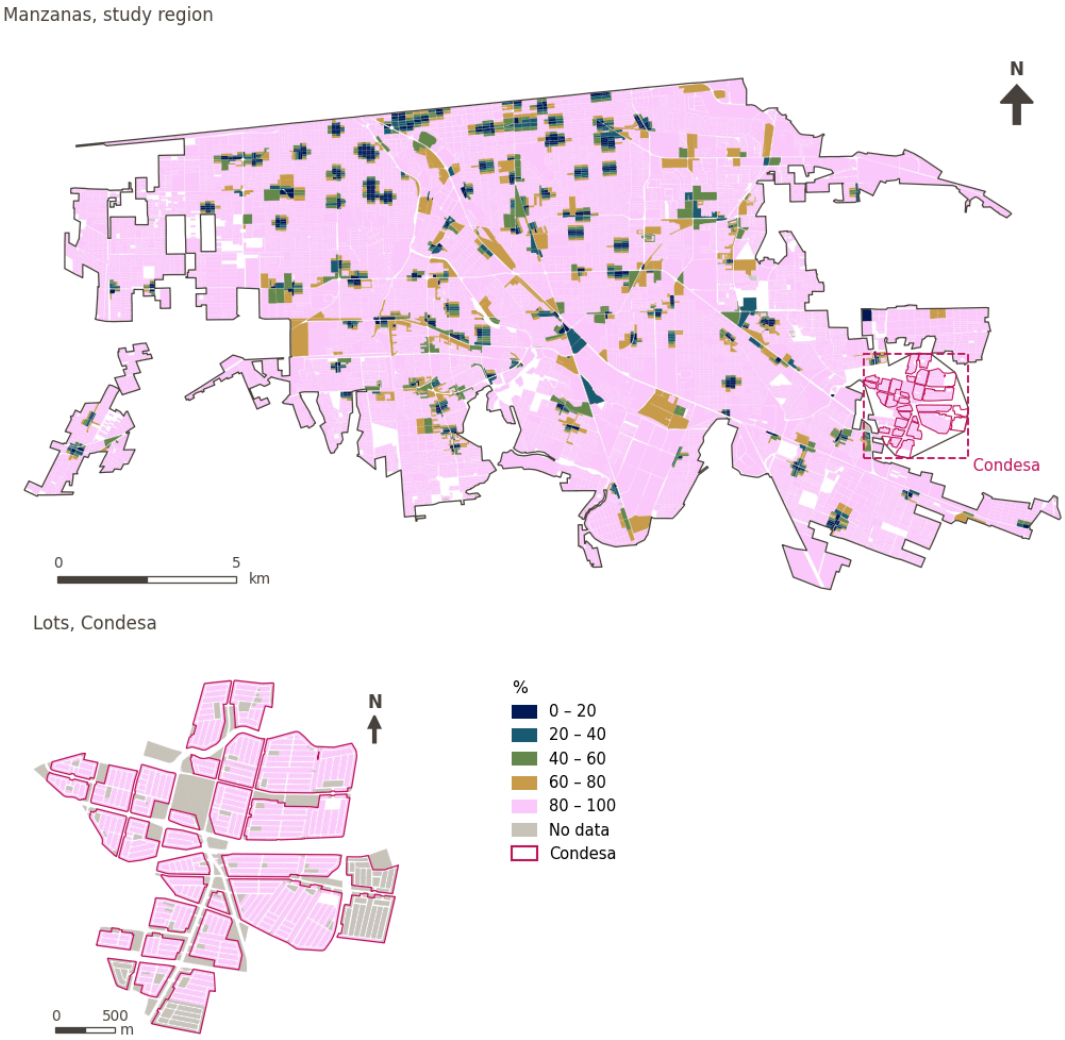
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	92.2%	100.0%
Region score (`uli`)	0.0	+4.7
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	100.0% (100.0% – 100.0%)	100.0% (100.0% – 100.0%)
Range	0.0% – 100.0%	100.0% – 100.0%
Score: median (P25–P75) (`uli`)	+4.7 (+4.7 – +4.7)	+4.7 (+4.7 – +4.7)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: 92.2% of the population live 250 m or more from a petrol station.
- Condesa: 100.0% of the population live 250 m or more from a petrol station.
- For the study region as a whole the value is 92.2%; in Condesa, 100.0%. Condesa's score lies 4.7 points above the study region average (0.0 for the study region).

Map



Map. 250 m or more from petrol stations: measured value (%) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- The sheet's notes cite an increased relative risk of childhood leukaemia within 250 m of a petrol station, and record doubts about relevance, as most people in Mexicali drive.

Convenience stores within 300 m

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> Walking distance to the nearest convenience store, scored with a 300 m soft threshold.

Identity

Identifier	`convenience`
Core measure (variable sheet)	Daily essential-services access
Sheet / register ID	132 · #132
Sample point variable	`sp_walk_nearest_node_denue_food_retail`
Area variable (measured value)	`avg_walk_dist_denue_food_retail` (m)
Normalised score	`index_uli__convenience`
Domains (share of weight)	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)
Subdomain	Social infrastructure: Public services; Mobility and transport: Active mobility; Economic development: Food environment
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 3.6% · `uli_plain` 3.6%
Status	Scored in the published index

Rationale and interpretation

> The availability of minimarts is an indicator of local food access, supporting the convenient provision of daily necessities that shape urban life quality.
> *— Indicator register #132: Access to minimarts (Economic Development / Food Environment)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'; Alderton et al., 2019, 'What is the meaning of urban liveability for a city in a low-to-middle-income country? Contextualising liveability for Bangkok, Thailand'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'

How to read it. … of the population live within 300 m of a convenience store; on average the nearest is … away (lower is better).

Access to convenience stores (Daily essential-services access; ID 132) - *How to calculate (variable sheet):* Walking distance access to convenience stores and minimart

Percentage of population with access to convenience stores (Daily essential-services access; ID 132) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Distance (m) along the pedestrian network to the nearest convenience store (source: OpenStreetMap or custom data))

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_denue_food_retail`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_food_retail` (Food retail, any (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"(codigo_act LIKE '4611%' AND codigo_act NOT IN ('461211','461212','461213','461220')) OR codigo_act LIKE '4621%'"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1506`):

```
- variable: sp_walk_nearest_node_denue_food_retail
  name: convenience
  domains: [social_infrastructure, mobility_transport, economic_development]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a servicios esenciales cotidianos', en: 'Daily essential-services access'}
  label: {es: 'Tiendas de conveniencia a menos de 300 m', en: 'Convenience stores within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de una tienda de conveniencia; en promedio, la más cercana está a {value}', en: '{access}
of the population live within 300 m of a convenience store; on average the nearest is {value} away'}
  subdomain:
    social_infrastructure: {es: 'Servicios públicos', en: 'Public services'}
    mobility_transport: {es: 'Movilidad activa', en: 'Active mobility'}
    economic_development: {es: 'Ambiente de alimentación', en: 'Food environment'}
  about:
    es: Distancia a pie a la tienda de conveniencia más cercana, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest convenience store, scored with a 300 m soft threshold.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:496`:

```
- {name: denue_food_retail, category: food_retail, variant: lenient, layer: destinations, where: "dest_name = 'denue_food_retail'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_denue_food_retail	soft access, 300 m threshold, k = 5	0.179 · 0.675 · 1.172	0.000 – 0.993	−0.94 / −0.77 / 0.5%	none
`uli_plain`	sp_walk_nearest_node_denue_food_retail	soft access, 300 m threshold, k = 5	0.179 · 0.675 · 1.172	0.000 – 0.993	−0.94 / −0.77 / 0.5%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

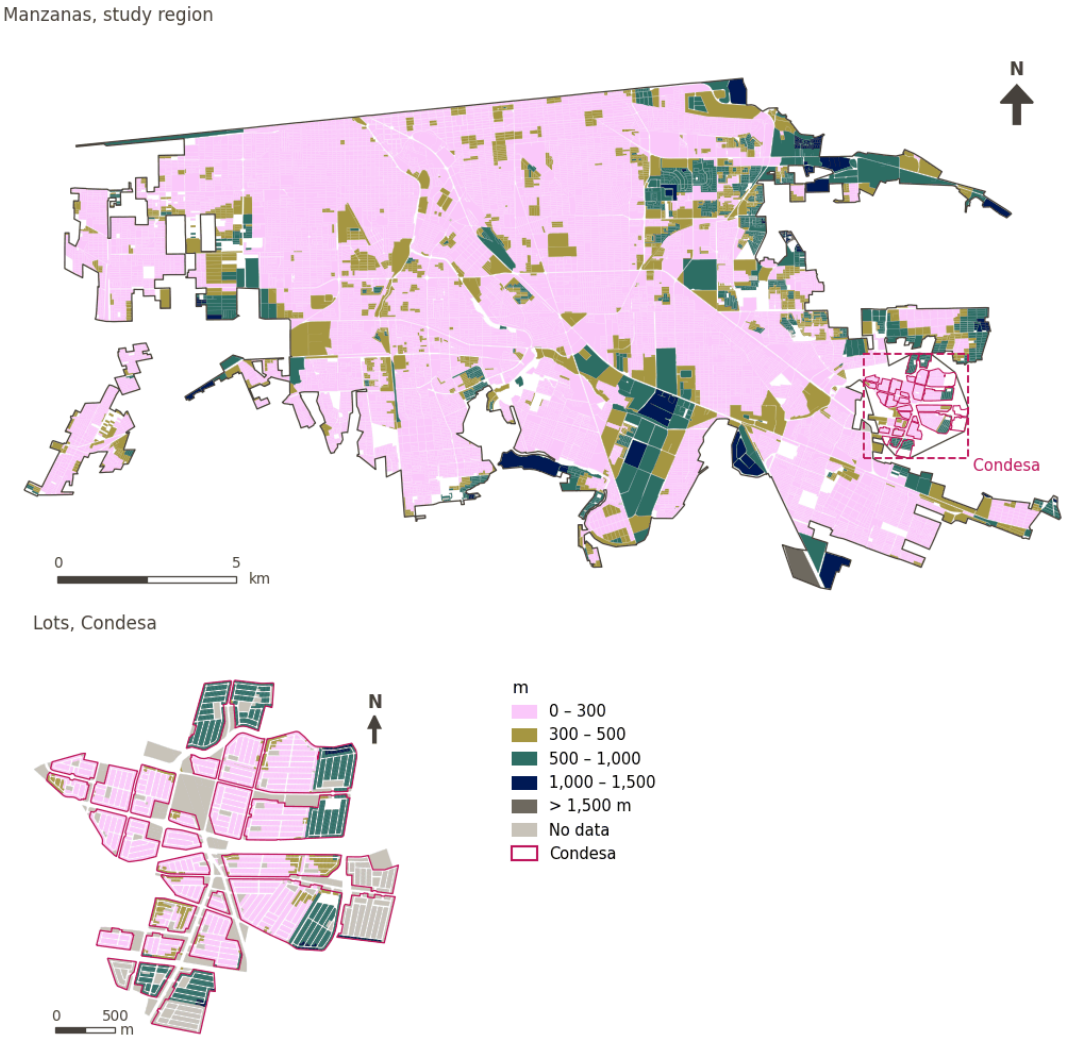
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	181 m	341 m
Region score (`uli`)	+8.0	−4.2
Areas with a value	14,209 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	136 m (99 m – 203 m)	210 m (140 m – 527 m)
Range	16 m – 1,594 m	10 m – 1,660 m
Nothing within the distance searched	0.0%	0.0%
Score: median (P25–P75) (`uli`)	+14.3 (+6.3 – +16.9)	+8.2 (−38.5 – +15.6)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 87.2% of the population live within 300 m of a convenience store; on average the nearest is 181 m away.
- Condesa: 67.4% of the population live within 300 m of a convenience store; on average the nearest is 341 m away.
- For the study region as a whole the value is 181 m; in Condesa, 341 m. Condesa's score lies 4.2 points below the study region average (+8.0 for the study region).

Map



Map. Convenience stores within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Labelled "convenience stores", the destination is DENUE food retail of any kind (abarrotes and misceláneas, confectionery, prepared food, self-service and department stores), which includes the fresh food retailers.

Fresh food diversity within 500 m

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> Diversity (Shannon entropy, 0-1) of the kinds of fresh food retail within a 500 m walk.

Identity

Identifier	`div_fresh_food_500m`
Core measure (variable sheet)	Daily essential-services access
Sheet / register ID	129, 11, 367 · #129, #11, #367
Sample point variable	`sp_walk_diversity_fresh_food_500m`
Area variable (measured value)	`avg_diversity_walk_fresh_food_500m` (0–1)
Normalised score	`index_uli__div_fresh_food_500m`
Domains (share of weight)	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)
Subdomain	Social infrastructure: Public services; Mobility and transport: Active mobility; Economic development: Food environment
Lens	Diversity
Better when	higher
Effective weight in the index	`uli` 3.6% · `uli_plain` 3.6%
Status	Scored in the published index

Rationale and interpretation

> Local markets provide residents with access to fresh food and opportunities for economic activity and social interaction within their neighborhoods.
> *— Indicator register #129: Access to markets (Economic Development / Food Environment)*

- Workbook citation: Ghasemi, 2024, 'Enhancing urban livability: Analyzing Tehran through equitable land use distribution'; Yang et al., 2021, 'Spatiotemporal trends and driving factors of urban livability in the Yangtze River Delta Agglomeration'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The presence of local bakeries and food services contributes to the convenience of urban life and supports the everyday functional needs of residents.
> *— Indicator register #11: Access to bakery (Economic Development / Food Environment)*

- Workbook citation: Ghasemi, 2024, 'Enhancing urban livability: Analyzing Tehran through equitable land use distribution'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The availability of supermarkets is a indicator of the formal food environment that influences residents' ability to access varied and nutritional food options.
> *— Indicator register #367: Access to supermarket (Economic Development / Food Environment)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Mean diversity of fresh food retail within a 500 m walk: … (0 = one kind or none, 1 = every kind equally).

Diversity of available fresh food (Daily essential-services access; ID 129, 11, 367) - *How to calculate (variable sheet):* Weighted share/number of supermarkets, markets, bakeries, abarroterías, tortillerías, pollerías, carnicerías, fruterías, verdulerías reachable within a walking distance - *Notes (sheet):* Using DENUE data

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_diversity_walk_fresh_food_500m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_fresh_food` (Fresh food retail (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('461121','461122','461123','461130','461140','461150','462111','462112')"`

- `denue_bakery_tortilleria` (Bakeries and tortillerias (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('311812','311813','311830')"`

Inputs: Diversity sets and their sub-types, DENUE destinations by SCIAN code

Method: theory

1. For each of the `k = 7` sub-types of the set "fresh_food" (`meat`, `produce`, `grains_and_spices`, `dairy`, `supermarket`, `minisuper`, `bakery_tortilleria`), the number of distinct destinations reachable within 500 m along the pedestrian network, `c_g` (R4).
2. Normalised Shannon entropy: `p_g = c_g / Σc`, `H = −Σ p_g·ln p_g / ln k`, from 0 (one sub-type or none) to 1 (every sub-type equally). `k` is the number of configured sub-types, not the number present.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R4. Diversity of what is reachable, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1491`):

```
- variable: sp_walk_diversity_fresh_food_500m
  name: div_fresh_food_500m
  domains: [social_infrastructure, mobility_transport, economic_development]
  group: {es: 'Acceso a servicios esenciales cotidianos', en: 'Daily essential-services access'}
  label: {es: 'Diversidad de alimentos frescos a 500 m', en: 'Fresh food diversity within 500 m'}
  reading: {es: 'Diversidad media de comercio de alimentos frescos a 500 m a pie: {value} (0 = un tipo o ninguno, 1 = todos los tipos por igual)', en:
'Mean diversity of fresh food retail within a 500 m walk: {value} (0 = one kind or none, 1 = every kind equally)'}
  subdomain:
    social_infrastructure: {es: 'Servicios públicos', en: 'Public services'}
    mobility_transport: {es: 'Movilidad activa', en: 'Active mobility'}
    economic_development: {es: 'Ambiente de alimentación', en: 'Food environment'}
  about:
    es: Diversidad (entropía de Shannon, 0-1) de los tipos de comercio de alimentos frescos a 500 m a pie.
    en: Diversity (Shannon entropy, 0-1) of the kinds of fresh food retail within a 500 m walk.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:929` `_banded_counts()`
- `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_diversity_fresh_food_500m	as measured	−0.167 · 0.269 · 0.705	0.000 – 0.871	0.19 / −1.36 / 39.9%	none
`uli_plain`	sp_walk_diversity_fresh_food_500m	as measured	−0.167 · 0.269 · 0.705	0.000 – 0.871	0.19 / −1.36 / 39.9%	none

Recalculation (Reimagina Urbano)

A new destination adds 1 to its sub-type's `c_g` at points within 500 m along the network; entropy is recomputed at those points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

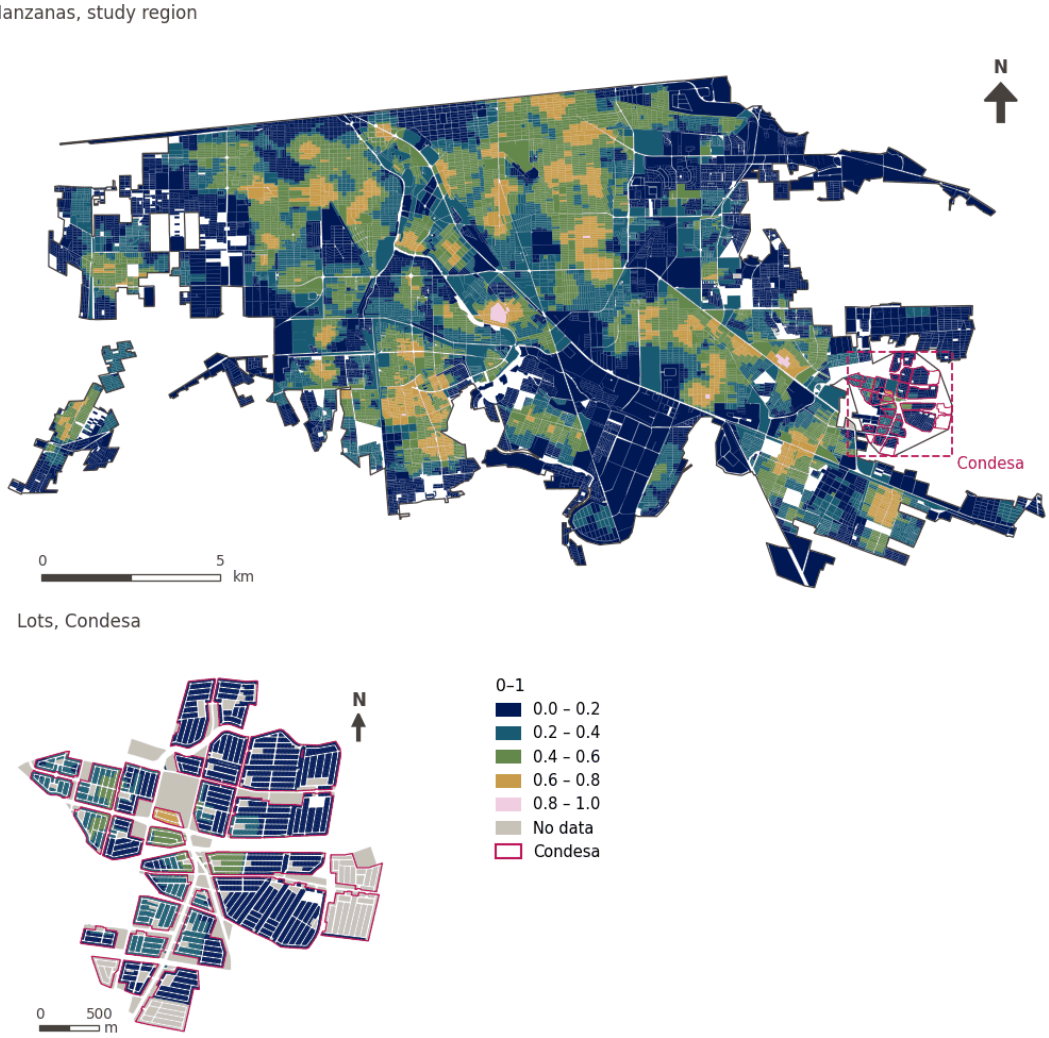
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.37	0.11
Region score (`uli`)	+7.0	−11.1
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.35 (0.18 – 0.51)	0.00 (0.00 – 0.27)
Range	0.00 – 0.85	0.00 – 0.70
Score: median (P25–P75) (`uli`)	+5.7 (−6.3 – +16.9)	−18.5 (−18.5 – +0.3)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean diversity of fresh food retail within a 500 m walk: 0.37 (0 = one kind or none, 1 = every kind equally).
- Condesa: Mean diversity of fresh food retail within a 500 m walk: 0.11 (0 = one kind or none, 1 = every kind equally).
- For the study region as a whole the value is 0.37; in Condesa, 0.11. Condesa's score lies 11.1 points below the study region average (+7.0 for the study region).

Map



Map. Fresh food diversity within 500 m: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Fresh food within 300 m

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> Walking distance to the nearest fresh food retailer, scored with a 300 m soft threshold.

Identity

Identifier	`fresh_food`
Core measure (variable sheet)	Daily essential-services access
Sheet / register ID	129, 11, 367 · #129, #11, #367
Sample point variable	`sp_walk_nearest_node_denue_fresh_food`
Area variable (measured value)	`avg_walk_dist_denue_fresh_food` (m)
Normalised score	`index_uli__fresh_food`
Domains (share of weight)	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)
Subdomain	Social infrastructure: Public services; Mobility and transport: Active mobility; Economic development: Food environment
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 3.6% · `uli_plain` 3.6%
Status	Scored in the published index

Rationale and interpretation

> Local markets provide residents with access to fresh food and opportunities for economic activity and social interaction within their neighborhoods.
> *— Indicator register #129: Access to markets (Economic Development / Food Environment)*

- Workbook citation: Ghasemi, 2024, 'Enhancing urban livability: Analyzing Tehran through equitable land use distribution'; Yang et al., 2021, 'Spatiotemporal trends and driving factors of urban livability in the Yangtze River Delta Agglomeration'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The presence of local bakeries and food services contributes to the convenience of urban life and supports the everyday functional needs of residents.
> *— Indicator register #11: Access to bakery (Economic Development / Food Environment)*

- Workbook citation: Ghasemi, 2024, 'Enhancing urban livability: Analyzing Tehran through equitable land use distribution'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The availability of supermarkets is a indicator of the formal food environment that influences residents' ability to access varied and nutritional food options.
> *— Indicator register #367: Access to supermarket (Economic Development / Food Environment)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 300 m of a fresh food retailer; on average the nearest is … away (lower is better).

Access to fresh food (Daily essential-services access; ID 129, 11, 367) - *How to calculate (variable sheet):* Walking distance access to supermarkets, markets, bakeries, abarroterías, tortillerías, pollerías, carnicerías, fruterías, pescaderías, verdulerías. SCIAN CODE: 4611 Comercio al por menor de abarrotes y alimentos - *Notes (sheet):* Using DENUE data

Percentage of population with access to fresh food (Daily essential-services access; ID 129, 11, 367) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Distance (m) along the pedestrian network to the nearest fresh food market / supermarket (source: OpenStreetMap or custom data)) - *Notes (sheet):* Use alongside—not instead of—proximity.

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_denue_fresh_food`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_fresh_food` (Fresh food retail (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('461121','461122','461123','461130','461140','461150','462111','462112')"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1475`):

```
- variable: sp_walk_nearest_node_denue_fresh_food
  name: fresh_food
  domains: [social_infrastructure, mobility_transport, economic_development]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a servicios esenciales cotidianos', en: 'Daily essential-services access'}
  label: {es: 'Alimentos frescos a menos de 300 m', en: 'Fresh food within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de un comercio de alimentos frescos; en promedio, el más cercano está a {value}', en:
'{access} of the population live within 300 m of a fresh food retailer; on average the nearest is {value} away'}
  subdomain:
    social_infrastructure: {es: 'Servicios públicos', en: 'Public services'}
    mobility_transport: {es: 'Movilidad activa', en: 'Active mobility'}
    economic_development: {es: 'Ambiente de alimentación', en: 'Food environment'}
  about:
    es: Distancia a pie al comercio de alimentos frescos más cercano, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest fresh food retailer, scored with a 300 m soft threshold.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:495`:

```
- {name: denue_fresh_food, category: food_retail, variant: strict, layer: destinations, where: "dest_name = 'denue_fresh_food'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_denue_fresh_food	soft access, 300 m threshold, k = 5	−0.091 · 0.405 · 0.902	0.000 – 0.993	0.30 / −1.57 / 1.5%	none
`uli_plain`	sp_walk_nearest_node_denue_fresh_food	soft access, 300 m threshold, k = 5	−0.091 · 0.405 · 0.902	0.000 – 0.993	0.30 / −1.57 / 1.5%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

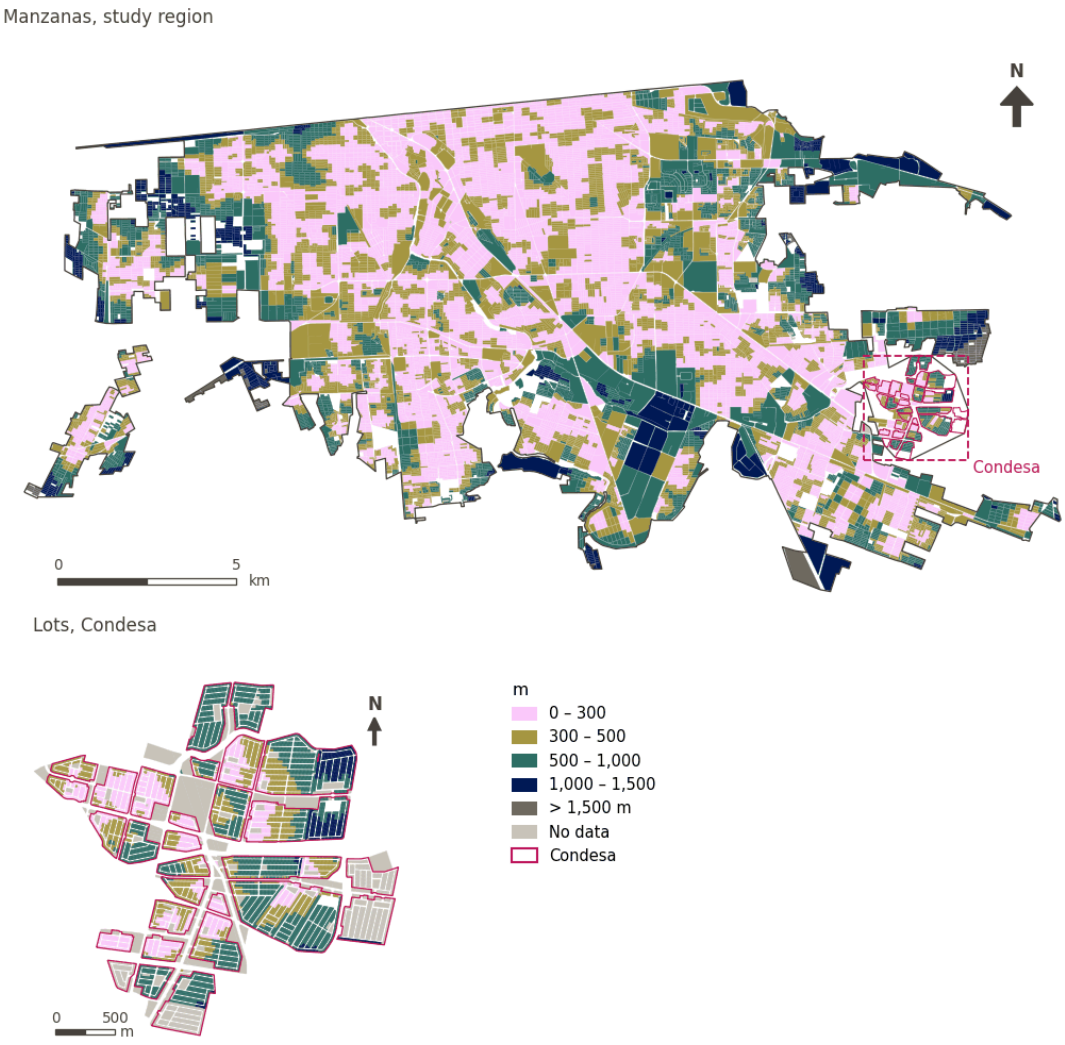
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	334 m	539 m
Region score (`uli`)	+5.3	−10.2
Areas with a value	14,120 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	303 m (208 m – 431 m)	501 m (313 m – 729 m)
Range	16 m – 2,754 m	10 m – 1,660 m
Nothing within the distance searched	0.1%	0.0%
Score: median (P25–P75) (`uli`)	+5.1 (−15.0 – +21.8)	−22.2 (−24.4 – +3.3)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 51.0% of the population live within 300 m of a fresh food retailer; on average the nearest is 334 m away.
- Condesa: 23.4% of the population live within 300 m of a fresh food retailer; on average the nearest is 539 m away.
- For the study region as a whole the value is 334 m; in Condesa, 539 m. Condesa's score lies 10.2 points below the study region average (+5.3 for the study region).

Map



Map. Fresh food within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Bakeries and tortillerías are classified by SCIAN as manufacturing and are not in this category; they enter fresh food diversity as a sub-type (configuration).

Job diversity

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> Diversity (0-1) of economic sectors around residential areas.

Identity

Identifier	`job_diversity`
Core measure (variable sheet)	Access to employment
Sheet / register ID	–
Sample point variable	`sp_ext_economic_sector_diversity`
Area variable (measured value)	`ext_economic_sector_diversity` (0–1)
Normalised score	`index_uli__job_diversity`
Domains (share of weight)	Economic development
Subdomain	Economy
Lens	Diversity
Better when	higher
Effective weight in the index	`uli` 5.0% · `uli_plain` 5.0%
Status	Scored in the published index

Rationale and interpretation

Pending: Not recorded in the project register.

How to read it. Mean diversity of economic sectors of … (0-1).

Job diversity (Access to employment; ID –) - *How to calculate (variable sheet):* Diversity of economic sectors within a XX buffer of residential areas. - *Doubts (sheet):* Ask Natalia about this one

Data sources

Sources declared in the configuration:

- Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).

Source recorded by GHSCI (data dictionary) (`ext_economic_sector_diversity`):

- Work package indicators: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)

Inputs: Work package indicators (linked)

Method: theory

1. A value prepared outside GHSCI by a work package and linked to the reference areas (R7: by geometry against the reference areas).
2. Replicated to each sample point from the area containing it (`grid`, `condesa_lotes (15 m)`, `manzanas (30 m)`).
3. AMPI normalisation (R11) of the value as measured, aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R7. Linking externally prepared indicators, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1656`):

```
- variable: sp_ext_economic_sector_diversity
  name: job_diversity
  domains: [economic_development]
  polarity: positive
  lens: diversity
  group: {es: 'Acceso al empleo', en: 'Access to employment'}
  label: {es: 'Diversidad de empleos', en: 'Job diversity'}
  reading: {es: 'Diversidad media de sectores económicos de {value} (0-1)', en: 'Mean diversity of economic sectors of {value} (0-1)'}
  subdomain: {es: 'Economía', en: 'Economy'}
  about:
    es: Diversidad (0-1) de los sectores económicos en torno a las áreas residenciales.
    en: Diversity (0-1) of economic sectors around residential areas.
  sources:
    - 'Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).'
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:403` `match_by_geometry()`
- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`

- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_ext_economic_sector_diversity	as measured	−0.236 · 0.264 · 0.764	0.000 – 1.000	1.00 / −0.96 / 71.9%	none
`uli_plain`	sp_ext_economic_sector_diversity	as measured	−0.236 · 0.264 · 0.764	0.000 – 1.000	1.00 / −0.96 / 71.9%	none

Recalculation (Reimagina Urbano)

The value comes from an external work package: a scenario (for example, more vegetation) replaces the area's value, which is replicated to its sample points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

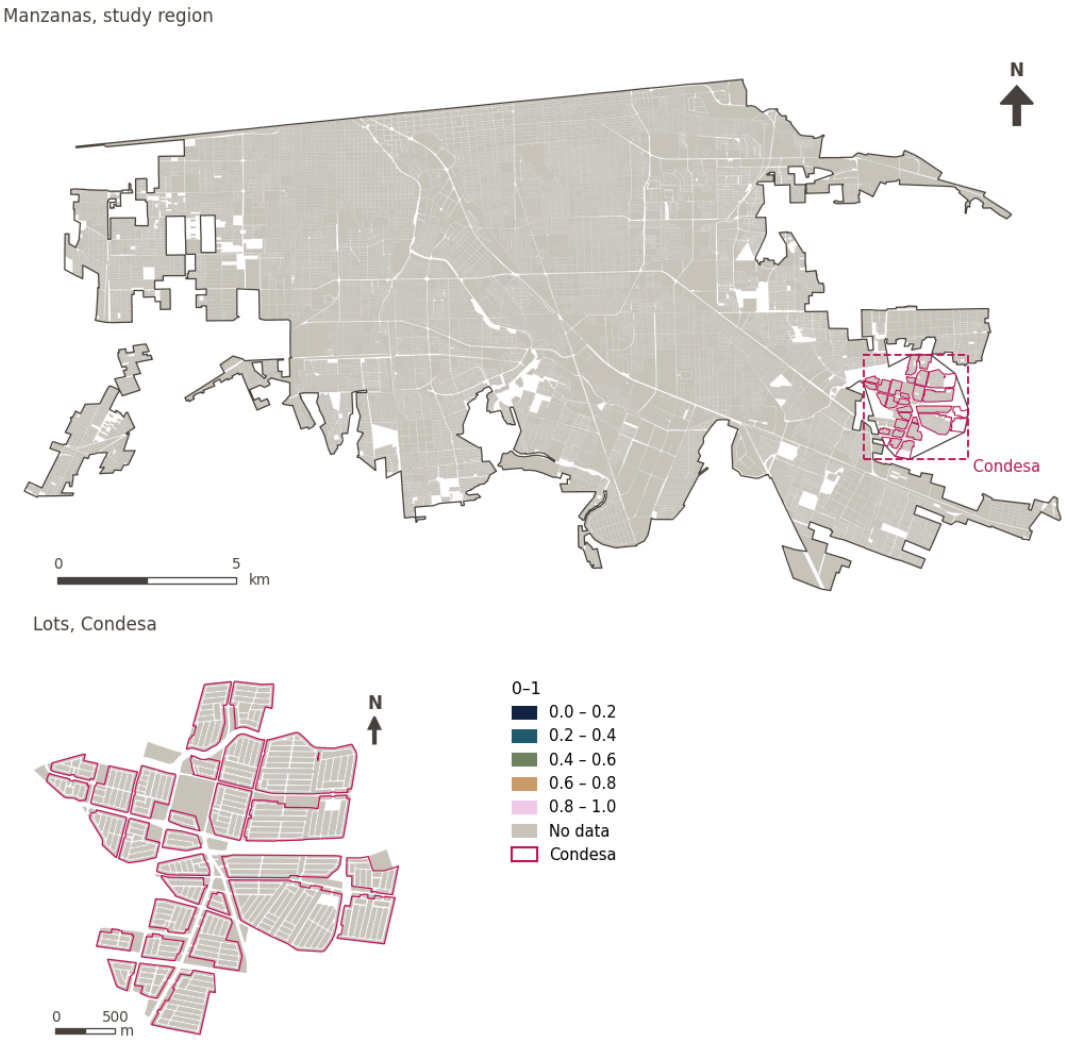
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.33	–
Region score (`uli`)	+3.7	−14.1
Areas with a value	0 of 14,252 manzanas	–
Median (P25–P75), weighted	–	–
Range	–	–
Score: median (P25–P75) (`uli`)	+2.1 (−13.4 – +15.1)	−15.8 (−15.8 – −15.8)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean diversity of economic sectors of 0.33 (0-1).
- Condesa: Mean diversity of economic sectors of – (0-1).
- For the study region as a whole the value is 0.33; in Condesa, –. Condesa's score lies 14.1 points below the study region average (+3.7 for the study region).

Map



Map. Job diversity: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- Delivered for the 100 m grid only, which tiles the city, so every sample point takes its cell's value (configuration). The sheet's buffer distance is unspecified ("XX").

Manufacturing within 1 km

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> Whether a manufacturing establishment lies within a 1 km walk, as a proxy for access to employment.

Identity

Identifier	`manufacturing_1000m`
Core measure (variable sheet)	Access to employment
Sheet / register ID	127 · #127
Sample point variable	`sp_walk_access_denue_manufacturing_1000m`
Area variable (measured value)	`pct_access_walk_denue_manufacturing_1000m` (%)
Normalised score	`index_uli__manufacturing_1000m`
Domains (share of weight)	Economic development
Subdomain	Economy
Lens	Accessibility
Better when	higher
Effective weight in the index	`uli` 5.0% · `uli_plain` 5.0%
Status	Scored in the published index

Rationale and interpretation

> Manufacturing centres are sources of industrial hazards and pollution; restricting residential proximity ensures physical safety and protects wellbeing.
> *— Indicator register #127: Access to manufacturing district / centre (Economic Development / Economy)*

- Workbook citation: Onnom et al., 2018, 'Development of a Liveable City Index (LCI) using multi criteria geospatial modelling for medium class cities in developing countries'
- Article list: #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within a 1 km walk of a manufacturing establishment (higher is better).

Access to manufacturing district/centre (Access to employment; ID 127) - *How to calculate (variable sheet):* Destination analysis using GHSCI tools with permutations of: distance to closest, access within threshold(s), and where applicable count per sqkm and per 1,000 persons - *Doubts (sheet):* Should this one be split into separate sub-variables?

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`pct_access_walk_denue_manufacturing_1000m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_manufacturing` (Manufacturing and industrial establishments (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act LIKE '31%' OR codigo_act LIKE '32%' OR codigo_act LIKE '33%'"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network to the nearest destination, `d` (R2).
2. Binary access (R3): `a = 1` if `d ≤ 1,000 m`, otherwise `a = 0` (including nothing found). Binary access is used because the 1,500 m search does not extend far enough beyond 1,000 m for a soft threshold.
3. For areas, the percentage of points with access. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R3. Banded access, and distance from a disamenity, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1642`):

```
- variable: sp_walk_access_denue_manufacturing_1000m
  name: manufacturing_1000m
  domains: [economic_development]
  group: {es: 'Acceso al empleo', en: 'Access to employment'}
  label: {es: 'Manufactura a menos de 1 km', en: 'Manufacturing within 1 km'}
  reading: {es: '{value} de la población vive a menos de 1 km a pie de un establecimiento manufacturero', en: '{value} of the population live within a 1 km
walk of a manufacturing establishment'}
  subdomain: {es: 'Economía', en: 'Economy'}
  about:
    es: Si hay un establecimiento manufacturero a 1 km a pie, como aproximación al acceso al empleo.
    en: Whether a manufacturing establishment lies within a 1 km walk, as a proxy for access to employment.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:524`:

```
- {name: denue_manufacturing, category: manufacturing, variant: any, layer: destinations, where: "dest_name = 'denue_manufacturing'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_accessibility_spec.py:1272` `sample_point_access()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_access_denue_manufacturing_1000m	as measured	0.435 · 0.935 · 1.435	0.000 – 1.000	−3.51 / 10.34 / 93.5%	skewed, tied
`uli_plain`	sp_walk_access_denue_manufacturing_1000m	as measured	0.435 · 0.935 · 1.435	0.000 – 1.000	−3.51 / 10.34 / 93.5%	skewed, tied

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

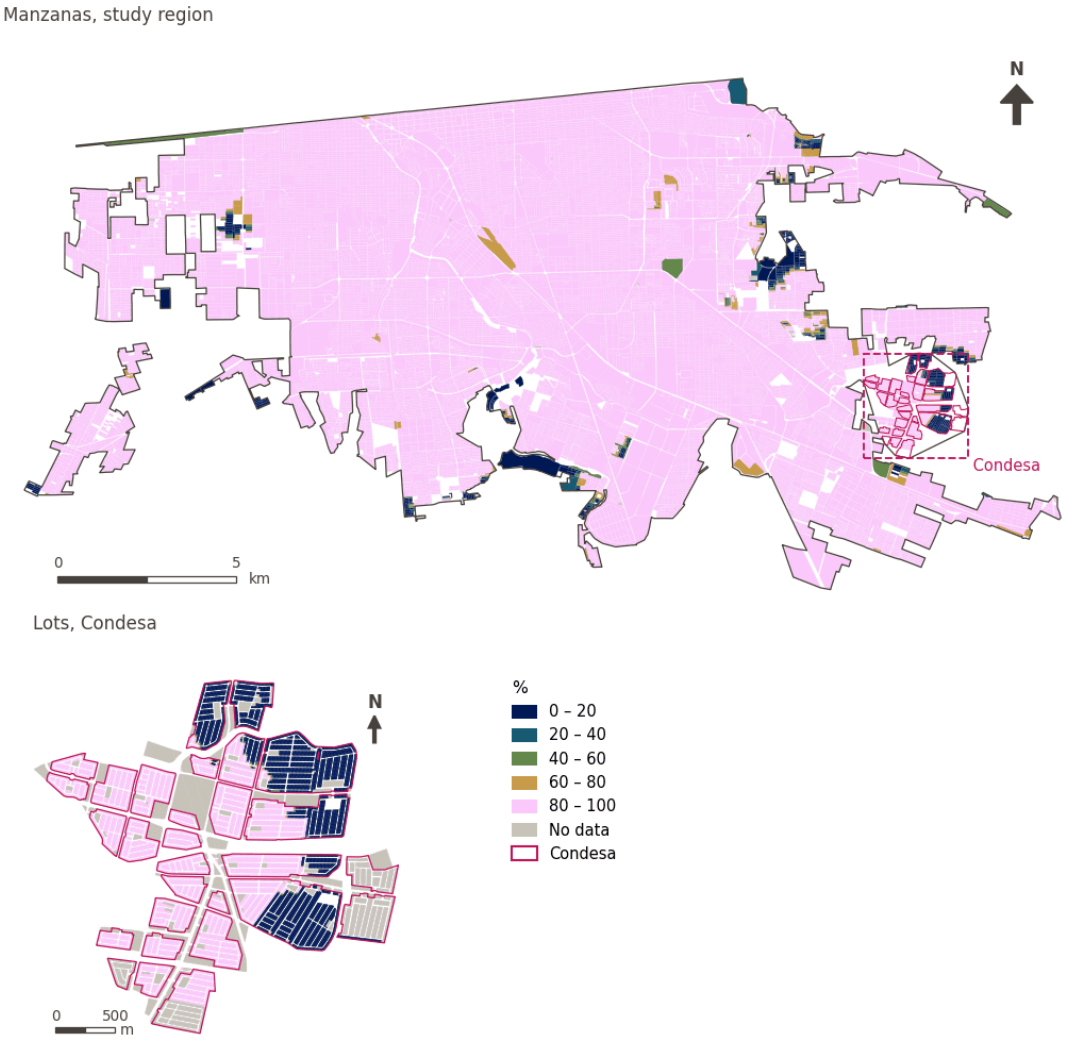
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	99.4%	62.6%
Region score (`uli`)	+3.6	−18.5
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	100.0% (100.0% – 100.0%)	100.0% (0.0% – 100.0%)
Range	0.0% – 100.0%	0.0% – 100.0%
Score: median (P25–P75) (`uli`)	+3.9 (+3.9 – +3.9)	+3.9 (−56.1 – +3.9)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: 99.4% of the population live within a 1 km walk of a manufacturing establishment.
- Condesa: 62.6% of the population live within a 1 km walk of a manufacturing establishment.
- For the study region as a whole the value is 99.4%; in Condesa, 62.6%. Condesa's score lies 18.5 points below the study region average (+3.6 for the study region).

Map



Map. Manufacturing within 1 km: measured value (%) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- Manufacturing is read here as a source of employment: the sheet places this sub-variable under the core measure "Access to employment", so being within reach of manufacturing counts as better. The register's rationale for the row comes from a source that treats manufacturing centres as industrial hazards; that reading is recorded above for completeness, but it is not the one the index applies.

Public transport within 300 m

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> Walking distance to the nearest public transport stop, scored with a 300 m soft threshold.

Identity

Identifier	`public_transport`
Core measure (variable sheet)	Public-transport access
Sheet / register ID	26 · #26, #223
Sample point variable	`sp_walk_nearest_node_pt_any`
Area variable (measured value)	`avg_walk_dist_pt_any` (m)
Normalised score	`index_uli__public_transport`
Domains (share of weight)	Mobility and transport
Subdomain	Public transport
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 4.2% · `uli_plain` 4.2%
Status	Scored in the published index

Rationale and interpretation

> Proximity to bus stops improves accessibility to city facilities and services, acting as a major engine of urban growth and resident liveability.
> *— Indicator register #26: Access to buses: stops (Mobility & Transport / Public Transport)*

- Workbook citation: Higgs et al., 2019, 'The Urban Liveability Index: developing a policy-relevant urban liveability composite measure and evaluating associations with transport mode choice'; Suptelo & Sehida, 2020, 'Factors of Post-Industrial Transformations of Urban Sociogeosystems (Case Kharkiv)'
- Article list: #5: Mahanta (2025), 'Constructing a city liveability index and evaluating the spatial pattern of liveability of Guwahati city'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Extensive public transit coverage ensures equitable access to city-wide opportunities and services, reducing health inequities related to mobility.
> *— Indicator register #223: Access to public transport (Mobility & Transport / Public Transport)*

- Workbook citation: Kashef, 2016, 'Urban liveability across disciplinary and professional boundaries'; Cervero, 2009, 'Transport infrastructure and global competitiveness'; Bassolas et al., 2019, 'Hierarchical organization of urban mobility and its connection with city liveability'; Vuchic, 2017, 'Transportation for Livable Cities'
- Article list: #27: Alzaim (2024), 'Integrated Framework for Enhancing Liveability and Ecological Sustainability in UAE Communities'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 300 m of a public transport stop; on average the nearest is … away (lower is better).

Access to public transport stops (Public-transport access; ID 26) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE

Data sources

Sources declared in the configuration:

- IMIP / CFC public transport stops (suburban transport routes).

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_pt_any`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; IMIP / CFC: 2026-05-18, Provided by IMIP via CFC, As per advice from Juan Ernesto Díaz Noguez

Destination filter:

- `pt_any` (pt_any): `MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb` — Destination filter: `"equipamiento_category='transp' AND TIPO != 'TERMINAL INTERIOR DE CARGA (CENTRAL DE SERVICIOS DE CARGA)'"`

Inputs: Public transport stops

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1460`):

```
- variable: sp_walk_nearest_node_pt_any
  name: public_transport
  domains: [mobility_transport]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso al transporte público', en: 'Public-transport access'}
  label: {es: 'Transporte público a menos de 300 m', en: 'Public transport within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de una parada de transporte público; en promedio, la más cercana está a {value}', en:
'{access} of the population live within 300 m of a public transport stop; on average the nearest is {value} away'}
  subdomain: {es: 'Transporte público', en: 'Public transport'}
  about:
    es: Distancia a pie a la parada de transporte público más cercana, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest public transport stop, scored with a 300 m soft threshold.
  sources:
    - 'IMIP / CFC public transport stops (suburban transport routes).'
```

`MX_Mexicali_2025_ULI.yml:492`:

```
- {name: pt_any,                category: pt,   variant: lenient, layer: destinations, where: "dest_name = 'pt_any'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_pt_any	soft access, 300 m threshold, k = 5	−0.433 · 0.064 · 0.561	0.000 – 0.993	3.44 / 10.89 / 41.0%	skewed
`uli_plain`	sp_walk_nearest_node_pt_any	soft access, 300 m threshold, k = 5	−0.433 · 0.064 · 0.561	0.000 – 0.993	3.44 / 10.89 / 41.0%	skewed

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

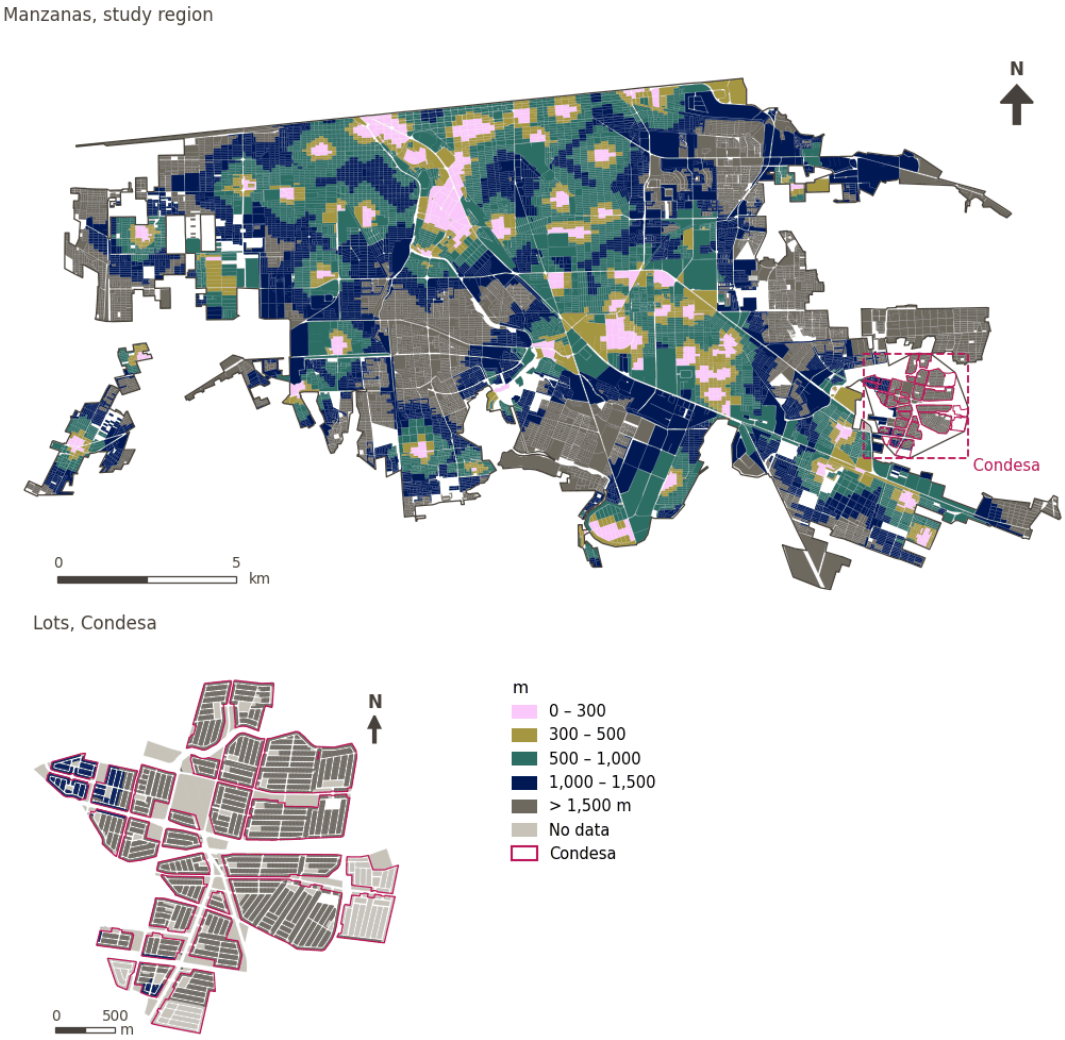
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	882 m	1,432 m
Region score (`uli`)	−0.2	−3.9
Areas with a value	9,959 of 14,252 manzanas	845 of 14,988 lots
Median (P25–P75), weighted	926 m (610 m – 1,259 m)	1,409 m (1,297 m – 1,540 m)
Range	38 m – 3,167 m	1,089 m – 1,962 m
Nothing within the distance searched	28.5%	93.6%
Score: median (P25–P75) (`uli`)	−3.9 (−3.9 – −3.8)	−3.9 (−3.9 – −3.9)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 5.5% of the population live within 300 m of a public transport stop; on average the nearest is 882 m away.
- Condesa: 0.0% of the population live within 300 m of a public transport stop; on average the nearest is 1,432 m away.
- For the study region as a whole the value is 882 m; in Condesa, 1,432 m. Condesa's score lies 3.9 points below the study region average (−0.2 for the study region).

Map



Map. Public transport within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Mexicali has no GTFS feed. IMIP stops are pooled with stops identified from OpenStreetMap, so a stop present in both is counted twice in any count of stops (it does not affect distances) (configuration, points_of_interest).

Community, culture and recreation diversity within 500 m

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> Diversity (Shannon entropy, 0-1) of community, cultural and recreational venues within a 500 m walk.

Identity

Identifier	`div_ccr_500m`
Core measure (variable sheet)	Community, culture and recreation access
Sheet / register ID	4, 46, 58, 121, 134, 225, 228, 257, 394 · #46, #58, #121, #134, #228, #225, #257, #394
Sample point variable	`sp_walk_diversity_community_culture_recreation_500m`
Area variable (measured value)	`avg_diversity_walk_community_culture_recreation_500m` (0–1)
Normalised score	`index_uli__div_ccr_500m`
Domains (share of weight)	Social infrastructure
Subdomain	Community, culture and recreation
Lens	Diversity
Better when	higher
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

> Multi-purpose community centres provide important opportunities for social interaction and community connection, which are vital for mental and social wellbeing.
>
> *— Indicator register #46: Access to community centres (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Saitluanga, 2014, 'Spatial pattern of urban livability in Himalayan region: a case of Aizawl City'
- Article list: #22: Alderton (2019), 'What is the meaning of urban liveability for a city in a low-to-middle-income country? Contextualising liveability for Bangkok, Thailand'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Cultural centers serve as important hubs for social and cultural activities that enhance community identity and promote the psychological well-being of residents.
>
> *— Indicator register #58: Access to cultural centers (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Kang et al., 2022, 'Can tourism development make cities more livable? Investigating 40 cities in China'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Easy access to cultural facilities like libraries is a key component of urban convenience, allowing residents to satisfy their fundamental information and educational requirements.
>
> *— Indicator register #121: Access to libraries (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Onnom et al., 2018, 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'; Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Museums provide important cultural infrastructure that supports social development, lifelong learning, and community social cohesion.
>
> *— Indicator register #134: Access to museums (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Access to religious centres contributes to the amenity value of urban areas, providing social and cultural comfort that enhances human wellbeing.
>
> *— Indicator register #228: Access to religious buildings (Social Infrastructure / Community, Culture & Leisure)*

- Workbook citation: Rehdanz and Maddison, 2008, 'Local environmental quality and life-satisfaction in Germany'
- Article list: #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Easy access to recreational facilities emphasizes the provision of spaces that offer pleasure and physical comfort, improving the overall livability of urban environments.
>
> *— Indicator register #225: Access to recreational areas (Social Infrastructure / Community, Culture & Leisure)*

- Workbook citation: Basu et al., 2021, 'Urban Livability Index Assessment Based on Land-Use Changes in an Indian Medium-Sized City (Raiganj)'; Fu et al., 2019, 'The Livable Urban Landscape: GIS and Remote Sensing Extracted Land Use Assessment for Urban Livability in Changchun Proper, China'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The provision of sport centres is critical for promoting physical activity, which reduces the risk of chronic diseases and improves mental health outcome.
>
> *— Indicator register #257: Access to sports centres (Social Infrastructure / Community, Culture & Leisure)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'; Sachs et al., 2019, 'Six transformations to achieve the sustainable development goals'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'; #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'; #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Access to youth centres provides essential social infrastructure that supports social cohesion, mental health, and community wellbeing for young residents.
>
> *— Indicator register #394: Access to youth centres (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Mean diversity of community, cultural and recreational venues within a 500 m walk: … (0 = one kind or none, 1 = every kind equally).

Diversity of community, culture and recreation services (Community, culture and recreation access; ID 4, 46, 58, 121,134,225,228,257,394) - *How to calculate (variable sheet):* Weighted share/number of cultural centers, libraries, museums, recreational areas, religious buildings, sports centers, youth centers, activity centers and community centers reachable within a walking distance - *Doubts (sheet):* Do we want to split this in 3: 1. Community 2. Recreational and 3. Cultural?

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_diversity_walk_community_culture_recreation_500m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf

Destination filter:

- `denue_community` (Community and social facilities (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('813210','813230','931610')"`
- `denue_culture` (Cultural facilities (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('712111','712112','519211','519212','512130','711311','711312')"`
- `denue_sport` (Sports and active recreation (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('713941','713942','713943','713944','713910','713950','611621','611622')"`
- `denue_recreation` (Public recreation venues (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('713111','713112','713113','712131','712132')"`

Inputs: Diversity sets and their sub-types, DENUE destinations by SCIAN code

Method: theory

1. For each of the `k = 4` sub-types of the set "community_culture_rec" (`community`, `culture`, `sport`, `recreation`), the number of distinct destinations reachable within 500 m along the pedestrian network, `c_g` (R4).
2. Normalised Shannon entropy: `p_g = c_g / Σc`, `H = −Σ p_g·ln p_g / ln k`, from 0 (one sub-type or none) to 1 (every sub-type equally). `k` is the number of configured sub-types, not the number present.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R4. Diversity of what is reachable, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1627`):

```
- variable: sp_walk_diversity_community_culture_rec_500m
  name: div_ccr_500m
  domains: [social_infrastructure]
  group: {es: 'Acceso a comunidad, cultura y recreación', en: 'Community, culture and recreation access'}
  label: {es: 'Diversidad de comunidad, cultura y recreación a 500 m', en: 'Community, culture and recreation diversity within 500 m'}
  reading: {es: 'Diversidad media de espacios comunitarios, culturales y recreativos a 500 m a pie: {value} (0 = un tipo o ninguno, 1 = todos los tipos por
igual)', en: 'Mean diversity of community, cultural and recreational venues within a 500 m walk: {value} (0 = one kind or none, 1 = every kind equally)'}
  subdomain: {es: 'Comunidad, cultura y recreación', en: 'Community, culture and recreation'}
  about:
    es: Diversidad (entropía de Shannon, 0-1) de los espacios comunitarios, culturales y recreativos a 500 m a pie.
    en: Diversity (Shannon entropy, 0-1) of community, cultural and recreational venues within a 500 m walk.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

The current configuration names this variable `sp_walk_diversity_community_culture_rec_500m`; the published results were produced with `sp_walk_diversity_community_culture_recreation_500m`, before the name was shortened to respect PostgreSQL's 63-character identifier limit. Re-running the analysis adopts the new name.

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:929` `_banded_counts()`
- `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_diversity_community_culture_recreation_500m	as measured	−0.331 · 0.163 · 0.656	0.000 – 0.987	1.01 / −0.42 / 64.3%	none
`uli_plain`	sp_walk_diversity_community_culture_recreation_500m	as measured	−0.331 · 0.163 · 0.656	0.000 – 0.987	1.01 / −0.42 / 64.3%	none

Recalculation (Reimagina Urbano)

A new destination adds 1 to its sub-type's `c_g` at points within 500 m along the network; entropy is recomputed at those points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

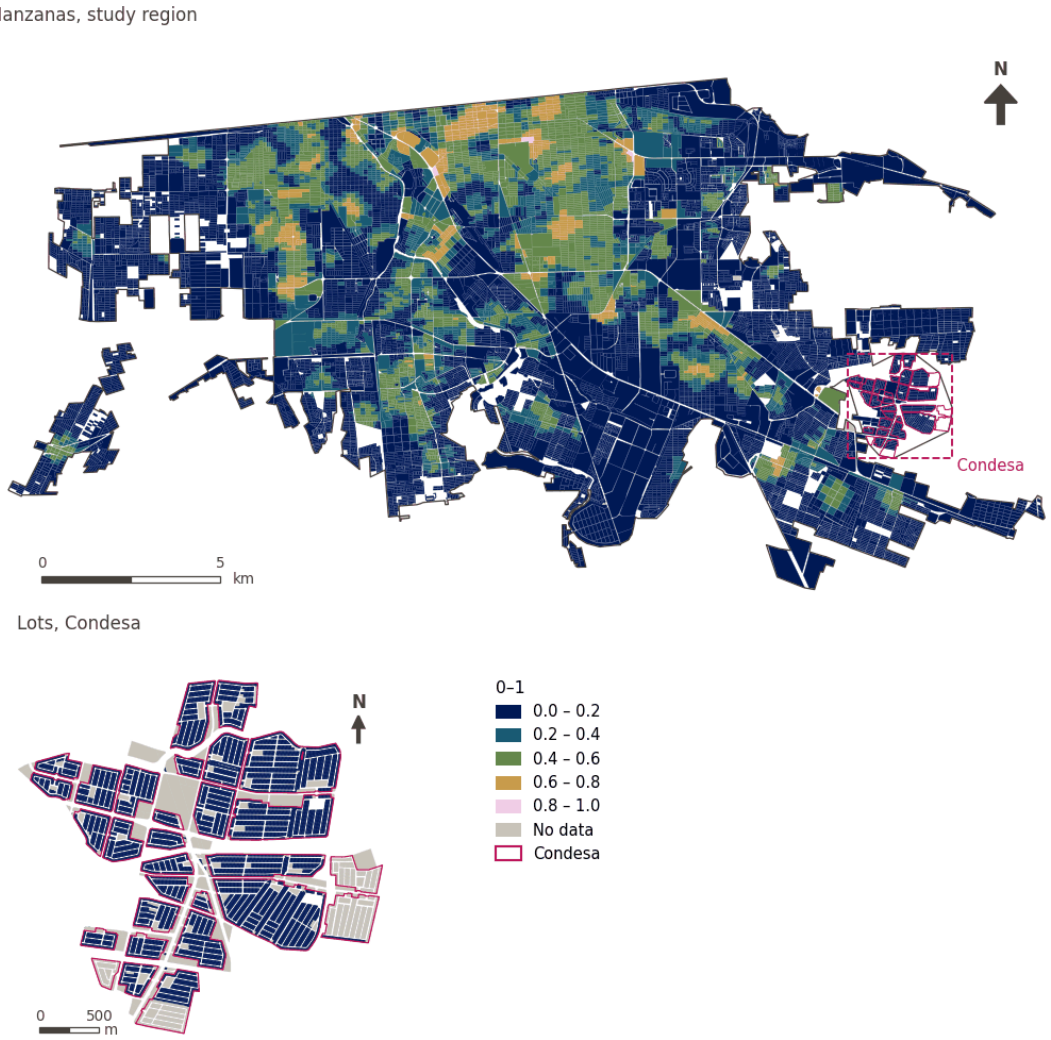
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.22	0.00
Region score (`uli`)	+3.4	−9.9
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.05 (0.00 – 0.39)	0.00 (0.00 – 0.00)
Range	0.00 – 0.83	0.00 – 0.00
Score: median (P25–P75) (`uli`)	−6.6 (−9.9 – +13.8)	−9.9 (−9.9 – −9.9)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean diversity of community, cultural and recreational venues within a 500 m walk: 0.22 (0 = one kind or none, 1 = every kind equally).
- Condesa: Mean diversity of community, cultural and recreational venues within a 500 m walk: 0.00 (0 = one kind or none, 1 = every kind equally).
- For the study region as a whole the value is 0.22; in Condesa, 0.00. Condesa's score lies 9.9 points below the study region average (+3.4 for the study region).

Map



Map. Community, culture and recreation diversity within 500 m: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Community, culture and recreation within 300 m

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> Walking distance to the nearest community, cultural or recreational venue, scored with a 300 m soft threshold.

Identity

Identifier	`ccr_any`
Core measure (variable sheet)	Community, culture and recreation access
Sheet / register ID	4, 46, 58, 121, 134, 225, 228, 257, 394 · #46, #58, #121, #134, #228, #225, #257, #394
Sample point variable	`sp_walk_nearest_node_denue_ccr_any`
Area variable (measured value)	`avg_walk_dist_denue_ccr_any` (m)
Normalised score	`index_uli__ccr_any`
Domains (share of weight)	Social infrastructure
Subdomain	Community, culture and recreation
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

> Multi-purpose community centres provide important opportunities for social interaction and community connection, which are vital for mental and social wellbeing.
>
> *— Indicator register #46: Access to community centres (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Saitluanga, 2014, 'Spatial pattern of urban livability in Himalayan region: a case of Aizawl City'
- Article list: #22: Alderton (2019), 'What is the meaning of urban liveability for a city in a low-to-middle-income country? Contextualising liveability for Bangkok, Thailand'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Cultural centers serve as important hubs for social and cultural activities that enhance community identity and promote the psychological well-being of residents.
>
> *— Indicator register #58: Access to cultural centers (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Kang et al., 2022, 'Can tourism development make cities more livable? Investigating 40 cities in China'
- Article list: #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Easy access to cultural facilities like libraries is a key component of urban convenience, allowing residents to satisfy their fundamental information and educational requirements.
>
> *— Indicator register #121: Access to libraries (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Onnom et al., 2018, 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'; Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Museums provide important cultural infrastructure that supports social development, lifelong learning, and community social cohesion.
>
> *— Indicator register #134: Access to museums (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Access to religious centres contributes to the amenity value of urban areas, providing social and cultural comfort that enhances human wellbeing.
>
> *— Indicator register #228: Access to religious buildings (Social Infrastructure / Community, Culture & Leisure)*

- Workbook citation: Rehdanz and Maddison, 2008, 'Local environmental quality and life-satisfaction in Germany'
- Article list: #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Easy access to recreational facilities emphasizes the provision of spaces that offer pleasure and physical comfort, improving the overall livability of urban environments.
>
> *— Indicator register #225: Access to recreational areas (Social Infrastructure / Community, Culture & Leisure)*

- Workbook citation: Basu et al., 2021, 'Urban Livability Index Assessment Based on Land-Use Changes in an Indian Medium-Sized City (Raiganj)'; Fu et al., 2019, 'The Livable Urban Landscape: GIS and Remote Sensing Extracted Land Use Assessment for Urban Livability in Changchun Proper, China'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The provision of sport centres is critical for promoting physical activity, which reduces the risk of chronic diseases and improves mental health outcome.
>
> *— Indicator register #257: Access to sports centres (Social Infrastructure / Community, Culture & Leisure)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'; Sachs et al., 2019, 'Six transformations to achieve the sustainable development goals'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'; #16: Kashi (2025), 'Spatial analysis and ranking of urban districts based on a comprehensive livability approach: the case of Tehran'; #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Access to youth centres provides essential social infrastructure that supports social cohesion, mental health, and community wellbeing for young residents.
>
> *— Indicator register #394: Access to youth centres (Social Infrastructure / Community, culture & leisure)*

- Workbook citation: Lowe et al., 2015, 'Planning healthy, liveable and sustainable cities: how can indicators inform policy?'
- Article list: #13: Alderton (2021), 'Measuring and monitoring liveability in a low-to-middle income country: a proof-of-concept for Bangkok, Thailand and lessons from an international partnership'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 300 m of a community, cultural or recreational venue; on average the nearest is … away (lower is better).

Access to community, culture and recreation services (Community, culture and recreation access; ID 4, 46, 58, 121,134,225,228,257,394) - *How to calculate (variable sheet):* Walking distance access to cultural centers, libraries, museums, recreational areas, religious buildings, sports centers, youth centers, activity centers and community centers. - *Doubts (sheet):* Do we want to split this in 3: 1. Community 2. Recreational and 3. Cultural?

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_denue_ccr_any`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_community` (Community and social facilities (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('813210','813230','931610')"`
- `denue_culture` (Cultural facilities (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('712111','712112','519211','519212','512130','711311','711312')"`
- `denue_sport` (Sports and active recreation (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('713941','713942','713943','713944','713910','713950','611621','611622')"`
- `denue_recreation` (Public recreation venues (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('713111','713112','713113','712131','712132')"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1614`):

```
- variable: sp_walk_nearest_node_denue_ccr_any
  name: ccr_any
  domains: [social_infrastructure]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a comunidad, cultura y recreación', en: 'Community, culture and recreation access'}
  label: {es: 'Comunidad, cultura y recreación a menos de 300 m', en: 'Community, culture and recreation within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de un espacio comunitario, cultural o recreativo; en promedio, el más cercano está a
{value}', en: '{access} of the population live within 300 m of a community, cultural or recreational venue; on average the nearest is {value} away'}
  subdomain: {es: 'Comunidad, cultura y recreación', en: 'Community, culture and recreation'}
  about:
    es: Distancia a pie al espacio comunitario, cultural o recreativo más cercano, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest community, cultural or recreational venue, scored with a 300 m soft threshold.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:517`:

```
- {name: denue_ccr_any, category: community_culture_recreation, variant: any, layer: destinations, where: "dest_name IN ('denue_community',
'denue_culture', 'denue_sport', 'denue_recreation')"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_denue_ccr_any	soft access, 300 m threshold, k = 5	−0.077 · 0.420 · 0.916	0.000 – 0.993	0.21 / −1.67 / 6.6%	none
`uli_plain`	sp_walk_nearest_node_denue_ccr_any	soft access, 300 m threshold, k = 5	−0.077 · 0.420 · 0.916	0.000 – 0.993	0.21 / −1.67 / 6.6%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

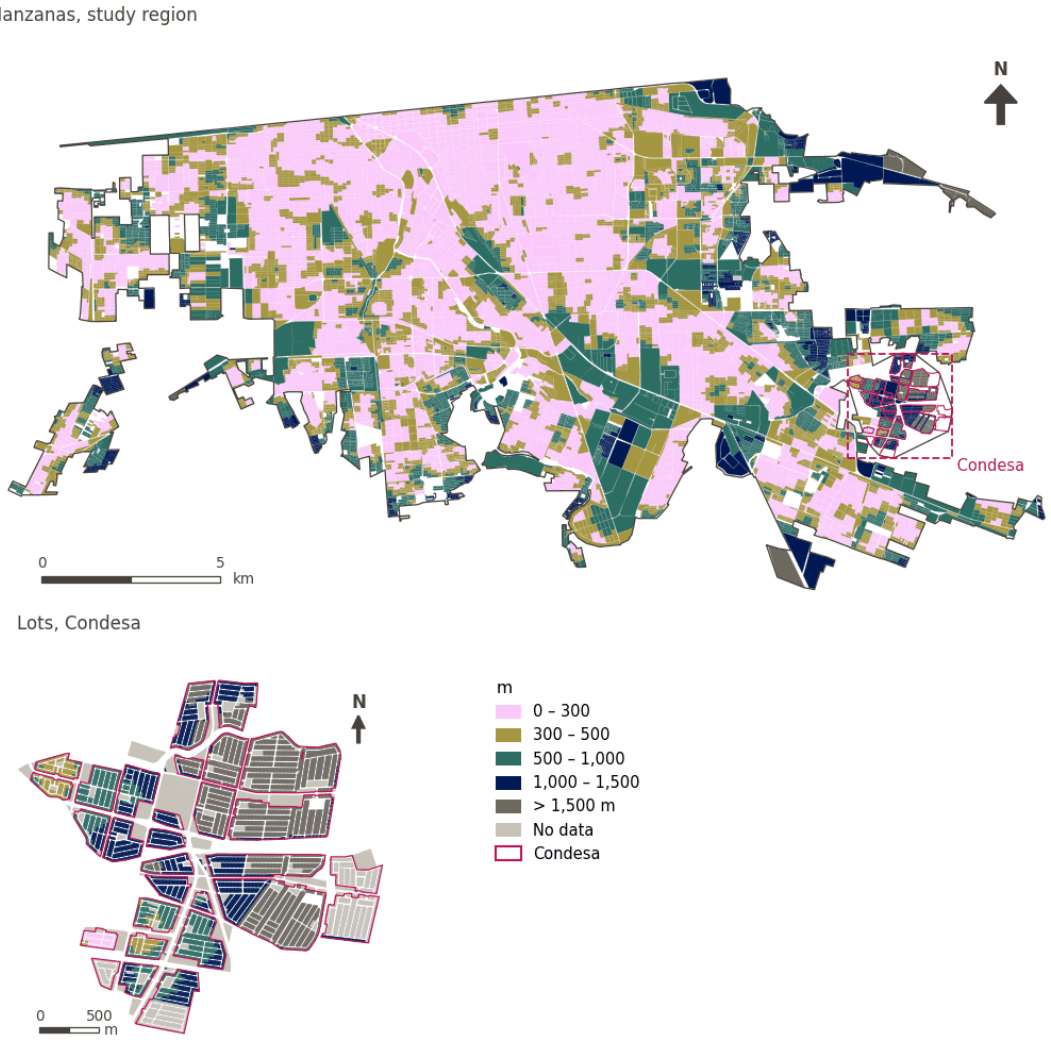
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	332 m	1,075 m
Region score (`uli`)	+6.5	−24.4
Areas with a value	14,110 of 14,252 manzanas	6,788 of 14,988 lots
Median (P25–P75), weighted	292 m (189 m – 458 m)	1,071 m (791 m – 1,417 m)
Range	17 m – 2,257 m	64 m – 2,412 m
Nothing within the distance searched	0.3%	48.8%
Score: median (P25–P75) (`uli`)	+6.5 (−18.1 – +24.0)	−25.3 (−25.3 – −25.3)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 55.7% of the population live within 300 m of a community, cultural or recreational venue; on average the nearest is 332 m away.
- Condesa: 1.4% of the population live within 300 m of a community, cultural or recreational venue; on average the nearest is 1,075 m away.
- For the study region as a whole the value is 332 m; in Condesa, 1,075 m. Condesa's score lies 24.4 points below the study region average (+6.5 for the study region).

Map



Map. Community, culture and recreation within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The sheet asks whether to split this into community, recreational and cultural access; one measure is used, and the split is reflected in its diversity.

Care service diversity within 500 m

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> Diversity (Shannon entropy, 0-1) of the kinds of care service within a 500 m walk.

Identity

Identifier	`div_care_500m`
Core measure (variable sheet)	Health, education and care access
Sheet / register ID	–
Sample point variable	`sp_walk_diversity_care_500m`
Area variable (measured value)	`avg_diversity_walk_care_500m` (0–1)
Normalised score	`index_uli__div_care_500m`
Domains (share of weight)	Social infrastructure
Subdomain	Public services
Lens	Diversity
Better when	higher
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

Pending: Not recorded in the project register.

How to read it. Mean diversity of care services within a 500 m walk: … (0 = one kind or none, 1 = every kind equally).

Diversity of care services (Health, education and care access; ID –) - *How to calculate (variable sheet):* ?

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_diversity_walk_care_500m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_care_support` (Care and social support services (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act LIKE '623%' OR codigo_act LIKE '624%'"`

Inputs: Diversity sets and their sub-types, DENUE destinations by SCIAN code

Method: theory

1. For each of the `k = 3` sub-types of the set "care" (`childcare`, `residential_care`, `social_assistance`), the number of distinct destinations reachable within 500 m along the pedestrian network, `c_g` (R4).
2. Normalised Shannon entropy: `p_g = c_g / Σc`, `H = −Σ p_g·ln p_g / ln k`, from 0 (one sub-type or none) to 1 (every sub-type equally). `k` is the number of configured sub-types, not the number present.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R4. Diversity of what is reachable, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1602`):

```
- variable: sp_walk_diversity_care_500m
  name: div_care_500m
  domains: [social_infrastructure]
  group: {es: 'Acceso a salud, educación y cuidados', en: 'Health, education and care access'}
  label: {es: 'Diversidad de servicios de cuidado a 500 m', en: 'Care service diversity within 500 m'}
  reading: {es: 'Diversidad media de servicios de cuidado a 500 m a pie: {value} (0 = un tipo o ninguno, 1 = todos los tipos por igual)', en: 'Mean
diversity of care services within a 500 m walk: {value} (0 = one kind or none, 1 = every kind equally)'}
  subdomain: {es: 'Servicios públicos', en: 'Public services'}
  about:
    es: Diversidad (entropía de Shannon, 0-1) de los tipos de servicio de cuidado a 500 m a pie.
    en: Diversity (Shannon entropy, 0-1) of the kinds of care service within a 500 m walk.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:929` `_banded_counts()`
- `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_diversity_care_500m	as measured	−0.440 · 0.060 · 0.560	0.000 – 1.000	2.91 / 7.16 / 89.8%	skewed
`uli_plain`	sp_walk_diversity_care_500m	as measured	−0.440 · 0.060 · 0.560	0.000 – 1.000	2.91 / 7.16 / 89.8%	skewed

Recalculation (Reimagina Urbano)

A new destination adds 1 to its sub-type's `c_g` at points within 500 m along the network; entropy is recomputed at those points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

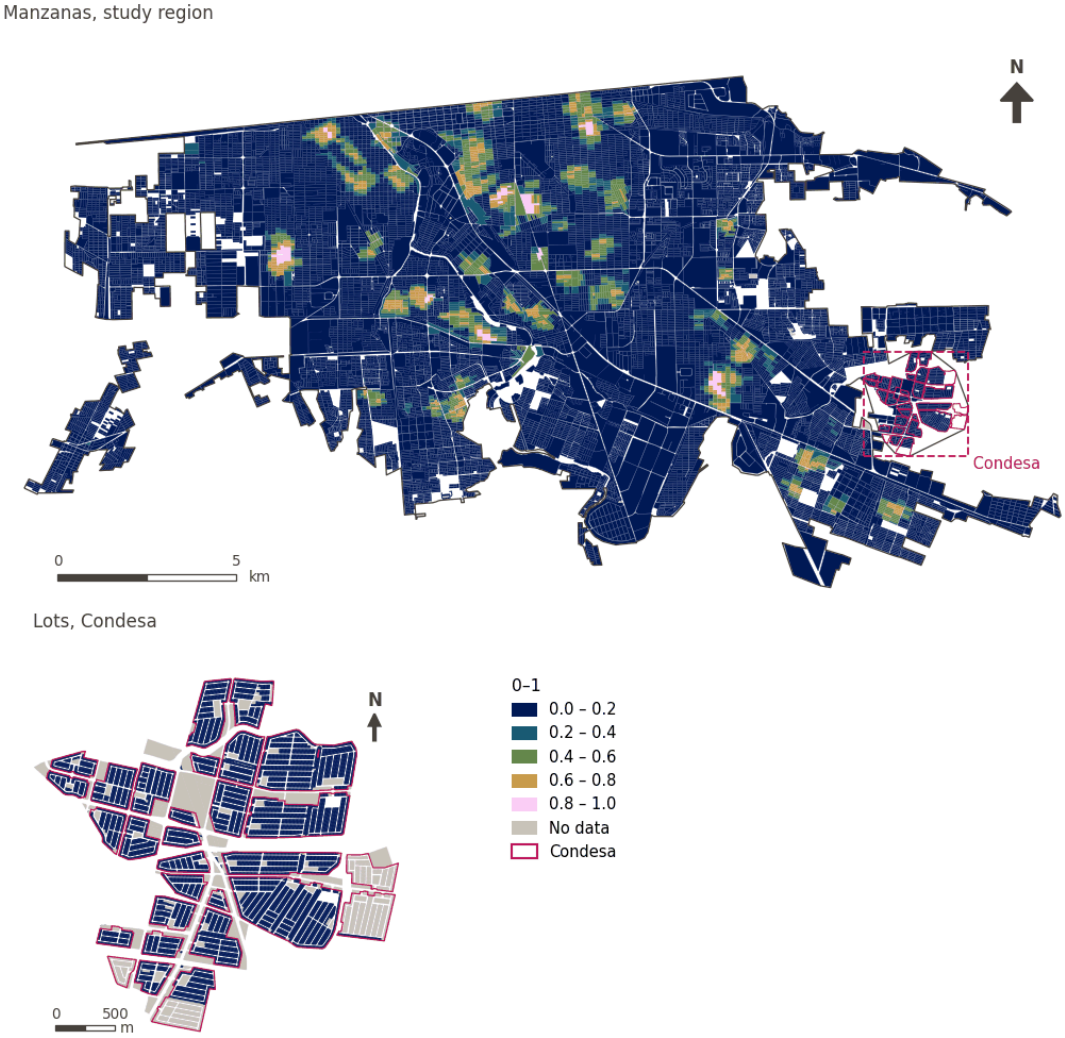
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.09	0.00
Region score (`uli`)	+1.7	−3.6
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.00 (0.00 – 0.00)	0.00 (0.00 – 0.00)
Range	0.00 – 1.00	0.00 – 0.00
Score: median (P25–P75) (`uli`)	−3.6 (−3.6 – −3.6)	−3.6 (−3.6 – −3.6)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean diversity of care services within a 500 m walk: 0.09 (0 = one kind or none, 1 = every kind equally).
- Condesa: Mean diversity of care services within a 500 m walk: 0.00 (0 = one kind or none, 1 = every kind equally).
- For the study region as a whole the value is 0.09; in Condesa, 0.00. Condesa's score lies 3.6 points below the study region average (+1.7 for the study region).

Map



Map. Care service diversity within 500 m: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Caveats and open questions

- The sheet's method for this row was not specified ("?"); the sub-types are the SCIAN groups within care and social support (configuration).

Care services within 300 m

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> Walking distance to the nearest care service, scored with a 300 m soft threshold.

Identity

Identifier	`care_support`
Core measure (variable sheet)	Health, education and care access
Sheet / register ID	–
Sample point variable	`sp_walk_nearest_node_denue_care_support`
Area variable (measured value)	`avg_walk_dist_denue_care_support` (m)
Normalised score	`index_uli__care_support`
Domains (share of weight)	Social infrastructure
Subdomain	Public services
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

Pending: Not recorded in the project register.

How to read it. … of the population live within 300 m of a care service; on average the nearest is … away (lower is better).

Access to care services (Health, education and care access; ID –) - *How to calculate (variable sheet):* USE GHSCI CALCULATION HERE (Average distance (m) along the pedestrian network to the nearest 'denue primary healthcare' (custom destination) (population weighted))

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_denue_care_support`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_care_support` (Care and social support services (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act LIKE '623%' OR codigo_act LIKE '624%'"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1589`):

```
- variable: sp_walk_nearest_node_denue_care_support
  name: care_support
  domains: [social_infrastructure]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a salud, educación y cuidados', en: 'Health, education and care access'}
  label: {es: 'Servicios de cuidado a menos de 300 m', en: 'Care services within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de un servicio de cuidado; en promedio, el más cercano está a {value}', en: '{access} of
the population live within 300 m of a care service; on average the nearest is {value} away'}
  subdomain: {es: 'Servicios públicos', en: 'Public services'}
  about:
    es: Distancia a pie al servicio de cuidado más cercano, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest care service, scored with a 300 m soft threshold.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:508`:

```
- {name: denue_care_support, category: care, variant: any, layer: destinations, where: "dest_name = 'denue_care_support'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_denue_care_support	soft access, 300 m threshold, k = 5	−0.336 · 0.161 · 0.657	0.000 – 0.993	1.75 / 1.58 / 19.2%	none
`uli_plain`	sp_walk_nearest_node_denue_care_support	soft access, 300 m threshold, k = 5	−0.336 · 0.161 · 0.657	0.000 – 0.993	1.75 / 1.58 / 19.2%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

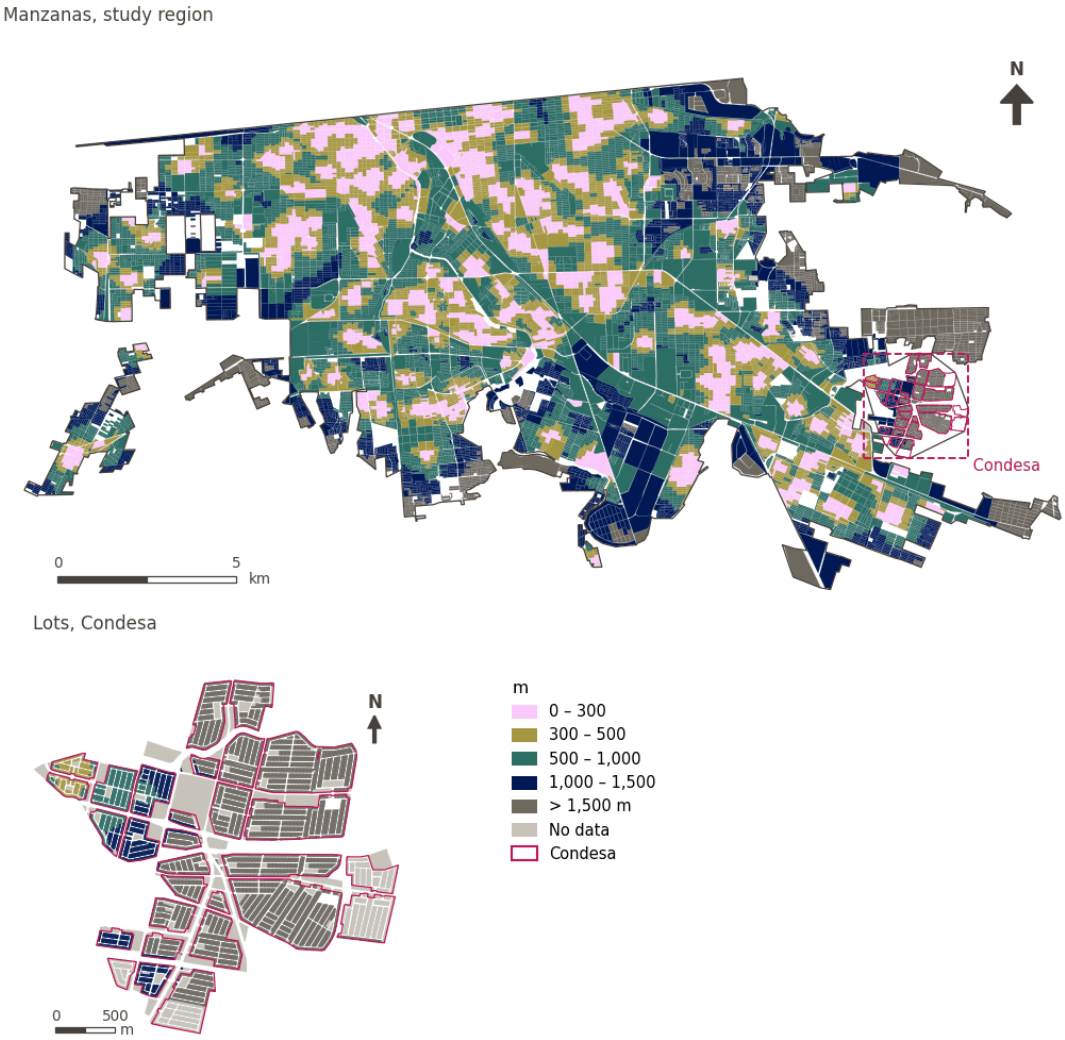
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	564 m	1,025 m
Region score (`uli`)	+4.1	−9.5
Areas with a value	12,813 of 14,252 manzanas	2,684 of 14,988 lots
Median (P25–P75), weighted	531 m (335 m – 820 m)	1,039 m (736 m – 1,374 m)
Range	31 m – 2,529 m	240 m – 1,639 m
Nothing within the distance searched	5.8%	79.8%
Score: median (P25–P75) (`uli`)	−8.0 (−9.7 – +12.5)	−9.7 (−9.7 – −9.7)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 22.0% of the population live within 300 m of a care service; on average the nearest is 564 m away.
- Condesa: 0.1% of the population live within 300 m of a care service; on average the nearest is 1,025 m away.
- For the study region as a whole the value is 564 m; in Condesa, 1,025 m. Condesa's score lies 9.5 points below the study region average (+4.1 for the study region).

Map



Map. Care services within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The sheet's "how to calculate" for this row describes primary healthcare; the care and social support category (SCIAN 623, 624) is used instead, matching the row's name.

Education diversity within 500 m

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> Diversity (Shannon entropy, 0-1) of the levels of education offered within a 500 m walk.

Identity

Identifier	`div_education_500m`
Core measure (variable sheet)	Health, education and care access
Sheet / register ID	180, 246, 379 · #180, #246, #379
Sample point variable	`sp_walk_diversity_education_500m`
Area variable (measured value)	`avg_diversity_walk_education_500m` (0–1)
Normalised score	`index_uli__div_education_500m`
Domains (share of weight)	Social infrastructure
Subdomain	Education
Lens	Diversity
Better when	higher
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

> Close proximity to primary schools supports family-oriented livability by facilitating easy and convenient access to essential basic education for young residents.
> *— Indicator register #180: Access to schools: primary (Social Infrastructure / Education)*

- Workbook citation: Onnom et al., 2018, 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'; Fu et al., 2019, 'The Livable Urban Landscape: GIS and Remote Sensing Extracted Land Use Assessment for Urban Livability in Changchun Proper, China'; Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Ensuring secondary schools are within a accessible distance enhances urban convenience and satisfies the educational requirements of residents, contributing to higher community satisfaction.
> *— Indicator register #246: Access to schools: secondary (Social Infrastructure / Education)*

- Workbook citation: Onnom et al., 2018, 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'; Fu et al., 2019, 'The Livable Urban Landscape: GIS and Remote Sensing Extracted Land Use Assessment for Urban Livability in Changchun Proper, China'; Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Universities provide essential social infrastructure; their accessibility contributes to the overall convenience and livability rating of urban environments.
> *— Indicator register #379: Access to schools: tertiary (universities) (Social Infrastructure / Education)*

- Workbook citation: Ibem and Amole, 2013, 'Residential satisfaction in public core housing in Abeokuta, Ogun State, Nigeria'; Huang and Du, 2015, 'Assessment and determinants of residential satisfaction with public housing in Hangzhou, China'
- Article list: #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Mean diversity of levels of education within a 500 m walk: … (0 = one kind or none, 1 = every kind equally).

Diversity of available educational services (Health, education and care access; ID 180, 246, 379) - *How to calculate (variable sheet):* Weighted share/number of preeschols, primary, secondary and tertiary educaitonal centers reachable within a walking distance

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_diversity_walk_education_500m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_preschool` (Preschools (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('611111','611112','611171','611172')"`
- `denue_primary_school` (Primary schools (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('611121','611122','611171','611172')"`
- `denue_secondary_school` (Secondary and upper secondary schools (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('611131','611132','611142','611151','611152','611161','611162','611171','611172')"`
- `denue_tertiary_education` (Tertiary and vocational education (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act LIKE '6112%' OR codigo_act LIKE '6113%' OR codigo_act IN ('611411','611412','611511','611512')"`

Inputs: Diversity sets and their sub-types, DENUE destinations by SCIAN code

Method: theory

1. For each of the `k = 4` sub-types of the set "education" (`preschool`, `primary`, `secondary`, `tertiary`), the number of distinct destinations reachable within 500 m along the pedestrian network, `c_g` (R4).
2. Normalised Shannon entropy: `p_g = c_g / Σc`, `H = −Σ p_g·ln p_g / ln k`, from 0 (one sub-type or none) to 1 (every sub-type equally). `k` is the number of configured sub-types, not the number present.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R4. Diversity of what is reachable, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1552`):

```
- variable: sp_walk_diversity_education_500m
  name: div_education_500m
  domains: [social_infrastructure]
  group: {es: 'Acceso a salud, educación y cuidados', en: 'Health, education and care access'}
  label: {es: 'Diversidad educativa a 500 m', en: 'Education diversity within 500 m'}
  reading: {es: 'Diversidad media de niveles educativos a 500 m a pie: {value} (0 = un tipo o ninguno, 1 = todos los tipos por igual)', en: 'Mean diversity
of levels of education within a 500 m walk: {value} (0 = one kind or none, 1 = every kind equally)'}
  subdomain: {es: 'Educación', en: 'Education'}
  about:
    es: Diversidad (entropía de Shannon, 0-1) de los niveles educativos ofrecidos a 500 m a pie.
    en: Diversity (Shannon entropy, 0-1) of the levels of education offered within a 500 m walk.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:929` `_banded_counts()`
- `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_diversity_education_500m	as measured	−0.232 · 0.268 · 0.768	0.000 – 1.000	0.67 / −1.15 / 57.3%	none
`uli_plain`	sp_walk_diversity_education_500m	as measured	−0.232 · 0.268 · 0.768	0.000 – 1.000	0.67 / −1.15 / 57.3%	none

Recalculation (Reimagina Urbano)

A new destination adds 1 to its sub-type's `c_g` at points within 500 m along the network; entropy is recomputed at those points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

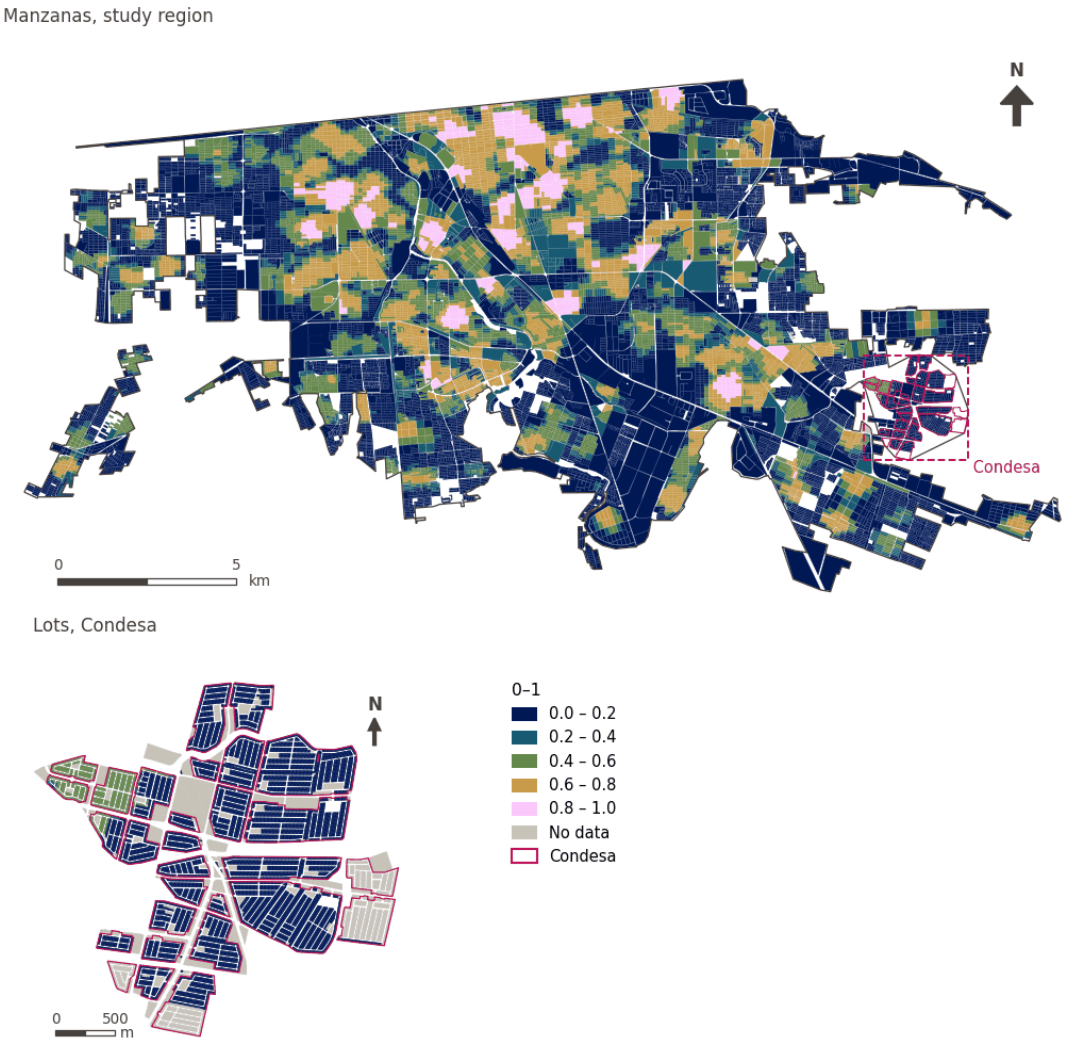
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.36	0.04
Region score (`uli`)	+5.8	−13.9
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.32 (0.00 – 0.55)	0.00 (0.00 – 0.00)
Range	0.00 – 1.00	0.00 – 0.74
Score: median (P25–P75) (`uli`)	+3.2 (−16.1 – +17.2)	−16.1 (−16.1 – −16.1)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean diversity of levels of education within a 500 m walk: 0.36 (0 = one kind or none, 1 = every kind equally).

- Condesa: Mean diversity of levels of education within a 500 m walk: 0.04 (0 = one kind or none, 1 = every kind equally).
- For the study region as a whole the value is 0.36; in Condesa, 0.04. Condesa's score lies 13.9 points below the study region average (+5.8 for the study region).

Map



Map. Education diversity within 500 m: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Health service diversity within 500 m

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> Diversity (Shannon entropy, 0-1) of the kinds of health service within a 500 m walk.

Identity

Identifier	`div_health_500m`
Core measure (variable sheet)	Health, education and care access
Sheet / register ID	97, 101, 170 · #97, #101, #170
Sample point variable	`sp_walk_diversity_health_care_500m`
Area variable (measured value)	`avg_diversity_walk_health_care_500m` (0–1)
Normalised score	`index_uli__div_health_500m`
Domains (share of weight)	Social infrastructure
Subdomain	Health and healthcare
Lens	Diversity
Better when	higher
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

> Proximity to healthcare facilities is essential for maintaining urban life quality and ensures residents have timely access to medical services.
> *— Indicator register #97: Access to health centres (Social Infrastructure / Health & Healthcare)*

- Workbook citation: McMichael, 2000, 'The urban environment and health in a world of increasing globalisation: issues for developing countries'
- Article list: #19: Wu (2022), 'Urban greenness, mixed land-use, and life satisfaction: Evidence from residential locations and workplace settings in Beijing'; #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> The physical availability and accessibility of healthcare facilities, such as hospitals, are essential determinants of health within the urban liveability framework.
> *— Indicator register #101: Access to hospitals (Social Infrastructure / Health & Healthcare)*

- Workbook citation: Mercer, 2017, 'Quality of Living Survey'; Economist Intelligence Unit (EIU), 2012, 'Quality of Life Index'
- Article list: #8: Onnom (2018), 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

> Access to medical stores is identified by residents as a fundamental amenity for liveability, ensuring the availability of healthcare commodities across the city.
> *— Indicator register #170: Access to pharmacies (Social Infrastructure / Health & Healthcare)*

- Workbook citation: Chi et al., 2021, 'From Comparative and Statistical Assessments of Liveability and Health Conditions of Districts in Hong Kong towards Future City Development'; Suptelo & Sehida, 2020, 'Factors of Post-Industrial Transformations of Urban Sociogeosystems (Case Kharkiv)'
- Article list: #5: Mahanta (2025), 'Constructing a city liveability index and evaluating the spatial pattern of liveability of Guwahati city'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. Mean diversity of health services within a 500 m walk: … (0 = one kind or none, 1 = every kind equally).

Diversity of available health & health care services (Health, education and care acc ess; ID 97,101, 170) - *How to calculate (variable sheet):* Weighted share/number of health centers, clinics, hospitals and pharmacies reachable within a walking distance

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_diversity_walk_health_care_500m`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_primary_healthcare` (Primary healthcare (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('621111','621112','621115','621116')"`

- `denue_medical_clinic` (Specialist and diagnostic clinics (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('621113','621211','621212','621511')"`
- `denue_hospital` (Hospitals (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act LIKE '622%'"`
- `denue_pharmacy` (Pharmacies and medicine retail (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('464111','464112','464113')"`
- `denue_care_support` (Care and social support services (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act LIKE '623%' OR codigo_act LIKE '624%'"`

Inputs: Diversity sets and their sub-types, DENUE destinations by SCIAN code

Method: theory

1. For each of the `k = 5` sub-types of the set "health_care" (`primary_healthcare`, `specialist_clinic`, `hospital`, `pharmacy`, `care_support`), the number of distinct destinations reachable within 500 m along the pedestrian network, `c_g` (R4).
2. Normalised Shannon entropy: `p_g = c_g / Σc`, `H = −Σ p_g·ln p_g / ln k`, from 0 (one sub-type or none) to 1 (every sub-type equally). `k` is the number of configured sub-types, not the number present.
3. AMPI normalisation (R11), aggregation (R12) and areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R4. Diversity of what is reachable, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1577`):

```
- variable: sp_walk_diversity_health_care_500m
  name: div_health_500m
  domains: [social_infrastructure]
  group: {es: 'Acceso a salud, educación y cuidados', en: 'Health, education and care access'}
  label: {es: 'Diversidad de servicios de salud a 500 m', en: 'Health service diversity within 500 m'}
  reading: {es: 'Diversidad media de servicios de salud a 500 m a pie: {value} (0 = un tipo o ninguno, 1 = todos los tipos por igual)', en: 'Mean diversity
of health services within a 500 m walk: {value} (0 = one kind or none, 1 = every kind equally)'}
  subdomain: {es: 'Salud y atención sanitaria', en: 'Health and healthcare'}
  about:
    es: Diversidad (entropía de Shannon, 0-1) de los tipos de servicio de salud a 500 m a pie.
    en: Diversity (Shannon entropy, 0-1) of the kinds of health service within a 500 m walk.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:929` `_banded_counts()`
- `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_diversity_health_care_500m	as measured	−0.190 · 0.310 · 0.810	0.000 – 1.000	0.32 / −1.53 / 48.1%	none
`uli_plain`	sp_walk_diversity_health_care_500m	as measured	−0.190 · 0.310 · 0.810	0.000 – 1.000	0.32 / −1.53 / 48.1%	none

Recalculation (Reimagina Urbano)

A new destination adds 1 to its sub-type's `c_g` at points within 500 m along the network; entropy is recomputed at those points.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

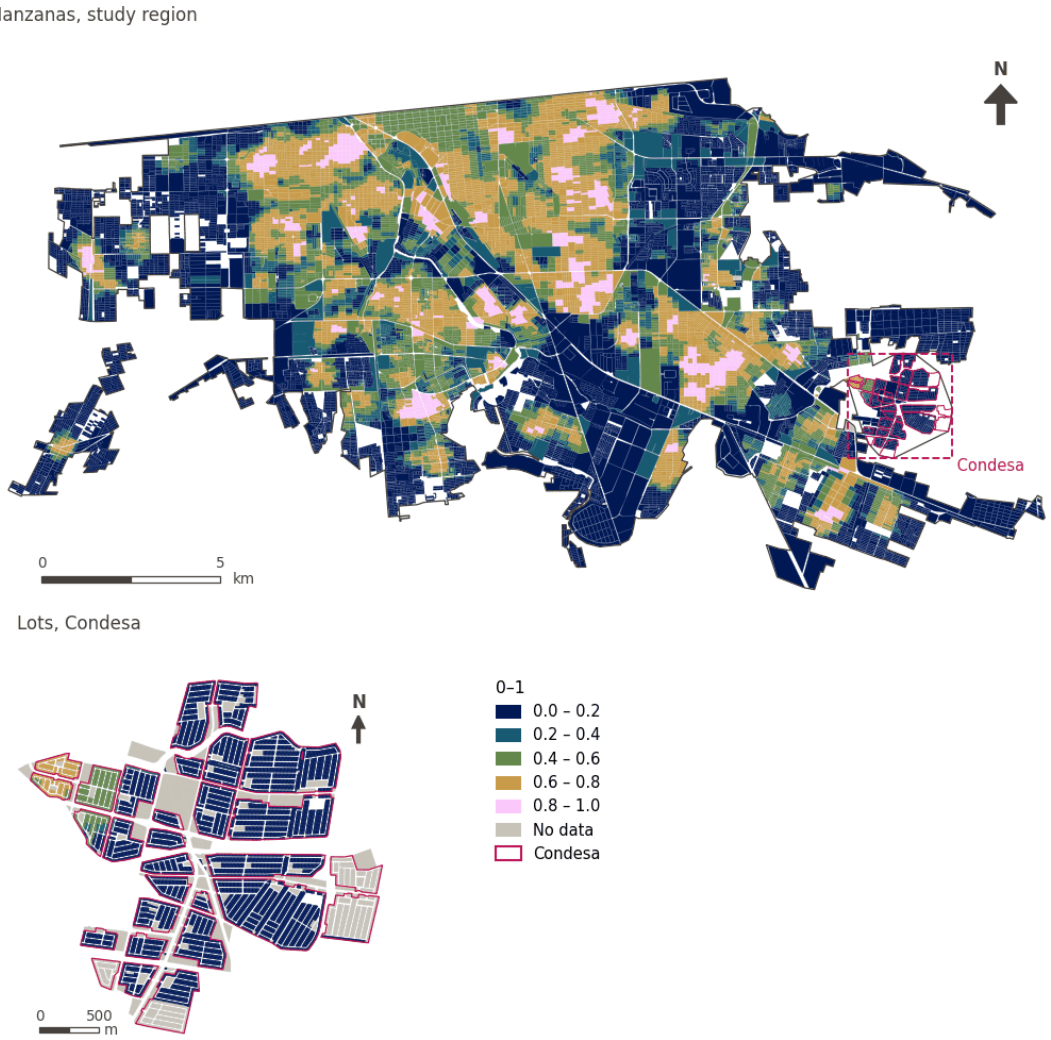
Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	0.43	0.04
Region score (`uli`)	+7.4	−15.9
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.42 (0.00 – 0.64)	0.00 (0.00 – 0.00)
Range	0.00 – 0.99	0.00 – 0.66
Score: median (P25–P75) (`uli`)	+6.4 (−18.6 – +19.6)	−18.6 (−18.6 – −18.6)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

- Study region: Mean diversity of health services within a 500 m walk: 0.43 (0 = one kind or none, 1 = every kind equally).
- Condesa: Mean diversity of health services within a 500 m walk: 0.04 (0 = one kind or none, 1 = every kind equally).
- For the study region as a whole the value is 0.43; in Condesa, 0.04. Condesa's score lies 15.9 points below the study region average (+7.4 for the study region).

Map



Map. Health service diversity within 500 m: measured value (0–1) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Primary healthcare within 300 m

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> Walking distance to the nearest primary healthcare service, scored with a 300 m soft threshold.

Identity

Identifier	`primary_healthcare`
Core measure (variable sheet)	Health, education and care access
Sheet / register ID	97, 101, 170 · #97
Sample point variable	`sp_walk_nearest_node_denue_primary_healthcare`
Area variable (measured value)	`avg_walk_dist_denue_primary_healthcare` (m)
Normalised score	`index_uli__primary_healthcare`
Domains (share of weight)	Social infrastructure
Subdomain	Health and healthcare
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

> Proximity to healthcare facilities is essential for maintaining urban life quality and ensures residents have timely access to medical services.
>
> *— Indicator register #97: Access to health centres (Social Infrastructure / Health & Healthcare)*

- Workbook citation: McMichael, 2000, 'The urban environment and health in a world of increasing globalisation: issues for developing countries'
- Article list: #19: Wu (2022), 'Urban greenness, mixed land-use, and life satisfaction: Evidence from residential locations and workplace settings in Beijing'; #20: Basu (2021), 'Urban livability index assessment based on land-use changes in an Indian medium-sized city (Raiganj)'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 300 m of primary healthcare; on average the nearest is … away (lower is better).

Access to health & health care services (Health, education and care access; ID 97,101, 170) - *How to calculate (variable sheet):* Walking distance access to health centers, clinics, hospitals and pharmacies.

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_denue_primary_healthcare`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_primary_healthcare` (Primary healthcare (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('621111','621112','621115','621116')"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1564`):

```
- variable: sp_walk_nearest_node_denue_primary_healthcare
  name: primary_healthcare
  domains: [social_infrastructure]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a salud, educación y cuidados', en: 'Health, education and care access'}
  label: {es: 'Atención primaria de salud a menos de 300 m', en: 'Primary healthcare within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de un servicio de atención primaria; en promedio, el más cercano está a {value}', en:
'{access} of the population live within 300 m of primary healthcare; on average the nearest is {value} away'}
  subdomain: {es: 'Salud y atención sanitaria', en: 'Health and healthcare'}
  about:
    es: Distancia a pie al servicio de atención primaria más cercano, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest primary healthcare service, scored with a 300 m soft threshold.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:505`:

```
- {name: denue_primary_healthcare, category: healthcare, variant: any, layer: destinations, where: "dest_name = 'denue_primary_healthcare'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_denue_primary_healthcare	soft access, 300 m threshold, k = 5	−0.358 · 0.139 · 0.636	0.000 – 0.993	1.99 / 2.46 / 26.8%	none
`uli_plain`	sp_walk_nearest_node_denue_primary_healthcare	soft access, 300 m threshold, k = 5	−0.358 · 0.139 · 0.636	0.000 – 0.993	1.99 / 2.46 / 26.8%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

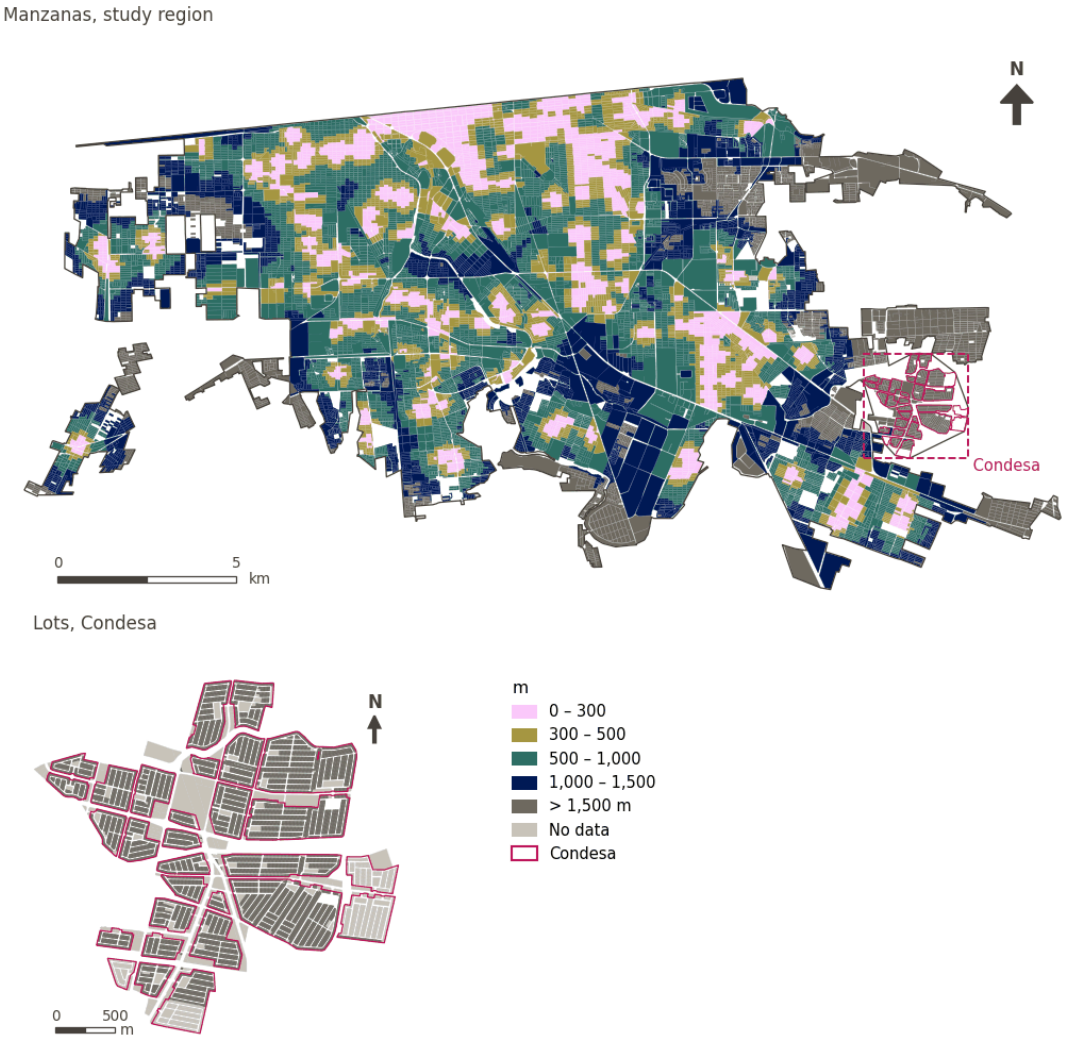
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	625 m	–
Region score (`uli`)	+3.2	−8.4
Areas with a value	12,068 of 14,252 manzanas	0 of 14,988 lots
Median (P25–P75), weighted	604 m (373 m – 917 m)	–
Range	42 m – 2,470 m	–
Nothing within the distance searched	10.7%	100.0%
Score: median (P25–P75) (`uli`)	−8.1 (−8.4 – +6.1)	−8.4 (−8.4 – −8.4)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 18.4% of the population live within 300 m of primary healthcare; on average the nearest is 625 m away.
- Condesa: 0.0% of the population live within 300 m of primary healthcare; on average the nearest is – away.
- For the study region as a whole the value is 625 m; in Condesa, –. Condesa's score lies 8.4 points below the study region average (+3.2 for the study region).

Map



Map. Primary healthcare within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- The ULI worksheet gives SCIAN 6221 for this row, but 6221 is general hospitals; general medical consulting rooms and clinics (6211xx) are used (configuration).

Primary schools within 300 m

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> Walking distance to the nearest primary school, scored with a 300 m soft threshold.

Identity

Identifier	`primary_school`
Core measure (variable sheet)	Health, education and care access
Sheet / register ID	180, 246, 379 · #180
Sample point variable	`sp_walk_nearest_node_denue_primary_school`
Area variable (measured value)	`avg_walk_dist_denue_primary_school` (m)
Normalised score	`index_uli__primary_school`
Domains (share of weight)	Social infrastructure
Subdomain	Education
Lens	Proximity
Better when	measured value lower (transformed score higher)
Effective weight in the index	`uli` 1.7% · `uli_plain` 1.7%
Status	Scored in the published index

Rationale and interpretation

> Close proximity to primary schools supports family-oriented livability by facilitating easy and convenient access to essential basic education for young residents.
> *— Indicator register #180: Access to schools: primary (Social Infrastructure / Education)*

- Workbook citation: Onnom et al., 2018, 'Development of a Liveable City Index (LCI) Using Multi Criteria Geospatial Modelling for Medium Class Cities in Developing Countries'; Fu et al., 2019, 'The Livable Urban Landscape: GIS and Remote Sensing Extracted Land Use Assessment for Urban Livability in Changchun Proper, China'; Liu et al., 2020, 'Using Multi-Source Data to Assess Livability in Hong Kong at the Community-Based Level: A Combined Subjective-Objective Approach'
- Article list: #18: Jodder (2024), 'Urban Livability in a Rapidly Urbanizing Mid-Size City: Lessons for Planning in the Global South'
- ⚠ the workbook citation names no author of the listed article; verify before citing (citation_audit.csv)

How to read it. … of the population live within 300 m of a primary school; on average the nearest is … away (lower is better).

Access to educational services (Health, education and care access; ID 180, 246, 379) - *How to calculate (variable sheet):* Walking distance access to preeschols, primary, secondary and tertiary educaitonal centers - *Notes (sheet):* Using DENUE data

Data sources

Sources declared in the configuration:

- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.

Source recorded by GHSCI (data dictionary) (`avg_walk_dist_denue_primary_school`):

- OpenStreetMap: Geofabrik GmbH, 20260815, ODbL, https://download.geofabrik.de/north-america/mexico-latest.osm.pbf, OpenStreetMap Contributors. Geofabrik GmbH (2026). https://download.geofabrik.de/north-america/mexico-latest.osm.pbf; INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE): 2026-05, Terminos de Libre Uso de la Informacion del INEGI, https://www.inegi.org.mx/app/mapa/denue/, INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Destination filter:

- `denue_primary_school` (Primary schools (DENUE)): `MX/INEGI/DENUE_2026/denue_inegi_02_.shp` — Destination filter: `"codigo_act IN ('611121','611122','611171','611172')"`

Inputs: DENUE destinations by SCIAN code

Method: theory

1. Distance along the pedestrian network from each sample point to the nearest destination, `d` (R2), censored at the largest band searched (1,500 m): a point with no destination within it has no value.
2. Soft access score (R10): `s(d) = 1 / (1 + exp(k·(d − t)/t))`, with `t = 300 m` and `k = 5`; `s = 0.5` at the threshold, ≈ 0.99 beside the destination, ≈ 0.007 at `2t`; no value → `s = 0`.
3. AMPI normalisation (R11) of `s` against the study region's goalposts, then aggregation to domains and index (R12) and to areas (R13).

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R10. Transforms: soft thresholds and stepped ladders, R11. Normalisation: the Adjusted Mazziotta–Pareto Index (AMPI), R12. Aggregation: domains, shared indicators and the index, R13. From sample points to areas, R14. Frozen goalposts and recalculating a scenario

Configuration (`MX_Mexicali_2025_ULI.yml:1539`):

```
- variable: sp_walk_nearest_node_denue_primary_school
  name: primary_school
  domains: [social_infrastructure]
  transform: {soft_threshold: 300}
  group: {es: 'Acceso a salud, educación y cuidados', en: 'Health, education and care access'}
  label: {es: 'Escuelas primarias a menos de 300 m', en: 'Primary schools within 300 m'}
  reading: {es: '{access} de la población vive a menos de 300 m de una escuela primaria; en promedio, la más cercana está a {value}', en: '{access} of the
population live within 300 m of a primary school; on average the nearest is {value} away'}
  subdomain: {es: 'Educación', en: 'Education'}
  about:
    es: Distancia a pie a la escuela primaria más cercana, puntuada con un umbral suave de 300 m.
    en: Walking distance to the nearest primary school, scored with a 300 m soft threshold.
  sources:
    - 'INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.'
```

`MX_Mexicali_2025_ULI.yml:502`:

```
- {name: denue_primary_school, category: primary_school, variant: any, layer: destinations, where: "dest_name = 'denue_primary_school'"}
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:806` `_banded_distances()`
- `process/subprocesses/setup_sp.py:888` `create_full_nodes()`
- `process/subprocesses/_composite_index.py:1457` `soft_threshold()`
- `process/subprocesses/_composite_index.py:1819` `normalise()`
- `process/subprocesses/_composite_index.py:1838` `mpi_aggregate()`

	Sample point variable	Transform	Goalposts (scale scored)	inf – sup	skew / kurt. / tied	Flags
`uli`	sp_walk_nearest_node_denue_primary_school	soft access, 300 m threshold, k = 5	−0.330 · 0.166 · 0.663	0.000 – 0.993	1.70 / 1.44 / 12.5%	none
`uli_plain`	sp_walk_nearest_node_denue_primary_school	soft access, 300 m threshold, k = 5	−0.330 · 0.166 · 0.663	0.000 – 0.993	1.70 / 1.44 / 12.5%	none

Recalculation (Reimagina Urbano)

A destination added or removed, or a change to the pedestrian network, changes `d` only at points whose nearest destination changes: a multi-source Dijkstra from the affected destinations, limited to 1,500 m, taking the minimum with the previous distance, suffices.

With the frozen goalposts of the parameters file (`Ref`, `Min`, `Max`, R14), a new score is `r = 100 ± 60·(x − Ref)/(Max − Min)` without re-resolving goalposts; the domain and index are recomputed by R12, less 100. A scenario is then compared with the baseline on the same scale.

Results

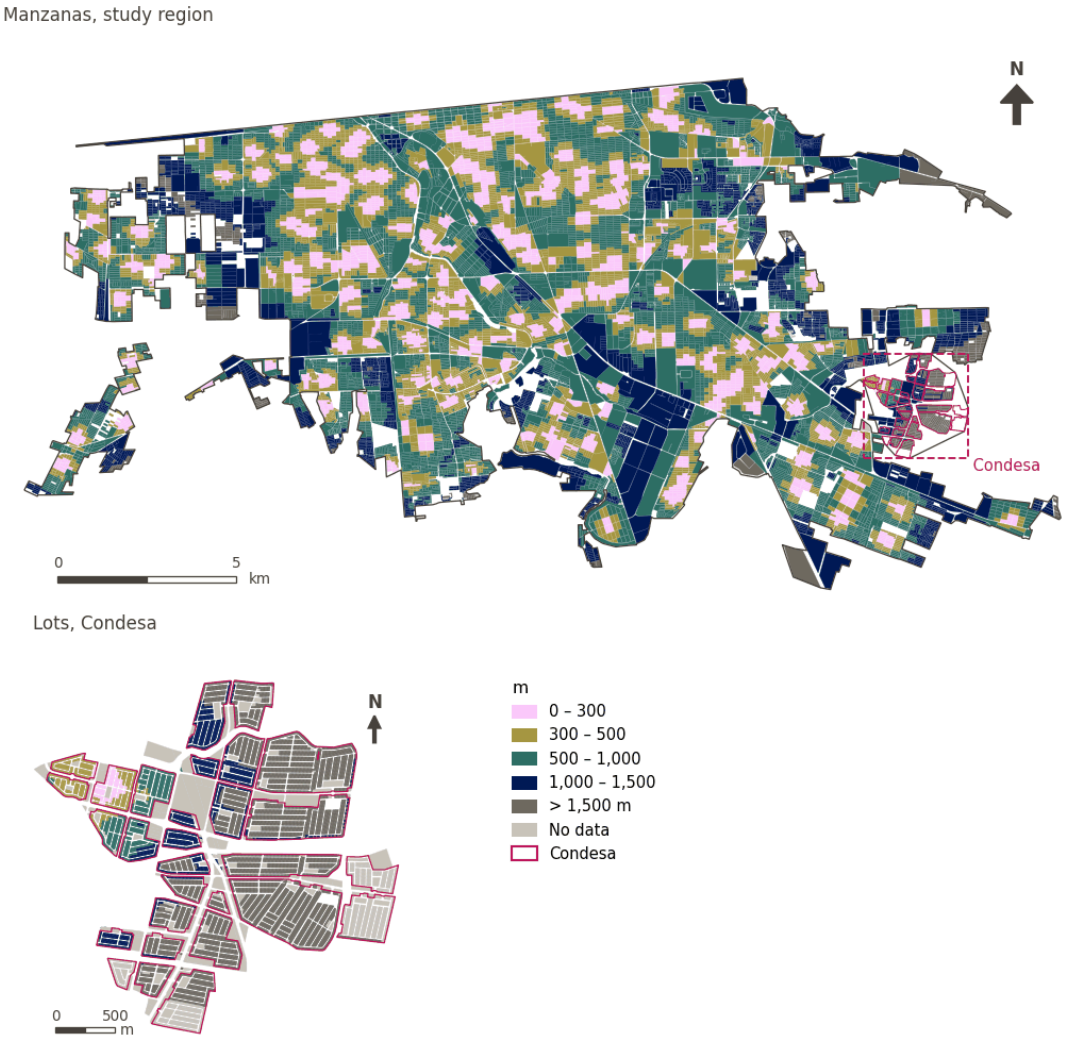
Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	525 m	1,013 m
Region score (`uli`)	+4.6	−8.6
Areas with a value	13,765 of 14,252 manzanas	4,084 of 14,988 lots
Median (P25–P75), weighted	500 m (341 m – 730 m)	1,039 m (518 m – 1,462 m)
Range	58 m – 2,777 m	33 m – 3,051 m
Nothing within the distance searched	1.0%	69.2%
Score: median (P25–P75) (`uli`)	−6.0 (−9.9 – +13.0)	−10.1 (−10.1 – −10.1)

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

A mean distance is taken over the sample points that found a destination within the distance searched; the row "Nothing within the distance searched" gives the share of population in areas that found none.

- Study region: 22.9% of the population live within 300 m of a primary school; on average the nearest is 525 m away.
- Condesa: 1.9% of the population live within 300 m of a primary school; on average the nearest is 1,013 m away.
- For the study region as a whole the value is 525 m; in Condesa, 1,013 m. Condesa's score lies 8.6 points below the study region average (+4.6 for the study region).

Map



Map. Primary schools within 300 m: measured value (m) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Caveats and open questions

- Schools combining several levels (SCIAN 611171/611172) are counted in each level they serve (configuration).

Walkability within 300 m and its components

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The ULI's walkability follows the GHSCI walkability index (Frank et al. 2010), with the daily living score judged within 300 m, the five-minute walk of Mexicali's walkability policy, rather than 500 m.

- **Daily living within 300 m** (`sp_walk_dl_300`): the sum of binary access within 300 m to a fresh food market, a convenience store (both from OpenStreetMap) and a public transport stop (IMIP or OpenStreetMap); 0 to 3. If any of the three is missing, the score is missing.
- **Neighbourhood population density** (`sp_local_nh_avg_pop_density`): for each node, the unweighted mean population density (persons/km²) of the 100 m cells containing any node reachable within 1,000 m along the network; interpolated to each point by its position along the edge.
- **Neighbourhood intersection density** (`sp_local_nh_avg_intersection_density`): likewise, with intersections per km² (OSMnx consolidated intersections, 12 m tolerance).
- **Walkability** (`sp_walk_idx_300`): `W = z(DL) + z(population density) + z(intersection density)`, z-scores using the sample standard deviation over all urban sample points with a value.
- **Walkability attenuated by thermal comfort** (`sp_walk_idx_300_tm`): `percentile_rank(W)·(1 − 0.5·h')`, where `h'` is the daytime mean UTCI re-scaled from 30 to 50 °C and clipped to 0–1. The percentile rank averages ties and runs from 0 to 1.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_walk_dl_300`	`walk_dl_300`	Average across sample points of the daily living score: the number of daily living destination types (fresh food market, convenience, pt any) with access within 300 m	0.300	0.016
`sp_local_nh_avg_pop_density`	`local_nh_population_density`	Average walkable neighbourhood population density (population per km²) of sample points in this area	4,274	131
`sp_local_nh_avg_intersection_density`	`local_nh_intersection_density`	Average walkable neighbourhood intersection density (intersections per km²) of sample points in this area	171	174
`sp_walk_idx_300`	`walk_idx_300`	Average across sample points of the walkability index with daily living access judged within 300 m: the sum of the z-scores of that daily living score, local population density and street intersection density	1.05	−1.94
`sp_walk_idx_300_tm`	`walk_idx_300_tm`	Average across sample points of the percentile rank (0-1) of the walkability index with daily living access judged within 300 m: the sum of the z-scores of that daily living score, local population density and street intersection density, multiplied by (1 - lambda x h), where h is thermal comfort re-scaled from 0 at 30 to 1 at 50 (a fixed range); lambda = 0.5 (attenuation)	0.41	0.12

Method: practice

Recipes: R1. Sample points and the pedestrian network, R2. Network distance to the nearest destination, censored, R3. Banded access, and distance from a disamenity, R9. Walkability, and its attenuation by thermal comfort

Configuration (`MX_Mexicali_2025_ULI.yml:909`):

```
walkability_variants:
  distances: [300]
  daily_living: [fresh_food_market, convenience, pt_any]
  heat:
    t: {variable: sp_utci_day_mean, range: [30, 50], label: {es: confort térmico, en: thermal comfort}}
  sets: [t]
  forms: [multiplicative]
  attenuation: 0.5
  percentiles: [5, 95]
```

Code (GHSCI):

- `process/subprocesses/_11_neighbourhood_analysis.py:115` `compute_nodes_pop_intersect_density()`
- `process/subprocesses/_walkability_variants.py:243` `zscore()`
- `process/subprocesses/_walkability_variants.py:270` `percentile_rank()`
- `process/subprocesses/_walkability_variants.py:261` `rescale()`
- `process/subprocesses/_walkability_variants.py:275` `compute_variants()`

Used by

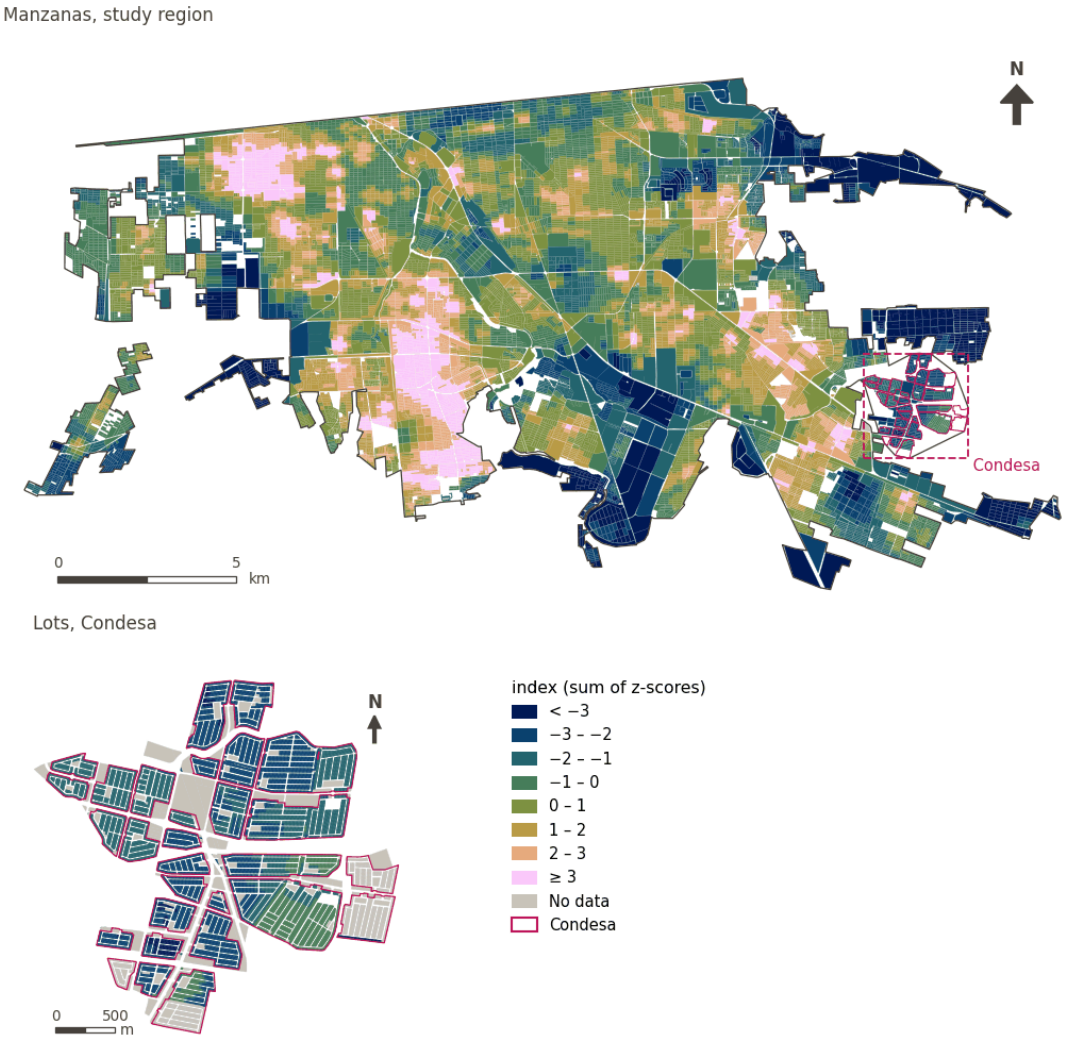
Walkability (300 m), attenuated by thermal comfort

Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	1.05	−1.94
Areas with a value	14,216 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	0.824 (−0.124 – 1.94)	−2.02 (−2.36 – −1.58)
Range	−5.59 – 7.01	−4.26 – −0.363

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

Map



Map. Walkability within 300 m and its components: measured value (index (sum of z-scores)) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (higher is better).

Outdoor thermal comfort (UTCI)

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Universal Thermal Climate Index modelled by work package 2 (Rossano Schifanella) with SOLWEIG for the hottest day of 2023 (21 July; maximum temperature 48.5 °C), forced by Mexicali airport observations, over pedestrian surfaces, and converted to UTCI with the operational polynomial (Bröde et al. 2012). It was delivered summarised for the reference areas (grid, AGEBs, manzanas, lots and subdivisions) and is linked by identifier. Units with fewer than 100 pedestrian pixels are masked (`reliable`).

- `utci_day_mean`: unit mean of the daytime mean UTCI (sunlit hours); the input to walkability's attenuation and the thermal comfort indicator.
- `utci_peak`: unit mean of the daily maximum UTCI (effectively 17:00); linked for reference, not used in the index.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_utci_day_mean`	`utci_day_mean`	Unit mean of the daytime mean Universal Thermal Climate Index (hours with sunlight), over pedestrian surfaces, on the hottest day of 2023.	45.1 °C	45.5 °C
`sp_utci_peak`	`utci_peak`	Unit mean of the daily maximum Universal Thermal Climate Index (effectively 17:00), over pedestrian surfaces, on the hottest day of 2023.	52.8 °C	52.9 °C

Column	Name	Units	Direction
`utci_day_mean`	Daytime thermal comfort (mean UTCI)	°C	lower_is_better
`utci_peak`	Peak UTCI (17:00)	°C	lower_is_better

- `data`: MX/UTCI - 2026-08-25/mexicali_utci_2023-07-21.gpkg
- `match`: id
- `id`: area_id
- `mask`: reliable
- `source`: Rossano Schifanella (WP02); SOLWEIG and the UTCI operational polynomial (Bröde et al. 2012)
- `licence`: Terrain from FABDEM v1 (CC BY-NC-SA 4.0, non-commercial); other inputs CC BY 4.0 or public domain
- `notes`: Modelled day 2023-07-21 (Tmax 48.5 °C). See README_mexicali_utci.md, sections 3.1 and 5, for forcing and limitations.

Method: practice

Recipes: R7. Linking externally prepared indicators

Configuration (`MX_Mexicali_2025_ULI.yml:770`):

```
utci:
  data: MX/UTCI - 2026-08-25/mexicali_utci_2023-07-21.gpkg
  match: id
  id: area_id
  mask: reliable
  label: {es: Confort térmico exterior (UTCI), en: Outdoor thermal comfort (UTCI)}
  source: Rossano Schifanella (WP02); SOLWEIG and the UTCI operational polynomial (Bröde et al. 2012)
  licence: Terrain from FABDEM v1 (CC BY-NC-SA 4.0, non-commercial); other inputs CC BY 4.0 or public domain
  notes: 'Modelled day 2023-07-21 (Tmax 48.5 °C).  See README_mexicali_utci.md, sections 3.1 and 5, for forcing and limitations.'
  columns:
    utci_day_mean:
      label: {es: 'Confort térmico diurno (UTCI medio)', en: 'Daytime thermal comfort (mean UTCI)'}
      units: °C
      direction: lower_is_better
      description: Unit mean of the daytime mean Universal Thermal Climate Index (hours with sunlight), over pedestrian surfaces, on the hottest day of
2023.
    utci_peak:
      label: {es: 'UTCI máximo (17:00)', en: 'Peak UTCI (17:00)'}
      units: °C
      direction: lower_is_better
      description: Unit mean of the daily maximum Universal Thermal Climate Index (effectively 17:00), over pedestrian surfaces, on the hottest day of
2023.
  areas:
    grid: grid_100m
    agebs: agebs
    manzanas: manzanas
    condesa_lotes: {layer: condesa_lotes, boundary_id: id}
    condesa_fraccionamientos: {layer: condesa_fraccionamientos, boundary_id: fid}
    region_mexicali: region
    region_condesa: region
  ## the 100 m grid tiles the study region: sample points lie on streets,
  ## between blocks, so a block layer would leave most of them unvalued
  sample_points: [grid_100m]
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:512` `apply_rules()`
- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`

Used by

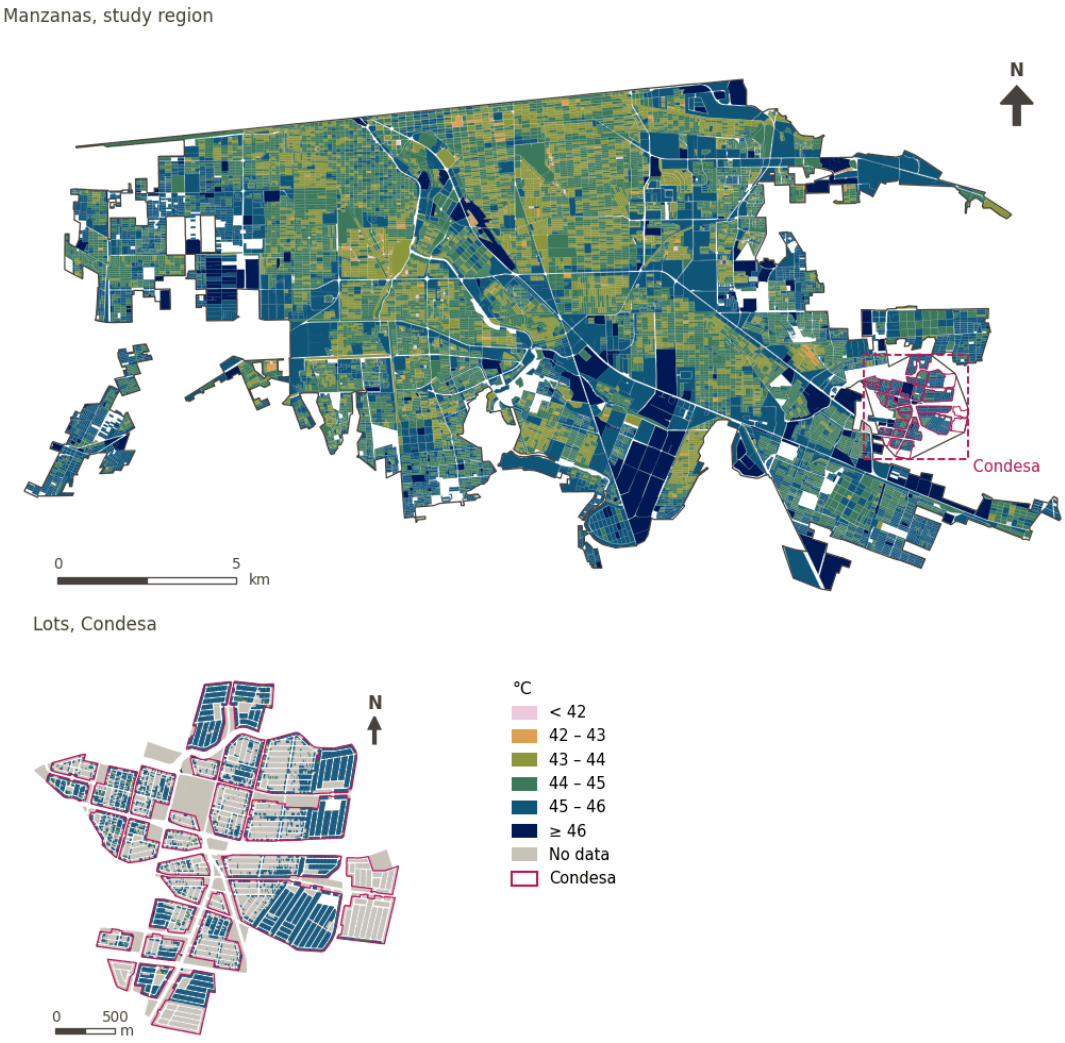
Daytime thermal comfort, Walkability (300 m), attenuated by thermal comfort

Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	45.1 °C	45.5 °C
Areas with a value	14,200 of 14,252 manzanas	7,766 of 14,988 lots
Median (P25–P75), weighted	44.3 °C (43.9 °C – 44.8 °C)	45.9 °C (45.6 °C – 46.0 °C)
Range	40.4 °C – 46.7 °C	40.4 °C – 46.4 °C

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

Map



Map. Outdoor thermal comfort (UTCI): measured value (°C) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Global Urban Heat Vulnerability Index (GUHVI) and its inputs

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The GUHVI (Turner et al. 2025) is computed in Google Earth Engine on an analysis grid. Each input is min–max normalised to 0–100 within the region (weighted by the share of the cell that is not water); the child dependency ratio, subnational HDI and infant mortality use a fixed 0–100 range.

- **Heat exposure (HEI)**: land surface temperature (Landsat 8, over the hottest four consecutive months identified from five years of MODIS).
- **Heat sensitivity (HSI)**: equally weighted mean of inverted albedo, negated NDVI, NDBI, local climate zone, population density and vulnerable population percentage.
- **Adaptive capability (ACI)**: equally weighted mean of the child dependency ratio, subnational HDI and infant mortality rate.
- **GUHVI**: the mean of the three; the GUHVI class is quintiles 1–5.

The ULI uses the three sub-indices, not the overall GUHVI. Each sample point takes the value of the cell containing it.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_urban_heat_exposure_index`	`urban_heat_exposure_index`	Heat Exposure Index (HEI): the heat exposure sub-index, derived from land surface temperature	49.0	33.3
`sp_urban_heat_sensitivity_index`	`urban_heat_sensitivity_index`	Heat Sensitivity Index (HSI): the equally weighted heat sensitivity sub-index of land surface albedo, NDVI, NDBI, local climate zone, population density and vulnerable population	59.7	33.8
`sp_urban_heat_adaptive_capability_index`	`urban_heat_adaptive_capability_index`	Adaptive Capability Index (ACI): the equally weighted adaptive capability sub-index of child dependency ratio, subnational Human Development Index and infant mortality rate; higher values indicate lower adaptive capability	22.2	15.0
`sp_urban_heat_guhvi`	`urban_heat_guhvi`	Global Urban Heat Vulnerability Index (GUHVI): the equally weighted composite of the Heat Exposure, Heat Sensitivity and Adaptive Capability indices, (HEI + HSI + ACI) / 3	43.6	27.4

Variable	Description	Study region	Condesa
`urban_heat_adaptive_capability_index`	Adaptive Capability Index (ACI): the equally weighted adaptive capability sub-index of child dependency ratio, subnational Human Development Index and infant mortality rate; higher values indicate lower adaptive capability	22.2	15.0
`urban_heat_exposure_index`	Heat Exposure Index (HEI): the heat exposure sub-index, derived from land surface temperature	49.0	33.3
`urban_heat_guhvi`	Global Urban Heat Vulnerability Index (GUHVI): the equally weighted composite of the Heat Exposure, Heat Sensitivity and Adaptive Capability indices, (HEI + HSI + ACI) / 3	43.6	27.4
`urban_heat_guhvi_class`	Global Urban Heat Vulnerability Index (GUHVI) class, from 1 (least vulnerable) to 5 (most vulnerable)	3.01	1.02
`urban_heat_sensitivity_index`	Heat Sensitivity Index (HSI): the equally weighted heat sensitivity sub-index of land surface albedo, NDVI, NDBI, local climate zone, population density and vulnerable population	59.7	33.8

Method: practice

Recipes: R8. Earth Engine: heat vulnerability and large green space

Configuration (`MX_Mexicali_2025_ULI.yml:706`):

```
gee: true
```

Code (GHSCI):

- `process/subprocesses/_earth_engine_indicators.py:846` `guhvi_analysis()`
- `process/subprocesses/_earth_engine_indicators.py:1105` `normalise_band()`

Used by

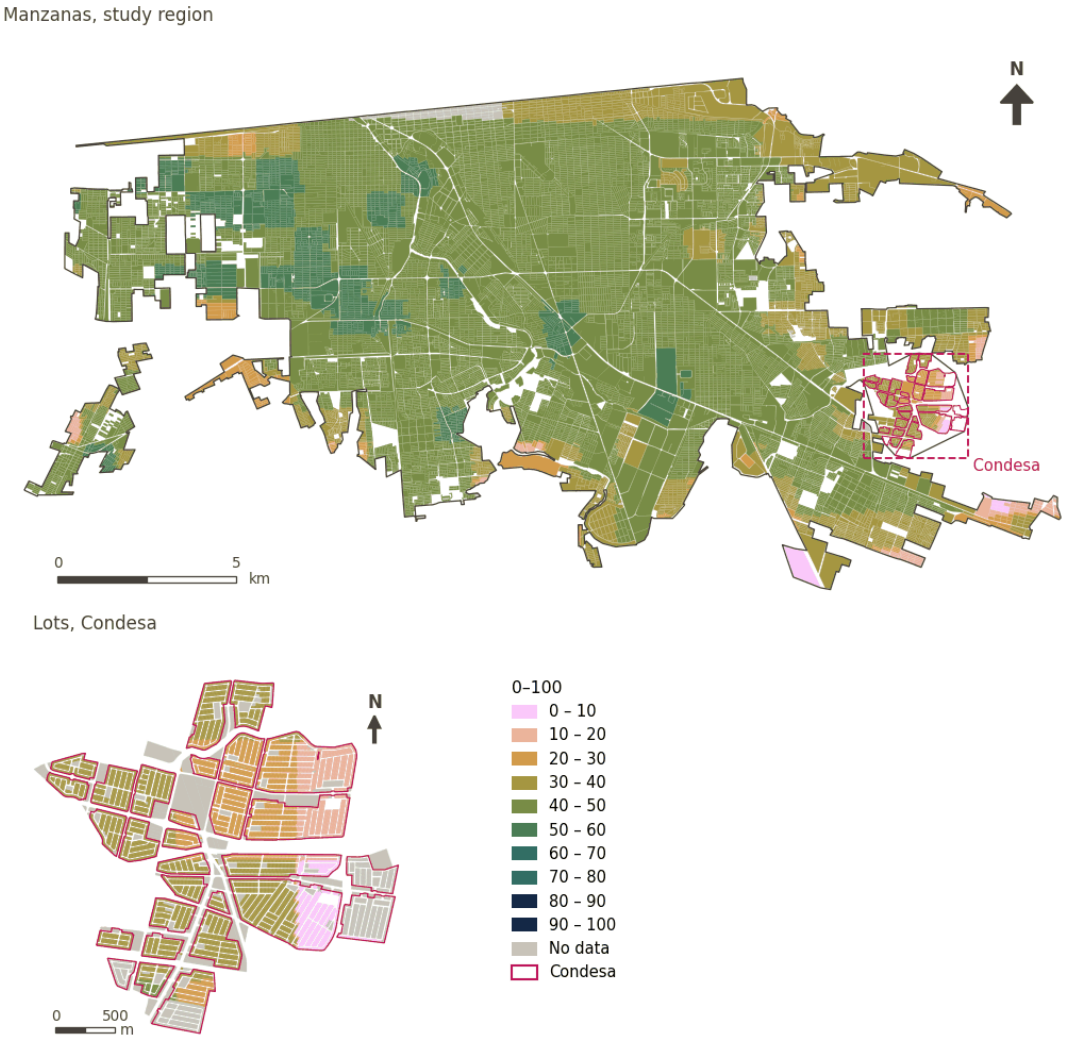
Heat adaptive capability (GUHVI), Heat exposure (GUHVI), Heat sensitivity (GUHVI)

Results

Measure	Study region (Mexicali incl. Condesa)	Condesa
Region value	43.6	27.4
Areas with a value	14,148 of 14,252 manzanas	13,265 of 14,988 lots
Median (P25–P75), weighted	46.0 (41.8 – 48.2)	30.6 (27.0 – 33.5)
Range	3.3 – 55.7	9.2 – 46.4

Distributions: study region manzanas weighted by census population (2020); Condesa lots weighted equally (3.2 residents assumed per lot). Region values are those GHSCI reports for each region as a whole.

Map



Map. Global Urban Heat Vulnerability Index (GUHVI) and its inputs: measured value (0–100) by manzana across the study region and by lot in Condesa (inset). Dashboard classes; the favourable end is the light end (lower is better).

Public open space and large public green space

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GHSCI identifies areas of open space from OpenStreetMap polygons (rules in `configuration/osm_open_space.yml`), clusters them into contiguous areas and retains those with public access. Water is excluded from public open space, so that a canal cannot chain contiguous parks into one enormous area. Access is measured to points every 20 m along the area's perimeter that a street passes within 30 m of (`aos_public_any_nodes_30m_line`).

A **large public green space** (LPUGS) is a public open space of at least 1 ha with annual mean NDVI ≥ 0.2 (Sentinel-2), identified in Google Earth Engine; its entry points are the subset of access points belonging to those areas.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_walk_nearest_node_public_open_space_any`	`avg_walk_dist_public_open_space_any`	Average distance (m) along the pedestrian network to the nearest public open space (source: OpenStreetMap)	469 m	849 m
`sp_access_large_public_green_space_score`	`pct_access_500m_large_public_green_space_score`	Percentage of population with access within 500 m of a large public green space of at least 1 hectare (source: OpenStreetMap, Google Earth Engine)	8.2%	0.0%

Method: practice

Recipes: R2. Network distance to the nearest destination, censored, R3. Banded access, and distance from a disamenity, R8. Earth Engine: heat vulnerability and large green space

Configuration (`MX_Mexicali_2025_ULI.yml:475`):

```
- {name: public_open_space_any,  category: pos,  variant: lenient, layer: aos_public_any_nodes_30m_line}
```

Code (GHSCI):

- `process/subprocesses/_06_open_space_areas_setup.py:43` `aos_setup_queries()`
- `process/subprocesses/_earth_engine_indicators.py:282` `lpugs_analysis()`
- `process/subprocesses/_earth_engine_indicators.py:664` `lpugs_indicators()`

Used by

Parks within 300 m, Large public green space within 500 m

Natural blue space

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Blue space is derived independently of public open space, from OpenStreetMap water tags on both **polygons and lines**: Mexicali's canals and drains are largely mapped as lines and would never enter a polygon-based process. Swimming pools, water parks and private features are excluded. Polygons are sampled along their exterior ring and lines along their length; points are generated every 20 m and those with a street within 30 m are retained (`blue_space_nodes_30m_line`), to which walking distance is measured.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_walk_nearest_node_blue_space`	`avg_walk_dist_blue_space`	Average distance (m) along the pedestrian network to the nearest natural blue space -- a river, canal, drain, lake, reservoir, wetland, beach or coast (source: OpenStreetMap or custom data)	797 m	590 m

Method: practice

Recipes: R2. Network distance to the nearest destination, censored

Configuration (`MX_Mexicali_2025_ULI.yml:489`):

```
- {name: blue_space, category: blue_space, variant: any, layer: blue_space_nodes_30m_line}
```

Code (GHSCI):

- `process/subprocesses/_06_open_space_areas_setup.py:365` `blue_space_criteria()`
- `process/subprocesses/_06_open_space_areas_setup.py:381` `blue_space_setup()`
- `process/subprocesses/_06_open_space_areas_setup.py:485` `blue_space_nodes_setup_query()`

Used by

Natural blue space nearby

DENUE destinations by SCIAN code

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Functional destination categories are read from the National Statistical Directory of Economic Units (DENUE, INEGI, May 2026) for Baja California, each with an OGR filter on the SCIAN activity code (`codigo_act`), and clipped to the buffered study region. The codes were derived from the dataset itself. Each establishment keeps its SCIAN code as a tag, so that a category can be subdivided for the diversity measures. The table reproduces the filters as configured.

Destination	Name	SCIAN filter (OGR)	Used by
`denue_fresh_food`	Fresh food retail (DENUE)	`"codigo_act IN ('461121','461122','461123','461130','461140','461150','462111','462112')"`	Fresh food within 300 m, Fresh food diversity within 500 m
`denue_food_retail`	Food retail, any (DENUE)	`"(codigo_act LIKE '4611%' AND codigo_act NOT IN ('461211','461212','461213','461220')) OR codigo_act LIKE '4621%'"`	Convenience stores within 300 m
`denue_bakery_tortilleria`	Bakeries and tortillerias (DENUE)	`"codigo_act IN ('311812','311813','311830')"`	Fresh food diversity within 500 m
`denue_household_goods`	Basic household goods (DENUE)	`"codigo_act IN ('465311','465914','466111','466112','466114','467111','467115')"`	
`denue_pharmacy`	Pharmacies and medicine retail (DENUE)	`"codigo_act IN ('464111','464112','464113')"`	Health service diversity within 500 m
`denue_finance_comms`	Financial and postal services (DENUE)	`"codigo_act IN ('522110','522210','522320','523122','491110')"`	
`denue_preschool`	Preschools (DENUE)	`"codigo_act IN ('611111','611112','611171','611172')"`	Education diversity within 500 m
`denue_primary_school`	Primary schools (DENUE)	`"codigo_act IN ('611121','611122','611171','611172')"`	Primary schools within 300 m, Education diversity within 500 m
`denue_secondary_school`	Secondary and upper secondary schools (DENUE)	`"codigo_act IN ('611131','611132','611142','611151','611152','611161','611162','611171','611172')"`	Education diversity within 500 m
`denue_tertiary_education`	Tertiary and vocational education (DENUE)	`"codigo_act LIKE '6112%' OR codigo_act LIKE '6113%' OR codigo_act IN ('611411','611412','611511','611512')"`	Education diversity within 500 m
`denue_primary_healthcare`	Primary healthcare (DENUE)	`"codigo_act IN ('621111','621112','621115','621116')"`	Primary healthcare within 300 m, Health service diversity within 500 m
`denue_medical_clinic`	Specialist and diagnostic clinics (DENUE)	`"codigo_act IN ('621113','621211','621212','621511')"`	Health service diversity within 500 m
`denue_hospital`	Hospitals (DENUE)	`"codigo_act LIKE '622%'"`	Health service diversity within 500 m
`denue_care_support`	Care and social support services (DENUE)	`"codigo_act LIKE '623%' OR codigo_act LIKE '624%'"`	Health service diversity within 500 m, Care services within 300 m, Care service diversity within 500 m
`denue_childcare`	Childcare (DENUE)	`"codigo_act IN ('624411','624412')"`	
`denue_community`	Community and social facilities (DENUE)	`"codigo_act IN ('813210','813230','931610')"`	Community, culture and recreation within 300 m, Community, culture and recreation diversity within 500 m
`denue_culture`	Cultural facilities (DENUE)	`"codigo_act IN ('712111','712112','519211','519212','512130','711311','711312')"`	Community, culture and recreation within 300 m, Community, culture and recreation diversity within 500 m
`denue_sport`	Sports and active recreation (DENUE)	`"codigo_act IN ('713941','713942','713943','713944','713910','713950','611621','611622')"`	Community, culture and recreation within 300 m, Community, culture and recreation diversity within 500 m
`denue_recreation`	Public recreation venues (DENUE)	`"codigo_act IN ('713111','713112','713113','712131','712132')"`	Community, culture and recreation within 300 m, Community, culture and recreation diversity within 500 m
`denue_employment_any`	Any economic establishment (DENUE)	`(all)`	
`denue_employer_medium`	Employers of 11 or more people (DENUE)	`"per_ocu NOT IN ('0 a 5 personas','6 a 10 personas')"`	
`denue_manufacturing`	Manufacturing and industrial establishments (DENUE)	`"codigo_act LIKE '31%' OR codigo_act LIKE '32%' OR codigo_act LIKE '33%'"`	Manufacturing within 1 km
`denue_petrol_station`	Petrol stations (DENUE)	`"codigo_act IN ('468411','468413')"`	250 m or more from petrol stations

Method: practice

Recipes: R2. Network distance to the nearest destination, censored

Configuration (`MX_Mexicali_2025_ULI.yml:192`):

```
denue_fresh_food: &denue
  data: MX/INEGI/DENUE_2026/denue_inegi_02_.shp -where "codigo_act IN ('461121','461122','461123','461130','461140','461150','462111','462112')"
  dest_name_full: Fresh food retail (DENUE)
  domain: Daily essential services
  ## Retain each establishment's SCIAN activity code as a destination tag, so
  ## that a category imported once can be subdivided into its sub-types for
  ## the diversity measures below without declaring a separate entry for each.
  ## Shared by every DENUE category through this anchor.
  keep_columns: codigo_act
  source: INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE)
  publication_date: 2026-05
  url: https://www.inegi.org.mx/app/mapa/denue/
  licence: Terminos de Libre Uso de la Informacion del INEGI
  citation: "INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia,
Mexico."
```

Code (GHSCI):

- `process/subprocesses/_05_compile_destinations.py:201` `compile_poi_destinations()`

Used by

250 m or more from petrol stations, Convenience stores within 300 m, Fresh food diversity within 500 m, Fresh food within 300 m, Manufacturing within 1 km, Community, culture and recreation diversity within 500 m, Community, culture and recreation within 300 m, Care service diversity within 500 m, Care services within 300 m, Education diversity within 500 m, Health service diversity within 500 m, Primary healthcare within 300 m, Primary schools within 300 m

Diversity sets and their sub-types

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Each diversity set declares its sub-types as SQL filters on the destinations layer. The number of sub-types `k` is fixed, so that scores are comparable between areas. The table reproduces the configuration.

Set	Sub-type	Filter (SQL)
`fresh_food` (k = 7)	`meat`	`dest_name = 'denue_fresh_food' AND tags->>'codigo_act' IN ('461121','461122','461123')`
`fresh_food` (k = 7)	`produce`	`dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '461130'`
`fresh_food` (k = 7)	`grains_and_spices`	`dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '461140'`
`fresh_food` (k = 7)	`dairy`	`dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '461150'`
`fresh_food` (k = 7)	`supermarket`	`dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '462111'`
`fresh_food` (k = 7)	`minisuper`	`dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '462112'`
`fresh_food` (k = 7)	`bakery_tortilleria`	`dest_name = 'denue_bakery_tortilleria'`
`education` (k = 4)	`preschool`	`dest_name = 'denue_preschool'`
`education` (k = 4)	`primary`	`dest_name = 'denue_primary_school'`
`education` (k = 4)	`secondary`	`dest_name = 'denue_secondary_school'`
`education` (k = 4)	`tertiary`	`dest_name = 'denue_tertiary_education'`
`health_care` (k = 5)	`primary_healthcare`	`dest_name = 'denue_primary_healthcare'`
`health_care` (k = 5)	`specialist_clinic`	`dest_name = 'denue_medical_clinic'`
`health_care` (k = 5)	`hospital`	`dest_name = 'denue_hospital'`
`health_care` (k = 5)	`pharmacy`	`dest_name = 'denue_pharmacy'`
`health_care` (k = 5)	`care_support`	`dest_name = 'denue_care_support'`
`care` (k = 3)	`childcare`	`dest_name = 'denue_care_support' AND tags->>'codigo_act' IN ('624411','624412')`
`care` (k = 3)	`residential_care`	`dest_name = 'denue_care_support' AND tags->>'codigo_act' LIKE '623%'`
`care` (k = 3)	`social_assistance`	`dest_name = 'denue_care_support' AND tags->>'codigo_act' LIKE '624%' AND tags->>'codigo_act' NOT IN ('624411','624412')`
`community_culture_rec` (k = 4)	`community`	`dest_name = 'denue_community'`
`community_culture_rec` (k = 4)	`culture`	`dest_name = 'denue_culture'`
`community_culture_rec` (k = 4)	`sport`	`dest_name = 'denue_sport'`
`community_culture_rec` (k = 4)	`recreation`	`dest_name = 'denue_recreation'`

Method: practice

Recipes: R4. Diversity of what is reachable

Configuration (`MX_Mexicali_2025_ULI.yml:595`):

```
diversity:
  ## Sub-types within one imported category, separated by activity code
  ## (hence keep_columns: codigo_act above), plus bakeries and tortillerias,
  ## which SCIAN classifies as manufacturing and which are therefore a
  ## separate category rather than a code subset.
  fresh_food:
    distances: [300, 500, 1000, 1500]
    groups:
      meat: "dest_name = 'denue_fresh_food' AND tags->>'codigo_act' IN ('461121','461122','461123')"
      produce: "dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '461130'"
      grains_and_spices: "dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '461140'"
      dairy: "dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '461150'"
      supermarket: "dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '462111'"
      minisuper: "dest_name = 'denue_fresh_food' AND tags->>'codigo_act' = '462112'"
      bakery_tortilleria: "dest_name = 'denue_bakery_tortilleria'"
  ## Sub-types that are already separate imported categories, so no code
  ## filtering is needed.  A school combining several levels (611171 /
  ## 611172) is imported under each level it serves and so counts towards
  ## each, which is the intended reading: a school serving that level is
  ## available.
  education:
    distances: [300, 500, 1000, 1500]
    groups:
      preschool: "dest_name = 'denue_preschool'"
      primary: "dest_name = 'denue_primary_school'"
      secondary: "dest_name = 'denue_secondary_school'"
      tertiary: "dest_name = 'denue_tertiary_education'"
  ## denue_childcare is a subset of denue_care_support, so only the broader
  ## category is used here; counting both would count a creche twice and
  ## overstate the evenness of what is available.
  health_care:
    distances: [300, 500, 1000, 1500]
    groups:
      primary_healthcare: "dest_name = 'denue_primary_healthcare'"
      specialist_clinic: "dest_name = 'denue_medical_clinic'"
      hospital: "dest_name = 'denue_hospital'"
      pharmacy: "dest_name = 'denue_pharmacy'"
      care_support: "dest_name = 'denue_care_support'"
  ## Diversity of care services (ULI sub-variable "Diversity of care
  ## services"): the SCIAN groups within care and social support.
  ## Childcare is taken from the care category by code, not from
  ## denue_childcare, so that no establishment is counted twice.
  care:
    distances: [300, 500, 1000, 1500]
    groups:
      childcare: "dest_name = 'denue_care_support' AND tags->>'codigo_act' IN ('624411','624412')"
      residential_care: "dest_name = 'denue_care_support' AND tags->>'codigo_act' LIKE '623%'"
      social_assistance: "dest_name = 'denue_care_support' AND tags->>'codigo_act' LIKE '624%' AND tags->>'codigo_act' NOT IN ('624411','624412')"
  community_culture_rec:
    distances: [300, 500, 1000, 1500]
    groups:
      community: "dest_name = 'denue_community'"
      culture: "dest_name = 'denue_culture'"
      sport: "dest_name = 'denue_sport'"
      recreation: "dest_name = 'denue_recreation'"
```

Code (GHSCI):

- `process/subprocesses/_accessibility_spec.py:299` `diversity_sets()`
- `process/subprocesses/_accessibility_spec.py:466` `normalised_entropy()`
- `process/subprocesses/_accessibility_spec.py:493` `richness()`

Used by

Fresh food diversity within 500 m, Community, culture and recreation diversity within 500 m, Care service diversity within 500 m, Education diversity within 500 m, Health service diversity within 500 m

Cycling network by level of traffic stress (LTS)

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Each OpenStreetMap network link is given a cycling facility type (bike path, shared path, separated lane, simple lane, shared street or none), a speed (`maxspeed`, or a default by road class) and an assumed average daily traffic (local 750, tertiary 3,000, secondary 10,000). LTS is assigned by cascade; the first rule met wins:

LTS	Rules (summary of `assign_lts`)
1	bike path or shared path; off-road classes; separated lane ≤ 50 km/h; simple lane on a local/tertiary/secondary road with ADT ≤ 10,000 and ≤ 30 km/h; local road with ADT ≤ 2,000 and ≤ 30 km/h
2	separated lane ≤ 60; simple lane ≤ 50 (ADT ≤ 10,000); simple lane on a primary road (or ADT > 10,000) ≤ 40; local road with ADT ≤ 750 and ≤ 50, or ADT ≤ 2,000 and ≤ 40; local/tertiary with ADT ≤ 3,000 and ≤ 30
3	simple lane ≤ 60; local with ADT ≤ 750 and ≤ 60; local/tertiary with ADT ≤ 3,000 and ≤ 50; secondary/primary (or ADT > 3,000) ≤ 40
4	anything else

The **low-stress** network is the links with LTS ≤ 2 where cycling, or dismounting to walk the bike, is permitted; its cost is length (tripled on links where one must dismount).

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_cycle_safe_nearest_node_fresh_food_pooled`	`avg_cycle_dist_safe_fresh_food_pooled`	Average distance (m) by cycling to the nearest fresh food market, supermarket or convenience store (pooled fresh food destinations), using a fully low-stress route (Level of Traffic Stress 1–2)	826 m	1,025 m

Method: practice

Recipes: R6. Level of traffic stress and the low-stress cycling network

Configuration (`MX_Mexicali_2025_ULI.yml:698`):

```
cycling_indicators: true
```

Code (GHSCI):

- `process/subprocesses/_cycling_lts_network.py:432` `classify_cycleway()`
- `process/subprocesses/_cycling_lts_network.py:491` `assign_adt()`
- `process/subprocesses/_cycling_lts_network.py:546` `assign_lts()`
- `process/subprocesses/_cycling_lts_network.py:607` `add_impedance()`
- `process/subprocesses/_cycling_accessibility.py:469` `cycling_poi_distance()`

Used by

Food within 1.5 km by low-stress cycling

Public transport stops

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Mexicali has no GTFS feed, so the IMIP facilities layer is the local source of stops (transport category, excluding the inland freight terminal). With `replace` at its default (false), they are pooled with stops identified in OpenStreetMap under the same destination name, `pt_any`.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_walk_nearest_node_pt_any`	`avg_walk_dist_pt_any`	Average distance (m) along the pedestrian network to the nearest public transport stop (best result of GTFS and OpenStreetMap/custom sources)	882 m	1,432 m

Method: practice

Recipes: R2. Network distance to the nearest destination, censored

Configuration (`MX_Mexicali_2025_ULI.yml:168`):

```
pt_any:
  data: MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "equipamiento_category='transp' AND TIPO !=
'TERMINAL INTERIOR DE CARGA (CENTRAL DE SERVICIOS DE CARGA)'"
  source: IMIP / CFC
  publication_date: 2026-05-18
  url: ""
  licence: "Provided by IMIP via CFC"
  citation: As per advice from Juan Ernesto Díaz Noguez
```

Used by

Public transport within 300 m

Police and fire stations (straight line)

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Stations are read from the IMIP facilities layer, each specification with its own file and filter, clipped to the study region buffered by its maximum distance and outside the exclusion region (California). Coverage is measured in a straight line because an emergency response does not follow footpaths.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_euclid_dist_imip_police_station`	`avg_euclid_dist_imip_police_station`	Average straight-line distance (m) to the nearest 'imip police station' (custom destination)	1,406 m	4,156 m
`sp_euclid_dist_imip_fire_station`	`avg_euclid_dist_imip_fire_station`	Average straight-line distance (m) to the nearest 'imip fire station' (custom destination)	2,297 m	6,371 m

- **`imip_police_station`** (Police stations (IMIP)): `MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "TIPO = 'INSTALACIONES POLICIALES'"`; distances [15000], max_distance 45000. 27 police facilities in the IMIP equipamiento layer (administración pública). (IMIP / CFC; Provided by IMIP via CFC)
- **`imip_fire_station`** (Municipal fire stations (IMIP)): `MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "TIPO = 'ESTACION DE BOMBEROS MUNICIPAL'"`; distances [75000]. 10 municipal fire stations in the IMIP equipamiento layer (administración pública). (IMIP / CFC; Provided by IMIP via CFC)

Method: practice

Recipes: R5. Straight-line catchments

Configuration (`MX_Mexicali_2025_ULI.yml:679`):

```
euclidean:
  destinations:
    - name: imip_police_station
      dest_name_full: Police stations (IMIP)
      data: MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "TIPO = 'INSTALACIONES POLICIALES'"
      distances: [15000]
      max_distance: 45000
      source: IMIP / CFC
      licence: Provided by IMIP via CFC
      notes: 27 police facilities in the IMIP equipamiento layer (administración pública).
    - name: imip_fire_station
      dest_name_full: Municipal fire stations (IMIP)
      data: MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "TIPO = 'ESTACION DE BOMBEROS MUNICIPAL'"
      distances: [75000]
      source: IMIP / CFC
      licence: Provided by IMIP via CFC
      notes: 10 municipal fire stations in the IMIP equipamiento layer (administración pública).
```

Code (GHSCI):

- `process/subprocesses/_euclidean_accessibility.py:66` `resolve_spec()`
- `process/subprocesses/_euclidean_accessibility.py:177` `import_destinations()`
- `process/subprocesses/_euclidean_accessibility.py:247` `nearest_distances()`

Used by

Fire station within 3 km, Police station within 2 km

Work package indicators (linked)

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Indicators prepared by the ULI work packages (`final_indicators.gpkg`, 22 September 2026). The file carries no identifiers, so features are identified by geometry against the reference areas: grid cells, AGEBs, lots and subdivisions are identical to them, and manzanas are matched by their largest overlap (several parts of one block are combined as an area-weighted mean). Values are brought down to sample points from the first layer holding one: the grid, then Condesa lots within 15 m and manzanas within 30 m.

Sample point variable	Area variable (measured value)	Description	Study region (Mexicali incl. Condesa)	Condesa
`sp_ext_ndvi`	`ext_ndvi`	Vegetation cover (NDVI)	0.08	0.05
`sp_ext_major_road_length_300m`	`ext_major_road_length_300m`	Major roads within 300 m	1,996 m	414 m
`sp_ext_block_perimeter_m`	`ext_block_perimeter_m`	Block size (perimeter)	677 m	712 m
`sp_ext_economic_sector_diversity`	`ext_economic_sector_diversity`	Job diversity	0.33	–

Column	Name	Units	Direction
`ext_ndvi`	Vegetation cover (NDVI)	NDVI	higher_is_better
`ext_tree_canopy_pct`	Tree canopy cover	%	higher_is_better
`ext_avg_walk_dist_park_m`	Distance to parks (work package)	m	lower_is_better
`ext_major_road_length_300m`	Major roads within 300 m	m	lower_is_better
`ext_pop_affected_flood`	Population exposed to flooding	persons	lower_is_better
`ext_flood_area_m2`	Flooded area on main roads	m^2	lower_is_better
`ext_block_perimeter_m`	Block size (perimeter)	m	lower_is_better
`ext_residential_area_ha`	Residential land use	ha	–
`ext_commercial_area_ha`	Commercial land use	ha	–
`ext_land_use_entropy`	Land use mix	index 0-1	higher_is_better
`ext_economic_sector_diversity`	Job diversity	index 0-1	higher_is_better
`ext_protected_cycling_street_pct`	Protected cycle lane coverage	%	higher_is_better

- `data`: MX/Update_on_ULI - 2026-09-22/final_indicators.gpkg
- `match`: geometry
- `reference`: MX/Mexicali Liveability/geography/mexicali_reference_areas.gpkg
- `null_as`: {'pop_affected_flood': 0, 'flood_area_m2': 0}
- `source`: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)

Method: practice

Recipes: R7. Linking externally prepared indicators

Configuration (`MX_Mexicali_2025_ULI.yml:814`):

```
external:
  data: MX/Update_on_ULI - 2026-09-22/final_indicators.gpkg
  match: geometry
  reference: MX/Mexicali Liveability/geography/mexicali_reference_areas.gpkg
  reference_layers: {grid: grid_100m}
  prefix: ext_
  label: {es: Indicadores de los paquetes de trabajo, en: Work package indicators}
  source: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)
  null_as: {pop_affected_flood: 0, flood_area_m2: 0}
  columns:
    ndvi:
      label: {es: Cobertura vegetal (NDVI), en: Vegetation cover (NDVI)}
      units: NDVI
      direction: higher_is_better
    tree_canopy_pct:
      label: {es: Cobertura arbórea, en: Tree canopy cover}
      units: '%'
      direction: higher_is_better
    avg_walk_dist_park_m:
      label: {es: Distancia a parques (paquete de trabajo), en: Distance to parks (work package)}
      units: m
      direction: lower_is_better
    major_road_length_300m:
      label: {es: Vialidades principales en 300 m, en: Major roads within 300 m}
      units: m
      direction: lower_is_better
    pop_affected_flood:
      label: {es: Población expuesta a inundaciones, en: Population exposed to flooding}
      units: persons
      direction: lower_is_better
    flood_area_m2:
      label: {es: Área de inundación en vialidades principales, en: Flooded area on main roads}
      units: m²
      direction: lower_is_better
    block_perimeter_m:
      label: {es: Tamaño de manzana (perímetro), en: Block size (perimeter)}
      units: m
      direction: lower_is_better
    residential_area_ha:
      label: {es: Uso de suelo residencial, en: Residential land use}
      units: ha
    commercial_area_ha:
      label: {es: Uso de suelo comercial, en: Commercial land use}
      units: ha
    land_use_entropy:
      label: {es: Mezcla de usos de suelo, en: Land use mix}
      units: index 0-1
      direction: higher_is_better
    economic_sector_diversity:
      label: {es: Diversidad de empleos, en: Job diversity}
      units: index 0-1
      direction: higher_is_better
    protected_cycling_street_pct:
      label: {es: Cobertura de ciclovías protegidas, en: Protected cycle lane coverage}
      units: '%'
      direction: higher_is_better
  areas:
    grid: grid
    agebs: agebs
    manzanas: manzanas
    condesa_lotes: {layer: condesa_lotes, boundary_id: id}
    condesa_fraccionamientos: {layer: condesa_fraccionamientos, boundary_id: fid}
  ## the grid first, as it tiles the region; flood values exist only for
  ## blocks and lots, so points take them from the nearest within the
  ## distance each is aggregated over
  sample_points:
    - {layer: grid}
    - {layer: condesa_lotes, within: 15}
    - {layer: manzanas, within: 30}
```

Code (GHSCI):

- `process/subprocesses/_linkage_indicators.py:403` `match_by_geometry()`
- `process/subprocesses/_linkage_indicators.py:479` `combine_matched()`
- `process/subprocesses/_linkage_indicators.py:512` `apply_rules()`
- `process/subprocesses/_linkage_indicators.py:537` `sample_point_values()`

Used by

Vegetation cover (NDVI), Proximity to major roads, Block size (perimeter), Job diversity

Appendix A. Variable index

Variable	Entry
avg_cycle_dist_safe_fresh_food_pooled	Food within 1.5 km by low-stress cycling
avg_diversity_walk_care_500m	Care service diversity within 500 m
avg_diversity_walk_community_culture_recreation_500m	Community, culture and recreation diversity within 500 m
avg_diversity_walk_education_500m	Education diversity within 500 m
avg_diversity_walk_fresh_food_500m	Fresh food diversity within 500 m
avg_diversity_walk_health_care_500m	Health service diversity within 500 m
avg_euclid_dist_imip_fire_station	Fire station within 3 km
avg_euclid_dist_imip_police_station	Police station within 2 km
avg_walk_dist_blue_space	Natural blue space nearby
avg_walk_dist_denue_care_support	Care services within 300 m
avg_walk_dist_denue_ccr_any	Community, culture and recreation within 300 m
avg_walk_dist_denue_food_retail	Convenience stores within 300 m
avg_walk_dist_denue_fresh_food	Fresh food within 300 m
avg_walk_dist_denue_primary_healthcare	Primary healthcare within 300 m
avg_walk_dist_denue_primary_school	Primary schools within 300 m
avg_walk_dist_pt_any	Public transport within 300 m
avg_walk_dist_public_open_space_any	Parks within 300 m
ext_block_perimeter_m	Block size (perimeter)
ext_economic_sector_diversity	Job diversity
ext_major_road_length_300m	Proximity to major roads
ext_ndvi	Vegetation cover (NDVI)
index_uli	Mexicali Urban Liveability Index
index_uli__adaptive_capability	Heat adaptive capability (GUHVI)
index_uli__ambient_environment	Ambient environment
index_uli__block_size	Block size (perimeter)
index_uli__blue_space	Natural blue space nearby
index_uli__built_environment	Built environment
index_uli__care_support	Care services within 300 m
index_uli__ccr_any	Community, culture and recreation within 300 m
index_uli__convenience	Convenience stores within 300 m
index_uli__cycle_low_stress_food	Food within 1.5 km by low-stress cycling
index_uli__div_care_500m	Care service diversity within 500 m
index_uli__div_ccr_500m	Community, culture and recreation diversity within 500 m
index_uli__div_education_500m	Education diversity within 500 m
index_uli__div_fresh_food_500m	Fresh food diversity within 500 m
index_uli__div_health_500m	Health service diversity within 500 m
index_uli__economic_development	Economic development
index_uli__fire_3km	Fire station within 3 km
index_uli__fresh_food	Fresh food within 300 m
index_uli__heat_exposure	Heat exposure (GUHVI)
index_uli__heat_sensitivity	Heat sensitivity (GUHVI)
index_uli__job_diversity	Job diversity
index_uli__lpugs	Large public green space within 500 m
index_uli__major_roads	Proximity to major roads
index_uli__manufacturing_1000m	Manufacturing within 1 km
index_uli__mobility_transport	Mobility and transport
index_uli__parks	Parks within 300 m
index_uli__petrol_250m	250 m or more from petrol stations
index_uli__police_2km	Police station within 2 km
index_uli__primary_healthcare	Primary healthcare within 300 m
index_uli__primary_school	Primary schools within 300 m
index_uli__public_transport	Public transport within 300 m

Variable	Entry
index_uli__safety	Safety and protection
index_uli__social_infrastructure	Social infrastructure
index_uli__vegetation	Vegetation cover (NDVI)
index_uli__walkability	Walkability (300 m), attenuated by thermal comfort
index_uli_mean	Mexicali Urban Liveability Index
index_uli_penalty	Mexicali Urban Liveability Index
index_uli_plain	Mexicali Urban Liveability Index: Walkability (300 m), without thermal comfort attenuation
index_uli_plain__ambient_environment	Ambient environment
index_uli_plain__built_environment	Built environment
index_uli_plain__economic_development	Economic development
index_uli_plain__mobility_transport	Mobility and transport
index_uli_plain__safety	Safety and protection
index_uli_plain__social_infrastructure	Social infrastructure
index_uli_plain__thermal_comfort	Daytime thermal comfort
index_uli_plain__walkability	Walkability (300 m), attenuated by thermal comfort
index_uli_plain_mean	Mexicali Urban Liveability Index: Walkability (300 m), without thermal comfort attenuation
index_uli_plain_penalty	Mexicali Urban Liveability Index: Walkability (300 m), without thermal comfort attenuation
local_nh_intersection_density	Walkability within 300 m and its components
local_nh_population_density	Walkability within 300 m and its components
pct_access_500m_large_public_green_space_score	Large public green space within 500 m
pct_access_walk_denue_manufacturing_1000m	Manufacturing within 1 km
pct_beyond_walk_denue_petrol_station_250m	250 m or more from petrol stations
sp_access_large_public_green_space_score	Large public green space within 500 m
sp_cycle_safe_nearest_node_fresh_food_pooled	Food within 1.5 km by low-stress cycling
sp_euclid_dist_imip_fire_station	Fire station within 3 km
sp_euclid_dist_imip_police_station	Police station within 2 km
sp_ext_block_perimeter_m	Block size (perimeter)
sp_ext_economic_sector_diversity	Job diversity
sp_ext_major_road_length_300m	Proximity to major roads
sp_ext_ndvi	Vegetation cover (NDVI)
sp_local_nh_avg_intersection_density	Walkability within 300 m and its components
sp_local_nh_avg_pop_density	Walkability within 300 m and its components
sp_urban_heat_adaptive_capability_index	Heat adaptive capability (GUHVI)
sp_urban_heat_exposure_index	Heat exposure (GUHVI)
sp_urban_heat_guhvi	Global Urban Heat Vulnerability Index (GUHVI) and its inputs
sp_urban_heat_sensitivity_index	Heat sensitivity (GUHVI)
sp_utci_day_mean	Daytime thermal comfort
sp_utci_peak	Outdoor thermal comfort (UTCI)
sp_walk_access_denue_manufacturing_1000m	Manufacturing within 1 km
sp_walk_beyond_denue_petrol_station_250m	250 m or more from petrol stations
sp_walk_diversity_care_500m	Care service diversity within 500 m
sp_walk_diversity_community_culture_recreation_500m	Community, culture and recreation diversity within 500 m
sp_walk_diversity_education_500m	Education diversity within 500 m
sp_walk_diversity_fresh_food_500m	Fresh food diversity within 500 m
sp_walk_diversity_health_care_500m	Health service diversity within 500 m
sp_walk_dl_300	Walkability within 300 m and its components
sp_walk_idx_300	Walkability within 300 m and its components
sp_walk_idx_300_tm	Walkability (300 m), attenuated by thermal comfort
sp_walk_nearest_node_blue_space	Natural blue space nearby
sp_walk_nearest_node_denue_care_support	Care services within 300 m
sp_walk_nearest_node_denue_ccr_any	Community, culture and recreation within 300 m
sp_walk_nearest_node_denue_food_retail	Convenience stores within 300 m
sp_walk_nearest_node_denue_fresh_food	Fresh food within 300 m
sp_walk_nearest_node_denue_primary_healthcare	Primary healthcare within 300 m

Variable	Entry
`sp_walk_nearest_node_denue_primary_school`	Primary schools within 300 m
`sp_walk_nearest_node_pt_any`	Public transport within 300 m
`sp_walk_nearest_node_public_open_space_any`	Parks within 300 m
`urban_heat_adaptive_capability_index`	Heat adaptive capability (GUHVI)
`urban_heat_exposure_index`	Heat exposure (GUHVI)
`urban_heat_guhvi`	Global Urban Heat Vulnerability Index (GUHVI) and its inputs
`urban_heat_sensitivity_index`	Heat sensitivity (GUHVI)
`utci_day_mean`	Daytime thermal comfort
`utci_peak`	Outdoor thermal comfort (UTCI)
`walk_dl_300`	Walkability within 300 m and its components
`walk_idx_300`	Walkability within 300 m and its components
`walk_idx_300_tm`	Walkability (300 m), attenuated by thermal comfort

Appendix B. General dashboard indicators

Every variable of the general dashboard and the results GeoPackage, from the data dictionary (`outputs/mexicali_uli_ghsci_data_dictionary.csv`), grouped by category. ULI variables have their own entries (Appendix A).

Derived study region statistics

Variable	Description	Units
`area_sqkm`	Area (km²)	km²
`intersection_count`	Intersection count (following consolidation based on intersection tolerance parameter in region configuration)	count
`intersections_per_sqkm`	Intersections per km²	count per km²
`pop_est`	Population estimate, as per configured population data source	persons
`pop_per_sqkm`	Population per km²	persons per km²

Indicator estimates: access (walking)

Variable	Description	Units
`avg_count_walk_care__childcare_1000m`	Average number of 'childcare' destinations, a sub-type of 'care', reachable within 1000 m along the pedestrian network	count
`avg_count_walk_care__childcare_1500m`	Average number of 'childcare' destinations, a sub-type of 'care', reachable within 1500 m along the pedestrian network	count
`avg_count_walk_care__childcare_300m`	Average number of 'childcare' destinations, a sub-type of 'care', reachable within 300 m along the pedestrian network	count
`avg_count_walk_care__childcare_500m`	Average number of 'childcare' destinations, a sub-type of 'care', reachable within 500 m along the pedestrian network	count
`avg_count_walk_care__residential_care_1000m`	Average number of 'residential care' destinations, a sub-type of 'care', reachable within 1000 m along the pedestrian network	count
`avg_count_walk_care__residential_care_1500m`	Average number of 'residential care' destinations, a sub-type of 'care', reachable within 1500 m along the pedestrian network	count
`avg_count_walk_care__residential_care_300m`	Average number of 'residential care' destinations, a sub-type of 'care', reachable within 300 m along the pedestrian network	count
`avg_count_walk_care__residential_care_500m`	Average number of 'residential care' destinations, a sub-type of 'care', reachable within 500 m along the pedestrian network	count
`avg_count_walk_care__social_assistance_1000m`	Average number of 'social assistance' destinations, a sub-type of 'care', reachable within 1000 m along the pedestrian network	count
`avg_count_walk_care__social_assistance_1500m`	Average number of 'social assistance' destinations, a sub-type of 'care', reachable within 1500 m along the pedestrian network	count
`avg_count_walk_care__social_assistance_300m`	Average number of 'social assistance' destinations, a sub-type of 'care', reachable within 300 m along the pedestrian network	count
`avg_count_walk_care__social_assistance_500m`	Average number of 'social assistance' destinations, a sub-type of 'care', reachable within 500 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__community_1000m`	Average number of 'community' destinations, a sub-type of 'community culture recreation', reachable within 1000 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__community_1500m`	Average number of 'community' destinations, a sub-type of 'community culture recreation', reachable within 1500 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__community_300m`	Average number of 'community' destinations, a sub-type of 'community culture recreation', reachable within 300 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__community_500m`	Average number of 'community' destinations, a sub-type of 'community culture recreation', reachable within 500 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__culture_1000m`	Average number of 'culture' destinations, a sub-type of 'community culture recreation', reachable within 1000 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__culture_1500m`	Average number of 'culture' destinations, a sub-type of 'community culture recreation', reachable within 1500 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__culture_300m`	Average number of 'culture' destinations, a sub-type of 'community culture recreation', reachable within 300 m along the pedestrian network	count
`avg_count_walk_community_culture_recreation__culture_500m`	Average number of 'culture' destinations, a sub-type of 'community culture recreation', reachable within 500 m along the pedestrian network	count

Variable	Description	Units
avg_count_walk_community_culture_recreation__recreation_1000m	Average number of 'recreation' destinations, a sub-type of 'community culture recreation', reachable within 1000 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__recreation_1500m	Average number of 'recreation' destinations, a sub-type of 'community culture recreation', reachable within 1500 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__recreation_300m	Average number of 'recreation' destinations, a sub-type of 'community culture recreation', reachable within 300 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__recreation_500m	Average number of 'recreation' destinations, a sub-type of 'community culture recreation', reachable within 500 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__sport_1000m	Average number of 'sport' destinations, a sub-type of 'community culture recreation', reachable within 1000 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__sport_1500m	Average number of 'sport' destinations, a sub-type of 'community culture recreation', reachable within 1500 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__sport_300m	Average number of 'sport' destinations, a sub-type of 'community culture recreation', reachable within 300 m along the pedestrian network	count
avg_count_walk_community_culture_recreation__sport_500m	Average number of 'sport' destinations, a sub-type of 'community culture recreation', reachable within 500 m along the pedestrian network	count
avg_count_walk_education__preschool_1000m	Average number of 'preschool' destinations, a sub-type of 'education', reachable within 1000 m along the pedestrian network	count
avg_count_walk_education__preschool_1500m	Average number of 'preschool' destinations, a sub-type of 'education', reachable within 1500 m along the pedestrian network	count
avg_count_walk_education__preschool_300m	Average number of 'preschool' destinations, a sub-type of 'education', reachable within 300 m along the pedestrian network	count
avg_count_walk_education__preschool_500m	Average number of 'preschool' destinations, a sub-type of 'education', reachable within 500 m along the pedestrian network	count
avg_count_walk_education__primary_1000m	Average number of 'primary' destinations, a sub-type of 'education', reachable within 1000 m along the pedestrian network	count
avg_count_walk_education__primary_1500m	Average number of 'primary' destinations, a sub-type of 'education', reachable within 1500 m along the pedestrian network	count
avg_count_walk_education__primary_300m	Average number of 'primary' destinations, a sub-type of 'education', reachable within 300 m along the pedestrian network	count
avg_count_walk_education__primary_500m	Average number of 'primary' destinations, a sub-type of 'education', reachable within 500 m along the pedestrian network	count
avg_count_walk_education__secondary_1000m	Average number of 'secondary' destinations, a sub-type of 'education', reachable within 1000 m along the pedestrian network	count
avg_count_walk_education__secondary_1500m	Average number of 'secondary' destinations, a sub-type of 'education', reachable within 1500 m along the pedestrian network	count
avg_count_walk_education__secondary_300m	Average number of 'secondary' destinations, a sub-type of 'education', reachable within 300 m along the pedestrian network	count
avg_count_walk_education__secondary_500m	Average number of 'secondary' destinations, a sub-type of 'education', reachable within 500 m along the pedestrian network	count
avg_count_walk_education__tertiary_1000m	Average number of 'tertiary' destinations, a sub-type of 'education', reachable within 1000 m along the pedestrian network	count
avg_count_walk_education__tertiary_1500m	Average number of 'tertiary' destinations, a sub-type of 'education', reachable within 1500 m along the pedestrian network	count
avg_count_walk_education__tertiary_300m	Average number of 'tertiary' destinations, a sub-type of 'education', reachable within 300 m along the pedestrian network	count
avg_count_walk_education__tertiary_500m	Average number of 'tertiary' destinations, a sub-type of 'education', reachable within 500 m along the pedestrian network	count
avg_count_walk_fresh_food__bakery_tortilleria_1000m	Average number of 'bakery tortilleria' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food__bakery_tortilleria_1500m	Average number of 'bakery tortilleria' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food__bakery_tortilleria_300m	Average number of 'bakery tortilleria' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food__bakery_tortilleria_500m	Average number of 'bakery tortilleria' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count

Variable	Description	Units
avg_count_walk_fresh_food_dairy_1000m	Average number of 'dairy' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food_dairy_1500m	Average number of 'dairy' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food_dairy_300m	Average number of 'dairy' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food_dairy_500m	Average number of 'dairy' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count
avg_count_walk_fresh_food_grains_and_spices_1000m	Average number of 'grains and spices' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food_grains_and_spices_1500m	Average number of 'grains and spices' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food_grains_and_spices_300m	Average number of 'grains and spices' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food_grains_and_spices_500m	Average number of 'grains and spices' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count
avg_count_walk_fresh_food_meat_1000m	Average number of 'meat' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food_meat_1500m	Average number of 'meat' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food_meat_300m	Average number of 'meat' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food_meat_500m	Average number of 'meat' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count
avg_count_walk_fresh_food_minisuper_1000m	Average number of 'minisuper' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food_minisuper_1500m	Average number of 'minisuper' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food_minisuper_300m	Average number of 'minisuper' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food_minisuper_500m	Average number of 'minisuper' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count
avg_count_walk_fresh_food_produce_1000m	Average number of 'produce' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food_produce_1500m	Average number of 'produce' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food_produce_300m	Average number of 'produce' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food_produce_500m	Average number of 'produce' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count
avg_count_walk_fresh_food_supermarket_1000m	Average number of 'supermarket' destinations, a sub-type of 'fresh food', reachable within 1000 m along the pedestrian network	count
avg_count_walk_fresh_food_supermarket_1500m	Average number of 'supermarket' destinations, a sub-type of 'fresh food', reachable within 1500 m along the pedestrian network	count
avg_count_walk_fresh_food_supermarket_300m	Average number of 'supermarket' destinations, a sub-type of 'fresh food', reachable within 300 m along the pedestrian network	count
avg_count_walk_fresh_food_supermarket_500m	Average number of 'supermarket' destinations, a sub-type of 'fresh food', reachable within 500 m along the pedestrian network	count
avg_count_walk_health_care__care_support_1000m	Average number of 'care support' destinations, a sub-type of 'health care', reachable within 1000 m along the pedestrian network	count
avg_count_walk_health_care__care_support_1500m	Average number of 'care support' destinations, a sub-type of 'health care', reachable within 1500 m along the pedestrian network	count
avg_count_walk_health_care__care_support_300m	Average number of 'care support' destinations, a sub-type of 'health care', reachable within 300 m along the pedestrian network	count
avg_count_walk_health_care__care_support_500m	Average number of 'care support' destinations, a sub-type of 'health care', reachable within 500 m along the pedestrian network	count
avg_count_walk_health_care__hospital_1000m	Average number of 'hospital' destinations, a sub-type of 'health care', reachable within 1000 m along the pedestrian network	count
avg_count_walk_health_care__hospital_1500m	Average number of 'hospital' destinations, a sub-type of 'health care', reachable within 1500 m along the pedestrian network	count
avg_count_walk_health_care__hospital_300m	Average number of 'hospital' destinations, a sub-type of 'health care', reachable within 300 m along the pedestrian network	count
avg_count_walk_health_care__hospital_500m	Average number of 'hospital' destinations, a sub-type of 'health care', reachable within 500 m along the pedestrian network	count

Variable	Description	Units
avg_count_walk_health_care__pharmacy_1000m	Average number of 'pharmacy' destinations, a sub-type of 'health care', reachable within 1000 m along the pedestrian network	count
avg_count_walk_health_care__pharmacy_1500m	Average number of 'pharmacy' destinations, a sub-type of 'health care', reachable within 1500 m along the pedestrian network	count
avg_count_walk_health_care__pharmacy_300m	Average number of 'pharmacy' destinations, a sub-type of 'health care', reachable within 300 m along the pedestrian network	count
avg_count_walk_health_care__pharmacy_500m	Average number of 'pharmacy' destinations, a sub-type of 'health care', reachable within 500 m along the pedestrian network	count
avg_count_walk_health_care__primary_healthcare_1000m	Average number of 'primary healthcare' destinations, a sub-type of 'health care', reachable within 1000 m along the pedestrian network	count
avg_count_walk_health_care__primary_healthcare_1500m	Average number of 'primary healthcare' destinations, a sub-type of 'health care', reachable within 1500 m along the pedestrian network	count
avg_count_walk_health_care__primary_healthcare_300m	Average number of 'primary healthcare' destinations, a sub-type of 'health care', reachable within 300 m along the pedestrian network	count
avg_count_walk_health_care__primary_healthcare_500m	Average number of 'primary healthcare' destinations, a sub-type of 'health care', reachable within 500 m along the pedestrian network	count
avg_count_walk_health_care__specialist_clinic_1000m	Average number of 'specialist clinic' destinations, a sub-type of 'health care', reachable within 1000 m along the pedestrian network	count
avg_count_walk_health_care__specialist_clinic_1500m	Average number of 'specialist clinic' destinations, a sub-type of 'health care', reachable within 1500 m along the pedestrian network	count
avg_count_walk_health_care__specialist_clinic_300m	Average number of 'specialist clinic' destinations, a sub-type of 'health care', reachable within 300 m along the pedestrian network	count
avg_count_walk_health_care__specialist_clinic_500m	Average number of 'specialist clinic' destinations, a sub-type of 'health care', reachable within 500 m along the pedestrian network	count
avg_diversity_walk_care_1000m	Average diversity (0-1) of the 'care' destination sub-types reachable within 1000 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_care_1500m	Average diversity (0-1) of the 'care' destination sub-types reachable within 1500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_care_300m	Average diversity (0-1) of the 'care' destination sub-types reachable within 300 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_care_500m	Average diversity (0-1) of the 'care' destination sub-types reachable within 500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_community_culture_recreation_1000m	Average diversity (0-1) of the 'community culture recreation' destination sub-types reachable within 1000 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_community_culture_recreation_1500m	Average diversity (0-1) of the 'community culture recreation' destination sub-types reachable within 1500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_community_culture_recreation_300m	Average diversity (0-1) of the 'community culture recreation' destination sub-types reachable within 300 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_community_culture_recreation_500m	Average diversity (0-1) of the 'community culture recreation' destination sub-types reachable within 500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_education_1000m	Average diversity (0-1) of the 'education' destination sub-types reachable within 1000 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_education_1500m	Average diversity (0-1) of the 'education' destination sub-types reachable within 1500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_education_300m	Average diversity (0-1) of the 'education' destination sub-types reachable within 300 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_education_500m	Average diversity (0-1) of the 'education' destination sub-types reachable within 500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_fresh_food_1000m	Average diversity (0-1) of the 'fresh food' destination sub-types reachable within 1000 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_fresh_food_1500m	Average diversity (0-1) of the 'fresh food' destination sub-types reachable within 1500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_fresh_food_300m	Average diversity (0-1) of the 'fresh food' destination sub-types reachable within 300 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1

Variable	Description	Units
avg_diversity_walk_fresh_food_500m	Average diversity (0-1) of the 'fresh food' destination sub-types reachable within 500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_health_care_1000m	Average diversity (0-1) of the 'health care' destination sub-types reachable within 1000 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_health_care_1500m	Average diversity (0-1) of the 'health care' destination sub-types reachable within 1500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_health_care_300m	Average diversity (0-1) of the 'health care' destination sub-types reachable within 300 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_diversity_walk_health_care_500m	Average diversity (0-1) of the 'health care' destination sub-types reachable within 500 m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined, so that 1 is an even spread across all of them and 0 is one sub-type only, or nothing reachable at all	index 0-1
avg_richness_walk_care_1000m	Average share (0-1) of the 'care' destination sub-types with at least one destination reachable within 1000 m along the pedestrian network	index 0-1
avg_richness_walk_care_1500m	Average share (0-1) of the 'care' destination sub-types with at least one destination reachable within 1500 m along the pedestrian network	index 0-1
avg_richness_walk_care_300m	Average share (0-1) of the 'care' destination sub-types with at least one destination reachable within 300 m along the pedestrian network	index 0-1
avg_richness_walk_care_500m	Average share (0-1) of the 'care' destination sub-types with at least one destination reachable within 500 m along the pedestrian network	index 0-1
avg_richness_walk_community_culture_recreation_1000m	Average share (0-1) of the 'community culture recreation' destination sub-types with at least one destination reachable within 1000 m along the pedestrian network	index 0-1
avg_richness_walk_community_culture_recreation_1500m	Average share (0-1) of the 'community culture recreation' destination sub-types with at least one destination reachable within 1500 m along the pedestrian network	index 0-1
avg_richness_walk_community_culture_recreation_300m	Average share (0-1) of the 'community culture recreation' destination sub-types with at least one destination reachable within 300 m along the pedestrian network	index 0-1
avg_richness_walk_community_culture_recreation_500m	Average share (0-1) of the 'community culture recreation' destination sub-types with at least one destination reachable within 500 m along the pedestrian network	index 0-1
avg_richness_walk_education_1000m	Average share (0-1) of the 'education' destination sub-types with at least one destination reachable within 1000 m along the pedestrian network	index 0-1
avg_richness_walk_education_1500m	Average share (0-1) of the 'education' destination sub-types with at least one destination reachable within 1500 m along the pedestrian network	index 0-1
avg_richness_walk_education_300m	Average share (0-1) of the 'education' destination sub-types with at least one destination reachable within 300 m along the pedestrian network	index 0-1
avg_richness_walk_education_500m	Average share (0-1) of the 'education' destination sub-types with at least one destination reachable within 500 m along the pedestrian network	index 0-1
avg_richness_walk_fresh_food_1000m	Average share (0-1) of the 'fresh food' destination sub-types with at least one destination reachable within 1000 m along the pedestrian network	index 0-1
avg_richness_walk_fresh_food_1500m	Average share (0-1) of the 'fresh food' destination sub-types with at least one destination reachable within 1500 m along the pedestrian network	index 0-1
avg_richness_walk_fresh_food_300m	Average share (0-1) of the 'fresh food' destination sub-types with at least one destination reachable within 300 m along the pedestrian network	index 0-1
avg_richness_walk_fresh_food_500m	Average share (0-1) of the 'fresh food' destination sub-types with at least one destination reachable within 500 m along the pedestrian network	index 0-1
avg_richness_walk_health_care_1000m	Average share (0-1) of the 'health care' destination sub-types with at least one destination reachable within 1000 m along the pedestrian network	index 0-1
avg_richness_walk_health_care_1500m	Average share (0-1) of the 'health care' destination sub-types with at least one destination reachable within 1500 m along the pedestrian network	index 0-1
avg_richness_walk_health_care_300m	Average share (0-1) of the 'health care' destination sub-types with at least one destination reachable within 300 m along the pedestrian network	index 0-1
avg_richness_walk_health_care_500m	Average share (0-1) of the 'health care' destination sub-types with at least one destination reachable within 500 m along the pedestrian network	index 0-1
avg_walk_dist_activity_centre_civic_all	Average distance (m) along the pedestrian network to the nearest activity centre of the 'civic' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_activity_centre_essentials_complete	Average distance (m) along the pedestrian network to the nearest activity centre of the 'essentials' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_activity_centre_essentials_local	Average distance (m) along the pedestrian network to the nearest activity centre of the 'essentials' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_activity_centre_health_all	Average distance (m) along the pedestrian network to the nearest activity centre of the 'health' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_activity_centre_jobs_complete	Average distance (m) along the pedestrian network to the nearest activity centre of the 'jobs' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	metres

Variable	Description	Units
avg_walk_dist_activity_centre_jobs_local	Average distance (m) along the pedestrian network to the nearest activity centre of the 'jobs' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_activity_centre_services_complete	Average distance (m) along the pedestrian network to the nearest activity centre of the 'services' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_activity_centre_services_local	Average distance (m) along the pedestrian network to the nearest activity centre of the 'services' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	metres
avg_walk_dist_blue_space	Average distance (m) along the pedestrian network to the nearest natural blue space -- a river, canal, drain, lake, reservoir, wetland, beach or coast (source: OpenStreetMap or custom data)	metres
avg_walk_dist_convenience	Average distance (m) along the pedestrian network to the nearest convenience store (source: OpenStreetMap or custom data)	metres
avg_walk_dist_denue_bakery_tortilleria	Average distance (m) along the pedestrian network to the nearest 'denue bakery tortilleria' (custom destination)	metres
avg_walk_dist_denue_care_support	Average distance (m) along the pedestrian network to the nearest 'denue care support' (custom destination)	metres
avg_walk_dist_denue_ccr_any	Average distance (m) along the pedestrian network to the nearest 'denue ccr any' (custom destination)	metres
avg_walk_dist_denue_childcare	Average distance (m) along the pedestrian network to the nearest 'denue childcare' (custom destination)	metres
avg_walk_dist_denue_community	Average distance (m) along the pedestrian network to the nearest 'denue community' (custom destination)	metres
avg_walk_dist_denue_culture	Average distance (m) along the pedestrian network to the nearest 'denue culture' (custom destination)	metres
avg_walk_dist_denue_employer_medium	Average distance (m) along the pedestrian network to the nearest 'denue employer medium' (custom destination)	metres
avg_walk_dist_denue_employment_any	Average distance (m) along the pedestrian network to the nearest 'denue employment any' (custom destination)	metres
avg_walk_dist_denue_finance_comms	Average distance (m) along the pedestrian network to the nearest 'denue finance comms' (custom destination)	metres
avg_walk_dist_denue_food_retail	Average distance (m) along the pedestrian network to the nearest 'denue food retail' (custom destination)	metres
avg_walk_dist_denue_fresh_food	Average distance (m) along the pedestrian network to the nearest 'denue fresh food' (custom destination)	metres
avg_walk_dist_denue_hospital	Average distance (m) along the pedestrian network to the nearest 'denue hospital' (custom destination)	metres
avg_walk_dist_denue_household_goods	Average distance (m) along the pedestrian network to the nearest 'denue household goods' (custom destination)	metres
avg_walk_dist_denue_manufacturing	Average distance (m) along the pedestrian network to the nearest 'denue manufacturing' (custom destination)	metres
avg_walk_dist_denue_medical_clinic	Average distance (m) along the pedestrian network to the nearest 'denue medical clinic' (custom destination)	metres
avg_walk_dist_denue_petrol_station	Average distance (m) along the pedestrian network to the nearest 'denue petrol station' (custom destination)	metres
avg_walk_dist_denue_pharmacy	Average distance (m) along the pedestrian network to the nearest 'denue pharmacy' (custom destination)	metres
avg_walk_dist_denue_preschool	Average distance (m) along the pedestrian network to the nearest 'denue preschool' (custom destination)	metres
avg_walk_dist_denue_primary_healthcare	Average distance (m) along the pedestrian network to the nearest 'denue primary healthcare' (custom destination)	metres
avg_walk_dist_denue_primary_school	Average distance (m) along the pedestrian network to the nearest 'denue primary school' (custom destination)	metres
avg_walk_dist_denue_recreation	Average distance (m) along the pedestrian network to the nearest 'denue recreation' (custom destination)	metres
avg_walk_dist_denue_secondary_school	Average distance (m) along the pedestrian network to the nearest 'denue secondary school' (custom destination)	metres
avg_walk_dist_denue_sport	Average distance (m) along the pedestrian network to the nearest 'denue sport' (custom destination)	metres
avg_walk_dist_denue_tertiary_education	Average distance (m) along the pedestrian network to the nearest 'denue tertiary education' (custom destination)	metres
avg_walk_dist_fresh_food_market	Average distance (m) along the pedestrian network to the nearest fresh food market / supermarket (source: OpenStreetMap or custom data)	metres
avg_walk_dist_fresh_food_pooled	Average distance (m) along the pedestrian network to the nearest fresh food market, supermarket or convenience store (pooled fresh food destinations)	metres
avg_walk_dist_pt_any	Average distance (m) along the pedestrian network to the nearest public transport stop (best result of GTFS and OpenStreetMap/custom sources)	metres

Variable	Description	Units
avg_walk_dist_public_open_space_any	Average distance (m) along the pedestrian network to the nearest public open space (source: OpenStreetMap)	metres
avg_walk_dist_public_open_space_large	Average distance (m) along the pedestrian network to the nearest public open space larger than 1.5 hectares (source: OpenStreetMap)	metres
avg_walk_dist_public_open_space_with_water	Average distance (m) along the pedestrian network to the nearest public open space containing water, such as a park with a lake (source: OpenStreetMap)	metres
pct_access_500m_convenience_score	Percentage of population with access within 500 m of a convenience store (source: OpenStreetMap or custom data)	percent
pct_access_500m_fresh_food_market_score	Percentage of population with access within 500 m of a fresh food market / supermarket (source: OpenStreetMap or custom data)	percent
pct_access_500m_large_public_green_space_score	Percentage of population with access within 500 m of a large public green space of at least 1 hectare (source: OpenStreetMap, Google Earth Engine)	percent
pct_access_500m_pt_any_score	Percentage of population with access within 500 m of any public transport stop (best result of GTFS and OpenStreetMap/custom sources)	percent
pct_access_500m_pt_osm_any_score	Percentage of population with access within 500 m of any public transport stop (source: OpenStreetMap or custom data)	percent
pct_access_500m_public_open_space_any_score	Percentage of population with access within 500 m of any public open space (source: OpenStreetMap)	percent
pct_access_500m_public_open_space_large_score	Percentage of population with access within 500 m of a public open space larger than 1.5 hectares (source: OpenStreetMap)	percent
pct_access_walk_activity_centre_civic_all_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'civic' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_civic_all_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'civic' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_civic_all_300m	Percentage of population with walking access within 300 m to an activity centre of the 'civic' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_civic_all_500m	Percentage of population with walking access within 500 m to an activity centre of the 'civic' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_complete_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'essentials' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_complete_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'essentials' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_complete_300m	Percentage of population with walking access within 300 m to an activity centre of the 'essentials' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_complete_500m	Percentage of population with walking access within 500 m to an activity centre of the 'essentials' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_local_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'essentials' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_local_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'essentials' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_local_300m	Percentage of population with walking access within 300 m to an activity centre of the 'essentials' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_essentials_local_500m	Percentage of population with walking access within 500 m to an activity centre of the 'essentials' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_health_all_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'health' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_health_all_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'health' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_health_all_300m	Percentage of population with walking access within 300 m to an activity centre of the 'health' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_health_all_500m	Percentage of population with walking access within 500 m to an activity centre of the 'health' definition ('all' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_complete_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'jobs' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_complete_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'jobs' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent

Variable	Description	Units
pct_access_walk_activity_centre_jobs_complete_300m	Percentage of population with walking access within 300 m to an activity centre of the 'jobs' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_complete_500m	Percentage of population with walking access within 500 m to an activity centre of the 'jobs' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_local_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'jobs' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_local_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'jobs' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_local_300m	Percentage of population with walking access within 300 m to an activity centre of the 'jobs' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_jobs_local_500m	Percentage of population with walking access within 500 m to an activity centre of the 'jobs' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_complete_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'services' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_complete_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'services' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_complete_300m	Percentage of population with walking access within 300 m to an activity centre of the 'services' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_complete_500m	Percentage of population with walking access within 500 m to an activity centre of the 'services' definition ('complete' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_local_1000m	Percentage of population with walking access within 1000 m to an activity centre of the 'services' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_local_1500m	Percentage of population with walking access within 1500 m to an activity centre of the 'services' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_local_300m	Percentage of population with walking access within 300 m to an activity centre of the 'services' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_activity_centre_services_local_500m	Percentage of population with walking access within 500 m to an activity centre of the 'services' definition ('local' tier): a network location whose walkable catchment contains at least one destination of each required category	percent
pct_access_walk_all_essentials_lenient_1000m	Percentage of population with walking access within 1000 m to all destination categories of the 'essentials' combined-access set (lenient variants)	percent
pct_access_walk_all_essentials_lenient_1500m	Percentage of population with walking access within 1500 m to all destination categories of the 'essentials' combined-access set (lenient variants)	percent
pct_access_walk_all_essentials_lenient_300m	Percentage of population with walking access within 300 m to all destination categories of the 'essentials' combined-access set (lenient variants)	percent
pct_access_walk_all_essentials_lenient_500m	Percentage of population with walking access within 500 m to all destination categories of the 'essentials' combined-access set (lenient variants)	percent
pct_access_walk_all_essentials_strict_1000m	Percentage of population with walking access within 1000 m to all destination categories of the 'essentials' combined-access set (strict variants)	percent
pct_access_walk_all_essentials_strict_1500m	Percentage of population with walking access within 1500 m to all destination categories of the 'essentials' combined-access set (strict variants)	percent
pct_access_walk_all_essentials_strict_300m	Percentage of population with walking access within 300 m to all destination categories of the 'essentials' combined-access set (strict variants)	percent
pct_access_walk_all_essentials_strict_500m	Percentage of population with walking access within 500 m to all destination categories of the 'essentials' combined-access set (strict variants)	percent
pct_access_walk_all_lenient_1000m	Percentage of population with walking access within 1000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport)	percent
pct_access_walk_all_lenient_1500m	Percentage of population with walking access within 1500 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport)	percent
pct_access_walk_all_lenient_300m	Percentage of population with walking access within 300 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport)	percent
pct_access_walk_all_lenient_500m	Percentage of population with walking access within 500 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport)	percent
pct_access_walk_all_services_lenient_1000m	Percentage of population with walking access within 1000 m to all destination categories of the 'services' combined-access set (lenient variants)	percent
pct_access_walk_all_services_lenient_1500m	Percentage of population with walking access within 1500 m to all destination categories of the 'services' combined-access set (lenient variants)	percent
pct_access_walk_all_services_lenient_300m	Percentage of population with walking access within 300 m to all destination categories of the 'services' combined-access set (lenient variants)	percent
pct_access_walk_all_services_lenient_500m	Percentage of population with walking access within 500 m to all destination categories of the 'services' combined-access set (lenient variants)	percent

Variable	Description	Units
pct_access_walk_all_services_strict_1000m	Percentage of population with walking access within 1000 m to all destination categories of the 'services' combined-access set (strict variants)	percent
pct_access_walk_all_services_strict_1500m	Percentage of population with walking access within 1500 m to all destination categories of the 'services' combined-access set (strict variants)	percent
pct_access_walk_all_services_strict_300m	Percentage of population with walking access within 300 m to all destination categories of the 'services' combined-access set (strict variants)	percent
pct_access_walk_all_services_strict_500m	Percentage of population with walking access within 500 m to all destination categories of the 'services' combined-access set (strict variants)	percent
pct_access_walk_all_strict_1000m	Percentage of population with walking access within 1000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport)	percent
pct_access_walk_all_strict_1500m	Percentage of population with walking access within 1500 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport)	percent
pct_access_walk_all_strict_300m	Percentage of population with walking access within 300 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport)	percent
pct_access_walk_all_strict_500m	Percentage of population with walking access within 500 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport)	percent
pct_access_walk_blue_space_1000m	Percentage of population with walking access within 1000 m to a natural blue space -- a river, canal, drain, lake, reservoir, wetland, beach or coast (source: OpenStreetMap or custom data)	percent
pct_access_walk_blue_space_1500m	Percentage of population with walking access within 1500 m to a natural blue space -- a river, canal, drain, lake, reservoir, wetland, beach or coast (source: OpenStreetMap or custom data)	percent
pct_access_walk_blue_space_300m	Percentage of population with walking access within 300 m to a natural blue space -- a river, canal, drain, lake, reservoir, wetland, beach or coast (source: OpenStreetMap or custom data)	percent
pct_access_walk_blue_space_500m	Percentage of population with walking access within 500 m to a natural blue space -- a river, canal, drain, lake, reservoir, wetland, beach or coast (source: OpenStreetMap or custom data)	percent
pct_access_walk_convenience_1000m	Percentage of population with walking access within 1000 m to a convenience store (source: OpenStreetMap or custom data)	percent
pct_access_walk_convenience_1500m	Percentage of population with walking access within 1500 m to a convenience store (source: OpenStreetMap or custom data)	percent
pct_access_walk_convenience_300m	Percentage of population with walking access within 300 m to a convenience store (source: OpenStreetMap or custom data)	percent
pct_access_walk_convenience_500m	Percentage of population with walking access within 500 m to a convenience store (source: OpenStreetMap or custom data)	percent
pct_access_walk_denue_bakery_tortilleria_1000m	Percentage of population with walking access within 1000 m to 'denue bakery tortilleria' (custom destination)	percent
pct_access_walk_denue_bakery_tortilleria_1500m	Percentage of population with walking access within 1500 m to 'denue bakery tortilleria' (custom destination)	percent
pct_access_walk_denue_bakery_tortilleria_300m	Percentage of population with walking access within 300 m to 'denue bakery tortilleria' (custom destination)	percent
pct_access_walk_denue_bakery_tortilleria_500m	Percentage of population with walking access within 500 m to 'denue bakery tortilleria' (custom destination)	percent
pct_access_walk_denue_care_support_1000m	Percentage of population with walking access within 1000 m to 'denue care support' (custom destination)	percent
pct_access_walk_denue_care_support_1500m	Percentage of population with walking access within 1500 m to 'denue care support' (custom destination)	percent
pct_access_walk_denue_care_support_300m	Percentage of population with walking access within 300 m to 'denue care support' (custom destination)	percent
pct_access_walk_denue_care_support_500m	Percentage of population with walking access within 500 m to 'denue care support' (custom destination)	percent
pct_access_walk_denue_ccr_any_1000m	Percentage of population with walking access within 1000 m to 'denue ccr any' (custom destination)	percent
pct_access_walk_denue_ccr_any_1500m	Percentage of population with walking access within 1500 m to 'denue ccr any' (custom destination)	percent
pct_access_walk_denue_ccr_any_300m	Percentage of population with walking access within 300 m to 'denue ccr any' (custom destination)	percent
pct_access_walk_denue_ccr_any_500m	Percentage of population with walking access within 500 m to 'denue ccr any' (custom destination)	percent
pct_access_walk_denue_childcare_1000m	Percentage of population with walking access within 1000 m to 'denue childcare' (custom destination)	percent
pct_access_walk_denue_childcare_1500m	Percentage of population with walking access within 1500 m to 'denue childcare' (custom destination)	percent
pct_access_walk_denue_childcare_300m	Percentage of population with walking access within 300 m to 'denue childcare' (custom destination)	percent
pct_access_walk_denue_childcare_500m	Percentage of population with walking access within 500 m to 'denue childcare' (custom destination)	percent

Variable	Description	Units
pct_access_walk_denue_community_1000m	Percentage of population with walking access within 1000 m to 'denue community' (custom destination)	percent
pct_access_walk_denue_community_1500m	Percentage of population with walking access within 1500 m to 'denue community' (custom destination)	percent
pct_access_walk_denue_community_300m	Percentage of population with walking access within 300 m to 'denue community' (custom destination)	percent
pct_access_walk_denue_community_500m	Percentage of population with walking access within 500 m to 'denue community' (custom destination)	percent
pct_access_walk_denue_culture_1000m	Percentage of population with walking access within 1000 m to 'denue culture' (custom destination)	percent
pct_access_walk_denue_culture_1500m	Percentage of population with walking access within 1500 m to 'denue culture' (custom destination)	percent
pct_access_walk_denue_culture_300m	Percentage of population with walking access within 300 m to 'denue culture' (custom destination)	percent
pct_access_walk_denue_culture_500m	Percentage of population with walking access within 500 m to 'denue culture' (custom destination)	percent
pct_access_walk_denue_employer_medium_1000m	Percentage of population with walking access within 1000 m to 'denue employer medium' (custom destination)	percent
pct_access_walk_denue_employer_medium_1500m	Percentage of population with walking access within 1500 m to 'denue employer medium' (custom destination)	percent
pct_access_walk_denue_employer_medium_300m	Percentage of population with walking access within 300 m to 'denue employer medium' (custom destination)	percent
pct_access_walk_denue_employer_medium_500m	Percentage of population with walking access within 500 m to 'denue employer medium' (custom destination)	percent
pct_access_walk_denue_employment_any_1000m	Percentage of population with walking access within 1000 m to 'denue employment any' (custom destination)	percent
pct_access_walk_denue_employment_any_1500m	Percentage of population with walking access within 1500 m to 'denue employment any' (custom destination)	percent
pct_access_walk_denue_employment_any_300m	Percentage of population with walking access within 300 m to 'denue employment any' (custom destination)	percent
pct_access_walk_denue_employment_any_500m	Percentage of population with walking access within 500 m to 'denue employment any' (custom destination)	percent
pct_access_walk_denue_finance_comms_1000m	Percentage of population with walking access within 1000 m to 'denue finance comms' (custom destination)	percent
pct_access_walk_denue_finance_comms_1500m	Percentage of population with walking access within 1500 m to 'denue finance comms' (custom destination)	percent
pct_access_walk_denue_finance_comms_300m	Percentage of population with walking access within 300 m to 'denue finance comms' (custom destination)	percent
pct_access_walk_denue_finance_comms_500m	Percentage of population with walking access within 500 m to 'denue finance comms' (custom destination)	percent
pct_access_walk_denue_food_retail_1000m	Percentage of population with walking access within 1000 m to 'denue food retail' (custom destination)	percent
pct_access_walk_denue_food_retail_1500m	Percentage of population with walking access within 1500 m to 'denue food retail' (custom destination)	percent
pct_access_walk_denue_food_retail_300m	Percentage of population with walking access within 300 m to 'denue food retail' (custom destination)	percent
pct_access_walk_denue_food_retail_500m	Percentage of population with walking access within 500 m to 'denue food retail' (custom destination)	percent
pct_access_walk_denue_fresh_food_1000m	Percentage of population with walking access within 1000 m to 'denue fresh food' (custom destination)	percent
pct_access_walk_denue_fresh_food_1500m	Percentage of population with walking access within 1500 m to 'denue fresh food' (custom destination)	percent
pct_access_walk_denue_fresh_food_300m	Percentage of population with walking access within 300 m to 'denue fresh food' (custom destination)	percent
pct_access_walk_denue_fresh_food_500m	Percentage of population with walking access within 500 m to 'denue fresh food' (custom destination)	percent
pct_access_walk_denue_hospital_1000m	Percentage of population with walking access within 1000 m to 'denue hospital' (custom destination)	percent
pct_access_walk_denue_hospital_1500m	Percentage of population with walking access within 1500 m to 'denue hospital' (custom destination)	percent
pct_access_walk_denue_hospital_300m	Percentage of population with walking access within 300 m to 'denue hospital' (custom destination)	percent
pct_access_walk_denue_hospital_500m	Percentage of population with walking access within 500 m to 'denue hospital' (custom destination)	percent

Variable	Description	Units
pct_access_walk_denue_household_goods_1000m	Percentage of population with walking access within 1000 m to 'denue household goods' (custom destination)	percent
pct_access_walk_denue_household_goods_1500m	Percentage of population with walking access within 1500 m to 'denue household goods' (custom destination)	percent
pct_access_walk_denue_household_goods_300m	Percentage of population with walking access within 300 m to 'denue household goods' (custom destination)	percent
pct_access_walk_denue_household_goods_500m	Percentage of population with walking access within 500 m to 'denue household goods' (custom destination)	percent
pct_access_walk_denue_manufacturing_1000m	Percentage of population with walking access within 1000 m to 'denue manufacturing' (custom destination)	percent
pct_access_walk_denue_manufacturing_1500m	Percentage of population with walking access within 1500 m to 'denue manufacturing' (custom destination)	percent
pct_access_walk_denue_manufacturing_300m	Percentage of population with walking access within 300 m to 'denue manufacturing' (custom destination)	percent
pct_access_walk_denue_manufacturing_500m	Percentage of population with walking access within 500 m to 'denue manufacturing' (custom destination)	percent
pct_access_walk_denue_medical_clinic_1000m	Percentage of population with walking access within 1000 m to 'denue medical clinic' (custom destination)	percent
pct_access_walk_denue_medical_clinic_1500m	Percentage of population with walking access within 1500 m to 'denue medical clinic' (custom destination)	percent
pct_access_walk_denue_medical_clinic_300m	Percentage of population with walking access within 300 m to 'denue medical clinic' (custom destination)	percent
pct_access_walk_denue_medical_clinic_500m	Percentage of population with walking access within 500 m to 'denue medical clinic' (custom destination)	percent
pct_access_walk_denue_pharmacy_1000m	Percentage of population with walking access within 1000 m to 'denue pharmacy' (custom destination)	percent
pct_access_walk_denue_pharmacy_1500m	Percentage of population with walking access within 1500 m to 'denue pharmacy' (custom destination)	percent
pct_access_walk_denue_pharmacy_300m	Percentage of population with walking access within 300 m to 'denue pharmacy' (custom destination)	percent
pct_access_walk_denue_pharmacy_500m	Percentage of population with walking access within 500 m to 'denue pharmacy' (custom destination)	percent
pct_access_walk_denue_preschool_1000m	Percentage of population with walking access within 1000 m to 'denue preschool' (custom destination)	percent
pct_access_walk_denue_preschool_1500m	Percentage of population with walking access within 1500 m to 'denue preschool' (custom destination)	percent
pct_access_walk_denue_preschool_300m	Percentage of population with walking access within 300 m to 'denue preschool' (custom destination)	percent
pct_access_walk_denue_preschool_500m	Percentage of population with walking access within 500 m to 'denue preschool' (custom destination)	percent
pct_access_walk_denue_primary_healthcare_1000m	Percentage of population with walking access within 1000 m to 'denue primary healthcare' (custom destination)	percent
pct_access_walk_denue_primary_healthcare_1500m	Percentage of population with walking access within 1500 m to 'denue primary healthcare' (custom destination)	percent
pct_access_walk_denue_primary_healthcare_300m	Percentage of population with walking access within 300 m to 'denue primary healthcare' (custom destination)	percent
pct_access_walk_denue_primary_healthcare_500m	Percentage of population with walking access within 500 m to 'denue primary healthcare' (custom destination)	percent
pct_access_walk_denue_primary_school_1000m	Percentage of population with walking access within 1000 m to 'denue primary school' (custom destination)	percent
pct_access_walk_denue_primary_school_1500m	Percentage of population with walking access within 1500 m to 'denue primary school' (custom destination)	percent
pct_access_walk_denue_primary_school_300m	Percentage of population with walking access within 300 m to 'denue primary school' (custom destination)	percent
pct_access_walk_denue_primary_school_500m	Percentage of population with walking access within 500 m to 'denue primary school' (custom destination)	percent
pct_access_walk_denue_recreation_1000m	Percentage of population with walking access within 1000 m to 'denue recreation' (custom destination)	percent
pct_access_walk_denue_recreation_1500m	Percentage of population with walking access within 1500 m to 'denue recreation' (custom destination)	percent
pct_access_walk_denue_recreation_300m	Percentage of population with walking access within 300 m to 'denue recreation' (custom destination)	percent
pct_access_walk_denue_recreation_500m	Percentage of population with walking access within 500 m to 'denue recreation' (custom destination)	percent

Variable	Description	Units
pct_access_walk_denue_secondary_school_1000m	Percentage of population with walking access within 1000 m to 'denue secondary school' (custom destination)	percent
pct_access_walk_denue_secondary_school_1500m	Percentage of population with walking access within 1500 m to 'denue secondary school' (custom destination)	percent
pct_access_walk_denue_secondary_school_300m	Percentage of population with walking access within 300 m to 'denue secondary school' (custom destination)	percent
pct_access_walk_denue_secondary_school_500m	Percentage of population with walking access within 500 m to 'denue secondary school' (custom destination)	percent
pct_access_walk_denue_sport_1000m	Percentage of population with walking access within 1000 m to 'denue sport' (custom destination)	percent
pct_access_walk_denue_sport_1500m	Percentage of population with walking access within 1500 m to 'denue sport' (custom destination)	percent
pct_access_walk_denue_sport_300m	Percentage of population with walking access within 300 m to 'denue sport' (custom destination)	percent
pct_access_walk_denue_sport_500m	Percentage of population with walking access within 500 m to 'denue sport' (custom destination)	percent
pct_access_walk_denue_tertiary_education_1000m	Percentage of population with walking access within 1000 m to 'denue tertiary education' (custom destination)	percent
pct_access_walk_denue_tertiary_education_1500m	Percentage of population with walking access within 1500 m to 'denue tertiary education' (custom destination)	percent
pct_access_walk_denue_tertiary_education_300m	Percentage of population with walking access within 300 m to 'denue tertiary education' (custom destination)	percent
pct_access_walk_denue_tertiary_education_500m	Percentage of population with walking access within 500 m to 'denue tertiary education' (custom destination)	percent
pct_access_walk_fresh_food_market_1000m	Percentage of population with walking access within 1000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data)	percent
pct_access_walk_fresh_food_market_1500m	Percentage of population with walking access within 1500 m to a fresh food market / supermarket (source: OpenStreetMap or custom data)	percent
pct_access_walk_fresh_food_market_300m	Percentage of population with walking access within 300 m to a fresh food market / supermarket (source: OpenStreetMap or custom data)	percent
pct_access_walk_fresh_food_market_500m	Percentage of population with walking access within 500 m to a fresh food market / supermarket (source: OpenStreetMap or custom data)	percent
pct_access_walk_fresh_food_pooled_1000m	Percentage of population with walking access within 1000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations)	percent
pct_access_walk_fresh_food_pooled_1500m	Percentage of population with walking access within 1500 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations)	percent
pct_access_walk_fresh_food_pooled_300m	Percentage of population with walking access within 300 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations)	percent
pct_access_walk_fresh_food_pooled_500m	Percentage of population with walking access within 500 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations)	percent
pct_access_walk_pt_any_1000m	Percentage of population with walking access within 1000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources)	percent
pct_access_walk_pt_any_1500m	Percentage of population with walking access within 1500 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources)	percent
pct_access_walk_pt_any_300m	Percentage of population with walking access within 300 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources)	percent
pct_access_walk_pt_any_500m	Percentage of population with walking access within 500 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources)	percent
pct_access_walk_public_open_space_any_1000m	Percentage of population with walking access within 1000 m to any public open space (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_any_1500m	Percentage of population with walking access within 1500 m to any public open space (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_any_300m	Percentage of population with walking access within 300 m to any public open space (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_any_500m	Percentage of population with walking access within 500 m to any public open space (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_large_1000m	Percentage of population with walking access within 1000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_large_1500m	Percentage of population with walking access within 1500 m to a public open space larger than 1.5 hectares (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_large_300m	Percentage of population with walking access within 300 m to a public open space larger than 1.5 hectares (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_large_500m	Percentage of population with walking access within 500 m to a public open space larger than 1.5 hectares (source: OpenStreetMap)	percent

Variable	Description	Units
pct_access_walk_public_open_space_with_water_1000m	Percentage of population with walking access within 1000 m to a public open space containing water, such as a park with a lake (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_with_water_1500m	Percentage of population with walking access within 1500 m to a public open space containing water, such as a park with a lake (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_with_water_300m	Percentage of population with walking access within 300 m to a public open space containing water, such as a park with a lake (source: OpenStreetMap)	percent
pct_access_walk_public_open_space_with_water_500m	Percentage of population with walking access within 500 m to a public open space containing water, such as a park with a lake (source: OpenStreetMap)	percent
pct_beyond_walk_denue_petrol_station_250m	Percentage of population living further than 250 m along the pedestrian network from the nearest 'denue petrol station' (custom destination); proximity to this destination is a disamenity, so higher is better	percent

Indicator estimates: cycling accessibility

Variable	Description	Units
avg_cycle_dist_activity_centre_complete	Average distance (m) by cycling to the nearest 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_activity_centre_local	Average distance (m) by cycling to the nearest 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_fresh_food_market	Average distance (m) by cycling to the nearest fresh food market / supermarket (source: OpenStreetMap or custom data), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_fresh_food_pooled	Average distance (m) by cycling to the nearest fresh food market, supermarket or convenience store (pooled fresh food destinations), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_pt_any	Average distance (m) by cycling to the nearest public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_public_open_space_any	Average distance (m) by cycling to the nearest public open space (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_public_open_space_large	Average distance (m) by cycling to the nearest public open space larger than 1.5 hectares (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	metres
avg_cycle_dist_safe_activity_centre_complete	Average distance (m) by cycling to the nearest 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
avg_cycle_dist_safe_activity_centre_local	Average distance (m) by cycling to the nearest 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
avg_cycle_dist_safe_fresh_food_market	Average distance (m) by cycling to the nearest fresh food market / supermarket (source: OpenStreetMap or custom data), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
avg_cycle_dist_safe_fresh_food_pooled	Average distance (m) by cycling to the nearest fresh food market, supermarket or convenience store (pooled fresh food destinations), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
avg_cycle_dist_safe_pt_any	Average distance (m) by cycling to the nearest public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
avg_cycle_dist_safe_public_open_space_any	Average distance (m) by cycling to the nearest public open space (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
avg_cycle_dist_safe_public_open_space_large	Average distance (m) by cycling to the nearest public open space larger than 1.5 hectares (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	metres
pct_access_cycle_activity_centre_complete_1000m	Percentage of population with cycling access within 1000 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_activity_centre_complete_2000m	Percentage of population with cycling access within 2000 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_activity_centre_complete_5000m	Percentage of population with cycling access within 5000 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_activity_centre_complete_500m	Percentage of population with cycling access within 500 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_activity_centre_local_1000m	Percentage of population with cycling access within 1000 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_activity_centre_local_2000m	Percentage of population with cycling access within 2000 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_activity_centre_local_5000m	Percentage of population with cycling access within 5000 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent

Variable	Description	Units
pct_access_cycle_activity_centre_local_500m	Percentage of population with cycling access within 500 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_lenient_1000m	Percentage of population with cycling access within 1000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_lenient_2000m	Percentage of population with cycling access within 2000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_lenient_5000m	Percentage of population with cycling access within 5000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_lenient_500m	Percentage of population with cycling access within 500 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_strict_1000m	Percentage of population with cycling access within 1000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_strict_2000m	Percentage of population with cycling access within 2000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_strict_5000m	Percentage of population with cycling access within 5000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_all_strict_500m	Percentage of population with cycling access within 500 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_market_1000m	Percentage of population with cycling access within 1000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_market_2000m	Percentage of population with cycling access within 2000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_market_5000m	Percentage of population with cycling access within 5000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_market_500m	Percentage of population with cycling access within 500 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_pooled_1000m	Percentage of population with cycling access within 1000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_pooled_2000m	Percentage of population with cycling access within 2000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_pooled_5000m	Percentage of population with cycling access within 5000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_fresh_food_pooled_500m	Percentage of population with cycling access within 500 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_pt_any_1000m	Percentage of population with cycling access within 1000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_pt_any_2000m	Percentage of population with cycling access within 2000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_pt_any_5000m	Percentage of population with cycling access within 5000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_pt_any_500m	Percentage of population with cycling access within 500 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_any_1000m	Percentage of population with cycling access within 1000 m to any public open space (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_any_2000m	Percentage of population with cycling access within 2000 m to any public open space (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_any_5000m	Percentage of population with cycling access within 5000 m to any public open space (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_any_500m	Percentage of population with cycling access within 500 m to any public open space (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_large_1000m	Percentage of population with cycling access within 1000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_large_2000m	Percentage of population with cycling access within 2000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_public_open_space_large_5000m	Percentage of population with cycling access within 5000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent

Variable	Description	Units
pct_access_cycle_public_open_space_large_500m	Percentage of population with cycling access within 500 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a danger-weighted route over the full cycling network (higher-stress links usable at a proportionate distance penalty)	percent
pct_access_cycle_safe_activity_centre_complete_1000m	Percentage of population with cycling access within 1000 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_complete_2000m	Percentage of population with cycling access within 2000 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_complete_5000m	Percentage of population with cycling access within 5000 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_complete_500m	Percentage of population with cycling access within 500 m to a 'complete' activity centre: a network location whose walkable catchment (default 400 m) contains at least one high-amenity (strict) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_local_1000m	Percentage of population with cycling access within 1000 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_local_2000m	Percentage of population with cycling access within 2000 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_local_5000m	Percentage of population with cycling access within 5000 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_activity_centre_local_500m	Percentage of population with cycling access within 500 m to a 'local' activity centre: a network location whose walkable catchment (default 400 m) contains at least one everyday (lenient) destination of each category (food, public open space, public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_lenient_1000m	Percentage of population with cycling access within 1000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_lenient_2000m	Percentage of population with cycling access within 2000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_lenient_5000m	Percentage of population with cycling access within 5000 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_lenient_500m	Percentage of population with cycling access within 500 m to all core destination categories at their lenient variant (pooled fresh food, any public open space, any public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_strict_1000m	Percentage of population with cycling access within 1000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_strict_2000m	Percentage of population with cycling access within 2000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_strict_5000m	Percentage of population with cycling access within 5000 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_all_strict_500m	Percentage of population with cycling access within 500 m to all core destination categories at their strict variant (fresh food market, large public open space, frequent public transport), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_market_1000m	Percentage of population with cycling access within 1000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_market_2000m	Percentage of population with cycling access within 2000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_market_5000m	Percentage of population with cycling access within 5000 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_market_500m	Percentage of population with cycling access within 500 m to a fresh food market / supermarket (source: OpenStreetMap or custom data), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_pooled_1000m	Percentage of population with cycling access within 1000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_pooled_2000m	Percentage of population with cycling access within 2000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_pooled_5000m	Percentage of population with cycling access within 5000 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_fresh_food_pooled_500m	Percentage of population with cycling access within 500 m to a fresh food market, supermarket or convenience store (pooled fresh food destinations), using a fully low-stress route (Level of Traffic Stress 1–2)	percent

Variable	Description	Units
pct_access_cycle_safe_pt_any_1000m	Percentage of population with cycling access within 1000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_pt_any_2000m	Percentage of population with cycling access within 2000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_pt_any_5000m	Percentage of population with cycling access within 5000 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_pt_any_500m	Percentage of population with cycling access within 500 m to any public transport stop (best result of GTFS and OpenStreetMap/custom sources), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_any_1000m	Percentage of population with cycling access within 1000 m to any public open space (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_any_2000m	Percentage of population with cycling access within 2000 m to any public open space (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_any_5000m	Percentage of population with cycling access within 5000 m to any public open space (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_any_500m	Percentage of population with cycling access within 500 m to any public open space (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_large_1000m	Percentage of population with cycling access within 1000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_large_2000m	Percentage of population with cycling access within 2000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_large_5000m	Percentage of population with cycling access within 5000 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent
pct_access_cycle_safe_public_open_space_large_500m	Percentage of population with cycling access within 500 m to a public open space larger than 1.5 hectares (source: OpenStreetMap), using a fully low-stress route (Level of Traffic Stress 1–2)	percent

Indicator estimates: access (straight-line catchment)

Variable	Description	Units
avg_euclid_dist_imip_fire_station	Average straight-line distance (m) to the nearest 'imip fire station' (custom destination)	metres
avg_euclid_dist_imip_police_station	Average straight-line distance (m) to the nearest 'imip police station' (custom destination)	metres
pct_access_euclid_imip_fire_station_75000m	Percentage of population within 75000 m in a straight line of 'imip fire station' (custom destination)	percent
pct_access_euclid_imip_police_station_15000m	Percentage of population within 15000 m in a straight line of 'imip police station' (custom destination)	percent

Indicator estimates: linked indicators

Variable	Description	Units
ext_avg_walk_dist_park_m	Distance to parks (work package)	m
ext_block_perimeter_m	Block size (perimeter)	m
ext_commercial_area_ha	Commercial land use	ha
ext_economic_sector_diversity	Job diversity	index 0-1
ext_flood_area_m2	Flooded area on main roads	m^2
ext_land_use_entropy	Land use mix	index 0-1
ext_major_road_length_300m	Major roads within 300 m	m
ext_ndvi	Vegetation cover (NDVI)	NDVI
ext_pop_affected_flood	Population exposed to flooding	persons
ext_protected_cycling_street_pct	Protected cycle lane coverage	%
ext_residential_area_ha	Residential land use	ha
ext_tree_canopy_pct	Tree canopy cover	%
utci_day_mean	Unit mean of the daytime mean Universal Thermal Climate Index (hours with sunlight), over pedestrian surfaces, on the hottest day of 2023.	°C
utci_peak	Unit mean of the daily maximum Universal Thermal Climate Index (effectively 17:00), over pedestrian surfaces, on the hottest day of 2023.	°C

Indicator estimates: composite indices

Variable	Description	Units
index_uli	Average across sample points of the composite index 'uli' (Mazziotta-Pareto Index): the mean of its normalised indicators (100 = reference) less a penalty for how unbalanced they are, so that balanced performance is rewarded	points from the reference (0 = study region average)
index_uli__adaptive_capability	Average across sample points of the composite index 'uli': the normalised score of its indicator 'adaptive capability' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__ambient_environment	Average across sample points of the composite index 'uli', domain 'ambient environment': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli__block_size	Average across sample points of the composite index 'uli': the normalised score of its indicator 'block size' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__blue_space	Average across sample points of the composite index 'uli': the normalised score of its indicator 'blue space' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__built_environment	Average across sample points of the composite index 'uli', domain 'built environment': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli__care_support	Average across sample points of the composite index 'uli': the normalised score of its indicator 'care support' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__ccr_any	Average across sample points of the composite index 'uli': the normalised score of its indicator 'ccr any' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__convenience	Average across sample points of the composite index 'uli': the normalised score of its indicator 'convenience' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__cycle_low_stress_food	Average across sample points of the composite index 'uli': the normalised score of its indicator 'cycle low stress food' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__div_care_500m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'div care 500m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__div_ccr_500m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'div ccr 500m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__div_education_500m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'div education 500m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__div_fresh_food_500m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'div fresh food 500m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__div_health_500m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'div health 500m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__economic_development	Average across sample points of the composite index 'uli', domain 'economic development': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli__fire_3km	Average across sample points of the composite index 'uli': the normalised score of its indicator 'fire 3km' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__fresh_food	Average across sample points of the composite index 'uli': the normalised score of its indicator 'fresh food' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__heat_exposure	Average across sample points of the composite index 'uli': the normalised score of its indicator 'heat exposure' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__heat_sensitivity	Average across sample points of the composite index 'uli': the normalised score of its indicator 'heat sensitivity' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__job_diversity	Average across sample points of the composite index 'uli': the normalised score of its indicator 'job diversity' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__lpugs	Average across sample points of the composite index 'uli': the normalised score of its indicator 'lpugs' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__major_roads	Average across sample points of the composite index 'uli': the normalised score of its indicator 'major roads' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__manufacturing_1000m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'manufacturing 1000m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__mobility_transport	Average across sample points of the composite index 'uli', domain 'mobility transport': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli__parks	Average across sample points of the composite index 'uli': the normalised score of its indicator 'parks' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__petrol_250m	Average across sample points of the composite index 'uli': the normalised score of its indicator 'petrol 250m' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__police_2km	Average across sample points of the composite index 'uli': the normalised score of its indicator 'police 2km' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__primary_healthcare	Average across sample points of the composite index 'uli': the normalised score of its indicator 'primary healthcare' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__primary_school	Average across sample points of the composite index 'uli': the normalised score of its indicator 'primary school' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__public_transport	Average across sample points of the composite index 'uli': the normalised score of its indicator 'public transport' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__safety	Average across sample points of the composite index 'uli', domain 'safety': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)

Variable	Description	Units
index_uli__social_infrastructure	Average across sample points of the composite index 'uli', domain 'social infrastructure': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli__vegetation	Average across sample points of the composite index 'uli': the normalised score of its indicator 'vegetation' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli__walkability	Average across sample points of the composite index 'uli': the normalised score of its indicator 'walkability' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli_mean	Average across sample points of the composite index 'uli': mean level of its normalised indicators or domain scores (100 = reference), before the penalty for imbalance	points from the reference (0 = study region average)
index_uli_penalty	Average across sample points of the composite index 'uli': penalty for imbalance among its indicators or domain scores (standard deviation multiplied by the coefficient of variation), applied to the mean level	index (100 = reference)
index_uli_plain	Average across sample points of the composite index 'uli plain' (Mazziotta-Pareto Index): the mean of its normalised indicators (100 = reference) less a penalty for how unbalanced they are, so that balanced performance is rewarded	points from the reference (0 = study region average)
index_uli_plain__ambient_environment	Average across sample points of the composite index 'uli plain', domain 'ambient environment': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli_plain__built_environment	Average across sample points of the composite index 'uli plain', domain 'built environment': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli_plain__economic_development	Average across sample points of the composite index 'uli plain', domain 'economic development': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli_plain__mobility_transport	Average across sample points of the composite index 'uli plain', domain 'mobility transport': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli_plain__safety	Average across sample points of the composite index 'uli plain', domain 'safety': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli_plain__social_infrastructure	Average across sample points of the composite index 'uli plain', domain 'social infrastructure': the mean of the domain's normalised indicators (100 = reference) less a penalty for how unbalanced they are	points from the reference (0 = study region average)
index_uli_plain__thermal_comfort	Average across sample points of the composite index 'uli plain': the normalised score of its indicator 'thermal comfort' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli_plain__walkability	Average across sample points of the composite index 'uli plain': the normalised score of its indicator 'walkability' (100 = reference), oriented so that higher is better	points from the reference (0 = study region average)
index_uli_plain_mean	Average across sample points of the composite index 'uli plain': mean level of its normalised indicators or domain scores (100 = reference), before the penalty for imbalance	points from the reference (0 = study region average)
index_uli_plain_penalty	Average across sample points of the composite index 'uli plain': penalty for imbalance among its indicators or domain scores (standard deviation multiplied by the coefficient of variation), applied to the mean level	index (100 = reference)

Indicator estimates: walkability

Variable	Description	Units
local_daily_living	Average daily living score (/3) of sample points in this area	score 0-3
local_nh_intersection_density	Average walkable neighbourhood intersection density (intersections per km²) of sample points in this area	count per km²
local_nh_population_density	Average walkable neighbourhood population density (population per km²) of sample points in this area	persons per km²
local_walkability	Average walkability index of sample points in this area	index (sum of z-scores)
walk_dl_300	Average across sample points of the daily living score: the number of daily living destination types (fresh food market, convenience, pt any) with access within 300 m	destinations
walk_idx_300	Average across sample points of the walkability index with daily living access judged within 300 m: the sum of the z-scores of that daily living score, local population density and street intersection density	index (sum of z-scores)
walk_idx_300_tm	Average across sample points of the percentile rank (0-1) of the walkability index with daily living access judged within 300 m: the sum of the z-scores of that daily living score, local population density and street intersection density, multiplied by (1 - lambda x h), where h is thermal comfort re-scaled from 0 at 30 to 1 at 50 (a fixed range); lambda = 0.5 (attenuation)	index 0-1

Indicator estimates: urban heat vulnerability

Variable	Description	Units
pct_urban_heat_guhvi_class_5_most_vulnerable	Percentage of population in Global Urban Heat Vulnerability Index (GUHVI) class 5, the most vulnerable class	percent
urban_heat_adaptive_capability_index	Adaptive Capability Index (ACI): the equally weighted adaptive capability sub-index of child dependency ratio, subnational Human Development Index and infant mortality rate; higher values indicate lower adaptive capability	index 0-100
urban_heat_exposure_index	Heat Exposure Index (HEI): the heat exposure sub-index, derived from land surface temperature	index 0-100
urban_heat_guhvi	Global Urban Heat Vulnerability Index (GUHVI): the equally weighted composite of the Heat Exposure, Heat Sensitivity and Adaptive Capability indices, (HEI + HSI + ACI) / 3	index 0-100
urban_heat_guhvi_class	Global Urban Heat Vulnerability Index (GUHVI) class, from 1 (least vulnerable) to 5 (most vulnerable)	class 1-5
urban_heat_sensitivity_index	Heat Sensitivity Index (HSI): the equally weighted heat sensitivity sub-index of land surface albedo, NDVI, NDBI, local climate zone, population density and vulnerable population	index 0-100

Appendix C. Goalposts and weights

Goalposts and effective weights of each index, as the analysis wrote them (`data/`). Goalposts are on the scale scored: the soft access score (0 to 1) where there is a threshold, the ladder where there is one, otherwise the value as measured. Ref is the study region average, and scores 0.

`uli`

Goalposts resolved on 2026-09-26 from 154,239 sample points; method `ampi`, centre 0.

Indicator	Variable	Domains (share of weight)	Scored as	Min · Ref · Max	Effective weight
Heat exposure (GUHVI)	`sp_urban_heat_exposure_index`	Ambient environment	as measured	0 · 47 · 100	2.8%
Heat sensitivity (GUHVI)	`sp_urban_heat_sensitivity_index`	Ambient environment	as measured	0 · 58 · 100	2.8%
Heat adaptive capability (GUHVI)	`sp_urban_heat_adaptive_capability_index`	Ambient environment	as measured	0 · 22 · 100	2.8%
Natural blue space nearby	`sp_walk_nearest_node_blue_space`	Ambient environment	stepped ladder	56.9 · 76.9 · 96.9	2.8%
Vegetation cover (NDVI)	`sp_ext_ndvi`	Ambient environment (50%), Built environment (50%)	as measured	−0.251 · 0.082 · 0.415	4.0%
Parks within 300 m	`sp_walk_nearest_node_public_open_space_any`	Ambient environment (50%), Built environment (50%)	soft access, 300 m threshold	−0.197 · 0.299 · 0.796	4.0%
Large public green space within 500 m	`sp_access_large_public_green_space_score`	Ambient environment (50%), Built environment (50%)	as measured	−0.419 · 0.081 · 0.581	4.0%
Proximity to major roads	`sp_ext_major_road_length_300m`	Ambient environment (50%), Safety and protection (50%)	as measured	−2,429 · 1,756 · 5,940	5.9%
Walkability (300 m), attenuated by thermal comfort	`sp_walk_idx_300_tm`	Built environment (33%), Mobility and transport (33%), Social infrastructure (33%)	as measured	−0.022 · 0.315 · 0.651	3.7%
Block size (perimeter)	`sp_ext_block_perimeter_m`	Built environment (33%), Mobility and transport (33%), Safety and protection (33%)	as measured	−3,860 · 856 · 5,572	6.2%
Police station within 2 km	`sp_euclid_dist_imip_police_station`	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 2,000 m threshold	0.115 · 0.611 · 1.108	6.2%
Fire station within 3 km	`sp_euclid_dist_imip_fire_station`	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 3,000 m threshold	0.096 · 0.592 · 1.089	6.2%
Food within 1.5 km by low-stress cycling	`sp_cycle_safe_nearest_node_fresh_food_pooled`	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 1,500 m threshold	0.059 · 0.555 · 1.052	6.2%
Public transport within 300 m	`sp_walk_nearest_node_pt_any`	Mobility and transport	soft access, 300 m threshold	−0.433 · 0.064 · 0.561	4.2%
Fresh food within 300 m	`sp_walk_nearest_node_denue_fresh_food`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	soft access, 300 m threshold	−0.091 · 0.405 · 0.902	3.6%
Fresh food diversity within 500 m	`sp_walk_diversity_fresh_food_500m`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	as measured	−0.167 · 0.269 · 0.705	3.6%
Convenience stores within 300 m	`sp_walk_nearest_node_denue_food_retail`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	soft access, 300 m threshold	0.179 · 0.675 · 1.172	3.6%
250 m or more from petrol stations	`sp_walk_beyond_denue_petrol_station_250m`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	as measured	0.421 · 0.921 · 1.421	3.6%
Primary schools within 300 m	`sp_walk_nearest_node_denue_primary_school`	Social infrastructure	soft access, 300 m threshold	−0.330 · 0.166 · 0.663	1.7%
Education diversity within 500 m	`sp_walk_diversity_education_500m`	Social infrastructure	as measured	−0.232 · 0.268 · 0.768	1.7%
Primary healthcare within 300 m	`sp_walk_nearest_node_denue_primary_healthcare`	Social infrastructure	soft access, 300 m threshold	−0.358 · 0.139 · 0.636	1.7%
Health service diversity within 500 m	`sp_walk_diversity_health_care_500m`	Social infrastructure	as measured	−0.190 · 0.310 · 0.810	1.7%
Care services within 300 m	`sp_walk_nearest_node_denue_care_support`	Social infrastructure	soft access, 300 m threshold	−0.336 · 0.161 · 0.657	1.7%
Care service diversity within 500 m	`sp_walk_diversity_care_500m`	Social infrastructure	as measured	−0.440 · 0.060 · 0.560	1.7%
Community, culture and recreation within 300 m	`sp_walk_nearest_node_denue_ccr_any`	Social infrastructure	soft access, 300 m threshold	−0.077 · 0.420 · 0.916	1.7%
Community, culture and recreation diversity within 500 m	`sp_walk_diversity_community_culture_recreation_500m`	Social infrastructure	as measured	−0.331 · 0.163 · 0.656	1.7%
Manufacturing within 1 km	`sp_walk_access_denue_manufacturing_1000m`	Economic development	as measured	0.435 · 0.935 · 1.435	5.0%
Job diversity	`sp_ext_economic_sector_diversity`	Economic development	as measured	−0.236 · 0.264 · 0.764	5.0%

`uli_plain`

Goalposts resolved on 2026-09-26 from 154,239 sample points; method `ampi`, centre 0.

Indicator	Variable	Domains (share of weight)	Scored as	Min · Ref · Max	Effective weight
Heat exposure (GUHVI)	`sp_urban_heat_exposure_index`	Ambient environment	as measured	0 · 47 · 100	2.4%
Heat sensitivity (GUHVI)	`sp_urban_heat_sensitivity_index`	Ambient environment	as measured	0 · 58 · 100	2.4%
Heat adaptive capability (GUHVI)	`sp_urban_heat_adaptive_capability_index`	Ambient environment	as measured	0 · 22 · 100	2.4%
Daytime thermal comfort	`sp_utci_day_mean`	Ambient environment	as measured	41.938 · 44.949 · 47.960	2.4%
Natural blue space nearby	`sp_walk_nearest_node_blue_space`	Ambient environment	stepped ladder	56.9 · 76.9 · 96.9	2.4%
Vegetation cover (NDVI)	`sp_ext_ndvi`	Ambient environment (50%), Built environment (50%)	as measured	−0.251 · 0.082 · 0.415	3.8%
Parks within 300 m	`sp_walk_nearest_node_public_open_space_any`	Ambient environment (50%), Built environment (50%)	soft access, 300 m threshold	−0.197 · 0.299 · 0.796	3.8%
Large public green space within 500 m	`sp_access_large_public_green_space_score`	Ambient environment (50%), Built environment (50%)	as measured	−0.419 · 0.081 · 0.581	3.8%
Proximity to major roads	`sp_ext_major_road_length_300m`	Ambient environment (50%), Safety and protection (50%)	as measured	−2,429 · 1,756 · 5,940	5.7%
Walkability (300 m), attenuated by thermal comfort	`sp_walk_idx_300`	Built environment (33%), Mobility and transport (33%), Social infrastructure (33%)	as measured	−10.0 · 0.0 · 10.0	3.7%
Block size (perimeter)	`sp_ext_block_perimeter_m`	Built environment (33%), Mobility and transport (33%), Safety and protection (33%)	as measured	−3,860 · 856 · 5,572	6.2%
Police station within 2 km	`sp_euclid_dist_imip_police_station`	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 2,000 m threshold	0.115 · 0.611 · 1.108	6.2%
Fire station within 3 km	`sp_euclid_dist_imip_fire_station`	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 3,000 m threshold	0.096 · 0.592 · 1.089	6.2%
Food within 1.5 km by low-stress cycling	`sp_cycle_safe_nearest_node_fresh_food_pooled`	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 1,500 m threshold	0.059 · 0.555 · 1.052	6.2%
Public transport within 300 m	`sp_walk_nearest_node_pt_any`	Mobility and transport	soft access, 300 m threshold	−0.433 · 0.064 · 0.561	4.2%
Fresh food within 300 m	`sp_walk_nearest_node_denue_fresh_food`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	soft access, 300 m threshold	−0.091 · 0.405 · 0.902	3.6%
Fresh food diversity within 500 m	`sp_walk_diversity_fresh_food_500m`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	as measured	−0.167 · 0.269 · 0.705	3.6%
Convenience stores within 300 m	`sp_walk_nearest_node_denue_food_retail`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	soft access, 300 m threshold	0.179 · 0.675 · 1.172	3.6%
250 m or more from petrol stations	`sp_walk_beyond_denue_petrol_station_250m`	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	as measured	0.421 · 0.921 · 1.421	3.6%
Primary schools within 300 m	`sp_walk_nearest_node_denue_primary_school`	Social infrastructure	soft access, 300 m threshold	−0.330 · 0.166 · 0.663	1.7%
Education diversity within 500 m	`sp_walk_diversity_education_500m`	Social infrastructure	as measured	−0.232 · 0.268 · 0.768	1.7%
Primary healthcare within 300 m	`sp_walk_nearest_node_denue_primary_healthcare`	Social infrastructure	soft access, 300 m threshold	−0.358 · 0.139 · 0.636	1.7%
Health service diversity within 500 m	`sp_walk_diversity_health_care_500m`	Social infrastructure	as measured	−0.190 · 0.310 · 0.810	1.7%
Care services within 300 m	`sp_walk_nearest_node_denue_care_support`	Social infrastructure	soft access, 300 m threshold	−0.336 · 0.161 · 0.657	1.7%
Care service diversity within 500 m	`sp_walk_diversity_care_500m`	Social infrastructure	as measured	−0.440 · 0.060 · 0.560	1.7%
Community, culture and recreation within 300 m	`sp_walk_nearest_node_denue_ccr_any`	Social infrastructure	soft access, 300 m threshold	−0.077 · 0.420 · 0.916	1.7%
Community, culture and recreation diversity within 500 m	`sp_walk_diversity_community_culture_recreation_500m`	Social infrastructure	as measured	−0.331 · 0.163 · 0.656	1.7%
Manufacturing within 1 km	`sp_walk_access_denue_manufacturing_1000m`	Economic development	as measured	0.435 · 0.935 · 1.435	5.0%
Job diversity	`sp_ext_economic_sector_diversity`	Economic development	as measured	−0.236 · 0.264 · 0.764	5.0%

Appendix D. Data source register

Dataset records as they stand in the region configuration, unaltered. Each dataset keeps its own licence.

Study region boundary

`MX_Mexicali_2025_ULI.yml:43`

- `data`: MX/INEGI-02l-NOMGEO-Mexicali-Condesa-Fraccionamientos.geojson
- `source`: Instituto Nacional de Estadística, Geografía e Informática (INEGI); Centro para el Futuro de las Ciudades (CFC)
- `publication_date`: 2025-07-31
- `url`: https://en.www.inegi.org.mx/contenidos/productos/prod_serv/contenidos/espanol/bvinegi/productos/geografia/marcogeo/794551163061/02_bajacalifornia.zip
- `licence`: Términos de Libre Uso de la Información del INEGI https://www.inegi.org.mx/inegi/terminos.html
- `citation`: INEGI (2025). Marco Geoestadístico 2025 - Baja California. https://en.www.inegi.org.mx/app/biblioteca/ficha.html?upc=794551163061; CFC (2026) Condesa development boundaries.
- `notes`: Dissolved union of MX/INEGI/02_bajacalifornia.zip/conjunto_de_datos/02l.shp -where "NOMGEO='Mexicali'" and convex hull of MX/CFC/Nvo desarrollo/_Fraccionamientos.geojson, so that the analysis covers 100% of the Condesa development.

Exclusion region

`MX_Mexicali_2025_ULI.yml:55`

- `data`: MX/exclusion/ca_state.zip
- `source`: https://data.ca.gov/dataset/e212e397-1277-4df3-8c22-40721b095f33/resource/3db1e426-fb51-44f5-82d5-a54d7c6e188b/download/ca_state.zip
- `licence`: public domain
- `citation`: California State boundary in shapefile format from the US Census Bureau's 2023 MAF/TIGER database.
- `notes`: Es necesario tener cuidado para garantizar que la zona de estudio delimitada no se extienda más allá de la frontera con Estados Unidos // Care must be taken to ensure the buffered study region does not expand across the border with the United States, so that road network and amenities outside Mexico are not included in accessibility calculations. Per https://github.com/healthysustainablecities/global-indicators/issues/610 the network setup is restricted to the main study region (not the buffered region).

Population

`MX_Mexicali_2025_ULI.yml:61`

- `name`: Global Human Settlements population data: 2025, Mollweide (EU JRC, 2023)
- `data_dir`: MX/CFC/GHS_POP_E2025_GLOBE_R2023A_54009_100_V1_0_R6_C8/
- `year_published`: 2023
- `source_url`: https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/GHSL/GHS_POP_GLOBE_R2023A/GHS_POP_E2025_GLOBE_R2023A_54009_100/V1-0/tiles/GHS_POP_E2025_GLOBE_R2023A_54009_100_V1_0_R6_C8.zip
- `licence`: CC BY 4.0
- `citation`: Schiavina, M., Freire, S., Carioli, A., MacManus, K. (2023). GHS-POP R2023A - GHS population grid multitemporal (1975-2030). European Commission, Joint Research Centre (JRC). https://doi.org/10.2905/2FF68A52-5B5B-4A22-8F40-C41DA8332CFE
- `notes`: Population is NOT redirected to a custom aggregation; the 100 m grid derived from this raster is a required output scale.

OpenStreetMap

`MX_Mexicali_2025_ULI.yml:396`

- `data_dir`: MX/mexico-260815.osm.pbf
- `source`: Geofabrik GmbH
- `publication_date`: 20260815
- `url`: https://download.geofabrik.de/north-america/mexico-latest.osm.pbf
- `licence`: ODbL
- `note`: Mexico OpenStreetMap excerpt

IMIP: public transport stops (pt_any)

`MX_Mexicali_2025_ULI.yml:168`

- `data`: MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "equipamiento_category='transp' AND TIPO != 'TERMINAL INTERIOR DE CARGA (CENTRAL DE SERVICIOS DE CARGA)'"
- `source`: IMIP / CFC
- `publication_date`: 2026-05-18
- `licence`: Provided by IMIP via CFC

- `citation`: As per advice from Juan Ernesto Díaz Noguez

INEGI DENUE (May 2026)

`MX_Mexicali_2025_ULI.yml:192`

- `data`: MX/INEGI/DENUE_2026/denue_inegi_02_.shp -where "codigo_act IN ('461121','461122','461123','461130','461140','461150','462111','462112')"
- `source`: INEGI, Directorio Estadistico Nacional de Unidades Economicas (DENUE)
- `publication_date`: 2026-05
- `url`: https://www.inegi.org.mx/app/mapa/denue/
- `licence`: Terminos de Libre Uso de la Informacion del INEGI
- `citation`: INEGI (2026). Directorio Estadistico Nacional de Unidades Economicas (DENUE), mayo 2026. Instituto Nacional de Estadistica y Geografia, Mexico.

Police stations (IMIP) (`imip_police_station`)

- `name`: imip_police_station
- `data`: MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "TIPO = 'INSTALACIONES POLICIALES'"
- `source`: IMIP / CFC
- `licence`: Provided by IMIP via CFC
- `notes`: 27 police facilities in the IMIP equipamiento layer (administración pública).

Municipal fire stations (IMIP) (`imip_fire_station`)

- `name`: imip_fire_station
- `data`: MX/CFC/cleaned/por_estado/02_baja_california/002_mexicali/imip/equipamiento_combined.fgb -where "TIPO = 'ESTACION DE BOMBEROS MUNICIPAL'"
- `source`: IMIP / CFC
- `licence`: Provided by IMIP via CFC
- `notes`: 10 municipal fire stations in the IMIP equipamiento layer (administración pública).

Outdoor thermal comfort (UTCI) (`utci`)

`MX_Mexicali_2025_ULI.yml:770`

- `data`: MX/UTCI - 2026-08-25/mexicali_utci_2023-07-21.gpkg
- `source`: Rossano Schifanella (WP02); SOLWEIG and the UTCI operational polynomial (Bröde et al. 2012)
- `licence`: Terrain from FABDEM v1 (CC BY-NC-SA 4.0, non-commercial); other inputs CC BY 4.0 or public domain
- `notes`: Modelled day 2023-07-21 (Tmax 48.5 °C). See README_mexicali_utci.md, sections 3.1 and 5, for forcing and limitations.

Work package indicators (`external`)

`MX_Mexicali_2025_ULI.yml:814`

- `data`: MX/Update_on_ULI - 2026-09-22/final_indicators.gpkg
- `source`: Mexicali ULI work packages (final_indicators.gpkg; see excel_work_sheet_key.xlsx)

Custom aggregations

- **`agebs`**: `MX/mexicali_AGEB_population.geojson` — INEGI 2020 AGEB boundaries with census population, as prepared in Mexicali_2025.ipynb.
- **`manzanas`**: `MX/CFC/mexicali_manzanas.gpkg:manzanas` — INEGI 2020 census manzanas with geometries updated by CFC (19,223 manzanas; 940,287 residents; 236.9 km2), prepared by MX/CFC/build_manzanas.py. Replaces MX/CFC/MXL_MZAPob.gpkg:02m, which covered the Condesa development only partially -- 6,274 of its 14,989 lots fell within a manzana, against 12,985 here. 3,497 manzanas have no recorded population (2,772 of them rural) and report a null pop_est rather than zero; they are excluded from population-weighted summaries. Sixteen newly mapped Condesa blocks had no manzana key and carry a synthetic identifier assigned by build_manzanas.py, recognisable by an 'S' where the manzana number would be.
- **`condesa_lotes`**: `MX/CFC/Nvo desarrollo/Lotes_desarrollo_EXE_cleaned_400sqm_or_less.geojson` — Individual lots in the Condesa development (median 120 m2). No population data; however we can assign an assumed value, assuming average household population of 3.2 for Mexicali (2020; https://inegi.org.mx/app/areasgeograficas/?ag=02002#tabMCcollapse-Indicadores). Note: this data appeared to include more than lots, larger block areas which appeared to be artifacts, so I filtered to those with area <= 400sqm (manually in QGIS).
- **`condesa_fraccionamientos`**: `MX/CFC/Nvo desarrollo/_Fraccionamientos.geojson` — Condesa subdivision polygons (median 7.9 ha). pop_est is summed from condesa_lotes (3.2 people per lot assumed).
- **`region_mexicali`**: `MX/CFC/mexicali_uli_boundaries.gpkg:region_mexicali` — The Mexicali urban area boundary, dissolved, giving a single summary row for the administrative area.
- **`region_condesa`**: `MX/CFC/mexicali_uli_boundaries.gpkg:region_condesa` — The Condesa development, dissolved, giving a single summary row for the development as a whole.

Google Earth Engine

The GUHVI and large public green space indicators are computed in Google Earth Engine (`gee: true`) from Landsat 8, MODIS and Sentinel-2 collections; the region configuration declares no dataset record for them. The sources GHSCI records are in the data dictionary.

Appendix E. Glossary

Term	Spanish	Definition
AGEB	AGEB (área geoestadística básica)	INEGI's basic census area; median 36 ha in Mexicali.
AMPI	AMPI (índice ajustado de Mazziotta–Pareto)	Composite index method re-scaling each indicator against goalposts and penalising imbalance between components (Mazziotta & Pareto 2018) (R11).
Censoring	Censura	Distances are searched only to the largest band (1,500 m walking); a point with nothing within it has no distance and no access.
Core measure	Medida central	The variable sheet's grouping of sub-variables (`group`), for example "Daily essential-services access".
DENUE	DENUE	INEGI's National Statistical Directory of Economic Units: establishments with SCIAN activity codes.
Domain	Dominio	One of the eight domains of the Adapted Urban Liveability Framework; six are scored.
Effective weight	Peso efectivo	An indicator's share of the index's mean level, after domain shares and domain weights (R12).
Fraccionamiento	Fraccionamiento	A subdivision, often gated; Condesa has 40.
Goalposts	Metas	Ref, Min and Max against which an indicator is re-scaled; frozen goalposts make scenarios comparable (R14).
GUHVI	GUHVI	Global Urban Heat Vulnerability Index (Turner et al. 2025): exposure, sensitivity and adaptive capability.
CFC	CFC	Centro para el Futuro de las Ciudades (Tecnológico de Monterrey); source of the Condesa boundaries, lots and updated manzana geometries.
IMIP	IMIP	Mexicali's municipal planning institute; source of the facilities layer (public transport stops, police and fire stations), provided via CFC.
Lens	Enfoque	The framework's view of an indicator: quality, proximity, diversity, accessibility or presence.
LPUGS	LPUGS (gran espacio verde público urbano)	Large public urban green space: public open space ≥ 1 ha with NDVI ≥ 0.2.
LTS	LTS (nivel de estrés del tráfico)	Level of traffic stress for cycling, 1 (all ages) to 4 (most stressful) (R6).
Manzana	Manzana	Census block; median 0.71 ha in Mexicali.
Penalty	Penalización	`S · cv`, subtracted from a composite's mean so that doing well on one component cannot wholly make up for doing badly on another.
Reference (Ref)	Referencia	The study region average of an indicator over all sample points; scores 0.
Sample point	Punto de muestra	A location every 30 m along the pedestrian network (and one per Condesa lot) at which indicators are computed (R1).
SCIAN	SCIAN	Sistema de Clasificación Industrial de América del Norte, the North American Industry Classification System as used in Mexico.
Soft threshold	Umbral suave	Logistic transform of a distance, 0.5 at the threshold (R10).
Study region	Región de estudio	The Mexicali urban area together with the Condesa development.
UTCI	UTCI	Universal Thermal Climate Index (°C): felt temperature from air temperature, radiant temperature, humidity and wind.

Appendix F. References

Methods

- Mazziotta, M. & Pareto, A. (2018) Measuring well-being over time: the Adjusted Mazziotta-Pareto Index versus other non-compensatory indices. Social Indicators Research 136, 967-976.
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- OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026).
- Turner et al. (2025) Urban Climate 64, 102716; computed in Google Earth Engine by GHSCI.
- Rossano Schifanella (WP02): SOLWEIG, forced by Mexicali airport observations for 21 July 2023; UTCI operational polynomial (Bröde et al. 2012).
- OpenStreetMap contributors (Geofabrik Mexico extract, 15 August 2026); public open space identified by GHSCI.
- GHSCI large public urban green space indicator (Google Earth Engine), with OpenStreetMap public open space.
- Mexicali ULI work packages (final_indicators.gpkg, 22 September 2026).
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- INEGI (2026) Directorio Estadístico Nacional de Unidades Económicas (DENUE), May 2026.
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Citations as given in the project workbook; several do not match the article list (citation_audit.csv) and should be verified before being cited.

Appendix G. Excluded and pending indicators

The ULI includes every sub-variable of the project's variable sheet (`excel_work_sheet_key.xlsx`, revised 25 September 2026) whose data have been delivered and vary across the city. The configuration records why the others are not scored (`MX_Mexicali_2025_ULI.yml:963`):

Sub-variable or core measure	Reason not scored
Flooding: population exposed; flooded area on main roads	Delivered values are missing for almost every area (none on the grid), pending the work package's reply on whether missing means no exposure. They are linked with missing taken as zero (`null_as`), for reference only.
Tree canopy cover	210 manzanas only, no grid values: too sparse to vary.
Protected cycle lane coverage	Almost all zero. INVI 2020 does not distinguish protected lanes, and OpenStreetMap was not available to the work package (sheet note).
Heat exposure (core measure 3): land surface temperature, extreme heat frequency, night-time heat retention	Awaiting delivery.
Sidewalk coverage, street lighting, bicycle lane coverage, reachable jobs, access to basic services (Housing)	Awaiting delivery.
Crime exposure	One municipal rate for every location, so it cannot distinguish between them.
Large public open space *availability*: the share of an area's land that is large public green space	Access to large public green space is calculated by the core GHSCI workflow and **is scored** in the index (Large public green space within 500 m); walking and cycling access to large public open space (over 1.5 ha) at each distance band is also calculated and shown in the general dashboard. Availability is the sheet's separate sub-variable. It describes an area's land, not a sample point's surroundings, so the index, which scores sample points, does not include it as it stands.
Net residential density, commercial land use, land use mix	Components of the walkability core measure, which walkability itself represents: scoring them too would count density twice. They remain in the general dashboard.
Night-time thermal comfort	Varies too little within the city (30.3–30.6 °C).
Fire station coverage within 75 km	Universal coverage: recorded for completeness; the 3 km soft threshold is scored instead.

Two domains, **Housing** and **Sociodemographic characteristics**, have no indicator yet. They are described but not scored, so the index is the composite of six domains.

Open questions recorded with the analysis

- Whether a missing flood value means no exposure (work package 5).
- Heat exposure core measure 3 (work package 2).
- The level of traffic stress indicator's destination and threshold (a provisional single choice from a suite).
- The Spanish names of the core measures (`group`), and the `about` and `sources` text of each indicator, drafted on 25 September 2026, await review.